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*2022 K-Innovation Partnership Program with Azerbaijan*

# Policy Consultation on the Operational Programs of HTP in Azerbaijan: Implementation and Capacity Building

(1st year): Improvement of the Technology Commercialization System in Azerbaijan and Development of the Technology Commercialization Program at HTP

## 2022 K-Innovation Partnership Program with Azerbaijan

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|-----------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Title                 | Policy Consultation on the Operational Programs of HTP in Azerbaijan: Implementation and Capacity Building (1st year): Improvement of the Technology Commercialization System in Azerbaijan and Development of the Technology Commercialization Program at HTP                                                                                                                                  |
| Institute             | Science and Technology Policy Institute (STEPI)                                                                                                                                                                                                                                                                                                                                                 |
| Partner Country       | Republic of Azerbaijan                                                                                                                                                                                                                                                                                                                                                                          |
| In Cooperation with   | High Technologies Park (HTP)                                                                                                                                                                                                                                                                                                                                                                    |
| Program Investigators | Ji Hyun Kim, Head, Office of SDGs Innovation Research, STEPI<br>Wangdong Kim, Senior Research Fellow, Office of SDGs Innovation Research, STEPI                                                                                                                                                                                                                                                 |
| Project Manager       | Eun Joo Kim, Principal Researcher, Division of Global Innovation Strategy, STEPI                                                                                                                                                                                                                                                                                                                |
| Policy Consultants    | Soo J. Sohn, Senior Research Fellow, Office of Innovation Law & Institution Research, STEPI<br>Youngtack Shim, Adjunct Professor, School of Law, Seoul National University<br>Gyeongsoo Shim, Professor, Seoul National University of Science and Technology                                                                                                                                    |
| Researchers           | Bünyamin Seyidov, Head, Incubation and Technology Transfer Center, HTP<br>Isa Gasimov, Director, Center for Innovative Technologies in Education<br>Vüqar Babayev, General Director, IQ Digital Solutions<br>Vusal Suleymanli, ANAS Innovation Consultant<br>Samir Veliev, Senior Economist, International Bank of Azerbaijan OJSC<br>Aysel Abdullayeva, Business Development Manager, MONT LLC |
| Project Coordinator   | Shinae Kang, Researcher, Division of Global Innovation Strategy, STEPI                                                                                                                                                                                                                                                                                                                          |

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### **Policy Consultation on the Operational Programs of HTP in Azerbaijan: Implementation and Capacity Building (1<sup>st</sup> year): Improvement of the Technology Commercialization System in Azerbaijan and Development of the Technology Commercialization Program at HTP**

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5<sup>th</sup>~7<sup>th</sup> Flrs., Building B, Sejong National Research Complex, 370 Sicheong-daero, Sejong-si,

30147. Republic of Korea

Tel: +82-44-287-2000 Fax: +82-44-287-2068

[www.stepi.re.kr](http://www.stepi.re.kr)

For more information, visit

[stepi.re.kr/global](http://stepi.re.kr/global)

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## Executive Summary

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Azerbaijan, located at a strategic point along the coast of the Caspian Sea, is the center of science and technology in the Eurasian region and has great importance for cooperation in the international energy market as a production and transit point for crude oil and natural gas. Azerbaijan is hugely reliant on the oil and gas industry, which accounts for more than 90% of its total exports, and has an economic structure trade that is contingent upon three neighbouring countries (Russia, Iran, and Turkey). In other words, it is a country that is economically and socially affected by unstable fluctuations in international oil prices and by changes in the circumstances of neighbouring countries.

Recently the Azerbaijan government resolved to move away from excessive dependence on the oil industry and make the transition to a knowledge-based economy, in a bid to avoid a financial crisis caused by unstable oil prices and to create a democratic political environment. It is in this context that the government established 'Azerbaijan 2020: Look into the Future' and 'Azerbaijan 2030: National Priorities for Socio-Economic Development' with the aim of achieving sustainable economic development based on innovation in the field of science and technology. According to the Azerbaijan National Cooperation Strategy (2016), it is shown that the government is making efforts to strengthen the national development plan-related science and technology innovation capacity and achieve industrial diversification policies in order to build national competitiveness. In addition, it emphasizes the need to increase investment in science and technology innovation and to improve the operation of its system by increasing the efficiency of knowledge creation, technology transfer, and the application and commercialization of technology through the 'Azerbaijan 2030 Strategic Roadmap'.

The High Technologies Park (HTP) was established in order to ensure the development of science in accordance with modern requirements, to form an innovative economic

ecosystem in the country and the region, to use scientific and research potential effectively, and to support business entrepreneurship. In accordance with its mission, the HTP aims to provide joint activities with leading local and international organizations and scientific institutions of various countries that can contribute to business entrepreneurship, innovative research and development, and the development of the knowledge economy in the country.

Since 2018 STEPI has supported the HTP in its efforts to develop its institutional capacity through the K-Innovation Partnership Program. This program plays an important role in the transfer and commercialization of technology and the incubation of start-ups, as well as other strategic activities of the HTP. This cooperation between the two organizations aims to commercialize the results of scientific research and promote their application, determine the role of academia-industry cooperation in the development of existing entrepreneurial activities, and foster a business environment that is conducive to adapting the scientific research works of the HTP and other scientific research institutions to the requirements of industry and real business, including the agricultural sector.

The 2022 project aims to provide policy consultation on improving the national system of technology transfer and commercialization in Azerbaijan and capacity building of the HTP. Moreover, it is intended to promote mutually beneficial economic cooperation between the two countries by contributing to the development of science and technology and improving the technology commercialization ecosystem in Azerbaijan.

With the activity of the project for the year 2022, the following expectations have been defined:

- Capacity building of the stakeholders of Azerbaijan's innovation ecosystem.
- Contribution to the formation of human capital in Azerbaijan.
- Expansion of Azerbaijan-Korea scientific, technology, and innovation (STI) cooperation.
- Improvement of the operation program of the HTP.
- Further improvement of the process of technology commercialization observed with the transformation of knowledge into economic value.

- Formation of a new entrepreneurial environment based on South Korea's experience of start-up and business incubation.

In the 1st year of the project, the goal is to increase Azerbaijan's capacity for technology transfer and commercialization services and, in the 2nd year, its capacity for start-up and incubation services. These include the transfer of knowledge, knowledge laboratories, and seminars, and the promotion of Korea-Azerbaijan cooperation projects, etc. In the 3rd year of the project, it is planned to implement the implementation process in Azerbaijan by strengthening its capacity in the fields of technology transfer and commercialization, start-up, and incubation. The project will offer services for *technology transfer - commercialization and start-up - incubation operational programs* as a continuation of the strategic plan and the road map developed as a result of the successful operation of STEPI with the HTP. We hope that this K-Innovation program will make a valuable contribution to fostering firm foundations for technology commercialization in Azerbaijan.

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# CHAPTER 1

## Project Overview



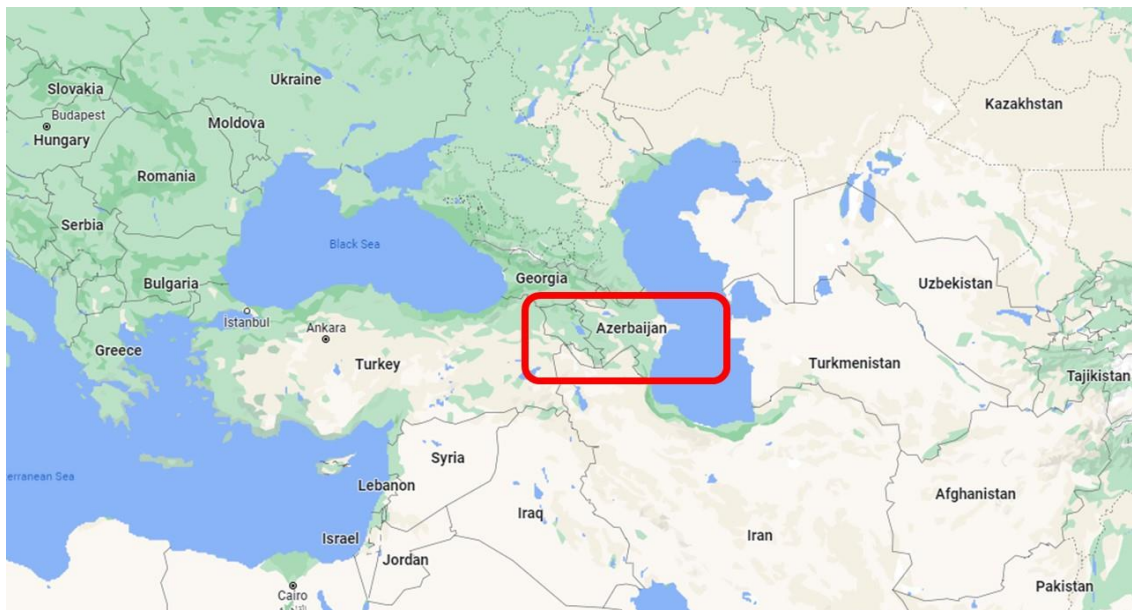


# Chapter 1. Project Overview

## 1. Project Background

Azerbaijan, located at a strategic point along the coast of the Caspian Sea, is the centre of science and technology in the Eurasian region and has great importance for cooperation in the international energy market as a production and transit point for crude oil and natural gas. Azerbaijan is hugely reliant on the oil and gas industry, which accounts for more than 90% of its total exports, and has an economic structure trade that is contingent upon three neighbouring countries (Russia, Iran, and Turkey). In other words, it is a country that is economically and socially affected by unstable fluctuations in international oil prices and by changes in the circumstances of neighbouring countries.

[Figure 1-1] Map of Azerbaijan



Source: Google map.

Recently the Azerbaijan government resolved to move away from excessive dependence on the oil industry and make the transition to a knowledge-based economy, in a bid to avoid a financial crisis caused by unstable oil prices and to create a democratic political environment. It is in this context that the government established 'Azerbaijan 2020: Look into the Future' and 'Azerbaijan 2030: National Priorities for Socio-Economic Development' with the aim of achieving sustainable economic development based on innovation in the field of science and technology. According to the Azerbaijan National Cooperation Strategy (2016), it is shown that the government is making efforts to strengthen the national development plan-related science and technology innovation (STI) capacity and achieve industrial diversification policies in order to build national competitiveness. In addition, it emphasizes the need to increase investment in science and technology innovation and to improve the operation of its system by increasing the efficiency of knowledge creation, technology transfer, and the application and commercialization of technology through the 'Azerbaijan 2030 Strategic Roadmap'.

[Table 1-1] Azerbaijan 2030: National Priorities for Socio-Economic Development

| Priority Goals                                                         | Objectives                                                                            |
|------------------------------------------------------------------------|---------------------------------------------------------------------------------------|
| 1. Establish a competitive economy based on steady growth              | Promote stable and high economic growth                                               |
|                                                                        | Resist internal and external influences                                               |
| 2. Establish a dynamic, inclusive society based on social justice      | Provide the benefits of development to every citizen                                  |
|                                                                        | Provide greater and fairer social security within an inclusive society                |
|                                                                        | Promote the balanced development of both the capital and the regions                  |
| 3. Establish areas of modern innovation and competitive human capital  | Provide education in accordance with the requirements of the 21 <sup>st</sup> century |
|                                                                        | Establish a creative and innovative society                                           |
|                                                                        | Promote healthy lifestyle of citizens                                                 |
| 4. Promote the “Great Return” to territories liberated from occupation | Pursue sustainable resettlement                                                       |
|                                                                        | Reintegrate liberated territories into economic activity                              |
| 5. Establish a clean environment and a country of “Green growth”       | A high-quality ecological environment                                                 |
|                                                                        | Create green energy spaces                                                            |

Source: Azerbaijan (2021), Azerbaijan 2030: National Priorities of Socio-Economic Development.



The Science and Technology Policy Institute (STEPI) received a request for continuous cooperation from the National Academy of Sciences (ANAS), an institution dedicated to implementing Azerbaijan's national science and technology innovation policy.

As a result, STEPI implemented two projects, namely 'STI Policy Development and Management & Operation of STI Parks in Azerbaijan' in 2018 and 'Promotion of Technology Transfer and Commercialization in Azerbaijan' in 2019-2020, under the K-Innovation Partnership Program.

According to the information collected during the execution of these projects in 2018-2020, Azerbaijan expects to acquire South Korea's ample experience and know-how of practical science and technology innovation policies including technology transfer and commercialization, STI park management and operation, STI human resource development, national R&D program establishment, technology management and STI strategies. More specifically, the requested agenda includes improving the national technology transfer and commercialization system and developing technology transfer and programs for the High Technologies Park (HTP).

Thus, 2022 project had anticipated to contribute to fulfil the demands of HTP and strategic partnership through knowledge sharing and policy consultation from Korean experiences.

Due to the COVID-19 pandemic, capacity-building programs, including invitational training in Korea and other workshops and seminars, were carried out online, and current status surveys were conducted in writing by local researchers.

## 2. Project Objectives

The 2022 project aims to provide policy consultation on improving the national system of technology transfer and commercialization in Azerbaijan and capacity-building for developing operational programs of the HTP. Furthermore, it is intended to promote mutually beneficial economic cooperation between the two countries by contributing to the development of STI in Azerbaijan according to the 『FRAMEWORK ACT ON INTERNATIONAL DEVELOPMENT COOPERATION』.

[Table 1-2] Framework Act on International Development Cooperation

| Article 3. Basic Ideas and Objectives                                                                                                                                                                                                                                                                                                                                                                                |                                                                                                      |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------|
| (1) The basic objectives of international development cooperation are to reduce poverty in developing nations, improve the human rights of women, children and people with disabilities, achieve gender equality, realize sustainable development and humanitarianism, promote mutually beneficial economic relationships with cooperation partners, and pursue peace and prosperity in the international community. |                                                                                                      |
| (2) The objectives of international development cooperation shall lie in achieving the following matters to pursue the basic ideas presented under paragraph (1):                                                                                                                                                                                                                                                    |                                                                                                      |
| The Basic Ideas of International Development Cooperation                                                                                                                                                                                                                                                                                                                                                             | Development of developing nations and improvement of the system and conditions for such development. |
|                                                                                                                                                                                                                                                                                                                                                                                                                      | Promotion of friendly relationships and mutual exchanges with developing nations.                    |
|                                                                                                                                                                                                                                                                                                                                                                                                                      | Contribution to the resolution of global problems related to international development cooperation.  |

Source: Framework Act on International Development Cooperation, No.17302(2020), Republic of Korea.

- Project Objectives: 1<sup>st</sup> year (2022) - 3<sup>rd</sup> year (2024)
  - (2022) Improvement of the TC System in Azerbaijan and Development of the TC Program at ANAS-HTP.
  - (2023) Improvement of the Start-up and Incubation System in Azerbaijan and Development of the Start-up and Incubation Program at HTP.
  - (2024) Implementation of the Application Process with the Implementation of Capacity Building of the TC Start-up and Incubation System in Azerbaijan.

### 3. Project Framework

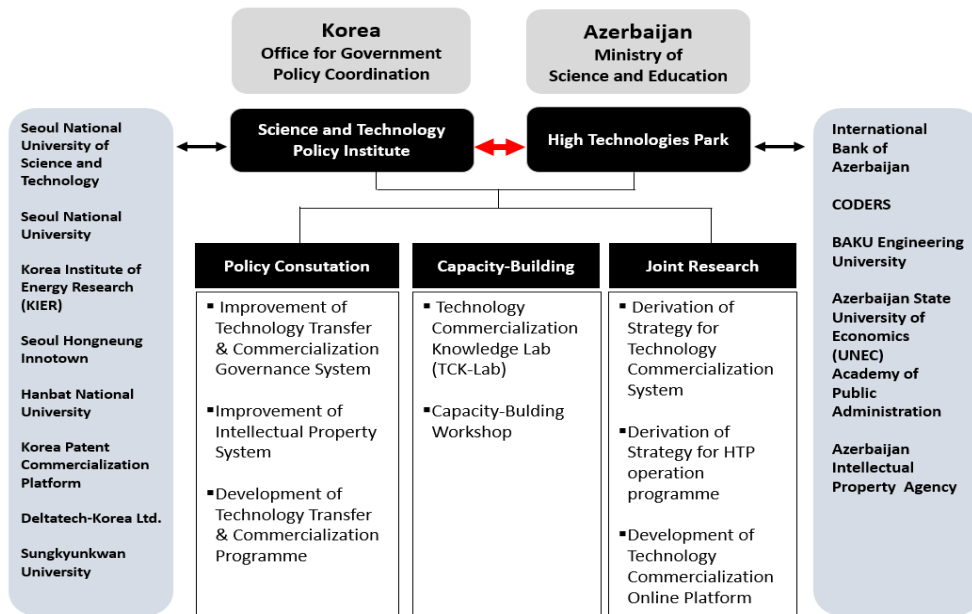
The fundamental concept of this project is to strengthen the capabilities of Azerbaijan's experts by sharing Korea's experience and through consultation. It aims to encourage them to develop the 'Technology Commercialization System Improvement Plan' and the 'HTP Operation Program' by themselves. Diverse approaches were taken to building their capacity and achieving outcomes.

First of all, a research team composed of local experts was formed to investigate the current status of technology commercialization systems and programs in Azerbaijan. The aim was to enable them to determine by themselves what needed to be improved and why before deriving strategic plans. Based on this study, the Korean consultants diagnosed the problems and challenges they were facing and identified the key points for policy recommendations.

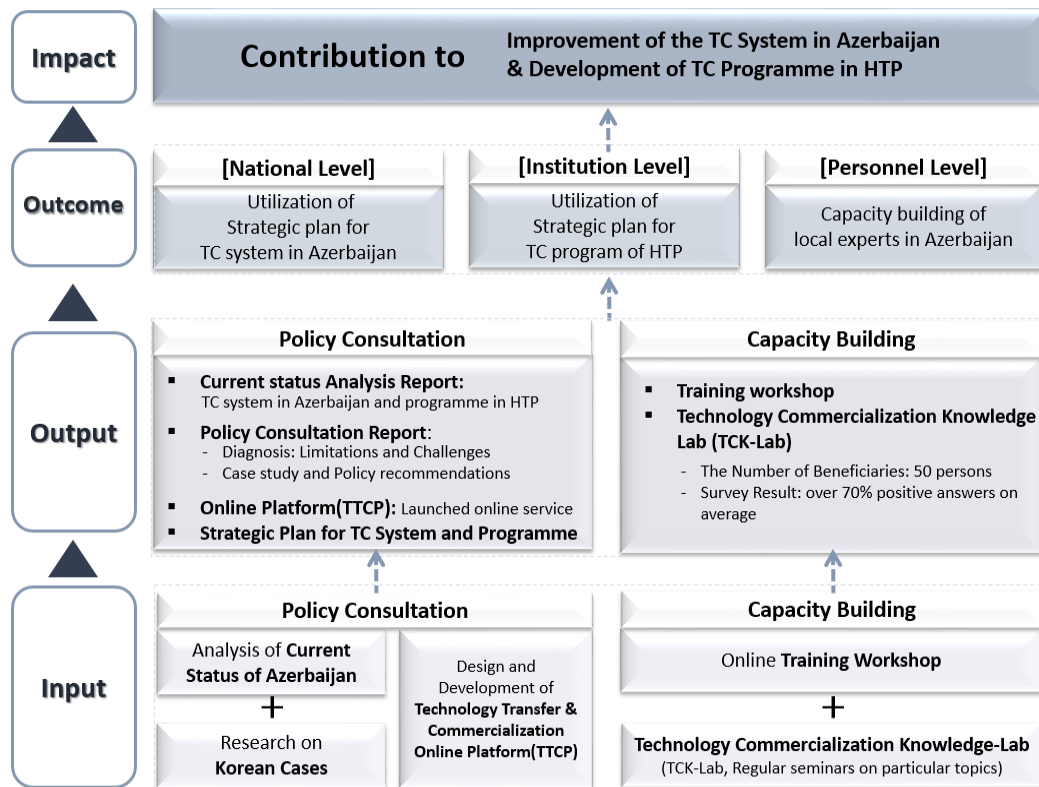
Secondly, the Technology Commercialization Knowledge Lab (TCK-Lab) and the Training Workshop were conducted to share Korea's experiences and methodologies. The TCK-Lab is a regular activity conducted in the form of a learning lab consisting of lectures and discussions by Korean experts in each specialized field related to technology transfer and commercialization. The training workshop was a one-off event that included a more comprehensive range of subjects. Both activities were opened in an on-offline hybrid.

Thirdly, the Technology Transfer and Commercialization Platform (TTCP) was developed to activate the technology commercialization ecosystem. The experts from both countries have jointly provided an advisory on this platform design. TTCP is expected to function which enables of archiving various sources and information, as well as collaboration and networking between all stakeholder in industry-university-research institutes.

[Figure 1-2] Schematization of the 2022 Project Structure



[Figure 1-3] Detailed Framework of the 2022 Project



## 4. Project Team

The project team consists of the following four groups: project management team, Korean consultants for policy recommendation, Korean experts for TCK-Lab, and Azerbaijan experts.

[Table 1-3] Project Team

| Type                     | Name              | Position/Affiliation                                                         | Role/Area of Expertise                                      |
|--------------------------|-------------------|------------------------------------------------------------------------------|-------------------------------------------------------------|
| Project Management       | Eun Joo Kim       | Principal Researcher/STEPI                                                   | Project Manager                                             |
|                          | Shinae Kang       | Researcher/STEPI                                                             | Project Coordinator                                         |
| Korean Consultants       | Soo J. Sohn       | Senior Researcher Fellow/STEPI                                               | TT&TC system and governance                                 |
|                          | Youngtack Shim    | Professor/Seoul Nat'l Univ.                                                  | Intellectual property system                                |
|                          | Gyeongsoo Shim    | Professor/Seoul Nat'l Univ. of S&T                                           | TT&TC programs                                              |
| Korean Experts (TCK-Lab) | Soo J. Sohn       | Senior Researcher Fellow/ STEPI                                              | TT&TC system in Korea                                       |
|                          | Youngtack Shim    | Professor/Seoul Nat'l Univ.                                                  | Intellectual property management                            |
|                          | Jong-Bok Park     | Professor/Gyeongsang Nat'l Univ.                                             | PRIs and TT&TC                                              |
|                          | Gyeongsoo Shim    | Professor/Seoul Nat'l Univ. of S&T                                           | Start-up policy and program                                 |
|                          | Lak-Kyoung Song   | Professor/Gwangju Institute of S&T (GIST)                                    | University start-up and TC                                  |
|                          | Jong-in Choi      | Professor/Hanbat Nat'l Univ.                                                 | Start-up and Death Valley                                   |
|                          | Seokeuy Hong      | Principal Technician/<br>Korea Institute of Geoscience and Mineral Resources | TT&TC governance                                            |
| Azerbaijan Experts       | Bünyamin Seyidov  | Head of Incubation and Technology Transfer Centre/HTP                        | Project Coordinator<br>Strategic planning for programme     |
|                          | Isa Gasimov       | Director/Centre for Innovative Technologies in Education(ITE)                | TC platform management,<br>Strategic planning for TC system |
|                          | Vusal Suleymanli  | Innovation Consultant/ANAS                                                   | Management of TCK-Labs                                      |
|                          | Samir Veliyev     | Senior Economist/<br>Azerbaijan International Bank                           | Current status research                                     |
|                          | Aysel Abdullayeva | Business Development Manager / MONT LLC                                      | Current status research                                     |
|                          | Vüqar Babayev     | Former Director/HTP                                                          | TC networking and PR activities                             |

## 5. Project Activities

### 5.1 Kick-off Seminar

- Date/Venue: 14:00-17:00, 29 March 2022, Online-offline hybrid
- Participants: 40 persons (STEPI and Korean consultants, Korea-Azerbaijan experts, Korean Embassy, Azerbaijan Embassy, ANAS and HTP officials)
- Purpose: Agreement on and mutual understanding of project topics and plans, and objectives between partner institutions
- Detailed contents:
  - Introduction of the K-Innovation Partnership Program
  - Project objectives and detailed plans
  - Roles and responsibilities of project teams
- Program:

[Table 1-4] Kick-off seminar program

| Time        | Contents                                                  | Presenters                                                                   |
|-------------|-----------------------------------------------------------|------------------------------------------------------------------------------|
| 14:00-14:20 | Opening Remarks                                           | Dr. Mi Ock Mun, President, STEPI                                             |
|             | Welcome Remarks                                           | Dr. Vasif Abbasov, General Director, ANAS-HTP                                |
|             | Congratulatory Remarks                                    | Prof. İrada Huseynova, Vice-president, ANAS                                  |
|             | Congratulatory Remarks                                    | Amb. Eun Yong Lee,<br>Embassy of the Republic of Korea in Azerbaijan         |
|             | Congratulatory Remarks                                    | Amb. Ramzi Teymurov,<br>Embassy of the Republic of Azerbaijan in Korea       |
| 14:20-14:30 | 2022-2024 K-Innovation program with Azerbaijan            | Dr. Eun Joo Kim, Principal Researcher, STEPI                                 |
| 14:30-14:40 | Cooperation performance of ANAS-HTP and STEPI             | Mr. Bunyamin Seyidov, Project Coordinator, ANAS-HTP                          |
| 14:40-14:55 | Status of TC in Azerbaijan                                | Dr. Vugar Babayev, General Director, IQ Digital Solutions                    |
| 14:55-15:25 | Role and Governance of TC                                 | Dr. Soo J. Sohn, Senior Research Fellow, STEPI                               |
| 15:25-15:55 | TC Programs at Universities/ Research Institutes          | Prof. Isa Gasimov, Director, Center for Innovative Technologies in Education |
| 15:55-16:10 | TC Programs at Universities/ Research Institutes of Korea | Prof. Jong-in Choi, Hanbat Nat'l University                                  |
| 16:10-17:00 | Q&A                                                       |                                                                              |

[Figure 1-4] Kick-off photos

|                                                                                                                                                                                                                                                                               |                                                                                                                                                                                                                           |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|  <p>2022 K-Innovation Partnership Program with Azerbaijan</p> <p>Dr. Mi Ock Mun<br/>President, Science and Technology Policy Institute, STEPI<br/>Azərbaycan Milli Elmlər Akademiyasının</p> |  <p>Kick-Off Seminar<br/>2022 K-Innovation Partnership Program with Azerbaijan</p> <p>Dr. Vasif Abbasov, General Director, ANAS-HTP</p> |
|  <p>Amb. Eun Yong Lee,<br/>Embassy of the Republic of Korea in Azerbaijan</p>                                                                                                               |  <p>Amb. Ramzi Teymurov,<br/>Embassy of the Republic of Azerbaijan in Korea</p>                                                        |
|  <p>2022 K-Innovation Partnership Program with Azerbaijan</p> <p>Dr. Eun Joo Kim<br/>Project Manager and Principal Researcher of STEPI</p>                                                 |  <p>Prof. İrada Huseynova, Vice-president, ANAS</p>                                                                                   |
|  <p>Participants of Kick-off seminar</p>                                                                                                                                                   |  <p>Korean Consultant Team</p>                                                                                                        |

## **5.2 Technology Commercialization Knowledge Lab (TCK-Lab)**

- Date/Venue: March-September 2022, Online session
- Participants: Local experts and TC-related stakeholders
  - (Group 1) Six selected experts from Azerbaijan
  - (Group 2) Open invitation to practitioners/learners relevant to the topics
- Purpose: Diffusion knowledge and promote the network among stakeholders via lab seminars in seven specialized fields
  - Designed to provide practical professional training to Azerbaijan TC officials by sharing Korea's experience and know-how
  - Derivation of ideas for promoting the ecosystem and the system of technology transfer and commercialization

## **5.3 Online Training Workshop**

- Date/Venue: 13-15 July 2022, Online-offline hybrid.
- Participants: Korean experts from universities, PRIs, and enterprises, and twenty experts in technology and innovation from Azerbaijan
- Purpose: Building the capacity of Azerbaijan's TC experts by sharing Korea's knowledge and experience

## **5.4 Technology Transfer & Commercialization Online Platform**

- Development duration: May to October 2022
- Purpose: Providing a platform service that enables the sharing of data and information on technology commercialization policies and programs, and activates networks and cooperation among stakeholders in Azerbaijan and the world



## **5.5 Current Status Analysis**

- Purpose:
  - Study and analysis of political and systematic circumstances
  - Establishment of strategic plans for improving the technology commercialization system and developing the HTP programs

## **5.6 Policy Recommendations**

- Purpose:
  - Diagnosis limitations and challenges in Azerbaijan
  - Conducting a case study of Korea's experiences and policy recommendations
  - Review and providing advice on strategic plans for the technology commercialization system and the HTP program

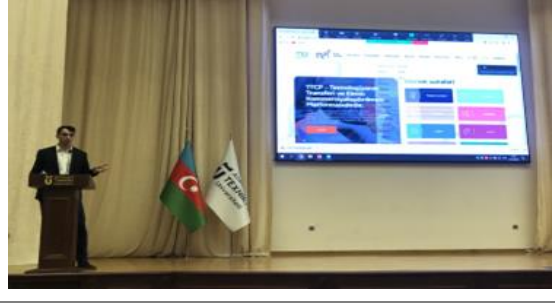
## **5.7 Dissemination Forum**

- Date/Venue: 25-26 October 2022, Baku Azerbaijan
- Participants: 50 stakeholders from the universities and ministries of Azerbaijan
- Purpose: Presenting the performances and results of projects in 2022

[Table 1-5] Dissemination forum program

| Date               | Time        | Program                                  | Presenters                                                                                  |
|--------------------|-------------|------------------------------------------|---------------------------------------------------------------------------------------------|
| Day 1<br>26<br>OCT | 10:00-10:30 | Opening Remarks                          | Dr. Eun Joo Kim, Principal Researcher, STEPI                                                |
|                    |             | Welcome Remarks                          | Dr. Vasif Abbasov, Director, HTP<br>Dr. Vilayet Veliyev, Rector, Azerbaijan Technical Univ. |
|                    |             | Congratulatory Remarks                   | Amb. Eun Yong Lee, Embassy of the Republic of Korea in Azerbaijan                           |
|                    | 10:30-11:30 | Introduction of the K-Innovation program | Dr. Eun Joo Kim, Principal Researcher, STEPI                                                |
|                    |             | Introduction of the 2022 project         | Mr. Bunyamin Seyidov, Project Coordinator, ANAS-HTP                                         |
|                    | 11:30-12:00 | Results of TCK-Lab                       | Mr. Vusal Suleymanlim, Innovation Consultant, ANAS                                          |
|                    | 14:00-14:40 | TC governance in Korea                   | Dr. Soo J. Sohn, Senior Research Fellow, STEPI                                              |
|                    | 14:40-15:20 | IP management in Korea                   | Prof. Youngtack Shim, Seoul Nat'l Univ.                                                     |
|                    | 15:20-16:00 | TC programs in Korea                     | Prof. Gyeongsoo Shim, Seoul Nat'l Univ. of Science and Technology                           |
|                    | 16:30-17:10 | University start-ups in Korea            | Prof. Jong-in Choi, Hanbat Nat'l Univ.                                                      |
| Day 2<br>27<br>OCT | 17:10-17:50 | HR development in Korea                  | Prof. Jin Kyu Lee, Head, Center for Future Human Resource Studies                           |
|                    | 09:30-10:00 | Strategic plan for the TC system         | Dr. Isa Gasimov, Director, ITE centre                                                       |
|                    | 10:00-10:30 | Strategic plan for the HTP program       | Mr. Bunyamin Seyidov, Project Coordinator, ANAS-HTP                                         |
|                    | 10:30-13:00 | Launch of the TC online platform         | Mr. Seymur Hamidov, Co-Founder, CODERS                                                      |
|                    | 13:00-15:00 | Certificate ceremony / Closing           | Dr. Eun Joo Kim, Principal Researcher, STEPI                                                |

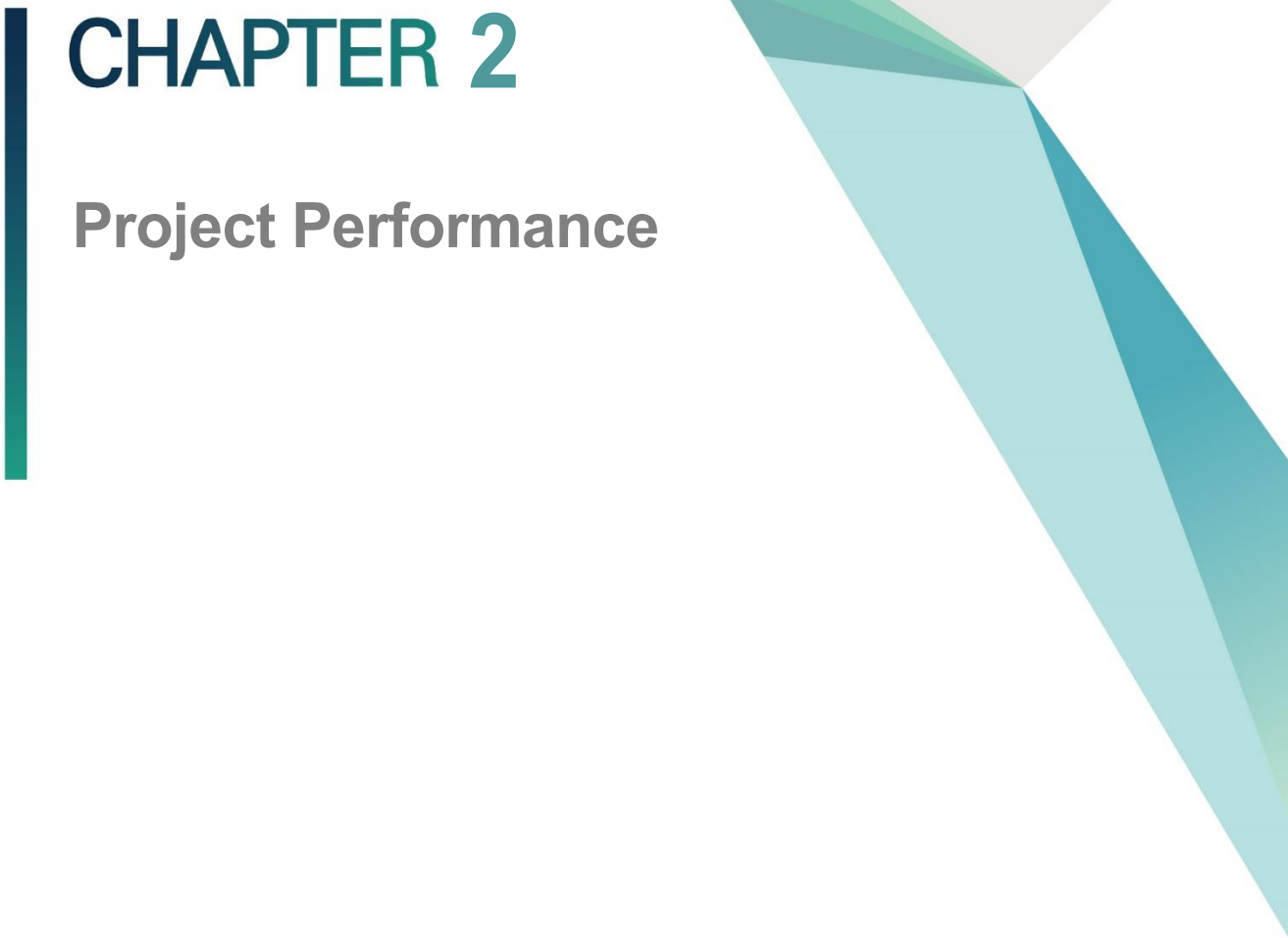
[Figure 1-5] Dissemination forum in Baku

|                                                                                     |                                                                                      |
|-------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------|
|    |    |
| <p>Dr. Eun Joo Kim, Principal Researcher, STEPI</p>                                 | <p>Dr. Vasif Abbasov, General Director, ANAS-HTP</p>                                 |
|    |    |
| <p>Amb. Eun Yong Lee,<br/>Embassy of the Republic of Korea in Azerbaijan</p>        | <p>Dr. Soo J. Sohn, Senior Research Fellow, STEPI</p>                                |
|  |  |
| <p>Participants of dissemination forum</p>                                          | <p>Closing Ceremony</p>                                                              |
|  |  |
| <p>Mr. Seymur Hamidov, Co-Founder, CODERS</p>                                       | <p>Core Stakeholders of TT&amp;TC field</p>                                          |

## 6. Project Schedule

[Table 1-6] Project Schedule

| Activity                         | 2022 |   |   |   |   |   |   |   |   |    |    |    | 2023 |
|----------------------------------|------|---|---|---|---|---|---|---|---|----|----|----|------|
|                                  | 1    | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 | 11 | 12 | 1    |
| Project planning                 |      |   |   |   |   |   |   |   |   |    |    |    |      |
| Design project framework         |      |   |   |   |   |   |   |   |   |    |    |    |      |
| Organizing project team          |      |   |   |   |   |   |   |   |   |    |    |    |      |
| Kick-off seminar                 |      |   |   |   |   |   |   |   |   |    |    |    |      |
| TCK-Lab                          |      |   |   |   |   |   |   |   |   |    |    |    |      |
| Development of TC platform       |      |   |   |   |   |   |   |   |   |    |    |    |      |
| Analysis of current status       |      |   |   |   |   |   |   |   |   |    |    |    |      |
| Diagnosis of current status      |      |   |   |   |   |   |   |   |   |    |    |    |      |
| Policy consultation              |      |   |   |   |   |   |   |   |   |    |    |    |      |
| Submittal of interim report      |      |   |   |   |   |   |   |   |   |    |    |    |      |
| Online training workshop         |      |   |   |   |   |   |   |   |   |    |    |    |      |
| Establishment of strategic plans |      |   |   |   |   |   |   |   |   |    |    |    |      |
| Dissemination forum              |      |   |   |   |   |   |   |   |   |    |    |    |      |
| Submittal of final report        |      |   |   |   |   |   |   |   |   |    |    |    |      |
| Presentation of final report     |      |   |   |   |   |   |   |   |   |    |    |    |      |
| Evaluation of final report       |      |   |   |   |   |   |   |   |   |    |    |    |      |
| Publication                      |      |   |   |   |   |   |   |   |   |    |    |    |      |



# CHAPTER 2

## Project Performance



## Chapter 2. Project Performance

### 1. Capacity Building

#### 1.1 Technology Commercialization Knowledge Lab (TCK-Lab)

The Technology Commercialization Knowledge Lab (TCK-Lab), which was held from March to September 2022, consisted of seven specialized topics in the field of technology commercialization. Korean experts delivered the lectures focusing on theory and Korean cases. In addition, the lectures emphasized the importance of governance and networking in establishing an ecosystem for technology commercialization, as well as methodologies for preparing the legal grounds and activation. Azerbaijan's experts explored strategies that could be applied to Azerbaijan's circumstances based on Korean cases, and engaged in-depth discussions based on a Q&A with Korean experts. To gather the results of the TCK-Labs, a survey on satisfaction has been conducted, in which the participants suggested the following three improvement factors: (1) needs for detailed scenario for Azerbaijan; (2) additional time for Q&A; and (3) a narrower range of topics and contents. The total cumulative number of learners in the 2022 TCK-Lab was 92, exceeding the initial target number of 50, ultimately achieving a performance score of 184%.

(\*Formulas for performance measurement:  $\frac{\text{Number of participants}}{\text{Number of target participants}} \times 100$  )

[Figure 2-1] Photos of the TCK-Lab



[Table 2-1] Results of Likert-scale questionnaire survey on TCK-Labs

| Questions                                                                                    | Strongly Agree | Agree  | Neutral | Disagree | Strongly Disagree |
|----------------------------------------------------------------------------------------------|----------------|--------|---------|----------|-------------------|
| The topics covered in the lecture were relevant to my career and work.                       | 65.84%         | 31.2%  | 12.95%  | -        | -                 |
| The contents were well-organized and easy to follow.                                         | 54.46%         | 45.54% | 5%      | -        | -                 |
| The lecture was helpful for understanding the field.                                         | 52.14%         | 46.9%  | 6.7%    | -        | -                 |
| The lecture would be helpful in strengthening the institutional capacity of my organization. | 48.96%         | 46.1%  | 8.68%   | -        | -                 |
| The time allotted for the lecture was sufficient.                                            | 39.94%         | 50.36% | 12.58%  | -        | -                 |
| The Q&A and discussion were adequate to enhance my understanding.                            | 40.69%         | 52.46% | 10.75%  | -        | -                 |
| I will be able to apply the knowledge I learnt to my work/practice.                          | 52.73%         | 41.69% | 11%     | -        | -                 |
| I will recommend this lecture to others.                                                     | 72.14%         | 36.12% | 6.57%   | -        | -                 |



[Table 2-2] TCK-Lab program

| Date         | Topics / Lecturers                                                                                                                                | Contents                                                                                                                                                                                                  |
|--------------|---------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 30 March     | <b>TT&amp;TC System in Korea</b><br>Dr. Soo Jeoung Sohn,<br>Senior Researcher Fellow, STEPI                                                       | <ul style="list-style-type: none"> <li>• Introduction to legislation and policy on TT&amp;TC, which was the one of the factors that enabled Korea to enter the top ranks of innovative nations</li> </ul> |
| 27-28 April  | <b>Intellectual Property Management</b><br>Prof. Youngtack Shim,<br>Seoul National University                                                     | <ul style="list-style-type: none"> <li>• Pros &amp; cons of patent laws</li> <li>• Patent rights in Korea</li> <li>• National strategy through patent rights</li> </ul>                                   |
| 18 May       | <b>Public Research Institutes and TT&amp;TC</b><br>Prof. Jong-Bok Park,<br>Gyeongsang National University                                         | <ul style="list-style-type: none"> <li>• TT&amp;TC in Korean government-funded research institutions</li> <li>• GRIs R&amp;D system and TC implications</li> </ul>                                        |
| 14 June      | <b>Start-up Policy and Programs</b><br>Prof. Gyeongsoo Shim,<br>Seoul National University of S&T                                                  | <ul style="list-style-type: none"> <li>• Case study: I-corps program in Korea</li> <li>• Lab-based start-up program utilizing public technology and IP</li> </ul>                                         |
| 1 August     | <b>University and Lab Start-ups</b><br>Prof. Lak-Kyoung Song,<br>Gwangju Institute of S&T                                                         | <ul style="list-style-type: none"> <li>• Case study: University start-ups in Korea.</li> <li>• Technology mismatches and the gap between the innovation system and the TC process</li> </ul>              |
| 30 August    | <b>Start-ups and Death Valley</b><br>Prof. Jong-in Choi,<br>Hanbat National University                                                            | <ul style="list-style-type: none"> <li>• Methods of overcoming the “death valley” with the INC model (Ideas, Needs, Capacity)</li> </ul>                                                                  |
| 28 September | <b>Support for TC Governance and Network</b><br>Dr. Seokeuy Hong,<br>Principal technician,<br>Korea Institute of Geoscience and Mineral Resources | <ul style="list-style-type: none"> <li>• Support program for TC network.</li> <li>• Case study: Technology exchange network in Korea</li> </ul>                                                           |

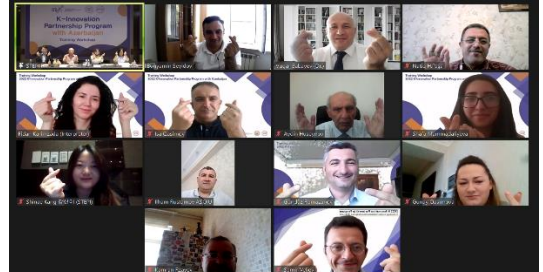
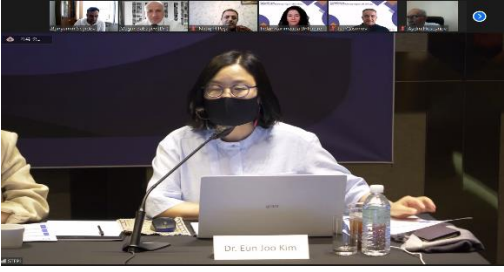
## 1.2 Online Training Workshop

The invitational training program was replaced with an online training workshop due to new restrictions on immigration between the two countries due to the continuous spread of COVID-19. This workshop included practical lectures with a focus on cases of technology commercialization cooperation between industry, academia, and research institutes. Furthermore, an expert review session was held on the first draft of the strategic plan for the technology commercialization system and the development of the HTP program, which are the final outputs of the 2022 project.

[Table 2-3] Training workshop program

| Date             | Time        | Topic                                                                          | Presenters                                                                                                                                           |
|------------------|-------------|--------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------|
| Day 1<br>13 July | 14:10-14:50 | Hongneung R&D Special Zone                                                     | Dr. Chiho Choi, Director, Seoul Hongneung Innotown                                                                                                   |
|                  | 14:50-15:30 | Industry-University-Research Collaboration                                     | Prof. Jong-in Choi, Hanbat National University                                                                                                       |
|                  | 15:30-16:10 | Technology Transfers of Research Institutes Sponsored by the Korean Government | Dr. Kidong Yoon, Korea Institute of Energy Research (KIER)                                                                                           |
|                  | 16:10-17:00 | Q&A                                                                            |                                                                                                                                                      |
| Day 2<br>14 July | 14:10-14:50 | Operation of the Corporate-University Linked Technology Platform               | Mr. Youshin Kim , CEO, Korea Patent Commercialization Platform                                                                                       |
|                  | 14:50-15:30 | Tech Start-up and Global Business                                              | Dr. Seungho Lee, CEO, Deltatech-Korea                                                                                                                |
|                  | 15:30-16:10 | Tech Valuation                                                                 | Prof. Kyungseon Cho, Sungkyunkwan University                                                                                                         |
|                  | 16:10-17:00 | Q&A                                                                            |                                                                                                                                                      |
| Day 3<br>15 July | 14:10-15:00 | Strategic Document on TC System Improvement                                    | Dr. Soo J. Sohn, Senior Research Fellow, STEPI<br>Dr. Vüqar Babayev, General Director, IQ Digital Solutions                                          |
|                  | 15:10-16:00 | Strategic Document on IP Management                                            | Prof. Youngtack Shim, Seoul Nat'l University<br>Prof. Isa Gasimov, Director, Center for Innovative Technologies in Education                         |
|                  | 16:10-17:00 | TC Program Development at ANAS-HTP                                             | Prof. Gyeongsoo Shim, Seoul Nat'l Univ. of Science and Technology<br>Mr. Bunyamin Seyidov, Head, Incubation and Technology Transfer Center, ANAS-HTP |

[Figure 2-2] Photos of the training workshop

|                                                                                     |                                                                                      |
|-------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------|
|    |    |
| <p>Workshop Day 1 Presenters</p>                                                    | <p>Workshop Day 2 Presenters</p>                                                     |
|    |    |
| <p>Participants of training workshop</p>                                            | <p>Dr. Eun Joo Kim, Principal Researcher, STEPI</p>                                  |
|  |  |
| <p>Dr. Soo J. Sohn, Senior Research Fellow, STEPI</p>                               | <p>Dr. Chiho Choi, Director,<br/>Seoul Hongneung Inn town</p>                        |
|  |  |
| <p>Prof. Gyeongsoo Shim,<br/>Seoul Nat'l Univ. of Science and Technology</p>        | <p>Prof. Youngtack Shim, Seoul Nat'l University</p>                                  |

## 2. Policy Consultation

### 2.1 Technology Transfer & Commercialization Platform (TTCP)

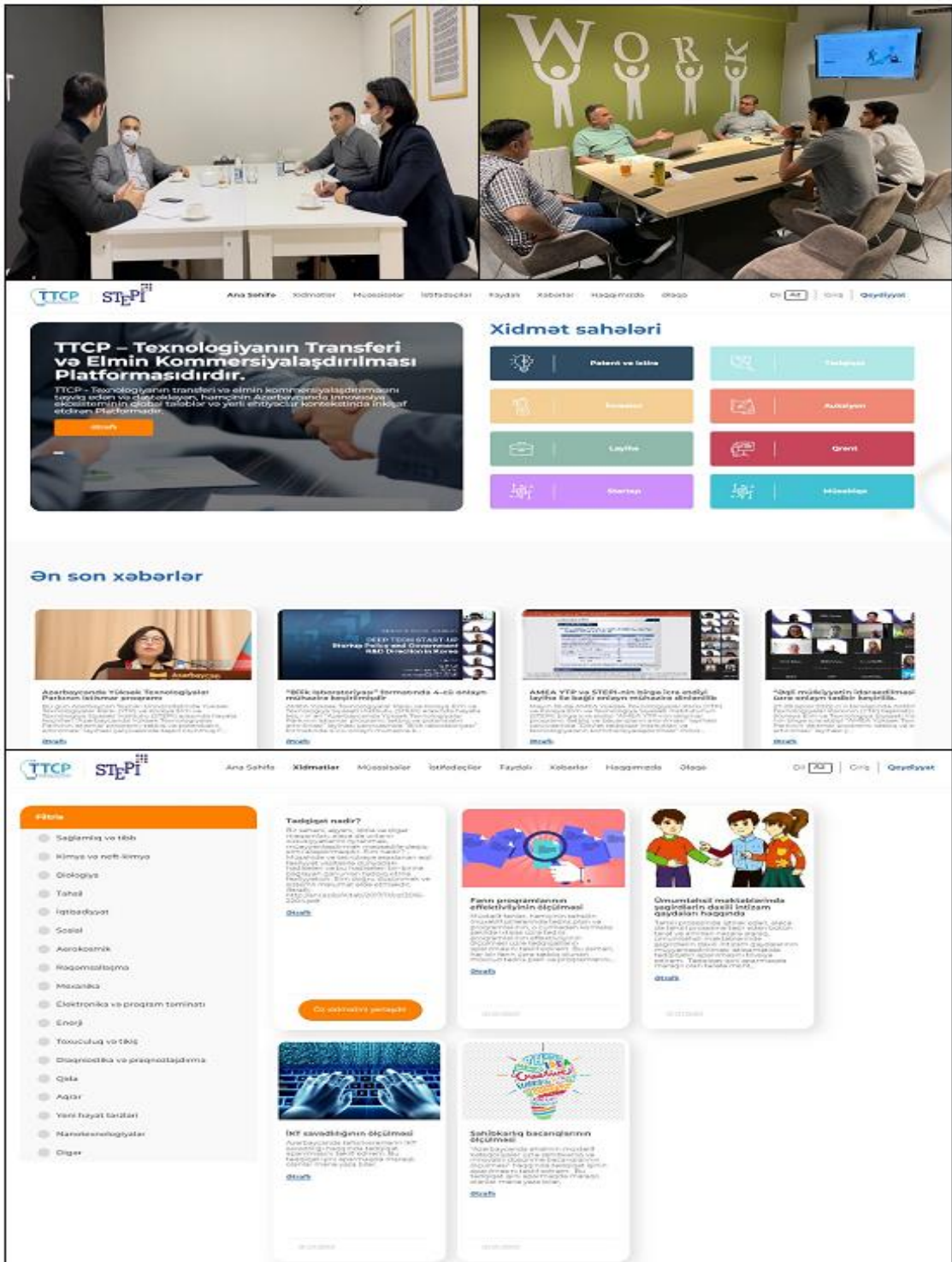
In order to activate the technology commercialization ecosystem in Azerbaijan, a platform is needed in which information and data can be easily accessed by diverse stakeholders including researchers, policymakers, technology developers, start-ups, SMEs, enterprises, research institutes, and Techno Parks. The main function of such a platform would be to provide solutions designed to address the needs of stakeholders. Moreover, networking and partnerships between diverse sectors could also be achieved by uploading the profiles of individuals and organizations on to this platform.

The process of platform development over the past six months is as follows:

- (May) Establishment of the basic plan for the platform: STEPI project team, Azerbaijan experts, and IT company 'CODERS' organized regular meetings to determine the contents and development plan
- (June-September) Starting with the UI/UX design, front-end and back-end development were carried out. Notably, the interface is designed to be easy for anyone to use
- (October) The open beta version service opened, and it has been collecting user feedback from technology commercialization officials to improve technical and qualitative issues
- (November-December) Based on the feedback collected by November, the upgraded version will be released in December with corrections and supplements

Sustainable management, including server maintenance and the construction of a database, was required, so collaboration with local ICT companies was essential. As a platform developer, the local company CODERS had a high level of understanding of the project and the technology commercialization field because it operates start-up acceleration programs and boot camps. Currently, the primary form of the platform has been completed, and the open beta service is in operation. Preparation of legal registration for the formalization of the platform, as well as app development, will proceed in 2023.

[Figure 2-3] Technology Transfer &amp; Commercialization Online Platform (TTCP)





## 2.2 Sharing the Knowledge Acquired from Korea's Experiences

Before sharing Korea's experience and know-how and success cases, it was necessary to review the results of a study on the current status of the technology commercialization system in Azerbaijan. In addition, to activate the technology commercialization ecosystem, it was necessary to examine the existing conditions, including the limitations and challenges..

The case studies of Korea's experiences were largely divided into the following three themes: Governance of Technology Commercialization, Intellectual Property Rights System, and Start-up Support Programs.

First, an overview of the technology commercialization system, the legal basis and significant policy were introduced, and the main projects of each department of the central government were classified in detail. Moreover, the support policy of the central government and performance compared to R&D investment in Azerbaijan were analysed to reveal any deficiencies and derive the implications.

Second, the history and process of development in establishing a virtuous cycle of the intellectual property ecosystem in Korea were shared. Regarding policy consultation on the systematic development of the intellectual property system. The history of development of the intellectual property system was introduced, and the importance of patent rights issues and inventions was emphasized.

Lastly, noteworthy start-up support programs in Korea were introduced for the development of the HTP's operating programs. As a specific case, information on Korean I-Corp, a start-up support program through the commercialization of science and technology, was shared. While accentuating the need for technology start-up education and training, practical implications were suggested for preparing HTP's start-up support program, based on Korean I-Corp experience, as a market-linked start-up exploration support project based on public technology.

## 2.3 Consultation on Strategic Plans

The ‘Strategic Plans’, one of the main outputs of the 2022 project, were drawn up by the Azerbaijani experts based on a study of the current situation and the advice of Korean consultants. The strategic plans were carried out in the following phases under the following two themes: 1) Improvement of the technology commercialization system at the national level; and 2) Development of the HTP operational program at the institutional level.

- ① First, the current system’s strengths, weaknesses, and problems were analysed based on a study on the current status of Azerbaijan’s experts.
- ② Then, Korean consultants diagnosed the circumstances and conditions based on the results of that study.
- ③ Furthermore, Korea’s methodologies, approaches, and experiences were shared in the form of various capacity-building activities, such as TCK-Labs and an online training workshop, to enable Azerbaijan’s experts to devise strategic plans by themselves.
- ④ Consequently, the final strategic plans were established based on the previous status diagnosis, the sharing of Korea’s knowledge and experiences, policy recommendations, and capacity-building activities.

[Table 2-4] Summary of strategic plans

| Level          | Strategic plans                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
|----------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| National Level | <ul style="list-style-type: none"> <li>• Reinforcement of expertise through education and training.</li> <li>• Mentoring services, access to resources, and business networking.</li> <li>• Formation of innovative venture capital.</li> <li>• Construction and activation of digital platforms.</li> <li>• Improvement of the intellectual property rights system.</li> <li>• Creation of free market opportunities and provision of access to global markets.</li> <li>• Establishment of a national policy and ecosystem to support the building mechanisms.</li> </ul> |
| HTP Level      | <ul style="list-style-type: none"> <li>• Education/programs aimed at raising of awareness of intellectual property rights and the related education service.</li> <li>• Project support services.</li> <li>• Industry-university-research cooperation service.</li> <li>• Intellectual property rights and licensing management services.</li> </ul>                                                                                                                                                                                                                        |



## 2022 K-Innovation Partnership Program with Azerbaijan





# CHAPTER 3

## Current Status of the Technology Commercialization System and Programs in Azerbaijan



# Chapter 3. Current Status of the Technology Commercialization System and Programs in Azerbaijan

## 1. Technology Commercialization

### 1.1 Introduction

The rapid development of technology and increasing competition are among the main factors that justify the commercialization of science, along with the actualization of innovative entrepreneurship. Gaining added value through technology transfer or the adoption of an innovative approach plays a key role in competition and development. One can observe, through the experience of developed societies, that the development of innovations and their application in real life are based on extensive, long-term and necessary resources, as well as being more productive with multidisciplinary approaches. In this process, the commercialization of science and technology transfer activities are considered to be the most important factor in building competitiveness, while the competitive advantages of innovation are becoming increasingly important for investors. Examples confirming these views can be seen in the market value of brands formed in accordance with the requirements of the information society. For example, all the activities of Apple and Amazon, which now have a collective market value of \$ 1 trillion (US), are based on knowledge, in other words, the application of new and high technologies. These and other examples show that well-designed educational and research processes that lead to high-tech production are the most important factors in making today's societies stronger.

### 1.2 Technology Policy in Economic Development

Technology policies are generally aimed at studying and monitoring technological changes and technological developments around the world, as well as increasing the economic

importance of scientific research (R&D). In other words, technology policies determine the forms of state intervention in the economy in knowledge-based approaches by influencing (or playing a role in) the process by which technological change takes place.<sup>1</sup> At present, countries that have more accurately identified the needs of the current era are pursuing medium- and long-term technology policies. As such, technology policy can be seen as a part of industrial policy. Increasing productivity through the application of technological innovations also affects technology policy, which is considered the most important factor in the industrial policy of many countries. Taking this into account, one can say that a country's industrial policy should be reorganized within the framework of the requirements arising from production capacity and technology transfer, based on the development and application of technological innovations.<sup>2</sup>

In a free market, the existence of and need for technological innovation are determined by the market participants. However, it should be noted that the state interferes in this process for a number of reasons. The creation of legal regulatory mechanisms necessary for the conditions of competition and organization of the competitive environment is enabled with the direct support of the state. In order to ensure a healthy development process and make it work fairly, it is necessary to guarantee products and services that have been developed on the basis of technological innovation in accordance with the law. Such legal regulations cover not only physical products, but also, in some cases, unrealized ideas. As a natural consequence of this situation, monopolies are inevitable in some cases. Potentially, in the defense sector and certain other areas, the fact that only the state is a customer, it has a long-term say in this area, as well as the need to further regulate this policy. In general, technology can be obtained in two ways: technology transfer and technology production. For a country or enterprise, the transfer method can only be considered in the short term, because while transfer is profitable in the short term. This is because continuous updates are required according to the technology cycle. In fact, it is possible to end up as a "technology consumer" rather than a "technology producer" through the transfer of technology. Meanwhile, technology production is an initiative that does not produce

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<sup>1</sup> Erol Taymaz, "Sanayi ve Teknoloji Politikaları: Amaçlar ve Araçlar", ODTÜ Gelişme Dergisi, 20(4), 1993, s. 550.

<sup>2</sup> İsa Qasımov, Fərid Cəbiyev, "İnnivasiya sahibkarlığı və texnoparklar", 2017, Bakı, s. 24.

immediate results in the short term, but can benefit a country in the long run. Technology production requires patience, time, and important scientific research. For example, it sometimes takes ten years or more for a technology park to produce the expected result or for a given investment to pay off. Although the policy of acquiring technology through production is a difficult and long-term process, it is more profitable than technology transfer.

In Azerbaijan there are complex infrastructure factors that regulate and ensure the process of technology transfer and commercialization. It should be noted that the work related to the transfer and commercialization of technology in this country is in the process of forming an ecosystem. While in some cases the infrastructure opportunities created in this direction are sufficient, there are problems in ensuring efficiency in the use of such infrastructure. This report provides information on existing infrastructure opportunities and their administrative management. The government is striving to ensure balanced and sustainable economic development in the country, improve the social welfare of the population, and create a qualitatively new model of development by creating organic interconnection and coordination between current, medium and long-term periods of socio-economic development, as well as accelerating progress in all areas. In this regard, in accordance with the Order of the President of the Republic of Azerbaijan No. 1862, dated November 29, 2011, the development concept titled "Azerbaijan 2020: Vision for the Future" has been developed with the participation of the relevant government agencies, scientific organizations and highly qualified specialists, and through effective cooperation with international organizations.<sup>3</sup> Support for scientific potential and innovation activities has a special place in the Development Concept. In accordance with the requirements of the Development Concept, it should be noted that a lot of work has been done in Azerbaijan to support scientific potential and innovation, especially in the field of infrastructure.

Paragraph 4.3 of the Concept for supporting scientific potential and innovation prescribes the following: One of the main directions will be the expansion of innovation activities to ensure long-term sustainable economic development, "to form a knowledge economy,

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<sup>3</sup> Arzu Hüseynova, Bilik iqtisadiyyatının inkişaf mərhələlərində innovasiya texnologiyalarının nəzəri-metodoloji aspektləri, [http://dspace.khazar.org/bitstream/20.500.12323/3003/1/Mazanova\\_Ofelya-meqale\\_1.pdf](http://dspace.khazar.org/bitstream/20.500.12323/3003/1/Mazanova_Ofelya-meqale_1.pdf), (26.01.2022).

accelerate the creation of science-based technology and products (works, services). In this regard, the development of science will be prioritized and progressive, and the process of its effective integration into global science will be continued. The scientific infrastructure will be improved, the material and technical base of science will be modernized, and the transfer of information support systems in this area to electronic form will be ensured.<sup>4</sup>

Paragraph 3.2 of the “Azerbaijan 2030: National Priorities for Socio-Economic Development”, which was approved by the Decree of the President of the Republic of Azerbaijan dated February 2, 2021 (№ 2469) under the heading “Creative and Innovative Society”, provides the following:<sup>5</sup>

"In the future, the digital society, which supports the latest technologies and innovations, will become a leading force of development. An ecosystem that stimulates creativity and innovation in society must be built to keep the country competitive. The ecosystem to be established in the new period (relevant incentives, incubation and acceleration centers, public-private partnership financing models, venture and crowdfunding institutions, etc.) should be aimed at increasing the technological capacity of the economy. Prospects must be created for Azerbaijan to become a space for technological exports by adapting to the rapid technological progress currently taking place all around the world. To this end, the development of the processing industry should be deepened, and the development of science-intensive and high-income medium and high-tech industries should be ensured. Furthermore, competitive funding mechanisms should be established to promote results-oriented research."

In general, a lot of work has been done to achieve high technological development, both in terms of organization and the formation of infrastructures, based on the existing legal framework and the requirements of state programs. In the activities of individual ministries and other law enforcement agencies, studies of the new challenges facing the world were taken into account and the relevant commitments were made. Such approaches have laid

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<sup>4</sup> <http://www.e-qanun.az/framework/25029>, 06.02.2022.

<sup>5</sup> <http://www.e-qanun.az/framework/25029>с 06.02.2022.

the foundation for the development of organizations and infrastructures alike.

### **Social-Economic Development Strategy of the Republic of Azerbaijan, 2022-2026 <sup>6</sup>**

The 2022-2026 Strategy adopted on July 22, 2022 underlines the essential points regarding the development of innovations and technologies. As the Strategy highlights, since Azerbaijan is entering a new period of strategic development in the post-pandemic and post-conflict era, its strategic aims include, but are not limited to, the following:

- Mobilization of modern and sustainable sources of economic growth through structural and institutional reforms;
- Strengthening the innovation-orientation and financial sustainability of the national economy; and
- Effective integration of the national economy into global value chains, thereby
- Achieving new targets in building a high-wealth society.

In order to achieve these aims, Azerbaijan will foster resources and provide services in order to boost exports, innovations, and private-public partnerships, etc. along with other strategic tools. One of the elements of the 2022-2026 Strategy that attracts attention is the government's intention to promote innovations and expand the implementation of modern technological solutions, and thereby ensure economic development based on effectiveness and competitiveness.

The 2022-2026 Strategy expects a number of important global trends, challenges and opportunities to arise in the next decade, including the following:

- Expansion of efforts to implement the outcomes of intellectual activities in major economic areas, and provision of support for innovation and creativity. (For this reason, the development of a creative economy dominated by intellectual property should be encouraged and supported.)

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<sup>6</sup> Social-Economic Development Strategy of the Republic of Azerbaijan, 2022-2026,  
<https://president.az/az/articles/view/56725>

- Ideological basis for the projected deep structural reforms in the new social-economic development model for until by 2030. (Entrepreneurial businesses, productive investments, intellectual properties that boost innovation, and competitive markets are at the core of this ideology.)

One of the five strategic frameworks of the 2022-2026 strategy concerns “competitive human capital and the destination of modern innovations”. This framework sets, but is not limited to, the following goals:

- Research in various fields of science will be expanded (attraction of modern-minded and creative youth to scientific research in the early stages; establishment of a mechanism for competitive and purposeful funding of scientific research; increase in the number of published articles in impact-factored journals by 40 percent).
- Commercialization of scientific research, technical-construction projects (in other words, R&D projects) will be supported (financing such research; developing such infrastructure; since commercializing technologies are a priority in innovative development) by providing an environment for transforming research findings into a product and eliminating obstacles to scientific research; thereby easing the transfer of innovations to the markets and increasing R&D expenditure to 0.1% of GDP.
- For the purposes of import-substitution and the localization of production, local production potential will be strengthened (by incentivizing the implementation of innovations in the private and public sectors, and promoting private sector investments in IT infrastructure, resulting in an increase in the R&D expenditures of transnational companies in the country).
- Quality and accessibility of education in innovative and digital skills will be expanded (by preparing professional leaders for talent development and re-education via a newly founded digital academy, and establishing cooperation with top international technology universities in IT education, thereby increasing the number of graduates of technical, technological and innovation-oriented study programs every year).
- A funding mechanism for all stages of start-up growth will be established (by developing a system of purposeful fiscal incentives for start-ups and investors, and



providing a sufficient pool of talents, resulting in an increase in the number of start-ups every year).

- The “Absheron Valley” Innovation Cluster will be established (by forming an innovation agenda and promoting it by government, and implementing innovations in priority fields of the economy, resulting in an increase in the number of residents of companies moving into the cluster every year).

The Action Plan of the 2022-2026 Strategy lists the tasks before which the relevant authorities need to achieve in the coming years. Some of the relevant tasks included within the scope of this report are as follows:

• **Goal 1.1. Sustainable and high economic growth**

- Increasing the effectiveness, targets and transparency of mechanisms to support entrepreneurs;
- Expanding opportunities for SMEs to access resources (including financial resources up to 30 million manats for project funding) and capacity-building and clustering, etc.
- Diversifying and increasing the volume of exports of non-oil goods and services (promoting the “Made in Azerbaijan” brand, etc.) and incentivizing FDI in non-oil industries (i.e. attracting at least six companies listed on the Fortune 500 and Fortune 2000 indices, etc.).
- Speeding up the transition to the digital economy (by adopting and promoting a strategy for the development of the digital economy, increasing implementation of Industry 4.0 technologies by 5% every year, etc.).
- Establishing industrial zones and increasing the potential of economic zones.
- Supporting the dynamic growth of the culture and creative industries and their export potential (increasing their share of GDP by 10% every year, their exports by 10% every year, etc.).
- Improving competition in the economy and strengthening the competitiveness of entrepreneurship (by adopting the draft Competition Code, etc.).

- **Goal 3.1. Education in accordance with the requirements of the 21st century**

- Improving the accessibility, competitiveness and quality of higher education (joint degree programs, etc.).
- Promoting the development of scientific-research and innovations, as well as integrating science and education (by establishing two research universities, creating mechanisms that strengthen scientific potential, increasing expenditure on scientific R&D projects with competitive funding schemes, etc.)

- **Goal 3.2. Creative and innovative society**

- Developing the innovation ecosystem (by establishing the “Absheron Valley” innovation cluster (attracting up to 300 new start-ups, attracting investments of up to 13 million USD by 2026); adopting legal acts on venture activities and crowdfunding; expanding tech diaspora networks; attracting talents from abroad and promoting “brain gain”; incentivizing the foreign registration of IP rights; simplifying the procedure for patent registration, assessing the country’s technology and research profile; attracting (at least 1 per year) start-ups to Azerbaijan; expanding incubation and acceleration programs with PPP mechanisms, etc.).
- Supporting and incentivizing innovations (improving government support mechanisms for innovative projects; 20 start-up certificates per year; financing of 30 projects per year); developing a subsidy mechanism for applying innovations in the private and public sector; developing venture financing schemes; forming the “Science-Industry Cluster 4.0); increasing private investments in R&D; establishing specialized innovation centers; developing smart city and smart village concepts; etc.).
- Expanding opportunities for the application of innovations to various sectors of the economy (innovations in transportation, logistics, energy, ICT, government services (e-government), etc.; adoption of the “Digital Transformation” concept; etc.).

As the 2022-2026 Strategy outlines, Azerbaijan’s ambitions in the coming years include making the country a far more innovation- and technology-oriented destination for key

economic players. In this strategic framework, ANAS, and particularly ANAS HTP should position itself at the core of the technology commercialization reforms along with other relevant agencies operating in these directions.

### 1.3 Agencies for Technology Transfer & Commercialization

At present, the following government agencies (or the relevant entities within these organizations) are actively operating in Azerbaijan to regulate and ensure the process of scientific research, technology transfer and commercialization:

- ① Azerbaijan National Academy of Sciences (ANAS)
- ② Science Development Fund under the President of the Republic of Azerbaijan
- ③ Supreme Attestation Commission under the President of the Republic of Azerbaijan
- ④ Intellectual Property Agency
- ⑤ National Aerospace Agency (NASA) of the Republic of Azerbaijan
- ⑥ State Agency for Public Service and Social Innovation under the President of the Republic of Azerbaijan
- ⑦ Small and Medium Business (SMB) Development Agency of the Republic of Azerbaijan
- ⑧ Azerbaijan Space Agency (Azercosmos) (a public legal entity)
- ⑨ Center for Analysis and Coordination of the Fourth Industrial Revolution (a public legal entity)
- ⑩ Azerbaijan Renewable Energy Agency
- ⑪ Economic Zones Development Agency
- ⑫ Center for Analysis of Economic Reforms and Communication
- ⑬ Agency for Innovation and Digital Development

### ① **Azerbaijan National Academy of Sciences (ANAS)**<sup>7</sup>

By decree of the Council of the People's Commissars of the USSR dated January 23, 1945, the Azerbaijan Affiliate Branch was transformed into the Azerbaijan Academy of Sciences. At that time the Academy comprised four sections, sixteen Scientific Research Institutes and Scientific Research Departments, three museums, the Central Scientific Library, and various scientific bases in Nakhchivan, Ganja, Khankendi and Guba. Within the same year. Thus, after more than two months of checks and discussions beginning on January 23, 1945, the first fifteen founders of the Academy of Sciences of the Azerbaijan SSR, and the first academicians, were determined appointed. Under the decree of the Council of People's Commissars of the Azerbaijan SSR on "The establishment of the Academy of Sciences of the Azerbaijan SSR" dated March 27, 1945, No. 316, the establishment of the Academy of Sciences of Azerbaijan was formalized and its Charter approved.

ANAS consists of the following six departments: Department of Physical-Mathematical and Technical Sciences (Institutes of Radiation Problems, Physics, Mathematics and Mechanics, Information Technology, Control Systems and Shamakhi Astrophysical Observatory, named after Nasraddin Tusi, Institute of Biophysics); Department of Chemical Sciences (Institute of Petrochemical Processes, named after the academician Yusif Mammadaliyev, Institute of Catalysis and Inorganic Chemistry, named after the academician Murtuza Naghiyev, Institute of Chemistry of Additives, named after the academician Ali Guliyev, Institute of Polymer Materials); Department of Biological and Medical Sciences (Institute of Botany, Institute of Molecular Biology and Biotechnologies, Institute of Zoology, Institute of Physiology, named after the academician Abdulla Garayev, Institute of Microbiology, Institute of Dendrology, Central Botanical Garden, Institute of Genetic Resources, Institute of Soilscience and Agrochemistry); Department of Humanitarian Sciences (Institute of Literature, named after Nizami Ganjavi, Institute of Linguistics, named after Nasimi, Institute of Folklore, Institute of Architecture and Art, National Museum of Azerbaijan Literature, named after Nizami Ganjavi, Huseyn Javid's Memorial Flat, Institute of Manuscripts, named after Mahammad Fuzuli); Department of Social Sciences (Institute of History, named after

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<sup>7</sup> <https://science.gov.az/en/pages/istoriya-osnovaniya>, 06.02.2022

Abbasgulu Agha Bakikhanov, Institute of Archaeology and Ethnography, Institute of Economy, Institute of Philosophy, Institute of Law and Human Rights, National Museum of Azerbaijan History, Institute of the History of Science, Institute of Caucasian study, Institute of Oriental Studies, named after the academician Ziya Bunyadov); Department of Earth Sciences (Institute of Geology and Geophysics, Institute of Geography, named after the academician Hasan Aliyev, Institute of Oil and Gas, Republican Seismic Survey Center).

ANAS also has two regional departments including the Nakhchivan Scientific Branch (Institutes of History, Ethnography and Archaeology, Art, Language and Literature, Natural Resources, Bioresources, Manuscripts Foundation, and Batabat Astrophysical Observatory) and the Ganja Branch (Nizami Ganjavi Center, Institute of Humanitarian Researches, Institute of Local-lore, Institute of Ecology and Natural Resources, Institute of Bioresources, Institute of Agrarian Problems and Botanical Garden);, as well as the Sheki and Lankaran Regional Scientific Centers.

In addition, a number of organizations are attached to the Presidium of ANAS, including the Central Scientific Library, Research Center for the Azerbaijan National Encyclopaedia, Museum of Natural History, "Cybernetics" Special Constructor Bureau, Republican Council for the Coordination of Scientific Research, Free Trade Union, Council of Veterans, Council of Young Scientists and Specialists, "Ashgar" Scientific-Production Association, "Nafta" Scientific-Production Association, "Science" publishing house, and a limited liability company (High Technologies Park of ANAS).

## ② Science Development Fund under the President of the Republic of Azerbaijan<sup>8</sup>

The Science Development Fund was established under the President of the Republic of Azerbaijan, by Decree No. 526 of the President of the Republic of Azerbaijan dated October 21, 2009, in order to stimulate fundamental scientific research in the Republic of Azerbaijan, strengthen research and studies on the country's natural resources, and its cultural and historical heritages, and ensure the proper representation of Azerbaijan's scientific efforts

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<sup>8</sup> <https://www.sdf.gov.az/en//generic/menu/Detail/101/menu/>, 06.02.2022.

and achievements in the global scientific arena.

The main purpose of the Fund is to preserve and maintain the country's scientific-technical potential, to effectively use this potential for the development of the national economy, and to enhance the role of science and technology in solving important social problems, as introduced by the scientific organizations of the Republic of Azerbaijan, scientific institutes, and other appropriate organizations and persons working in natural, technical, humanitarian and social fields, bearing importance for the state and society, fundamental, practical and research-innovation scientific-research, programs, projects and other scientific events, and to stimulate the scientific activities of the country's scientists and create additional opportunities.

The other important goals of the Science Development Fund are to enlarge the scientific information and innovation resources of the state, promote the exploration and investigation of the country's natural resources and its cultural and historical heritages, stimulate the productivity of research, support the representation of Azerbaijan in the global scientific arena, deepen the relationships between foreign and international organizations that operate in the science-research field, learn and implement foreign practices, build collaboration with non-state and other organizations, exchange scientific information, and obtain mass character of the science.

Starting from the first of day of its operation, the Science Development Fund set out to actively support the development of science, disseminate information on the scientific results achieved in Azerbaijan to other countries, renovate or strengthen on a very contemporary level the scientific researches logistical devices and gadgets based in scientific organizations, and, at the same time, place a special emphasis on fostering young cadres, organizing different scientific events, enabling scientists to acquire experience abroad, and maintaining scientific editions and the solutions or other issues like that as a goal. In this way, the Foundation serves to provide an additional and effective stimulus, agility, and new approaches, and also paves the way for collaboration between young scientists working within academic and educational systems and other organizations in the country.

### ③ **Supreme Attestation Commission under the President of the Republic of Azerbaijan**

The Higher Attestation Commission was established by the Decree of the President of the Republic of Azerbaijan dated April 23, 1992 under the Cabinet of Ministers of the Republic of Azerbaijan, and was later subordinated to the State Committee for Science and Technology of the Republic of Azerbaijan. By the Decree of the President of the Republic of Azerbaijan Heydar Aliyev, dated January 6, 1994, the Supreme Attestation Commission was removed from the State Committee for Science and Technology of the Republic of Azerbaijan and subordinated to the President of the Republic of Azerbaijan and renamed as the “Supreme Attestation Commission under the President of the Republic of Azerbaijan”.

The Higher Attestation Commission under the President of the Republic of Azerbaijan (hereinafter, the “Commission”) shall certify highly qualified scientific and scientific-pedagogical personnel in the Republic, award them the degrees of Doctor of Philosophy and Doctor of Sciences, and the titles of associate professor and professor in all fields of science, technology, education, culture and art. It is a state body that provides official recognition of the documents of scientific degrees and scientific titles awarded in foreign countries, and accreditation of scientific institutions (organizations), state policy and regulation in this field, and participates in the formation of scientific potential. The Commission consists of the Presidium and the Board. The Presidium is composed of full and corresponding members of ANAS, doctors of sciences, as well as leading specialists in the fields of science, technology, education and culture. The Board is composed of the Chairman, Deputy Chairmen, Chief Scientific Secretary, and the heads of the relevant structural units. The Chairman of the Commission chairs both the Presidium and the Board of the Commission. Approved by Decree No. 857 of the President of the Republic of Azerbaijan, dated February 21, 2003, the “Regulations on the Supreme Attestation Commission under the President of the Republic of Azerbaijan”, Regulation No. 769 of June 5, 2008, Amendments and Additions, were promulgated by Decrees No. 381 of February 15, 2011, No. 1544 of July 21, 2017, and No. 727 of June 11, 2019. In order to ensure the implementation of Decree No. 728 of the President of the Republic of Azerbaijan dated June 11, 2019, "On approval of the" Regulations on the procedure for awarding scientific degrees "and" Regulations on the

procedure for awarding scientific titles " Decisions No. 434, 435 and 436 of November 4, "Forms of Diplomas for Doctor of Philosophy and Doctor of Sciences", "Form of Certificates for Associate Professor and Professor" and "Nostrification of documents on scientific degrees and scientific titles issued in foreign countries (determination of equivalence) and the form of certificates of recognition on the basis of re-attestation". The Commission makes decisions independently based on the principles of legality, objectivity, transparency and compliance with professional and ethical norms.<sup>9</sup>

#### ④ Intellectual Property Agency

The operation and roles and responsibilities of the Intellectual Property Agency of the Republic of Azerbaijan are prescribed by Decree No. 222 of the President of the Republic of Azerbaijan dated July 30, 2018. The Intellectual Property Agency of the Republic of Azerbaijan was established in accordance with Part 2 of Order No. 5 of the President of the Republic of Azerbaijan dated April 20, 2018, "On measures to improve management in the field of consumer market control, standardization, metrology and the protection of intellectual property rights". the related rights, including rights to data sets and integrated circuit topologies, rights to inventions, utility models, industrial designs, trademarks and geographical indications, rights to samples of Azerbaijani folklore (traditional cultural samples), traditional knowledge and other intangible cultural heritage assets (hereinafter, "intellectual property rights"), constitute a public legal entity that provides for development, unified regulation and control in this area and coordinates its activities.<sup>10</sup>

#### ⑤ National Aerospace Agency (NASA) of Azerbaijan Republic

The National Aerospace Agency (NAA) was established in 1974 as the Caspian Scientific Center. In 1978, Heydar Aliyev visited the Caspian Scientific Center, and in 1980, he visited the Special Design Bureau for Radiophysical Equipment in Ordubad, Nakhchivan Autonomous Republic. On September 29, 1981, the Space Research Scientific-Production Union (SPU) under the National Academy of Sciences was established on the basis of the

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<sup>9</sup> <http://www.aak.gov.az/single/273>, 07.02.2022.

<sup>10</sup> <https://copat.gov.az/docs/Qanunvericilik/NIZAMNAME.pdf?t=1539330594>, 07.06.2022.



Caspian Scientific Center. The SPU was operated under the Ministry of General Mechanical Engineering of the USSR in 1985-1992. Under the Decree of the President of the Republic of Azerbaijan dated February 21, 1992, the Azerbaijan National Aerospace Agency was established on the basis of the Space Research Scientific-Production Union. Until 2003, NAA continued its activities under the auspices of the State Committee for Special Machine-Building and Conversion of the Republic of Azerbaijan, and was again subordinated to the Azerbaijan National Academy of Sciences, and continued operating within that structure until 2006. By the Decree of the President of the Republic of Azerbaijan dated September 27, 2006, however, MAKKA was transferred to the Ministry of Defense Industry (MODI) of the Republic of Azerbaijan and currently operates under that ministry.<sup>11</sup>

#### **⑥ State Agency for Public Service and Social Innovations under the President of the Republic of Azerbaijan**

The State Agency for Public Service and Social Innovations under the President of the Republic of Azerbaijan was established by Decree No. 685 of the President of the Republic of Azerbaijan dated 13 July 2012.<sup>12</sup> The State Agency is the central body of executive power that, according to the direction of its activities, conducts unified management of the “ASAN” service centers, coordinates the functions of the employees of the state agencies to act at those centers, has control of and conducts assessments, manages the integration of the databases of state agencies, accelerates the organization of electronic services and improves the management system in this area. The main tasks of the State Agency are as follows:

- Unified management of centers and coordination of the activities of employees of state entities who work at these centers.
- Control of implementation of services provided by the centers based on the principles of efficiency, transparency, kindness and civility, responsibility and comfort.

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<sup>11</sup> <https://nasa.az/tariximiz/>, 07.02.2022

<sup>12</sup> <http://vxsida.gov.az/en/page/agentlik-haqqinda>, 07.02.2022

- Assessment of the relevance to the law and the quality and transparency of each service rendered by the centers.
- Control of adherence to the rules of ethical behaviour by the employees of state entities who act at these centers; of ethical courses and training sessions aimed at developing skills related with the management of citizen flow, meetings with citizens, and listening and explaining.
- Analysis of each service rendered by state entities and implementation of measures for creating and upgrading the relevant programs, information systems, databases, and the registry of electronic services in order to ensure the provision of these services in electronic form. The State Agency ensures control over the provision of services to all citizens on a daily basis, based on the principles of efficiency, transparency, kindness and civility, responsibility and comfort, irrespective of the registration of the citizen.
- The State Agency ensures the activities of the “Call” center, which is equipped with a unified phone number to serve the citizens.
- The State Agency carries out the following activities in order to electronic arrangement and improve the services rendered by state agencies:
  - It studies the information systems held by state entities, and takes joint measures with the relevant agencies for the electronic organization of the services provided;
  - It prepares proposals for the determination of technical norms, standards and requirements for the provision of electronic services;
  - It assesses the organization and provision of electronic services in state entities and prepares notes and reports on the results;
  - It provides methodical and practical assistance to state entities for the provision of electronic services;
- The “ASAN service” centers are designed to establish a new approach to ensuring public satisfaction with state servants’ performance of their functions in relation to the general population, and to enrich the quality of relations between state servants and the general public.

- The functioning of the ASAN service centers is based on the principles of operability, transparency, ethical behaviour, responsibility, and comfort. The ASAN service centers are bodies that ensure that the services rendered by state entities are delivered in a uniform and coordinated manner.

#### **⑦ Small and Medium Business (SMB) Development Agency of the Republic of Azerbaijan**

The Small and Medium Business (SMB) Development Agency of the Republic of Azerbaijan was established in accordance with the Decree of the President of the Republic of Azerbaijan dated December 28, 2017. The Agency Charter and structure were endorsed by the Decree of the President of Azerbaijan dated June 26, 2018.

As a legal entity under the Ministry of Economy, the Agency is authorized to support the development of small and medium businesses (SMBs) in Azerbaijan by providing them with a number of services and by ensuring that the services rendered by government entities to SMBs are properly coordinated and regulated. The Agency leadership comprises seven members of the Supervision Council, which in turn is composed of the deputy ministers of Finance, Labor and Social Protection of the Population, and Agriculture, representatives of the State Tax Service under the Ministry of Economy of the Republic of Azerbaijan, and the Deputy Chairman of the State Agency for Citizens' Services and Social Innovation under the President of the Republic of Azerbaijan and the President of the National Confederation of Entrepreneurs (Employers) Organizations of Azerbaijan. The Council is chaired by the Minister of Economy of the Republic of Azerbaijan.<sup>13</sup>

#### **⑧ Azerbaijan Space Agency (Azercosmos) public legal entity**

Established on May 3, 2010 under the full ownership of the government of the Republic of Azerbaijan, "Azercosmos" is the only satellite operator in the Caucasus. Having launched Azerspace-1, the first-ever satellite of Azerbaijan, on February 8, 2013, the company provides highly reliable, satellite-based communication services to enterprises and

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<sup>13</sup> <https://smb.gov.az/en/nav/about-us>, 07.02.2022

governmental customers for DTH, voice, video, data and mobility applications.

With the advanced design and flexible switching capabilities of the Azerspace-1 telecommunication satellite, millions of viewers are able to gain access to information and communications services worldwide. Azerspace-1 plays a vital role as a gateway in joint space that connects more than fifty countries across Central Asia, Europe, Africa and the Middle East.

In December 2014, Azercosmos took over the rights to operate and commercialize Azersky, a high-resolution (1.5m imagery products) optical Earth observation satellite, and entered the commercial business of Earth observation services, including geo-information services. With its highly skilled personnel and advanced technical capacity, “Azercosmos” has succeeded in becoming one of the driving forces of the ICT sector in the country and the region in the short period since its inception. Its resources have allowed the company to become well-equipped and hence handle emerging technologies, as well as creating added value for the general public, that is beyond the reach of the terrestrial networks in benefiting from the satellite-based services.

The company’s strategic development plans include enhancing the coverage area and spectrum of its services. To this end, on September 25, 2018, Azercosmos launched its second telecommunications satellite, Azerspace-2, in a geostationary orbit of 45° East longitude.

“Azercosmos” holds an ISO 9001:2008 certificate issued by Bureau Veritas in January 2013. Azercosmos has focused its Quality and Information Security Policy in order to maintain the quality of its services and increase information security. This document may be submitted for acquaintance at the request of a customer or other interested parties.<sup>14</sup> The Research and Development Center at Azercosmos has been established for the purpose of conducting scientific and experimental research in the field of aerospace, as well as to support the development and effective exploitation of the space industry in Azerbaijan. The Research and Development Center is working on the development of software to efficiently operate

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<sup>14</sup> <https://azercosmos.az/about-us-azercosmos?lang=en>, 08.02.2022

satellites, automatic recognition of changes in satellite images, AI solutions to recognize objects on satellite images in a system built on neural networks, and the initial design of small satellites and analysis of the stability of their orbits among other projects.<sup>15</sup>

#### ⑨ **Center for Analysis and Coordination of the Fourth Industrial Revolution, public legal entity**

By the Decree of the President of the Republic of Azerbaijan dated January 6, 2021, the Center for Analysis and Coordination of the Fourth Industrial Revolution was established as a public legal entity under the Ministry of Economy of the Republic of Azerbaijan.

The Center for Analysis and Coordination of the Fourth Industrial Revolution cooperates and coordinates with the Republic of Azerbaijan and international organizations operating in the field of the Fourth Industrial Revolution, and also analyzes and coordinates challenges, initiatives, strategies and projects in the digital economy.<sup>16</sup>

#### ⑩ **Azerbaijan Renewable Energy Agency**

The Azerbaijan Renewable Energy Agency was established under the Ministry of Energy of the Republic of Azerbaijan by Decree No. 1159 of the President of the Republic of Azerbaijan dated September 22, 2020, and its Charter was duly approved.

The Agency's remit is to oversee the arrangement and regulation of activities in the field of renewable energy sources and ensure their efficient use in the Republic of Azerbaijan. It is also involved in the implementation of state policy and is included in the structure of the Ministry of Energy.<sup>17</sup>

#### ⑪ **Economic Zones Development Agency**

The Economic Zones Development Agency was established by the Decree of the President of the Republic of Azerbaijan dated January 22, 2021 and tasked with improving the

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<sup>15</sup> <https://azercosmos.az/innovative-projects/research-and-development-center?lang=en>, 08.02.2022

<sup>16</sup> <https://president.az/az/articles/view/49932>, 08.02.2022.

<sup>17</sup> <https://minenergy.gov.az/en/ministry/nazirliyin-tabeliyinde-olan-qurumlar>, 08.02.2022.

management of industrial parks, industrial districts and agro-parks. The Agency was established on the basis of the Sumgayit Chemical Industrial Park Limited Liability Company under the Ministry of Economy of the Republic of Azerbaijan.

It prepares and implements measures for the organization, management and development of industrial parks (except for the Balakhani Industrial Park), industrial districts (except for the industrial districts established in the Nakhchivan Autonomous Republic) and agro-parks, as well as in this direction. As a public legal entity it participates in specified state support measures<sup>18</sup>

### **12 Center for Analysis Economic Reforms and Communication**

The Center for Analysis Economic Reforms and Communication was set up by Decree No. 879 of the President of Azerbaijan Republic dated April 20 2016. The Center is independent in its activities, and is governed by the Constitution of the Republic of Azerbaijan, the international treaties to which Azerbaijan is a party, the law “on public legal persons” and other laws of the Republic of Azerbaijan, the decrees and orders of the President of Azerbaijan, the resolutions and orders of the Cabinet of Ministers of Azerbaijan, other regulatory and legal acts, and this Charter.<sup>19</sup> The Center’s main mission is to ensure expansion of the scope of economic reforms and the efficient coordination of works in this area.<sup>20</sup>

### **13 Agency for Innovation and Digital Development**

The Agency for Innovation and Digital Development was established by reorganizing the center in the form of a merger.<sup>21</sup> The Information and Communication Technologies Agency is a public legal entity that carries out certification, accounting, regulation and control in the field of information and communication technologies and communications (including quality control), as well as regulation of interconnection between telecommunications

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<sup>18</sup> <https://president.az/az/articles/view/50330>, 08.02.2022.

<sup>19</sup> <https://ereforms.gov.az/en/page/nizamname-4>, 08.02.2022.

<sup>20</sup> <https://ereforms.gov.az/en/page/missiya-2>, 08.02.2022.

<sup>21</sup> <https://ereforms.gov.az/en/page/missiya-2>, 08.02.2022.

operators and radio spectrum management. It also oversees the organization, coordination, and implementation (including control and regulation) of activities in the field of digital transformation, implements complex measures in the fields of high technology, nuclear science, nuclear technology and nuclear energy, scientific, scientific-technical, innovation activities. It is also a public legal entity that conducts experiments, assists individuals and legal entities in acquiring modern technologies and technological solutions, promotes innovation-oriented research and innovative projects (including start-ups) and provides them with financial support, and promotes innovation initiatives.<sup>22</sup>

I-Corps is a globally-known US approach to commercializing research outputs. Unfortunately, there are no similar educational programs in Azerbaijan at present. Instructors and mentors work with start-ups under an ad-hoc based approach. Fortunately, “Azerbaijan 2026: Social and Economic Development Strategy of the Republic of Azerbaijan 2022-2026” covers certain areas of research commercialization which resemble particular aspects of NSF I-Corps (though not necessarily similar programs). For example, the third column of the Strategy aims to support the commercialization of scientific research and R&D projects and increase the related expenditure to 0.1 percent of GDP, and to increase the added-value of high- and medium technology industries by up to 15% in terms of aggregate national added-value by 2026. Moreover, the Strategy will form the “Science-Industry Cluster 4.0” and increase the participation by 15 percent per year .

## 1.4 Free Economic Zones, Techno Parks, Industrial Parks

### *Economic Zones Development Agency*

#### **“Sumgayit Chemical Industrial Park” LLC (SCIP)<sup>23</sup>**

Industrial parks offer a modern and extensive infrastructure to their residents according to the principle of "Plug & Play". Such infrastructures include sewage works, supply networks for rainwater, potable water, processing and fire water, chemical waste disposal, natural gas supply lines, power lines, substations, transmitters, distribution points, roads and

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<sup>22</sup> <https://innovationagency.az/en/>, 08.02.2022.

<sup>23</sup> <https://scip.az/en>

railways, and fiberoptic cable lines. The construction of modern infrastructure is subsidized by the state, which enables entrepreneurs to focus more on investing in projects.<sup>24</sup>

### ***Industrial parks***

#### **Balakhani Industrial Park**<sup>25</sup>

“The Complex Plan of Measures for Improvement of the Ecological Situation in the Republic of Azerbaijan, 2006-2010” was approved by the President of the Republic of Azerbaijan on September 28, 2006 with the purpose of systematically resolving the existing ecological problems in the Absheron region. This complex plan of measures has great importance for enhancing the natural environment around Baku and the Absheron peninsula.

The Complex Plan reflects all major activities aimed at improving the current environmental situation. As such, it covers all measures and other issues related to improvement of the ecological situation in Baku Bay, the Bibi-Heybat area, the surrounding area of the Heydar Aliyev International Airport, the lakes of the Absheron peninsula, oil-contaminated lands, groundwater-flooded areas and other lands polluted with industrial wastes. The resolution of issues related to the collection, transportation and placement of solid household waste in Absheron are among the principal tasks approved in the Complex Plan of Measures.

According to the Decree of President of the Republic of Azerbaijan on “Improvement of the Management of Solid Household Waste in Baku City” dated August 6, 2008, the collection and transportation of all kinds of solid household wastes, regardless of their origins, remains under the competences of Baku City’s executive power. According to the Decree, the newly established Tamiz Shahar JSC shall “carry out the works related to improvement of the city’s ecological situation through the placement and disposal of solid household wastes in accordance with modern standards”.

The Tamiz Shahar JSC was entered in the state registration on March 12, 2009, whereupon it commenced its activities. As a state enterprise, it has built its administrative activity independently, while the supervision of its management and overall activities is carried out

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<sup>24</sup> <https://scip.az/en/page/ustunlukler/infrastruktur>, 08.02.2022.

<sup>25</sup> <https://tamizshahar.az/az/bsp>



by the State Committee for Property Issues and the Ministry of Economic Development of the Republic of Azerbaijan, as well as by a Supervisory Board functioning under the auspices of the Ministry of Economic Development.

In accordance with the order of the President dated September 22, 2021, Tamiz Shahar JSC was transferred under the management of Azerbaijan Investment Holding and the following composition of the Supervisory Board of Tamiz Shahar JSC was approved.<sup>26</sup>

Tamiz Shahar JSC was entrusted with the management, placement and disposal of household wastes in accordance with modern standards and in an organized manner, as well as the task of improving the environmental situation of the city and the development of this field based on the principles of the market economy.

### **Sumgait Technologies Park (STP<sup>27</sup>)**

- STP is one of the largest industrial production centers in the region in terms of surface area.
- Twelve factories constitute an unparalleled complex of specialized production bases, each one unique in terms of product assortment.
- More than 30 production sites are based within the 250 hectares of the enterprise.
- STP is resident member of the Sumgait Chemical Industrial Park (SCIP).
- A 50-hectare indoor production site is based within the 250 hectares of the enterprise.
- Twelve plants with more than 30 production sites employ up to 2,000 workers.
- The enterprise stocks more than 40 products not manufactured in the country.
- STP products are well known in Russia, Georgia, Kazakhstan, Kyrgyzstan, and Turkmenistan, as well as Turkey, for their high quality, and are sold under the brand “Made in Azerbaijan”.

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<sup>26</sup> <https://tamizshahar.az/en/about-us>, 08.02.2022.

<sup>27</sup> <https://www.stp.az/en/facts-and-figures>, 08.02.2022.

- For the first time in its history, STP's products have begun to be exported to Werden and Cologne (Germany), rightfully considered to be the leaders of European industry.
- STP is the biggest manufacturer in the oil sector of Azerbaijan.
- STP is one of the biggest taxpayers in the Republic of Azerbaijan, contributing to the sustainable development of the national economy.

### **Aghdam Industrial Park<sup>28</sup>**

Aghdam Industrial Park was established by the Decree of the President of the Republic of Azerbaijan dated May 28, 2021.<sup>29</sup>

- Large-scale construction and restoration work is being carried out to ensure the rapid integration of the liberated territories of the Republic of Azerbaijan into the national economy, to promote their sustainable development, and to achieve decent living conditions.
- Of particular importance are the successful implementation of construction and landscaping works in the liberated territories, the formation of a competitive and sustainable economy, the development of public-private partnerships in these areas, and the application of necessary incentives.
- In order to develop the region's economy in accordance with the long-term socio-economic development directions and national priorities, it is essential to create a modern and effective production, trade and service infrastructure, attract investments in innovative technologies, realize the region's industrial potential, and develop the necessary incentives and entrepreneurship. Furthermore, favourable conditions must be created to improve the overall environment.
- The favourable geographical position of the liberated Aghdam region, opportunities for socio-economic development, future concentration of cargo flows, and the

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<sup>28</sup> <https://scip.az/az/page/senaye-parklari/agdam>

<sup>29</sup> <https://president.az/az/articles/view/51773>, 08.02.2022.

existence of other important logistics potentials make it necessary to create an industrial park with a fully modern infrastructure in the region.

- The Aghdam Industrial Park is situated in an area equipped with the necessary infrastructures and management structures for entrepreneurship, which can be used in the production, processing (recycling) and sale of competitive products in the industrial sector through the application of modern technologies, along with the provision of services, thereby promoting the effective operation and development of entrepreneurs
- Measures related to the organization, management and development of the activity of the Aghdam Industrial Park are carried out by the Agency for the Development of Economic Zones (hereinafter, the Agency) under the Ministry of Economy of the Republic of Azerbaijan.
- In order to organize the activities of the Aghdam Industrial Park and create the required infrastructure (electricity and heat, gas, water, sewerage, communications, transport, fire protection, industrial, administrative, social and other infrastructure facilities) within its territory in the 2021 state budget of the Republic of Azerbaijan, five hundred thousand manats were initially allocated to the Ministry of Economy of the Republic of Azerbaijan from the funds provided for the reconstruction and restoration of the liberated territories.

### **“Araz Valley Economic Zone” Industrial Park<sup>30</sup>**

the “Araz Valley Economic Zone” Industrial Park<sup>31</sup> has established in the Decree of the President of the Republic of Azerbaijan. After ensuring the territorial integrity of the country, there are promising opportunities for the restoration of transportation and communication lines with the Nakhchivan Autonomous Republic and the implementation of transit traffic in this direction. Sustainable settlement in the liberated territories, a revival of economic activity, and the use of transport and logistics opportunities are expected to expand regional economic and trade relations.

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<sup>30</sup> <https://president.az/az/articles/view/51773>, 08.02.2022

<sup>31</sup> <https://president.az/az/articles/view/51773>, 08.02.2022

To ensure the successful implementation of the planned strategic directions, it is crucial to provide logistics, trade and other services in the region, as well as support for competitive and innovative industries based on advanced technologies and the concept of “green energy,” including construction materials.

The fact that the Arazboyu area of the liberated East Zangazur economic region consists mainly of plains, and serves as a gateway in the southern part of the region, means that it has favorable conditions for the establishment of an industrial park. The opening of a border checkpoint in the south of the East Zangazur economic region – in the direction of the historic Khudafar bridges in the Jabrayil region – will be more effective with the launch of the industrial park. At the same time, the establishment of a logistics center, various services and innovative industries, as well as enterprises for the production of construction materials in the Jabrayil region, will be useful in the restoration of other liberated regions.

### **Agroparks**

The “Model Regulations on Agroparks” were approved by the Decree No. 1257 of the President of the Republic of Azerbaijan, dated December 21, 2021. This decree aims to set forth “Measures for improving the management of industrial parks, industrial districts and agricultural parks.”<sup>32</sup> An event on “The Role of Agro-parks in Agricultural Development” organized by the Ministry of Economy and Agriculture on September 30 2022 was attended by Minister of Economy Mikayil Jabbarov, Minister of Agriculture Inam Karimov, and Chairman of “Amelioration and Water Management” Zaur Mikayilov, as well as the heads of relevant government agencies and leading companies in the agricultural sector. At present, work is underway to establish fifty-one agro-parks for an investment of 2.1 billion manat on 240,000 hectares in thirty-two regions of the country.

Forty-three such agro-parks have already been launched, and others are being designed. State support is provided for organization of the agro-parks, the relevant infrastructure is being created, and soft loans are also being provided. So far, the Entrepreneurship Development Fund of the Ministry of Economy has allocated 173 million manat for the

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<sup>32</sup> <http://www.e-qanun.az/framework/48328>, 08.02.2022

establishment of twenty-three agro-parks, as well as AZN soft loans worth 1 billion manats to twenty-seven agro-parks.

Twenty-five agro-parks that received investment promotion documents were issued an array of supporting documents for the import of machinery, technological equipment and facilities, and according to these supporting documents, these agro-parks received 32.8 million manats. Received a discount in the amount of manat. To date, 1 bln. More than manat has been invested. The Minister noted that the agro-parks will also play an important role in increasing employment. So far, a total of 5,000 people have been provided with permanent jobs in agricultural parks, while more than 4,500 people have been given seasonal jobs. There are also plans to create more than 4,300 additional jobs in existing and yet-to-be-opened agro-parks.<sup>33</sup>

[Table 3-1] Industrial Parks and Economic Zones in Azerbaijan

|                                   | Year of Est. | Sector                                              | No. of Residents | Strengths & Weaknesses                                                                                                                                                                                                                                                                                                                |
|-----------------------------------|--------------|-----------------------------------------------------|------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Balakhani Industrial Park         | 2011         | Recycling industry                                  | 25               | <ul style="list-style-type: none"> <li>• Successful public-private partnerships.</li> <li>• Focused on issues related to improvement of the ecological situation in Baku and its surroundings.</li> <li>• Promotion of industrial development in Azerbaijan.</li> <li>• Lack of professional staff.</li> </ul>                        |
| Sumgayit Chemical Industrial Park | 2011         | Agro, medicine, chemistry, and industrial equipment | 25               | <ul style="list-style-type: none"> <li>• Sustainable development of non-oil sectors.</li> <li>• Extensive infrastructure according to the principle of "Plug &amp; Play".</li> <li>• Residents are eligible for tax and customs incentives for 10 years.</li> <li>• Residents are exempted from customs duties on imported</li> </ul> |

<sup>33</sup> <https://www.taxes.gov.az/az/post/1568>

|                           | Year of Est. | Sector                                                          | No. of Residents                          | Strengths & Weaknesses                                                                                                                                                                                                                                                                                                                               |
|---------------------------|--------------|-----------------------------------------------------------------|-------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                           |              |                                                                 |                                           | <p>machinery, technological equipment and plant for 7 years.</p> <ul style="list-style-type: none"> <li>• Logistic opportunities.</li> <li>• Availability of bilateral Free Trade Agreements of Azerbaijan with other countries.</li> </ul>                                                                                                          |
|                           |              |                                                                 |                                           | <ul style="list-style-type: none"> <li>• Lack of professional industry staff.</li> <li>• Lack of research and development labs.</li> </ul>                                                                                                                                                                                                           |
| Sumgait Technology Park   | 2009         | Oil and gas, construction, agro and energy                      | 12 plants & more than 30 production sites | <ul style="list-style-type: none"> <li>• Production quality (“Made in Azerbaijan” brand).</li> <li>• Application of the latest, most advanced technologies.</li> <li>• Competitive products that meet international standards.</li> <li>• Use of high-quality feedstock.</li> <li>• Reasonable prices.</li> <li>• Highly qualified staff.</li> </ul> |
|                           |              |                                                                 |                                           | <ul style="list-style-type: none"> <li>• Lack of investment funding.</li> <li>• Lack of protection of IP rights.</li> </ul>                                                                                                                                                                                                                          |
| Pirallahi Industrial Park | 2016         | Non-oil sector Pharmaceuticals, medical items                   | 6 residents                               | <ul style="list-style-type: none"> <li>• Logistic opportunities</li> <li>• Fiscal and administrative incentives</li> <li>• International partnerships</li> </ul>                                                                                                                                                                                     |
|                           |              |                                                                 |                                           | <ul style="list-style-type: none"> <li>• Lack of industrial experiences in the priority sectors</li> <li>• Small scale basic production items</li> <li>• Challenges of attracting high-skill labor</li> </ul>                                                                                                                                        |
| Aghdam Industrial Park    | 2021         | Revival of Karabakh and development of its industrial potential | 9                                         | <ul style="list-style-type: none"> <li>• Business environment.</li> <li>• Applicable tax and customs benefits.</li> <li>• Investment in the production of construction materials, packaging of agricultural products, production and processing of canned fruits and vegetables, meat and dairy products, feed, and fertilizers.</li> </ul>          |

|                           | Year of Est. | Sector                                                                           | No. of Residents | Strengths & Weaknesses                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
|---------------------------|--------------|----------------------------------------------------------------------------------|------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                           |              |                                                                                  |                  | <ul style="list-style-type: none"> <li>• Negative consequences of the Karabakh War: The territory of the industrial park, which occupies 190 hectares of fields, has been completely cleared of mines and unexploded ordnance.</li> <li>• Lack of professional staff.</li> <li>• Lack of infrastructure.</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
| Araz Valley Economic Zone | 2022         | Restoration and reconstruction of territories liberated from Armenian occupation | 1                | <ul style="list-style-type: none"> <li>• Business environment and investment.</li> <li>• Applicable tax and customs benefits for residents.</li> <li>• Up-to-date infrastructure.</li> </ul> <hr/> <ul style="list-style-type: none"> <li>• Recently established.</li> <li>• Negative consequences of the Karabakh War: The territory of the Araz Valley Economic Zone Industrial Park has been fully cleared of mines and unexploded ordnance.</li> <li>• Lack of professional staff.</li> <li>• Lack of infrastructure.</li> <li>• Negative consequences of the Karabakh War: The territory of the "Araz Valley Economic Zone" Industrial Park has been fully cleared of mines and unexploded ordnance.</li> <li>• Lack of professional staff.</li> <li>• Lack of infrastructure.</li> </ul> |
| Agro parks                | 2021         | Agriculture                                                                      | 43               | <ul style="list-style-type: none"> <li>• Business environment focusing on agriculture.</li> <li>• Private and public funding.</li> <li>• Applicable tax and customs benefits.</li> </ul> <hr/> <ul style="list-style-type: none"> <li>• Lack of professional staff.</li> <li>• Lack of infrastructure.</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |

Source: Industrial parks, The Economic Zones Development Agency, <http://economiczones.gov.az/en/iqtisadi-zonalar/senaye-parklari>

## 1.5 Business Incubators and Accelerators

Generally, business incubator and accelerator centers are the kind of venture support organizations whose main objective is to nurture new businesses and start-ups for further commercialization in local and international markets. The centers provide all tangible (space, shared equipment, administrative services) and intangible (knowledge, networking, investment opportunities) business support services for young entrepreneurs. The Centers could be funded by the government or the private sector, or they could fund themselves by charging rental to incubated start-ups and young entrepreneurs.

### **Techno park of Baku Engineering University <sup>34</sup>**

The techno park has been operated as part of the Baku Engineering University (BMU) since 2016. So far, about fifty start-ups have moved in here and received developmental support, and there are ten start-ups in the Incubation Centre.

#### **Purpose of establishment and working principle of the Incubation Center:**

The start-up ecosystem in the BMU, including the BMU Techno park; was designed and established to encourage and support the development of business ideas of students and teachers, as well as other stakeholders. Although the Incubation Center operates primarily on the principles of the University (including the BMU Techno park), it is also open to approaches that meet the needs and requirements of the market and users, as well as stakeholders.

#### **1) Types of support for accepted start-ups:**

Mentoring, networks, seed investment and investors, office conditions (devices, internet, furniture, etc.), laboratories, libraries, seminars and meeting rooms, access to foreign markets, training and academic support services, as well as the social opportunities provided by BMU use, etc.

#### **2) Selection criteria, selection process and sectors:**

Innovation and practical significance of the project, the existence and determination of the

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<sup>34</sup> <https://beu.edu.az/az/page/tecnopark-30>, 09.02.2022.



team, the potential value-added potential of the claim, market access potential, etc.

3) Involvement of start-ups in the Incubation Center is carried out in two directions:

- New ideas are proposed for the development of start-up competition winners.
- Start-ups are accepted on the basis of direct applications.

### **ADA Innovation Center<sup>35</sup>**

The ADA Innovation Center (ADA Innovation Center) was officially launched as a type of educational institution on April 15, 2017, with the aim of contributing to the development of Start-ups and Small Business in Azerbaijan. The center will first provide services to ADA students and then to other students. Currently, there is an office area where five teams can work. Since 2017, the ADA Ideas Competition has been held each year and prizes are awarded for 1st, 2nd and 3rd places. Teams for the Innovation Center are also selected through the Ideas Competition.

1) Purpose of establishment and working principle of the Incubation Center:

The Incubation Center's remit is to strengthen the practical potential and activities of students, create opportunities for the application of knowledge, prepare students for the real business and industrial sector, and help them to build their education and relationships for the purpose of starting a business. During the academic year, training programs are held to develop students' business, IT and other practical skills; the Ideas competition is announced and the winners are announced in the 2nd semester of the academic year; and students are introduced to sponsors and provided with opportunities to share ideas through the ADA Foundation.

2) Types of support for accepted start-ups:

Winners of the Ideas Competition receive diverse benefits including cash prizes, assistance with preparing for competitions at home and abroad, opportunities to participate in various training programs free of charge or at low rates, access to office facilities at the Innovation Center, direct access to key members of the Start-up and Entrepreneurship ecosystem at

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<sup>35</sup> <https://www.ada.edu.az/en>, 09.02.2022

home and abroad, networking events/media support, a one-on-one presentation to ADA University teachers as needed, and recognition of the Center's work as an official internship to recruit resident teams registered at the Innovation Center, etc.

### 3) Selection criteria, selection process and sectors:

Selection is generally carried out through the Ideas Competition. However, teams that do not participate in the competition but show good results at partner centers are accepted. The criteria are mainly based on Business (Ability to Bring Money) and Innovation, and include selection from various fields (Sales, Marketing, IT, Networking, Design, etc.) One Academic Jury and one Industrial Jury are appointed to review the performances of the teams and determine the winners by secret ballot. There is no private sector option.

## **INNOLAND<sup>36</sup>**

INNOLAND is an incubation, acceleration and research center that was established to develop the private sector, promote innovation and expand the start-up movement in Azerbaijan and beyond its borders. INNOLAND performs its activities under the State Agency for Services to Citizens and Social Innovations (ASAN Service) under the President of the republic of Azerbaijan.

“INNOLAND” consists of an incubator, an accelerator, a co-working space and an IT Training and Education Center. The IT Training and Education Center is a place where young people acquire programming, IT and coding skills.

The mission and working principle of the Center are as follows:

- Foster a spirit of entrepreneurial innovation and promote the start-up ecosystem.
- Create an environment that inspires people to think differently.
- Establish a role model and empower entrepreneurs with the tools and community needed to grow and master their innovative businesses.

INNOLAND, an Incubation and Acceleration Center, is designed to allow anyone who has a

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<sup>36</sup> <http://innoland.az/en/home/activity>

good idea to realize their idea, as well as to develop the private sector, encourage innovation and expand the start-up movement. The types of support available for start-ups are as follows:

- Incubation stage: Residents receive the international experience of our mentors from the early stages of their formation and development.
- Acceleration stage: Residents undergo an intensive acceleration program designed to grow and develop start-ups in Azerbaijan and help them to enter international markets.
- Co-working stage: Tangible and intangible services are provided to physical or virtual residents. (The co-working environment is available 24/7 under the Virtual Residency program.)
- Education stage: Residents receive training sessions and programs on IT skills, such as coding, programming, and information technologies.

The partners of INNOLAND Center are active members of the innovation ecosystem of Azerbaijan, namely, the Next Step Innovation Center, SUP VC, Tech Academy, Start-up Azerbaijan, Start-up Estonia, Youth Inc. Entrepreneurship Program, etc. All the mentioned partners provide intensive contributions at different stages of the entire innovation process of nurturing start-ups and young entrepreneurs.

In 2021, at the initiative of the Innovations Center of the ASAN Service and AREA (Azerbaijan Robotics Engineering Academy), the INNOLAND AREA center was introduced to the innovation ecosystem of Azerbaijan with the aim of supporting the formation of an ecosystem that will ensure the development of start-ups in the regions and of strengthening the favourable conditions in order to increase regional potential.

The key activity consists in organizing training sessions in the fields of engineering and robotics for children and young people aged 4-19 living in the regions, and in developing their competences in the fields of entrepreneurship, business, innovation, start-up, project management, etc.

[Table 3-2] Incubation and Acceleration centers in Azerbaijan

|                                            | Year of Est. | Purpose                                                                                                                                                                                   | Strengths & Weaknesses                                                                                                                                                                                                                                                                                                                                                                                                                      |
|--------------------------------------------|--------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Techno park of Baku Engineering University | 2016         | To create an ecosystem that can provide modern infrastructure and research opportunities for companies, researchers and academic staff through university-industry collaboration          | <ul style="list-style-type: none"> <li>• Co-working center with free facilities.</li> <li>• Opportunities to participate in start-up competitions for business investment.</li> <li>• Mentoring and training opportunities with professional University staff.</li> <li>• Lack of financial support.</li> </ul>                                                                                                                             |
| ADA Innovation Center                      | 2017         | To contribute to the development of Start-ups and Small Business in Azerbaijan as an educational institution focusing first on ADA students and then on students from other universities. | <ul style="list-style-type: none"> <li>• Co-working center with free facilities.</li> <li>• Availability of development programs for start-ups with a final pitch Demo Day event for attracting business investment.</li> <li>• Initial focus on the ADA community.</li> </ul>                                                                                                                                                              |
| INNOLAND                                   | 2018         | To support and realize novel ideas from scratch, develop the private sector, encourage innovation and expand the start-up movement in Azerbaijan.                                         | <ul style="list-style-type: none"> <li>• Full support during the incubation and acceleration process of developing start-ups and residents of the Center with further local and international investment opportunities.</li> <li>• Co-working and Virtual Residency opportunities.</li> <li>• Availability of IT training and education programs.</li> <li>• Lack of enthusiastic young people with good novel ideas to succeed.</li> </ul> |

Source: Authors' elaboration

Moreover, one of the contemporary directions of the development of the innovation ecosystem is the availability of internal emerging innovation centres/units within huge business corporations, namely, Pasha Holding and the International Bank of Azerbaijan OJSC.

The main purpose of the creation and funding of such centres and innovation units is to attract start-ups and young people with smart ideas, mainly with a view to addressing corporate and internal needs and requirements. Their innovation programs can be categorized as follows:

- Intrapreneurship in which they attract ideas internally (close innovation development model).
- Entrepreneurship in which they attract ideas from outside the organization (open innovation development model).

More specifically, these major players in the corporate sector could be actively engaged in technology commercialization processes with significant outcomes for the integration of Azerbaijan into a global innovation hub. For instance, the ABB Innovation Centre's vision consists in - but is not limited to - developing high-level innovative FinTech solutions and enhancing digital platforms for corporations and individuals. As such, the ABB Innovation Centre is actively participating in the formation of an advanced and sustainable fintech ecosystem in Azerbaijan by developing structured innovation programs and community activities for advanced technologies and supporting innovative entrepreneurship.

In addition, there are some small innovation-related departments in local public institutions whose activities are described in the table below:

[Table 3-3] Technology Commercialization Programs by Department

| Ministry    | Azerbaijan National Academy of Sciences                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
|-------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Legal Basis | <b>DECISIONS</b> <ul style="list-style-type: none"> <li>• Form for "Regulations on the registration of residents of the technology business incubator, as well as registration of residents of technology parks and residents of technology business incubators"; Form for "Registration certificates of technology parks and technology business incubator"; "Application form for registration as a resident of a technology park and attached documents. Decision No. 257 of the Cabinet of Ministers of the Republic of Azerbaijan dated July 8, 2015 made upon approval of the "List."</li> </ul> |

|                                       |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
|---------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                       | <ul style="list-style-type: none"> <li>• Resolution of the Cabinet of Ministers of the Republic of Azerbaijan No. 191 dated November 29, 2007, made upon approval of the “Minimum amount of rent for the lease of state property (except for housing stock).”</li> <li>• Resolution No. 214 of the Cabinet of Ministers of the Republic of Azerbaijan dated May 10, 2019, made upon approval of the “Rules for placing state orders for products produced in the High Technology Park of the Azerbaijan National Academy of Sciences.”</li> <li>• Resolution of the Cabinet of Ministers of the Republic of Azerbaijan No. 192, dated May 15, 2015, made upon approval of the “Regulations on the Council of Experts of the Technology Park.”</li> <li>• Resolution No. 10/3 of the Presidium of the Azerbaijan National Academy of Sciences, dated April 13, 2021, made upon on approval of the “Report on the activities of the residents of the High Technology Park of ANAS, receipt of statistical information, monitoring of their activities.”</li> </ul> <p><b>LAWS</b></p> <ul style="list-style-type: none"> <li>• Science law of the Republic of Azerbaijan.</li> <li>• Land lease law of the Republic of Azerbaijan.</li> <li>• Lease law of the Republic of Azerbaijan.</li> <li>• Patent law of the Republic of Azerbaijan.</li> </ul> <p><b>PROGRAMS</b></p> <p>Azerbaijan National Academy of Sciences Development Program for 2020-2025</p> |
| Projects, Events and Other Activities | <p><b>PROJECTS</b></p> <ul style="list-style-type: none"> <li>• 2017 Training Workshop on ‘Science Technology Innovation Policy and Industry’ for Azerbaijan.</li> <li>• 2018 ANAS-HTP Technology Transfer Capacity Development Workshop..</li> <li>• STI Policy Development and Management &amp; Operation of STI Parks in Azerbaijan.</li> <li>• Policy Consultation on the Promotion of Technology Transfer and Commercialization in Azerbaijan.</li> <li>• Policy Consultation on the Promotion of Technology Transfer and Commercialization in Azerbaijan (2nd Year): Development of the Strategic Plan for ANAS HTP.</li> </ul> <p><b>ORGANIZATIONAL ACTIVITIES</b></p> <p>Creation of the Incubator and Technology Transfer Center.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
| Ministry                              | Ministry of Digital Development and Transport                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
| Legal Basis                           | <p><b>RESOLUTIONS</b></p> <p>Resolution of the Cabinet of Ministers of the Republic of Azerbaijan “on approval of the “Criteria for determining start-ups.”</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |

|                                       |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
|---------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                       | Resolution of the Cabinet of Ministers of the Republic of Azerbaijan on approval of the "Rules for maintaining a public register of innovative projects and the issuance of Start-up Passports."                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
| Ministry                              | Innovation and Digital Development Agency                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| Legal Basis                           | Ensuring implementation of Decree No. 1464 of the President of the Republic of Azerbaijan, dated October 11, 2021 on "measures for improving governance in the field of digitalization, innovation, high technology and communications in the Republic of Azerbaijan."                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
| Ministry                              | Intellectual Property Agency of the Republic of Azerbaijan                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
| Legal Basis                           | <ul style="list-style-type: none"> <li>• Law of the Republic of Azerbaijan on copyright and other related rights.</li> <li>• Law of the Republic of Azerbaijan on the legal protection of integrated circuit topologies.</li> <li>• Law of the Republic of Azerbaijan on the legal protection of samples of Azerbaijani folklore.</li> <li>• Law of the Republic of Azerbaijan on the legal protection of data sets"</li> <li>• Law of the Republic of Azerbaijan on ensuring intellectual property rights and combating piracy.</li> </ul>                                                                                                                                                                                                                                                                     |
| Projects, Events and Other Activities | <ul style="list-style-type: none"> <li>• "The role of the Copyright Agency in supporting intellectual property and research activities in the universities of the Republic of Azerbaijan (Ways of promoting innovative development and the commercialization of technologies in universities [scientific institutions])."</li> <li>• "Knowledge and Technology + Investment and Intellectual Property = University and Industry Human Capital Ecosystem.</li> <li>• "Centers for the Commercialization and Transfer of Technologies in Innovative Development (Recommendations for Creation and Operation).</li> <li>• "Commercialization of intellectual properties and research."</li> </ul> <p><b>ORGANIZATIONAL ACTIVITIES</b></p> <p>Creation of the Technology Commercialization and Transfer Center.</p> |
| Ministry                              | Ministry of Economy                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
| Legal Basis                           | <ul style="list-style-type: none"> <li>• Decree of the President of the Republic of Azerbaijan on improving management in the field of state support for the development of entrepreneurship.</li> <li>• Decree of the President of the Republic of Azerbaijan on measures for improving the management of industrial parks, industrial districts and agro-parks.</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                    |
| Projects, Events and Other Activities | Support for the Government of Azerbaijan in creating a more favorable environment for small and medium enterprises and increasing their competitiveness.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |

Source: Authors' elaboration

## 2. Intellectual Property System

### 2.1 Basic Legislation

There is no specific legislation on technology transfer and commercialization in Azerbaijan; however, the actual legal basis on IP<sup>37</sup> consists of the following laws of the Republic of Azerbaijan:

- ① Law on Patents, Law on Trademarks and Geographical Indications;
- ② Law on the Legal Protection of Topographies of Integrated Circuits;
- ③ Law on Copyrights and Other Related Rights; and
- ④ Law on the Legal Protection of Data Compilations.

Although the activities of different organizations (ANAS, Intellectual Property Agency, Ministry of Digital Development and Communication, Ministry of Economy, etc.) are governed by different legislative acts, there are no specific provisions on technology transfer and commercialization – a lacuna which creates certain difficulties in the field of technology transfer and commercialization.

The following legislative acts can be regarded as legislative acts on the partial commercialization of technologies:

- ① Decree of the President of the Republic of Azerbaijan on ensuring coordination in the field of innovative development in the Republic of Azerbaijan;
- ② Law of the Republic of Azerbaijan on science;
- ③ Rules for maintaining the registration of residents of technology business incubators, as well as the register of residents of technology parks and residents of technology business incubators ;
- ④ Form for registration certificates of technology parks and technology business incubators;

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<sup>37</sup> <https://www.ip-coster.com/IPGuides/c/intellectual-property-azerbaijan>



- ⑤ Resolution of the Cabinet of Ministers of the Republic of Azerbaijan on approval of the “Application form for registration as a resident of a technology park and the list of attached documents”;
- ⑥ Resolution of the Cabinet of Ministers of the Republic of Azerbaijan on approval of the “Criteria for determining start-ups”;
- ⑦ Azerbaijan 2030: National Priorities for Socio-Economic Development; and
- ⑧ Decree of the President of the Republic of Azerbaijan on ensuring the activity of the Innovation Agency (current name: Agency for Innovation and Digital Development) under the Ministry of Transport, Communications and High Technologies of the Republic of Azerbaijan (Ministry of Digital Development and Transport).

Social-Economic Development Strategy of the Republic of Azerbaijan: 2022-2026  
The decree of the President of the Republic of Azerbaijan on ensuring the activities of the Agency for Innovation and the Charter of the Agency for Innovation and Digital Development approved by this decree may be regarded as a central legislative act.

[Table 3-4] Legal notes on the activities of the Agency for Innovation and the Charter of the Agency for Innovation and Digital Development

| Articles                            | Contents                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
|-------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Article 1.<br>General<br>Provisions | <p>The Innovations Agency (currently, the Agency for Innovation and Digital Transformation) under the Ministry of Transport, Communications and High Technologies of the Republic of Azerbaijan (hereinafter, the Ministry) engages in innovation activities, provides assistance to local businesses in obtaining modern technologies and technological solutions and their transfer, and supports innovative research. It is a public legal entity operating in the field of project promotion (including start-ups), financing with grants, investment (60% / 40%, respectively) with authorized capital (including venture financing), innovation initiatives (hereinafter, the relevant field), and control and regulation.</p> <p>As the Innovations Agency has been dissolved and merged with the Agency for Innovation and Digital Transformation, there is no indicative budget for grants or investments. Such information will soon be available upon approval of its legal mandate and establishment. So far, the Agency has established a venture capital fund with the private sector and its indicative budget is about 10 million US dollars.</p> |

Source: the Agency for Innovation and Digital Development, Agency for Innovation and Digital Development, <https://www.digital.gov.az/az/page/haqqimizda>

## 2.2 Legislation of Intellectual Property

### 2.2.1 Current status of intellectual property

The Constitution of the Republic of Azerbaijan enshrines copyrights, the right of invention and other types of intellectual property rights (Article 30 of the Constitution), and provides full legal guarantees of creative freedom (Article 51 of the Constitution). At the same time, serious reforms have been carried out to ensure those rights and, in general, copyrights and other intellectual property rights have been accepted as a priority of the state. During this same period, the government has established the Copyright Agency, whose legal status is equal to that of a central executive body, and passed the following five laws providing legal regulation in the field of copyrights, related rights and other similar intellectual property rights (as well as taking important steps to ensure collective management of the property rights of authors):

- Law on Copyrights and Other Related Rights;
- Law on the Legal Protection of Integrated Circuit Topologies;
- Law on the Legal Protection of Data Sets";
- Law on the Legal Protection of Samples of Azerbaijani Folklore;
- Law on the Protection of Intellectual Property Rights and the Fight against Piracy .
- Law on Copyrights and Other Related Rights;
- Law on the Legal Protection of Integrated Circuit Topologies;
- Law on the Legal Protection of Data Sets";
- Law on the Legal Protection of Samples of Azerbaijani Folklore;
- Law on the protection of intellectual property rights and fight against piracy adopted, important steps have been taken to ensure the collective management of property rights of authors.

Azerbaijan, which has successfully entered the international intellectual property system, is now also a party to the Berne Convention for the Protection of Literary and Artistic Works under the auspices of the World Intellectual Property Organization (WIPO), as well as to

Conventions on the Protection of the Rights of Performers, Phonogram Producers, and Broadcasting Organizations, the WIPO Internet Contracts on Copyright and the Records of Performances and Phonograms. It has also acceded to the UNESCO "World Authorship Agreement" and the "Convention for the Safeguarding of the Intangible Cultural Heritage".

At the same time, serious steps have been taken to form a digital copyright system within a "knowledge-based economy" that meets the ultra-modern digital challenges facing the country. The structure of the system's technical support has been changed, especially in the information society, i.e. the "knowledge-based creative economy", while the situation within the digital copyright space has been analysed and studied, and a digital rights management system and e-services have all been established. Initially adopted by the Cabinet of Ministers of the Republic of Azerbaijan in December 2011 as a part of the formation of a more comprehensive copyrights system, the Center for Ensuring Intellectual Property Rights was established under the auspices of the Copyright Agency, and began to operate within the WIPO Academy.

The results of the successful economic reforms of our country have created the need to calculate the share of industry (creative industry) in terms of gross domestic product (GDP) based on copyrights and other related rights. Calculated using a special methodology developed by the Agency and based on international experience, this share rose from 2.1% in 2003 to 5% at the beginning of 2015, and is now at the level of developed countries. It should be noted that Azerbaijan is one of the first countries in the CIS to adopt the voluntary official registration of creative works. In order to automate the registration of works, the Agency has built an electronic database "Registration" service, as well as a digital online registration system and a digital library of Azerbaijani folklore.

As of the beginning of 2016, some 10,350 works were registered with the Agency, and information about those works and their authors (right holders) was loaded into the "Registration" electronic database. The creation of a national information system in the field of copyrights, ensuring its international exchange, registration of works and objects of related rights, data sets protected by the right of special protection, integrated circuit topologies and electronic database of right holders application and regular updates are

provided. In order to ensure the implementation of the Decree of the President of the Republic of Azerbaijan "On measures for organization of the e-services of government agencies", dated May 23, 2011, a special section named "Electronic Services" was created on the Agency's website, on which samples of application forms were posted, and administrative regulations for the provision of services were prepared and approved. Works are registered online via electronic services.

[Table 3-5] Ministries and institutions responsible for each copyright law

| Laws                                                                     | Ministry                                                   | Institution                                                                                                                                                       |
|--------------------------------------------------------------------------|------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Law of the Republic of Azerbaijan on copyrights and other related rights | Intellectual Property Agency of the Republic of Azerbaijan | <ul style="list-style-type: none"> <li>• Subordinated to the Intellectual Property Agency.</li> <li>• Intellectual Property Rights Enforcement Center.</li> </ul> |

Source: Authors' elaboration

## 2.2.2 Expected results in the field of intellectual property

- Adaptation of copyright protection, enforcement and protection to the level of developed countries.
- Its main mission is to stimulate creativity and innovation and further enhance the importance of intellectual property in supporting market regulation.
- Improvement of copyright legislation, creation of new mechanisms to ensure the protection of rights, and the imposition of effective measures and sanctions against piracy and counterfeiting.
- Effective preservation and dissemination of knowledge and the cultural heritage, including intangible cultural heritage, due to the potential of intellectual property, through the intellectual property of creativity and intellectual activity.
- Creation of additional opportunities and more favorable conditions for motivation and the acquisition of new products of intellectual activity.
- Promotion of the role of intellectual property rights in the "knowledge-based economy" and transformation of intellectual property into a powerful tool for

economic development, sustainable development of the creative economy, and implementation of measures to increase its share of the country's GDP.

- (Management) the growing proportion of intangible intellectual property assets in economic entities and investment attraction.
- As a result of the protection of information objects and technologies as objects of copyright, the management of rights in digital networks will be ensured and e-commerce will be further expanded, taking into account the entry of many works protected by intellectual property rights into the global digital network.
- Creation of a digital rights management system based on licensing and the provision of services in the format of one-stop-shops.
- Support for the wider dissemination of copyrighted information assets and technologies, in addition to formation of the information society.
- Creation of incentives, provision of electronic services to the general public on functional responsibilities.
- Improvement of the conceptual and theoretical framework for the transformation of copyrighted intellectual properties into a powerful tool for economic development and support for the formation of the information society.
- Increase of state support for authors and other copyright holders, strengthening of the functional base in the field of intellectual property, and liberalization, regulation, modernization and structural reforms.
- Improvement of the legal framework of the management system in the field of intellectual property rights, creation of an executive body operating in this field with a modern, flexible and improved system for managing human resources.
- Formation of a culture of intellectual property in Azerbaijani society and strengthening of respect for intellectual properties.
- Information technology - strengthening and development of intellectual property (IT-IP) connections.
- Establishment of an area of activity for the assessment of intellectual properties.

- Modernization of intellectual property information resources.
- Increase in the production of competitive products, expansion of the application of complex organizational and legal measures by government agencies and international and non-governmental organizations, including grant funds.

“Intellectual property right” refers to the exclusive right of a citizen or a legal entity to the results of its intellectual creativity, or equivalent products, types of work and services (firm name, trademark, service mark, etc.) in the cases and in the manner prescribed by law.

“Intellectual property rights” refers to the rights to copyrighted works, performances, phonograms, broadcasts, integrated circuit topologies, data sets, folklore (traditional cultural samples), inventions, utility models, industrial designs, trademarks, and geographical indications.

The Law of the Republic of Azerbaijan on the Enforcement of Intellectual Property Rights and Combating Piracy regulates the relations related to guaranteeing the protection of intellectual property rights, the protection of the interests of holders of intellectual property rights, and the prevention of all forms of illegal production and the imposition of sanctions and penalties in this field, including the copying of intellectual properties .

## 2.3 Legislation of Industrial Property Rights

THE LAW OF THE REPUBLIC OF AZERBAIJAN ON PATENTS regulates property relations and personal non-property relations arising in connection with the creation, legal protection and use of inventions, utility models and industrial designs in the territory of the Republic of Azerbaijan. The main provisions of the law are as follows:

### Article 13. Employer and author

- ① The owner of the enterprise or the head of the enterprise, his authorized body, as well as engaged in entrepreneurial activity not prohibited by the Law of the Azerbaijan Republic “On Entrepreneurial Activity”, regardless of the organizational

legal form in which the author's employee (hereinafter - the employee) works; Individuals who are considered employers. [32]

- ② The right to obtain a patent for an invention, utility model or industrial design created by an employee in connection with the performance of official duties or a written assignment (production, research, artistic-graphic and other type) given by the employer, the contract between the employer and the employee if not, it belongs to the employer. The relationship between the employer and the employee regarding the newly created invention, utility model and industrial design shall be regulated according to the terms of the contract.
- ③ An employee(s) who has created an invention, utility model or industrial design in accordance with the foregoing paragraph 2 of this Article shall submit a written notice to the employer within 1 month of its creation.
- ④ In the event that an employer refuses to file a claim for an invention, utility model or industrial design, or does not notify the employee concerned of his or her intention to exercise this right within 3 months from the date of receipt of the notification, then this right shall pass to the employee. Prior to filing the claim, the employer and the employee shall not disclose the nature of the invention, utility model or industrial design.
- ⑤ In accordance with the foregoing paragraph 4 of this article, within 6 months from the date of receipt of the patent, the employer shall have the pre-emptive right to use the patent under the license agreement.
- ⑥ In all other cases, except for those cases provided in paragraph 2 of this Article, the right to obtain a patent for an invention, utility model or industrial design created by an employee shall belong to the employee.
- ⑦ The employer shall pay the employee the appropriate fee in accordance with the terms of the contract concluded between them for the creation of an invention, utility model or industrial design. In the event that the employer assigns the right to use such invention, utility model or industrial design to another person, the employee shall have the right to demand an increase in the fee if the income received by the employer does not correspond to the amount of remuneration

paid to the employee, or if the employer continues to participate in the use of the facility. An employee has the right to receive an additional payment if the use of an invention, utility model or industrial design yields more economic benefits than provided for in the contract concluded between the employer and the employee. If the employer decides to waive the rights to a patent for an invention, utility model or industrial design, the employer shall offer the employee in advance a contract for the transfer of rights to the invention, utility model and industrial design.

- ⑧ When the relevant executive authority is a customer of special research or design work related to the national security of the state, the relationships pertaining to inventions, utility models or industrial designs created by the executor in connection with the performance of these works shall be regulated by the Law on State Secrets of the Azerbaijan Republic.[33]
- ⑨ In the event of a dispute between an employer and an employee concerning an invention, utility model or industrial design, it shall be settled administratively and in court. [34]

[Table 3-6] Laws on industrial patent rights

| Laws                | Ministry                     | Institution                                                                                                                                                                                                                                                                                                                                                                                                           |
|---------------------|------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Patent law          | Intellectual Property Agency | Patent and Trademark Examination Center<br>Any legal entity or individual may file a claim for a patent for an invention, utility model or industrial design. The claim must be submitted electronically or by mail by the applicant, his/her authorized representative or a patent attorney registered with the Agency, or should be submitted directly to the Agency.                                               |
| Utility model right | Intellectual Property Agency | Patent and Trademark Examination Center<br>An author's contract must be concluded between the Author(s) and the Applicant or the patent owner (non-employer) for the creation and use of an invention, utility model or industrial design.<br>The right to obtain a patent for an invention, utility model or industrial design created by the author(s) belongs to the Applicant or the patent owner (non-employer). |



| Laws                    | Ministry                     | Institution                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
|-------------------------|------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Trademark right         | Intellectual Property Agency | <p>Patent and Trademark Examination Center</p> <p>The application for registration of trademarks shall be submitted by the applicant to the Intellectual Property Agency of the Republic of Azerbaijan.</p> <p>The bidder may submit the claim to the Agency in the following order:</p> <ul style="list-style-type: none"> <li>- Directly;</li> <li>- Through a patent attorney registered with the Agency.</li> </ul> <p>Unless otherwise provided by interstate agreements to which the Republic of Azerbaijan is a party, foreign legal entities and individuals, excluding those engaged in entrepreneurial activities in the Republic of Azerbaijan, may file a claim with the Agency only through a patent attorney.</p> <p>The claim shall belong to a trademark, consisting of an application with the required documents attached.</p> |
| Design protection right | Intellectual Property Agency | Patent and Trademark Examination Center                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |

Source: Intellectual Property Agency of the Republic of Azerbaijan; <https://copat.gov.az/az/nav/qanunvericilik>

## 2.4 Legislation of Copyright

The main directions of the state policy in the field of copyright and other related rights are as follows:

- ① Creation of works of science, literature and art, stimulation of activities aimed at enhancing the moral values of the people;
- ② Creation of a legislative framework that defines and protects copyrights in accordance with advanced international practices;
- ③ Organization of public administration in the field of copyright and related rights;
- ④ Development of international relations with a view to protecting the rights of authors in the territories of other states.

The minimum amount of the author's fee shall be determined by the relevant executive authority. The development and integration of Azerbaijan's national patent system to the international level, and the strengthening of international cooperation in this field, are prioritized by the state policy. As a member of the World Intellectual Property Organization,

Azerbaijan actively cooperates with EU countries in the field of IP rights, and has signed intergovernmental agreements and MoUs for bilateral agreements with EU countries.

## 2.5 Intellectual Property Network

The role of IP rights is fostering innovations and tech commercialization. However, the idea that a stronger IP system leads to greater innovation and creativity could benefit from more data and a quantitative analysis.

In Azerbaijan, businesses are often unaware that their IP is valuable business asset that could formulate their competitive advantage in the marketplace. Therefore, the first thing that one should do is to protect one's business IP in a country where one is currently doing, or is planning to do, business in the future.

The Intellectual Property Agency of Azerbaijan is responsible for formulating a unified state policy in the field of IP rights and ensuring its implementation, as well as guaranteeing legal protection of IP rights, in order to prevent their appropriation and the violation of rights related to their use, and to take appropriate measures in this regard. It is also responsible for organizing the protection and management of IP rights in digital networks to support the development of the information society, including digital information technologies and electronic commerce, and for taking measures to support the formulation of IP policy in scientific research institutions, and commercialization of the results of intellectual activities and innovative technologies related to the creation of infrastructure for commercialization and technology transfer.

[Table 3-7] Intellectual Property Network

| Title of Platform                                                                                                                                             | Contents                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Patent and Trademark Examination Center under the Intellectual Property Agency<br><br><a href="https://patent.copat.gov.az/">https://patent.copat.gov.az/</a> | The Patent and Trademark Examination Center under the Intellectual Property Agency of the Republic of Azerbaijan considers applications for inventions, utility models, industrial designs, trademarks and geographical indications, prepares expert opinions for the issuance of patents and certificates, inventions, utility models, and industrial designs. It is a public legal entity subordinate to the Intellectual Property Agency of the Republic of Azerbaijan, which, in accordance with the law, handles other issues in the field of examination of trademarks and geographical indications, participates in the attestation of patent attorneys, and is involved in the education and intellectual property fields and the formation of an intellectual property culture. |
| Enforcement Center Intellectual Property Rights<br><a href="http://copag.az/">http://copag.az/</a>                                                            | The center under the Copyright Agency of the Republic of Azerbaijan is a public legal entity involved in the implementation of copyright and other related rights, the collective management of property rights, intellectual property education, and the formation of a robust intellectual property culture.                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |

Source: Intellectual Property Agency of the Republic of Azerbaijan; <https://copat.gov.az/en/nav/international-cooperation>

## 2.6 Requirements for Patent Application

The requirements for applications for patent registration are as follows:

- Obtain a patent for an invention, utility model or industrial design any legal or natural person can file a claim for. Claim the document is the applicant himself, his authorized representative or electronically through a patent attorney registered with the Agency in order, or by mail or directly to the Agency must be submitted.
- Pursuant to Part 1 of Article 27 of the Law, Patents and Trademarks of the Agency and its subordinates Officials of the Center of Expertise perform their duties and termination of their employment in those institutions patent application within one year after issuance they cannot give.
- Claim document by foreign legal entities and individuals in international agreements to which the Republic of Azerbaijan is a party unless otherwise provided, only register with the Agency obtained patents are issued through attorneys.

- When a claim is filed through a patent attorney, his/her powers are confirmed by the power of attorney issued by the applicant. A certified copy of a translation into Azerbaijani of a power of attorney written in another language must be attached to the original.
- A power of attorney for several claims of the same plaintiff, as well as any claim that a person is expected to file in the future, may be attributed to documents. If the power of attorney is not submitted, any actions on the part of the representative (patent attorney) concerning the claim will be considered invalid.
- Given by the plaintiff to the representative (or patent attorney) may terminate the power of attorney at any time. Termination of power of attorney notarized application of the bidder It is terminated from the day of entering the agency. Termination of power of attorney Representative (or patent attorney) of the Agency in connection with the issuance informed in writing by. The bidder applied directly
- When submitting to the agency, it requires notarization is not. Representative (or.) From the date of termination of the power of attorney patent attorney) any legal actions on behalf of the applicant cannot perform.
- The plaintiff is new on the claim document based on the power of attorney when appointing a representative or patent attorney, to the same claim document the actions performed with a previously issued power of attorney on their own maintains its force.
- An application must be submitted to obtain a patent. Of the application Sample, Completion and Submission Procedure Agency is determined by.
- If the applicant is a group of individuals, the application is one of them or an authorized representative, or their joint identification must be signed by an authorized representative. To the application attached claim document to an invention or utility model, or its own a group of inventions or utility models that unite them should be attributed.
- The statement of a claim includes the following:
- Complete the essence of the invention or utility model disclosing and creating sufficient clarity for implementation description;

- Based on a description of an invention or utility model and formula expressing the essence;
- To understand the nature of the object of the claim necessary drawings and other materials;
- Appearance of the product on the industrial design a set of pictures of the product, which creates a detailed picture of;
- Necessary to explain the nature of the industrial design if any, drawings of the general appearance of the product, ergonomic scheme, conference map;
- Reflecting important features of the industrial design description;
- Appearance of the product on the industrial design including photographs, images performed by computer graphics, reproduction and other methods pictures;

#### **Abstract**

- The primacy of the invention, utility model and industrial design submit to the Agency the required documents for identification and related procedures to Article 28 of the Law carried out in accordance with.
- The claimant of the invention, utility model and industrial design are two and when there are more natural persons, they are entitled to use the patent between the author's contract concluded between the plaintiff and the employer with the authors who are the creators of the object of patent law employment contract concluded between the employer or the industry service assignment on creation of object of property rights a copy of the contract must be attached to the claim document.
- Creation of an invention, utility model and industrial design and the author's contract and service assignment regarding its use the sample contract is determined by the Agency Requirements of Articles 13, 14 and 15 of the Law in contracts should be taken into account.
- Within 1 month from the date of filing the claim, invention, utility model and industry to the materials of the claim without changing the essence of the sample

have the right to make corrections and additions. Changes expert can also be carried out with the proposal.

- The invention, utility model and changing the nature of the industrial design and the previous of the claim document claim additional materials that do not have features in the materials is not taken into account when considering the document and is independent by the plaintiff can be drawn up as a claim document.
- An application for a patent shall be prepared in the Azerbaijani language. Other materials of the claim document in the Azerbaijani language or can be given in another language. The claim is filed in another language translation of materials into the Azerbaijani language in which the claim is filed must be submitted within 2 months from the date.
- Improve information retrieval and international search in order to integrate into the systems of the Agency "Industrial Property" invention and utility model for publication in the official bulletin the formula and abstract are compiled in Azerbaijani and Russian.
- Full name of the bidder (bidders), author (s) surname, name, patronymic presented in Azerbaijani, Russian and English languages is done.
- International with the participation of the Republic of Azerbaijan by agreements, by a foreign legal entity or individual include an international claim document sent (submitted) to the Agency within 2 months from the date of submission, the applicant shall submit his / her formula, abstract and provide a translation of the description into Azerbaijani.
- A claim drawn up using an electronic signature except for the materials of the document, the invention and the utility model description, formula, invention and useful, compiled in the Azerbaijani language drawing (sketches) and abstract 2 (two) indicating the content of the model in copies, and other documents in 1 (one) copy.
- In the materials included in the claim documents (description, formula, drawing, scheme, picture, abstract, etc. materials) contrary to the interests of society, humanism and moral principles phrases, slang, non-scientific, non-standard and clearly understood the use of incomprehensible terms is not allowed.

- Definitions used in these Requirements World Intellectual Property Organization Industrial Property in the Paris Convention for the Protection of Human Rights, “Patent Cooperation Agreement for the purposes of the Patent Procedure international recognition of microbial deposit on” in the Budapest Treaty, “Industrial Designs The Hague Agreement on International Registration and Eurasia Patent Convention ”.

## 2.7 Status of Patent Experts

The term IP encompasses key fields such as copyright, patents, and trademarks. The links between patents and innovations depend on the related industry and the influenced markets. Importantly, patents have a critical impact on market growth and could either foster or hinder innovation and tech commercialization, which should be taken into consideration when developing and reorganizing the National IP System of Azerbaijan.

IP experts are key actors of the National IP System and include lawyers, scholars, and policymakers. Specifically, a patent attorney is a specialist who is qualified to advise the market on patents and other intellectual property rights, and who assists with the acquisition of patents granted by patent offices around the world. Cooperation with individuals, universities, start-up companies and large corporations is one of the most essential activities of a patent attorney in adapting to the different market needs of the country.

Moreover, there are various IP Law Firms in Azerbaijan staffed with professional patent attorneys and experts who can provide the full range of IP protection services in the Azerbaijan market to diverse companies ranging from small businesses to large corporations.

[Table 3-8] Status of Patent Experts

| Experts                  | Affiliation and Roles                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
|--------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Patent attorney [lawyer] | <p>On the instructions of the principal, a patent attorney prepares, issues, obtains, maintains and performs other legally significant works related to inventions, utility models, industrial designs, trademarks and geographical indications;</p> <p>A patent attorney represents the power of attorney in the implementation of rights to inventions, utility models, industrial designs, trademarks and geographical indications, as well as in the issuance of rights and the conclusion of license agreements;</p> <p>A patent attorney represents the Authorized Property Agency of the Republic of Azerbaijan and its subordinate organizations, courts, and other bodies as a plaintiff or defendant within its powers; and advises the proxy in the field of protection of industrial property.</p> <p>The Patent attorney has the following rights:</p> <ul style="list-style-type: none"> <li>• to receive remuneration from the principal while serving him/her;</li> <li>• to request documents from departments and organizations on his/her behalf in connection with the performance of his/her powers within the scope of his/her powers, and to submit these documents or their copies to the principal;</li> <li>• to establish a patent service enterprise in accordance with the established rules or to engage in professional activities without a legal entity.</li> </ul> |

Source: About Patent Attorneys REGULATION.



### 3. Start-up Policy and Programs

#### 3.1 Business Conditions and the Direction of Economic Policy in Azerbaijan

Based on the World Bank's "Doing Business" ranking, from an external perspective, Azerbaijan's business environment and start-up ecosystem continue to rise in the rankings and have been evaluated as "Good".

Azerbaijan announced 'Azerbaijan 2030: National Priorities for Socio-Economic Development', which presents the country's vision for economic development over the next ten years, emphasizing the development of the non-oil sector and manufacturing industry, and private-led growth to promote it. In order to respond to the new strategic 'Post-COVID19' stage of development etc, the following four priority tasks related to the environment and green growth have been proposed: create a competitive economy that continues to grow; establish a dynamic and inclusive society based on social justice; foster innovative and competitive human capital; and promote return to resettlement in the restored territories. In particular, the non-oil sector should be placed at the center of economic growth in terms of discovering new engines for sustainable growth, deepening economic diversification and export-oriented industrialization, and transforming Azerbaijan into an economy that creates high added value through innovative and creative private initiatives.

As regards the national start-up fund, there is a million dollar question for Azerbaijan, namely, "Which industries should be prioritized for fostering and promoting at the national level?" From a research perspective, it is arguable that the best way would be to pick the winners based on their industry rather than providing a "level playing field" for all start-ups in the country. In reality, however, hardware start-ups face far greater challenges compared to the software and service industries. For example, fintech is growing slowly, but there are few start-ups (e.g. BankOff, Payriff, OnePoint, etc.). Start-ups in fintech usually offer loyalty programs, financial aggregators, etc. E-commerce is also one of the most attractive fields

for certain start-ups (e.g. Nextsale, 11delivery, etc.). The challenge facing e-commerce and marketplaces in Azerbaijan largely originates from the weak local manufacturing industries, compared to larger economies. Turkey's organized its first "Teknofest" event in Azerbaijan in 2022 as the Teknofest 2022: Aviation, Space and Technology Festival. Considering the success of the Turkish defence industry, the same founding team's interest in Azerbaijan could also foster defence-tech industry start-ups in the near future.

From a financial perspective, it is difficult for start-ups to obtain or attract game-changing investments. Therefore, incubation and acceleration centers usually work with industry-leading companies to organize joint programs so that their start-ups can obtain funding from companies at the end of their journeys. For example, INNOLAND organized such a program with Azergold, a local gold-mining agency, in a bid to find start-ups in mining, geology, and similar fields. However, such programs usually have a limited budget and do not distribute the actual amount of funds raised to the winning start-ups.

Regarding the total number of start-ups in Azerbaijan, it is important to note that there is not a single database which encompasses all the statistics about all start-up companies and their staff. Small and Medium Business Development Agency (KOBIA), under the Ministry of Economy, grants start-up certificates and lists them on their webpage. So far, forty-three start-up companies have been granted such certificates. In reality, however, this figure is selective and does not cover the whole picture. As such, the relevant authorities plan to collect such statistics in the recent future. The main obstacles facing start-ups in Azerbaijan can be listed as follows:

[Table 3-9] Difficulties Facing Start-ups in Azerbaijan

| Factor  | Limitations                                                                                                                                                                                                                                                                                                                                    |
|---------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Culture | <ul style="list-style-type: none"> <li>• Lack of family encourages and supports the development of entrepreneurship (a large number of approaches hinder entrepreneurial behavior).</li> <li>• Low need for innovation.</li> <li>• Insufficient integration of the country's population into the information dynamics of the world.</li> </ul> |

| Factor          | Limitations                                                                                                                                                                                                                                                                                                                                                                            |
|-----------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                 | <ul style="list-style-type: none"> <li>• the safer jobs and social benefits of the public sector still play a significant role in taking risks among youth (the unwillingness of youth take risks).</li> </ul>                                                                                                                                                                         |
| Human Resources | <ul style="list-style-type: none"> <li>• Lack of young people with sufficient knowledge, skills and abilities.</li> <li>• Weak approach to teamwork.</li> <li>• Little emphasis on specialist management.</li> </ul>                                                                                                                                                                   |
| Technology      | <ul style="list-style-type: none"> <li>• Weaknesses of strategic approaches to the use of potential resources, which have a scientific basis, and unwillingness to follow global technology trends.</li> </ul>                                                                                                                                                                         |
| Market          | <ul style="list-style-type: none"> <li>• Small domestic market.</li> <li>• Poor development of value-added sectors.</li> <li>• Lack of attention to long-term strategic approaches in big business.</li> <li>• Lack of environment to support the development of export potential products.</li> </ul>                                                                                 |
| Finance         | <ul style="list-style-type: none"> <li>• Preference for traditional methods in the investment environment.</li> <li>• Weaker domestic financial markets in which start-ups and innovation projects can aim to get listed</li> <li>• Lack of funds and legal basis (no venture fund until 2022).</li> </ul>                                                                             |
| Environment     | <ul style="list-style-type: none"> <li>• Lack of comprehensive incentives and supportive legislation.</li> <li>• Poor development of business partnerships.</li> <li>• Preference for non-professional approaches to business management.</li> <li>• Low interest in mass production among manufacturing enterprises.</li> <li>• Low interest in deep business development.</li> </ul> |

Source: Bunyamin Seyidov and Isa Gasimov (2022) "Current Status of Azerbaijan"

In addition, according to KOTRA's 2018 report on overseas market research, start-up and incubator programs are being operated by the government, but in many cases they are still operated as exhibitions due to the poor business environment. It was found that very few start-ups are discovered through this program. The local circumstances that hinder the development of start-ups are as follows:

- ① (Closed economy) Due to Azerbaijan's close geographical proximity to Russia and Turkey and its small market, foreign companies show little desire to enter the market. It has a closed economic structure in which capital liquidity is insufficient in the private sector and capital is concentrated in only a few companies.

- ② (Talent outflow) As the vested interests are very dominant and there is no background in a society centered on blood ties and delays, it is difficult to be incorporated into the vested interests, so many excellent talents head overseas.
- ③ (Conglomerate-oriented economy) Since high-ranking public officials implicitly tolerate company ownership, large corporations owned by high-ranking officials have greater dominance in the industry, making it difficult for new start-ups to emerge.
- ④ (Tax burden) Tax audits of general private companies are frequent and taxation is high. In many cases, companies are run by voice.
- ⑤ (Education system) The overall quality of education is low due to the low salaries of faculty members and poor educational facilities.

As a result, while the transition to a sustainable growth structure in terms of the industry and export structure is still considered insignificant, the diversification of manufactured goods and the rapid increase in exports of non-petroleum products are positive in terms of economic diversification.

### 3.2 Azerbaijan's SEM support policy

Azerbaijan launched the National Fund for Entrepreneurship Support in 1992 as an agency affiliated with the Ministry of Economy, Trade and Industry. Its core function is to provide necessary information or technology to founders, and its main goal is to provide financial support for start-ups at low interest rates. (The interest rate is 6%, which is very low compared to the market interest rate of 17-18%) The fund is mainly invested in greenhouse complexes, agricultural product distribution centers, grain storage and processing facilities, and livestock product distribution centers. Zero percent is invested in the agricultural sector. The fund amounted to approximately US \$1.5 billion from its establishment until 2014, but it is now known to amount to approximately \$350 million per year.

Azerbaijan's industry has developed with the main focus on the oil and gas industry, as befits the country's nickname, 'Land of Fire'. As of 2019, the oil and gas industry accounted for

about 38% of GDP and 90% of exports. However, due to the drop in oil prices since 2015, Azerbaijan's currency, the Manat, has been devalued more than twice, and the limitations of the resource-dependent economic structure have been sharply felt. As such, various policy changes were attempted with an emphasis on fostering start-ups and small and medium-sized enterprises (SMEs).

In this regard, the Azerbaijani government announced the "Strategic Road Maps for the National Economy and the Main Economic Sectors" in December 2016, which are focused on the development of the non-oil industry and the fostering of small and medium-sized enterprises (SMEs). The main aim is to strengthen the country's capacity for economic diversification, including the downstream development of the oil and gas industry from a simple resource export, the development of the agricultural, medical, pharmaceutical, and tourism industries, and the fostering of small and medium-sized enterprises (SMEs).

Among them, the Agency for Small and Medium Business Development of Azerbaijan was established in December 2017 as a measure for fostering small and medium-sized enterprises, and along with the ASAN Center. Azerbaijan has also established a business center for small and medium-sized enterprises (SMEs) to provide education, consulting, and financial and administrative services to small and medium-sized enterprises (SMEs).

In March 2018, financial support for start-ups was provided through the Youth Fund, while the tax law was amended to exempt companies from income tax for three years (i.e. companies that receive a Start-up Certificate).

In August 2018, the Entrepreneurship Development Fund, established under the Ministry of Economy, became independent, and a budget of 1.1 billion manat (about US\$ 645 million/KRW 807.6 billion) was allocated to private companies in the non-oil industry. This fund is being used in a variety of ways, from educating entrepreneurs to providing financial support for promoted projects. The industries receiving the most by order of priority are the agriculture, tourism, information and communication, and manufacturing industries. In addition, the Ministry of Transport and Information and Communication supports the holding of a start-up contest jointly with the Heydar Aliyev Foundation.

### 3.3 Azerbaijan's start-up programs and potential by sector

The majority of start-up and incubator support programs are led by the government, and the representative programs of the following institutions are currently being executed.

Moreover, there are newly emerging innovation centers and units within corporations, such as Pasha Holding and the International Bank of Azerbaijan OJSC. Such corporations organize and support incubation programs and attract start-ups mainly, but not only, for their own corporate needs. Their innovation programs can be categorized as follows:

- Intrapreneurship by which they attract ideas internally;
- Entrepreneurship by which they attract ideas from outside the organization.

However, these corporate sector players have not actively engaged in technology commercialization or made any significant attempts to do so.

- ① Start-up Azerbaijan: An organization established by the Ministry of Transport and Information and Communication to promote start-ups and strengthen links with new start-ups. [www.start-up.az](http://www.start-up.az)
- ② Azerbaijan Start-up Ecosystem: A site that collects comprehensive information about the start-up ecology for young entrepreneurs. [www.start-upazerbaijan.az](http://www.start-upazerbaijan.az)
- ③ Next Step Innovation Center ([www.icnextstep.com](http://www.icnextstep.com)): An organization that provides funds to the founders of start-ups who have completed the education provided by the organization ranging from product concept to prototype development. With an office in Baku City, office space is provided to selected prospective founders.
- ④ Social Innovation Lab SIL (Social Innovation Lab SIL): A research institute established to nurture fifty technology start-ups every year until 2025, providing founder education and overseas market entry services. [www.socialinnovationlab.az](http://www.socialinnovationlab.az)
- ⑤ SUP ([www.sup.az](http://www.sup.az)): An intensive training program for nurturing start-ups. It provides a 3-month education course on product invention and marketing practices.
- ⑥ Support for start-up programs of local telecommunications companies. Barama ([www.barama.az](http://www.barama.az)): Start-up support project of Azercell Telecom, a telecom operator, Bakcell AppLab ([www.applab.bakcell.com](http://www.applab.bakcell.com)): A program by Bakcell, a telecom operator, to support start-up founders' mobile app development.

- ⑦ University Start-up Program: Start-up support centers are based in Azerbaijan National Petroleum University, ADA University, etc.

Accurate information on the number of start-ups is not recorded in official figures. <https://www.smb.gov.az/en/nav/startap-sehadetnamesi> provides a list of start-ups that have received a "Start-up Certificate". The list includes start-up certificates for thirty-six organizations covering the period 2021-2022.

Additional information on the number of start-ups: The number of start-up projects registered in the "New Ideas" start-up competition held to support start-ups is as follows. Thus, 977 start-up projects have so far applied to participate in the "New Ideas" start-up competition, which has been held six times since 2013 (including 87 start-up projects in 2013, 212 in 2014, 196 in 2015, 199 in 2016, 148 in 2017, 135 in 2019.<sup>38</sup>) The Innovation Agency, which operates under the Ministry of Transport, Communications and High Technologies of Azerbaijan, conducted an average distribution of 100 start-ups in the fields of education, e-commerce, transport, medicine, tourism, defence industry, and others.

[Table 3-10] Percentage of Start-ups by Sector

| No. | Branches                | Average interest rate |
|-----|-------------------------|-----------------------|
| 1   | Education               | 12.3%                 |
| 2   | e-Trade                 | 11.4%                 |
| 3   | Social media            | 10.2%                 |
| 4   | Social                  | 8.9%                  |
| 5   | Media and advertisement | 7.6%                  |
| 6   | Entertainment           | 6.4%                  |
| 7   | Transport/logistics     | 5.1%                  |
| 8   | Finance                 | 5.1%                  |
| 9   | Agrarian                | 5.1%                  |
| 10  | Smart city              | 3.8%                  |

<sup>38</sup> <https://yenifikir.az/en>

| No. | Branches                               | Average interest rate |
|-----|----------------------------------------|-----------------------|
| 11  | Design/art                             | 3.4%                  |
| 12  | Productivity and security              | 2.5%                  |
| 13  | Infrastructure and security            | 1.7%                  |
| 14  | Medical                                | 1.7%                  |
| 15  | Tourism                                | 1.3%                  |
| 16  | Alternative energy                     | 0.8%                  |
| 17  | Data                                   | 0.8%                  |
| 18  | Automobiles and mechanical engineering | 0.8%                  |
| 19  | Defense industry                       | 0.4%                  |
| 20  | Others                                 | 10.6%                 |

Source: Bunyamin Seyidov and Isa Gasimov (2022) "Current Status of Azerbaijan"

## 3.4 Start-up Activities in Azerbaijan

### 3.4.1 New Ideas Contest<sup>39</sup>

#### 3.4.1.1 What is the New Ideas Contest?

The New Ideas Contest is a project organized by the former Qafqaz University Techno park (QU Techno park). Originally established at QU Techno park in 2013, the contest is currently held once a year in the Techno park of Baku Engineering University to support the development of ecosystems and the development of innovative entrepreneurship. At the same time, it aims to encourage entrepreneurship among young people and support and assist them with the development of their start-up projects. The New Idea LLC acts as a real bridge between the world of work and enterprising young people, as well as investors. The competition is open not only to members of the Baku Engineering University, but also to anyone living in the Republic of Azerbaijan.

<sup>39</sup> <http://yenifikir.az/>



### **3.4.1.2 Mission Objectives**

- To support innovative entrepreneurship.
- To prepare the groundwork for the formation of an ecosystem in the Techno park, which is important for the development of high technologies.
- To support the sustainable socio-economic development of the country.
- To support the development of ICT.
- To assist young entrepreneurs in setting up their own companies.
- To develop an entrepreneurial approach.

### **3.4.1.3 Purpose**

The purpose of this competition is to strengthen the innovation economy by reinforcing the ecosystem in the BMU Techno park and contributing to the emergence and application of innovative ideas in the field of technology. Its main targets are as follows: Start-ups, mobile applications, internet, software, games, software as a service, corporate software, new business models, services, robotics, information technology and consumer electronics, and health (health) information technology, etc.

### **3.4.1.4 Target**

- Formation of internationally competitive companies that can play a role in the development of the country and the region.
- Development of high technologies.
- Development of entrepreneurial approaches and innovation.
- Incorporation of business ideas that can turn a product into money.
- Support for the development of intellectual properties.

### **3.4.1.5 Forms of support for project development**

- Initial financial support (at least 40 start-ups received grants of approx. 1,000 to 10,000 AZN. Moreover, the total amount of financial support, including grants and prizes, was about 150,000 AZN).

- Support for finding seed investment. (The investment fund of the New Ideas Contest was arranged with the financial support of its stakeholders.)
- Free use of the BMU Techno park Incubation Center for 1 year.
- Mentoring support and training needed to promote an idea.
- Support to find investors for companies with a promising business idea.
- Study of international experience of companies with a promising business idea.
- International market integration.
- Media and lobbying support.

#### **3.4.1.6 Who is eligible to participate in the competition?**

Those who want to build a business with a good business idea.

#### **3.4.1.7 Evaluation Criteria**

- Innovation and technological innovation: What are the technologically innovative features, efficiency, new and technological advantages of an idea or project?
- Market strategy and competitiveness: Buyers of a product or service, reasons for buying, characteristics of the target market, market size and development potential, competitors and market valuation approaches.
- Teamwork and management: The skills and experience of the project team, the role of each member in the project, any staff shortcomings related to the project and how to overcome them.
- Presentation quality: Preparation and skills required to present a project (or business idea) in a short time (3-7 minutes).
- Business plan: Specificity, work opportunities and chronology, work methodology, financial planning, reliability of analysis and forecasts based on research results.
- Supported projects.

So far, 694 projects have been submitted to the New Ideas Contest, which has been held

four times so far. The development of twenty-nine of these projects was supported by the QU techno park (2013-2016). Although most of the projects collapsed after the competition, the supported teams were able to obtain the desired investment by establishing new start-ups in the same name in local - and several of them in foreign - markets. For instance, "WASCO", HypoLife, Trainer for Accountants, Braille Pad, SOTEQA LAB, "11delivery.com", Spinal and other projects managed to attract investments of 100,000 dollars or more in the local and global markets.

#### **3.4.1.8 What is the Start-up Azerbaijan Competition?**

The main goal of the Start-up Azerbaijan project is to improve the investment climate in the country's ICT sector, identify the best Internet and IT projects, and help transform innovative ideas into successful businesses. An innovative idea is an activity aimed at presenting a product or service through the effective use of information technology. The ultimate goal of the submitted projects should be to build a real business, provide services, including social services, and attract customers. It is possible to propose projects related to any existing business ideas, in which case the proposed project should be based on more effective innovations. Projects should not be of the research type. Such projects are not accepted for competitions and other contests held within the "Start-up Azerbaijan" initiative. Business projects can be submitted by a team of one or more people. Projects that can be used both globally and locally can be submitted to the competition. Expert evaluations of start-ups participating in the competitions is organized within the framework of "Start-up Azerbaijan". Start-up Azerbaijan is not just a competition but rather a project in which training is arranged for start-ups and meetings are held with investors. Start-ups are helped to form a team. For those who want to start an Internet business, training is held within the framework of the project in order to provide opportunities to gain knowledge and experience of starting and running a successful business.

The Start-up Azerbaijan team assists selected start-ups in presenting successful projects to investors and in finding investors, as well as providing investors with access to attractive projects. It also organizes start-up meetings and discussions to achieve the set goals. Although Start-up Azerbaijan presents symbolic prizes to the winners, the projects are

directly funded by presentations to investors and various funds. Start-up Azerbaijan provides financial and support for successful internet projects, meetings with investors, and internships with the support of local and foreign experts. Start-up Azerbaijan aims to help those who want to create a new Internet business to realize their dreams, attract investment, and manage their business.

### **3.4.1.9 Evaluation criteria**

- How attractive is the project financially and realistically?
- How well does the team know the market and business features?
- How clear and understandable is the business plan or review?
- Can the proposed structure-project be a market leader?
- Was the team able to create a prototype of the project?
- How successful and experienced is the team?
- Do one or more team members have the skills needed to promote ideas or services?
- Does the team have experience of working with investments or any specific plans in this regard?
- Approximately how long can a project survive in the market?

### **3.4.1.10 Latest activities**

Currently, the “Yeni Fikir” start-up competition has an experienced team of professionals. In addition, two Incubation Centers operate independently for the development of promising start-ups (one is located in the Techno park of Baku Engineering University, the other at Baku, Nakhchivani Str., 10), and have established a business and social network with a wide audience. The last “Yeni Fikir” start-up competition was held in 2019 with the financial support of the company BP Caspian, and in a partnership with the Ministry of Economy, Baku Engineering University, and Asan Innovation Center (Innoland).

Generally speaking, various organizations, including Qafqaz University (2013-2016), Baku Engineering University (2017-2019), Ministry of Education, Ministry of Economy, NRYTN,

ASAN (including Asan Innovation Center and Innoland), ANAS YTP, and private companies and organizations (as Pasha Bank OJSC, Baku Holiday Travel, Özsüt, Vitam, etc.), act as investors and promoters of the competition in different years. Nowadays, “Yeni Fikir” LLC cooperates with various university and scientific research institutions, as well as the Free Economic Zones Agency, KOBIA, and private companies. Moreover, the «New Idea» Incubation Center <sup>40</sup> was formed to offer intellectual services and incubation services, including training, mentoring, research, analysis, investment, competition, networking, etc. for the development of the innovation ecosystem in Azerbaijan.

### **3.4.2 About the Barama Innovation and Entrepreneurship Center<sup>41</sup>**

#### **3.4.2.1 What is the Barama project?**

The Barama Innovation and Entrepreneurship Center, which has been operating since 2009, was the first center to be established in the country by a private telecommunication company, Azercell. By helping local start-ups to emerge, it creates favourable conditions for the development of both technological literacy and entrepreneurial skills among young people.

#### **3.4.2.2 Mission Objectives**

- Support the development of small and medium enterprises by young people.
- Discover new ideas in the field of high technology.
- Strengthen the position of Azerbaijan in the world in the field of innovation.
- Form a start-up environment.
- Support projects in the ICT sector.

#### **3.4.2.3 Purpose**

The main goal of the Barama Incubation Center, which has achieved many successes during its six years of operation, is to increase the number of interesting projects by strengthening the development of the start-up ecosystem.

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<sup>40</sup> <https://yenifikir.az/en/>, 10.02.2022

<sup>41</sup> <http://barama.az/>

#### **3.4.2.4 Goals**

- Development of high technologies.
- Creation of a favourable business environment for new projects.
- Transformation of projects into profitable companies.
- Support for new public-private partnership projects.

#### **3.4.2.5 Forms of support for project development**

- Operation of innovation offices on a 24/7 basis.
- Types of technical support needed for computers, internet and projects.
- Individual incubation program designed specifically to turn a start-up into a business.
- ICT, law, finance, business development, strategy, partnership and marketing consultations by Azercell.
- Marketing consultations.
- Telecommunication support by Azercell.
- Close cooperation with local and foreign mentoring teams from Azerbaijan, the USA, Canada, Great Britain, Russia and Turkey.
- Barama Business Alliance.
- Organization of partnership meetings for projects that have completed the incubation program.
- Organization of meetings of investors with projects that have completed the incubation program.
- Formation of close relationships with our business incubator partners from around the world.

#### **3.4.2.6 Evaluation Criteria**

- The purpose of the project and how innovative the work will be.
- The difference between a product or service and a project that is currently on the market and doing similar work.

- The project solves a current problem that people face.
- Explanation of business model.
- Project members, the role of each member in the project, a few sentences to express the greatest success of the project members in their pre-start-up activities.
- Project presentation.

#### **3.4.2.7 Supported projects**

Thus far, more than thirty projects have been launched at the center, of which twenty have been successfully completed and four successful companies have been established, while one project has been established in the state of Virginia, United States of America. (Yavar 2016)

#### **3.4.2.8 Successful projects**

- |                                  |                                                  |
|----------------------------------|--------------------------------------------------|
| • Mysim.az                       | • StrikePlagiarism.com (anti-plagiarism program) |
| • HiAzerbaijan App               | • MobiStar                                       |
| • Sumaks (Smart home technology) | • Technollland.com                               |
| • Podcast app                    | • CyberLegion                                    |
| • NeWo                           | • Yollan.com                                     |
| • Architect Stone                | • Technote                                       |
| • Lisdo                          | • WoWoman                                        |
| • Sound book                     | • Mrs. Tech-Tech                                 |
| • Barama Media                   | • E-tap                                          |
| • Konullu.net                    | • Mr. Bee                                        |
| • Binar Studio                   | • Education.Online                               |
| • Adbox.az                       |                                                  |

#### **3.4.2.9 Latest activity**

Currently, the Barama Innovation and Entrepreneurship Center mainly focuses on intrapreneurship with the application of successful internal start-ups in AzerCell activity or their commercialization overseas.

### 3.4.3 Won and Realize Your Idea Competition<sup>42</sup>

The idea won the start of a business TV contest on television, which gave start-ups a colourful life in Azerbaijan. The main partners of the project, which is implemented by the Azerbaijan-Turkey Youth Friendship Society, are the Ministry of Youth and Sports of the Republic of Azerbaijan, the Youth Foundation under the President of the Republic of Azerbaijan, and the Youth INC program. Although there have been some TV shows and scandals in the competition, it can be considered a successful step in the development of start-ups and new businesses in general. Given that a large part of the public watches TV programs rather than reading news and blogs, one can say that the competition will help motivate young people in this way, while quickly and easily conveying the logic of start-ups to people.<sup>43</sup> "Realize your idea" is a project competition organized by the "Next Step" Innovation Center, which was established on May 15, 2015, and the first "Realize Your Idea" project competition was launched at that time. This project aims to encourage and support the development of innovative entrepreneurship, particularly among young people. The Center also provides access to the world market for enterprising young people, and serves as a channel of communication with the business world. Held twice a year, the "Realize Your Idea" Project Competition is open to anyone living in the Republic of Azerbaijan, regardless of one's nationality.

#### 3.4.3.1 Mission

- The Next Step Innovation Center supports innovative projects to develop ecosystems and increase local production in Azerbaijan.
- To support the development of information and communication technologies in the country and the education of people in this field.
- To create conditions for young people who want to turn their innovative ideas into a product or service.

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<sup>42</sup> <https://www.facebook.com/icnextstep/?pnref=lhc>, 19.01.2016.

<sup>43</sup> <http://ideyaqazandi.az/>, 26.05.2014.



### **3.4.3.2 Purpose**

The purpose of NEXT STEP Innovation Center's "Realize Your Idea" project competition is to provide material, technical, legal and other forms of assistance to people who have innovative ideas and want to produce a new product or service and increase the production capacity of Azerbaijan. The Center provides citizens with fully equipped study rooms, an electronics laboratory, a meeting room, and other facilities.

### **3.4.3.3 Forms of support for project development**

- Financial support (scholarship).
- Office conditions (office, furniture, internet, tea and coffee) during three months of acceleration at the Next Step Innovation Center.
- Mentoring support and training required to promote an idea.
- Support to find an investor in a company with a promising business idea.
- Study of international experience of companies with a strong business idea.
- Integration into international markets.
- Media and lobbying support.

### **3.4.3.4 Who is eligible to participate in the competition?**

Anyone who is a resident of Azerbaijan and has an innovative mind-set.

### **3.4.3.5 Evaluation Criteria**

- Innovativeness: An innovative product or service which a promising idea or project could bring to the market.
- Market research and competitiveness: Market demand for a product or service that will emerge at the end of the project, the mass of buyers, the approximate price, and the target income. Basically, competitiveness against alternative products on the market.

- Team & teamwork: Detailed description of the education, experience, skills and work to be done by the project executors, including other members such as the supervisor.
- Quality of presentation: Complete and successful delivery of the project and the goals and plan to the judges within the 10 minutes allotted for the presentation.

#### **3.4.3.6 Supported projects**

So far, 125 projects have applied to the Next Step Innovation Centre, of which five projects have received developmental supported from the Centre.

#### **3.4.3.7 Winning projects**

Winner of the 1<sup>st</sup> "Realize Your Idea" Project Competition:

- BakuParking

Winners of the 2<sup>nd</sup> "Realize Your Idea" Project Competition:

- Smart Wristlet
- WibeMe
- USTAM
- FoodLook

### **3.4.4 IdealHub Innovation and Technology Center<sup>44</sup>**

In addition, the IdealHub Innovation and Technology Center, which was established at the beginning of 2022, is a private innovation and technological center that aims to become a key player in creating and supporting the innovation ecosystem in Azerbaijan.

#### **3.4.4.1 Mission**

- Support and development of innovative projects.
- Assistance for the global growth of innovative projects.
- Development of the start-up environment.

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<sup>44</sup> <https://www.facebook.com/idealhubitcenter/>

- Assistance for start-ups and innovative small companies with identifying ways to help them to grow and innovate.

#### **3.4.4.2 Vision**

The vision of the IdealHub is to become the number one launch platform for technology projects seeking to enter the global market.

#### **3.4.4.3 Target**

IdealHub helps start-ups and researchers to realize their dreams, and provides clients with high-quality products and services that surpass their expectations, thereby contributing to the creation of a better society and innovation ecosystem for our partners and other private and public stakeholders.

#### **3.4.4.4 Forms of support for project development**

- Individual approach to training on IT skills.
- Support for emerging local start-ups including opportunities for access to international markets.
- Development and transformation of new ideas into actual products.
- Technical skills training and consultative assistance on IT development.
- Mentorship opportunities with prominent players in the industry.
- Investment attraction to guarantee growth and commercialization.
- Commercialization of local innovative ideas in local and international markets.

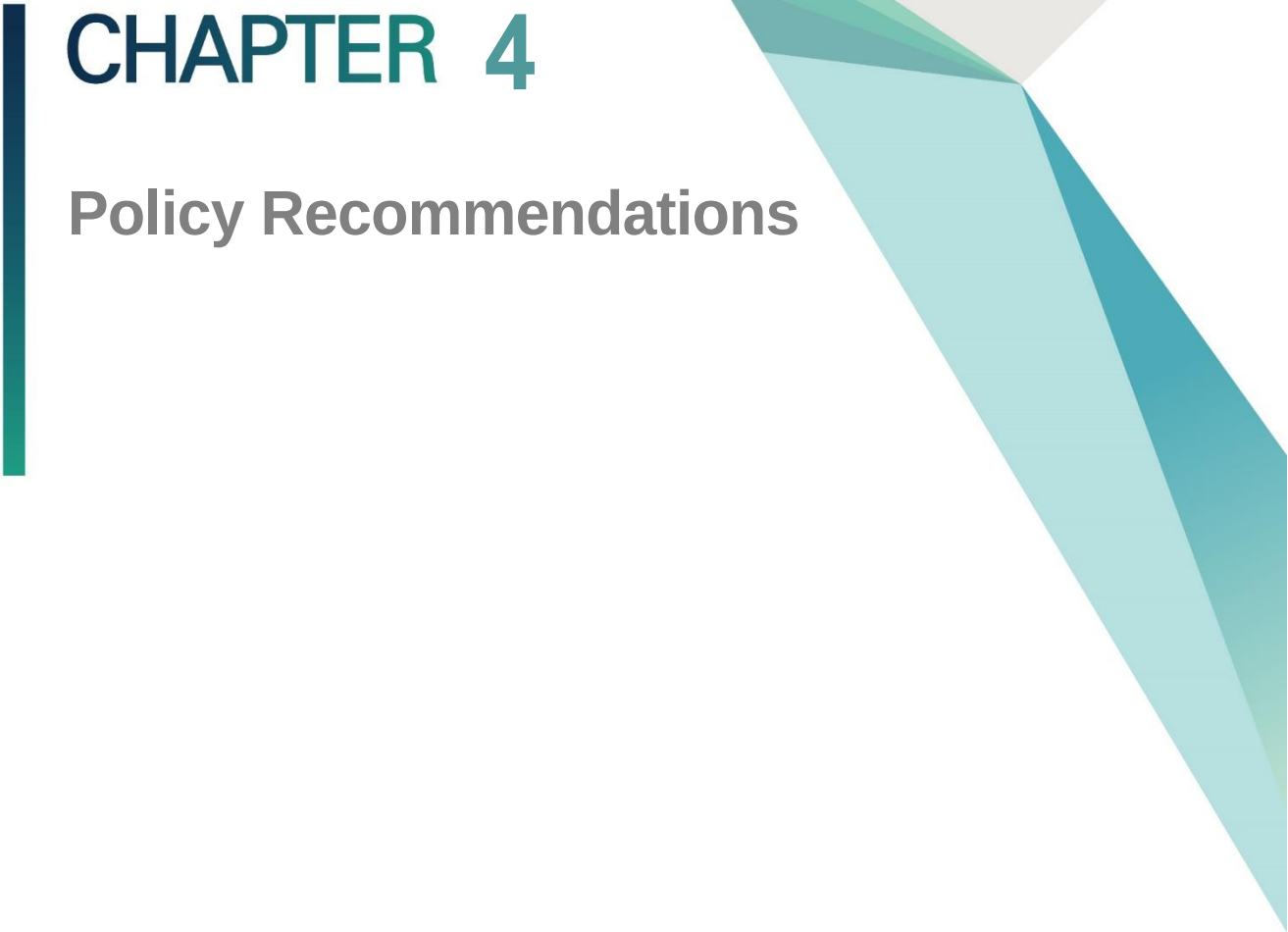
#### **3.4.4.5 Activity**

The **Learn & Earn program** offered by the IdealHub Innovation and Technology Center consists of global IT courses taught by international experts. The program allows successful graduates to build their future careers at EPAM, an American company with more than 60,000 employees worldwide, without having to leave Azerbaijan. The courses consist of various IT platforms, testing methodologies, and training for programs.

In addition, there is a number of newly emerging innovation centers/units based within corporations, such as Pasha Holding and the International Bank of Azerbaijan OJSC. Such corporations organize and support incubation programs and attract start-ups mainly to meet their own corporate needs (but not only). As such, their innovation programs can be categorized as follows:

- Intrapreneurship in which they attract ideas internally.
- Entrepreneurship in which they attract ideas from outside the organization.

However, these corporate sector players have yet to engage actively in technology commercialization or make any significant attempts to do so.



# CHAPTER 4

## Policy Recommendations





## Chapter 4. Policy Recommendations

### 1. Technology Commercialization Ecosystem

#### 1.1 Mid-Long Term Scenario

Technology commercialization is the process of connecting technologies, products or services, and markets. In other terms, it is a process by which various obstacles are overcome in order to introduce the immature technologies produced by a laboratory into the market.

Therefore, the commercialization of a technology basically requires technologies, financial resources, and human resources, but also needs to be harmonized specific models must be designed differently according to a given environment (e.g. governance, institutions, consensus, etc.).

From this perspective, two scenarios are conceivable based on the existing technology commercialization environment in Azerbaijan: One is the main platform model for national-level integrated support by pan-ministries, and the other is the specialized organization model dedicated to technology commercialization by the HTP under the Ministry of Science & Education.

##### 1.1.1 Scenario A: AZ InnoCom (tentative name)

AZ InnoCom is the main control tower of, and serves as the central axis for, national technology commercialization in Azerbaijan. In reality, it is a pan-ministerial promotion committee for technology commercialization, and thus forms part of the current 'state companies and state services' (including the State Migration Service, State Border Service,

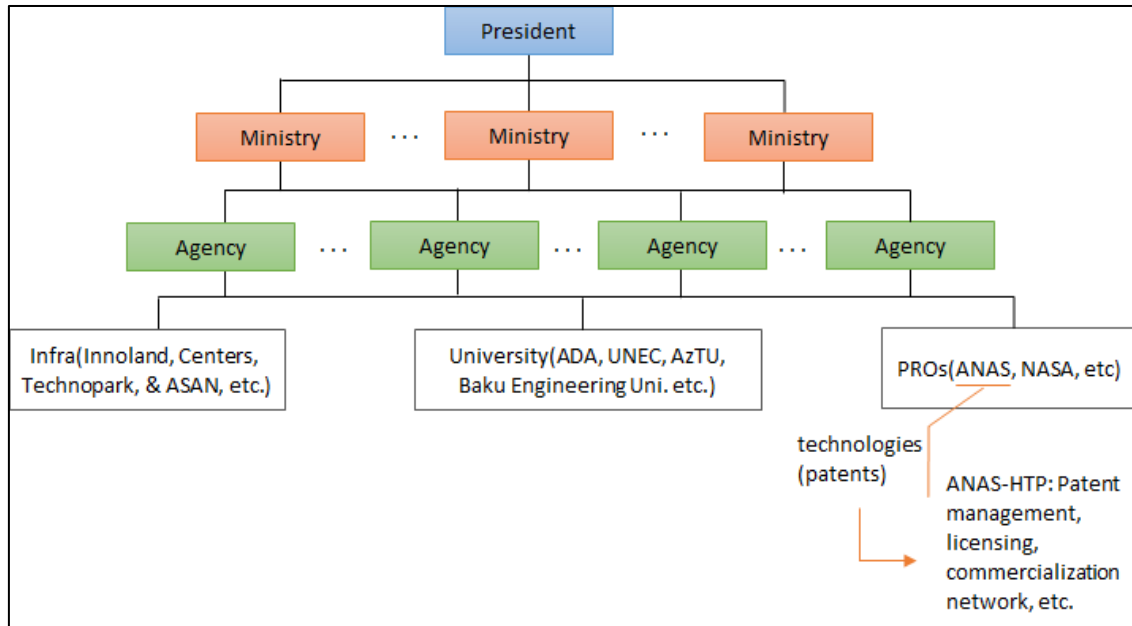
State Security Service, Foreign Intelligence Service, State Mobilization and Conscription Service, Food Safety Agency, and State Tourism Agency).

### **1.1.1.1 Current Governance for Technology Commercialization**

Azerbaijan is already striving to create a favorable environment for innovation. For example, it has ministries that support innovation and provide governmental R&D support, as well as various related agencies (e.g. IDDA) and innovation clusters (e.g. Innoland), etc. Furthermore, on April 6, 2021, the President of the Republic of Azerbaijan signed the Decree on the “Establishment of a Center for Commercialization and Technology Transfer” under the Intellectual Property Agency of the Republic of Azerbaijan. Thus, the Center has been established as a legal entity of public law whose remit is to establish communication, promote mutual order and coordinate proposals between science and industry; and promote technology transfer to ensure the qualitative growth of inventive and innovative activities. In addition, its adoption of an applied nature and commercialization of the results of intellectual activity serve to support innovative development, including start-up projects aimed at transferring the results of scientific research for use for commercial and non-commercial purposes, as well as generating income from these results, providing clients with services for the supply of technologies and projects to their target markets, and coordinating activities in this area (from a report drawn up by Azerbaijan local experts). The HTP was established to transfer and commercialize ANAS’s technology, while universities such as ADA, UNEC, etc. have set up organizations dedicated to technology commercialization.

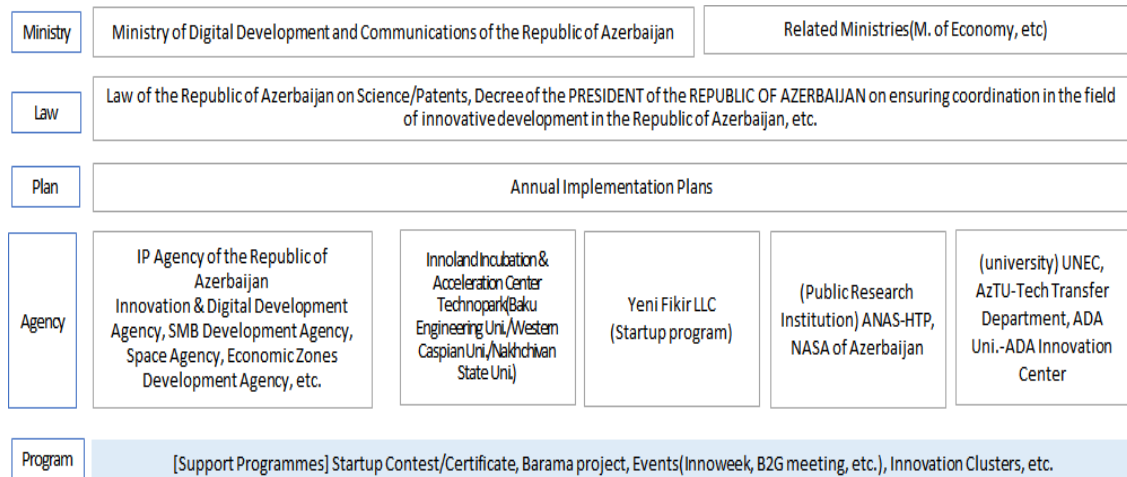


[Figure 4-1] Governance for Innovation (AS-IS)



Source: Author's elaboration

[Figure 4-2] Current governance for technology commercialization in Azerbaijan



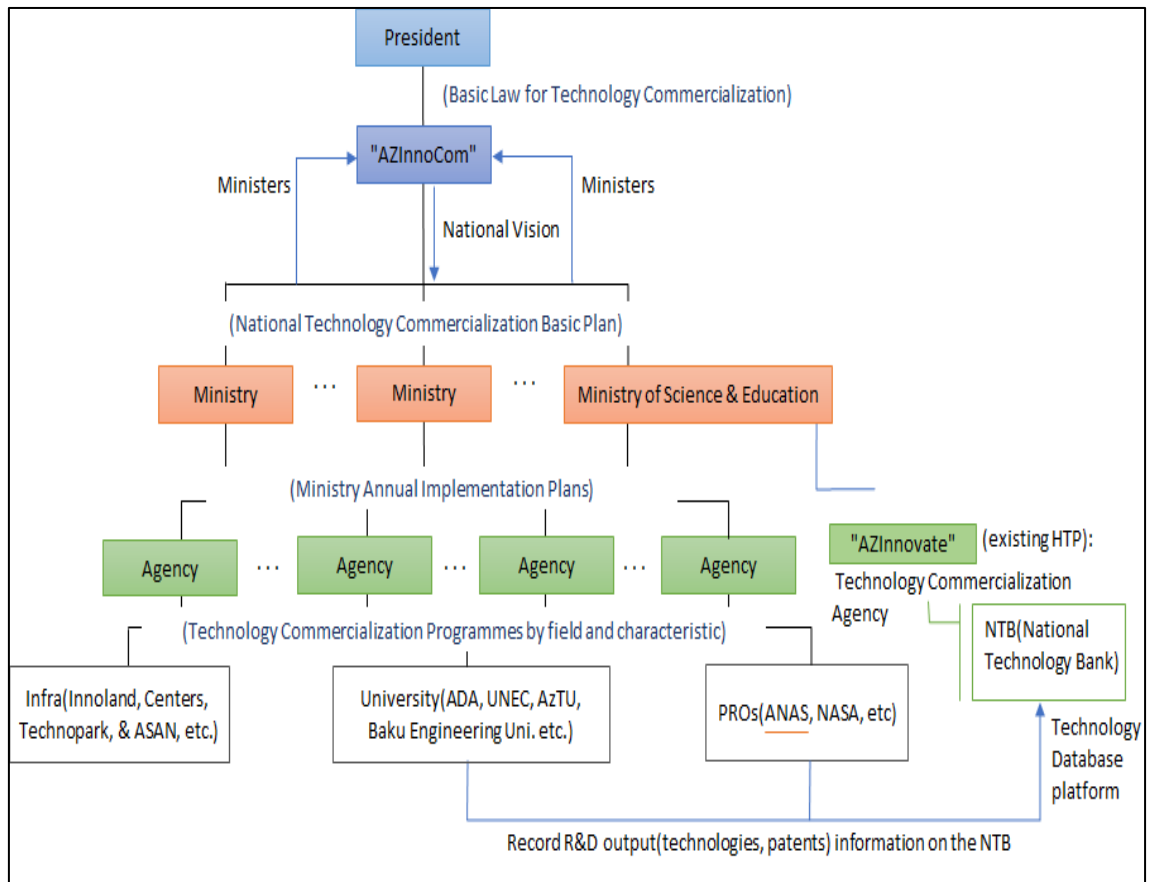
Source: Author's elaboration

### 1.1.1.2 Governance reorganization for Technology Commercialization

The basic pillars of the policy are based on the relevant laws, as follows:

- **(Law)** Technology Commercialization Act or Innovation Act. The competent ministries of this law are economic and industrial-related ministries, such as the Ministry of Economy and the Ministry of Digital Development and Communications of the Republic of Azerbaijan.
- **(Control Tower)** Policy governance is structured around the control tower created by pan-governmental linkage. The control tower, a decision-making body, is a joint committee of the relevant ministries, named AZ InnoCom. The basic law will specify the establishment of the committee. The membership of the committee is composed of ministers from each department. Its members design national strategies and set the roles, tasks and budgets of each ministry for the next five years, and each ministry in turn establishes its annual action plans and projects.
- **(Plan)** The committee sets the government's mid-term strategy for technology commercialization and adjusts its direction every five years. An expert group is established to develop an appropriate plan. Each ministry recommends two experts – including foreign experts - with a wealth of experience in such fields as markets, enterprises, research, investment, brokerage, and incubation (scale-up, demonstration, etc.). In addition, an internal and external innovation trend research team has been established as a permanent planning body. Each ministry checks the implementation of its plan every two years, and submits a report on its progress and performance to AZ InnoCom, which in turn reviews and revises the plan if necessary.

[Figure 4-3] Governance for Innovation (TO-BE)



Source: Author's elaboration

[Figure 4-4] Governance proposal for technology commercialization in Azerbaijan

|               |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |                                                                                                                    |                                                                                                                                          |                                                                                                                           |                                                                                            |
|---------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------|
| Law           | 『Technology Commercialization Act』 or 『Innovation Act』                                                                                                                                                                                                                                                                                                                                                                                                                                                         |                                                                                                                    |                                                                                                                                          |                                                                                                                           |                                                                                            |
| Control Tower | Pan-Ministries Committee for Innovation of Azerbaijan (AZ InnoCom)                                                                                                                                                                                                                                                                                                                                                                                                                                             |                                                                                                                    |                                                                                                                                          |                                                                                                                           |                                                                                            |
| Plan          | National Technology Commercialization Basic Plan (every 5 years)                                                                                                                                                                                                                                                                                                                                                                                                                                               |                                                                                                                    |                                                                                                                                          |                                                                                                                           |                                                                                            |
| Ministry      | Ministry of Digital Development and Communications of the Republic of Azerbaijan, M of Economy, etc                                                                                                                                                                                                                                                                                                                                                                                                            |                                                                                                                    |                                                                                                                                          |                                                                                                                           |                                                                                            |
| Plan          | Ministry Annual Implementation Plans, Implementation & Performance evaluation (Every 2 years)                                                                                                                                                                                                                                                                                                                                                                                                                  |                                                                                                                    |                                                                                                                                          |                                                                                                                           |                                                                                            |
| Agency        | IP Agency of the Republic of Azerbaijan<br>Space Agency,<br>Economic Zones<br>Development Agency,<br>SMB Agency<br>Innovation & Digital<br>Development Agency,<br>etc.                                                                                                                                                                                                                                                                                                                                         | AZInnovate<br>(Technology<br>commercialization<br>Policy: Transfer &<br>Startup program,<br>Validation Funds, Tax) | Innoland Incubation &<br>Acceleration Center<br>Technopark(Baku<br>Engineering<br>Uni./Western Caspian<br>Uni./Nakhchivan State<br>Uni.) | (Public Research<br>Institution: ANAS, etc.)<br>TLO(IP management,<br>Licensing Contract,<br>etc.),<br>NASA of Azerbaijan | (university) UNEC,<br>AzTU-Tech Transfer<br>Department, ADA Uni.-<br>ADA Innovation Center |
| Program       | <ol style="list-style-type: none"> <li>1) Technology Incubation: IP management, IP Packaging, Lab2Markets(Prototyping, Piloting),</li> <li>2) Startup/Venture Incubation: Technology Connect &amp; transfer, Startup Contest/Camp (Companybuilder, Consultant, Biz manager, etc.), Investment(GVC, VC, angel, etc),</li> <li>3) Innovation Clusters : Inno Town, Technopark, Industry Complex, etc,</li> <li>4) Infrastructure: Experts training programmes, Global Networks, Database system, etc.</li> </ol> |                                                                                                                    |                                                                                                                                          |                                                                                                                           |                                                                                            |

Source: Author's elaboration

**(Programs)** The basic axis of the national support project is divided into four main areas: Technology Incubation (Technology Scale-UP), Venture Incubation, Innovation Clusters, and Infrastructure.

The relevant policies are planned in the role of ministries, after which a dedicated agency carries them out. The ministries draw up the ministerial regulations and related forms for this purpose and use them as management standards.

[Table 4-1] Policies by the axis

| Axis                  | Required Policy                                                                                                                                                                                                                                                                                                                                 |
|-----------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Technology Incubation | <ul style="list-style-type: none"> <li>• IP management(creation-protection-diffusion).</li> <li>• Patent packaging-prototyping-piloting.</li> <li>• Lab2Mkt(matching technologies, bridging gaps).</li> <li>• Training for technicians (human resources).</li> </ul>                                                                            |
| Venture Incubation    | <ul style="list-style-type: none"> <li>• Seed technology incubation(post R&amp;D).</li> <li>• National start-up camp and contests, training for coordinators and consultants.</li> <li>• Government venture funds/investment funds .</li> <li>• Incubation centers (start-up/incubation centers at universities/PROs).</li> </ul>               |
| Innovation Clusters   | <ul style="list-style-type: none"> <li>• InnoTowns near universities(for young entrepreneurs), Techno parks, industrial complexes, etc.</li> <li>• All clusters have organizations that provide administrative and technical support.</li> <li>• A network project that supports meetings/communication of multidisciplinary actors.</li> </ul> |
| Infrastructure        | <ul style="list-style-type: none"> <li>• Expert training programs.</li> <li>• Certification system for technology trading/evaluation/brokerage.</li> <li>• Regulatory improvements, tax incentives, global networks, data systems, basic guidelines for technology transfer/cooperative R&amp;D, etc.</li> </ul>                                |

Source: Author's elaboration

### 1.1.2 Scenario B: 'AZ Innovate' (tentative name)

AZInnovate has been established as a more extensive and specialized organization under Azerbaijan's Ministry of Science and Education than ANAS-HTP in order to support start-ups as well as technology transfer. Therefore, the functions and roles of the existing ANAS-HTP as an affiliated organization of the ANAS will be further strengthened as an agency under the Ministry. AZInnovate basically needs four departments, namely, IP management, Technology Incubation, Enterprise, and Cooperation.

#### 1.1.2.1 IP Management Department

The IP Management Department manages the IP life cycle and designs policies for the planning, protection and dissemination of IP created in laboratories. For example, there is

an IP R&D system that investigates the IP required by a target product and reflects it in R&D planning, thus enabling the necessary IP development to be performed. This process also makes it possible to improve patent analysis techniques or expertise. In order to operate such a system, however, it is necessary to be able to access the patent information network and global databases such as PatSnap, Espacenet, and Patentscope, and, above all else, it is essential to accumulate patent information internally. Furthermore, information on the R&D output of universities and research institutes should be integrated into one site. To that end, a country-level data platform, i.e. the NTB(National Technology Bank<sup>45</sup>), is required. The obligation to register R&D output (patents) information should be specified in the provisions of the Basic Law for Technology Commercialization. In this way, the national patent capability can be checked, and technology consumers can search for required technologies. In addition, such information can be used in R&D planning.

### 1.1.2.2 Technology Incubation Department

First, the department needs seed finding, which is the process of finding technologies that can be used for technology transfer or start-ups based on an analysis of accumulated technical information, i.e. the NTB. It is also the process of finding technologies that can overcome the technological limitations of SMEs.

It bundles technologies required for target products or services and plans additional R&D for missing parts. Additional R&D is carried out through industry-university-PRO (Public Research Organization) collaboration, and the final technology is transferred to the company concerned. For this kind of R&D, the R&D budget is funded by the government or by corporate matching. It evaluates the technology value for calculating prices and royalties required for technology trade and licensing contracts. Valuation is approached via an objective evaluation system and/or subjective expert judgment. In Korea, public institutions have developed and used evaluation models such as SMART (KIPO), TTRS (KIBO), and Star-value (KISTI). Banks and private companies also have their own valuation systems. Generally, it is necessary to design a guideline for a technology transfer contract so that a standard

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<sup>45</sup> NTB is a technology data platform in Korea. <https://www.ntb.kr/eng/main/mainViewCall.do>

contract form can be used by universities, PROs and companies. In Korea, the 'Guidelines on Technology Transfer by Public Research Institutions' are prepared by the Ministry of Trade, Industry and Energy.

### **1.1.2.3 Enterprise Department**

It is important to create a venture ecosystem for innovative growth in which scale-up technology can be used as a seed for start-ups. The Enterprise Department offers support for start-ups, such as providing space, matching initial capital, connecting technology and experts, and providing regulatory improvement, tax incentives. In particular, when the domestic market is small, support for overseas expansion and public procurement is important. It is also important to train market experts, accelerators, investors, company builders, coordinators, and tax accountants. In particular, cooperation with VCs, the main players in the venture ecosystem, is also important. Therefore, AZInnovate should form the Azerbaijan VC Association and support its activities, and share NTB and start-up information with it. This process will serve as an opportunity for VCs to be more active in the venture ecosystem. Technology and enterprise incubation should be linked with the national innovation infrastructure (Innoland, Incubation centers, Techno park, etc.), the legal basis for which should be specified in the Basic Law for Technology Commercialization.

### **1.1.2.4 Cooperation Department**

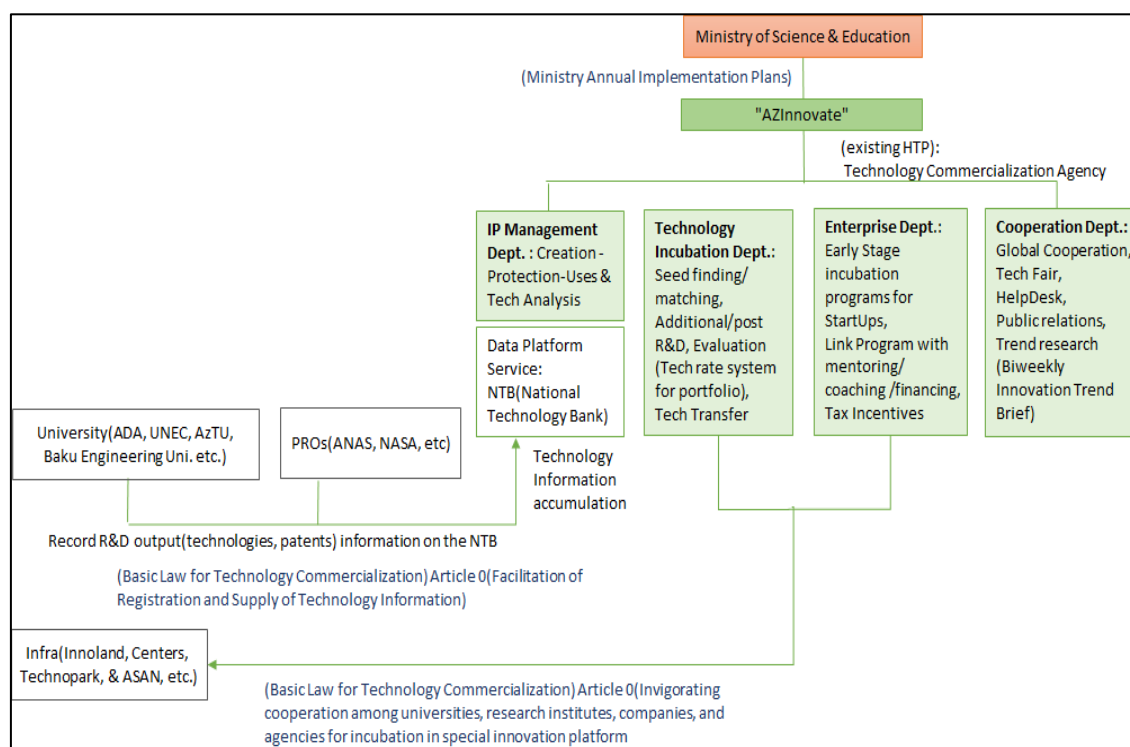
The Cooperative Department has various roles and functions related domestic and overseas matters.

First, its domestic programs include the Help Desk, public relations, Tech Fair and trend briefs among other things. Help Desk is a corporate counseling center that serves as a window for companies to receive consultation on technical difficulties or needs. PR includes activities designed to convey the importance of technology commercialization to the public, researchers, and enterprises, etc. Tech Fair is a large-scale annual event at which people involved in technology commercialization gather and promote their achievements and prototypes and new products are exhibited. There are also awards for best practices.

Global cooperation on approaches to forming the value chain necessary for technology commercialization is also underway, including a supply chain approach and a market

perspective approach. The department also prepares briefs on various global trends related to technology commercialization and presents them to governments, universities, research institutes, investors (VCs), and companies.

[Figure 4-5] Roles and Functions of AZInnovate (Existing ANAS-HTP)



Source: Author's elaboration



## 1.2 Short-Term Scenario

### 1.2.1 Governance

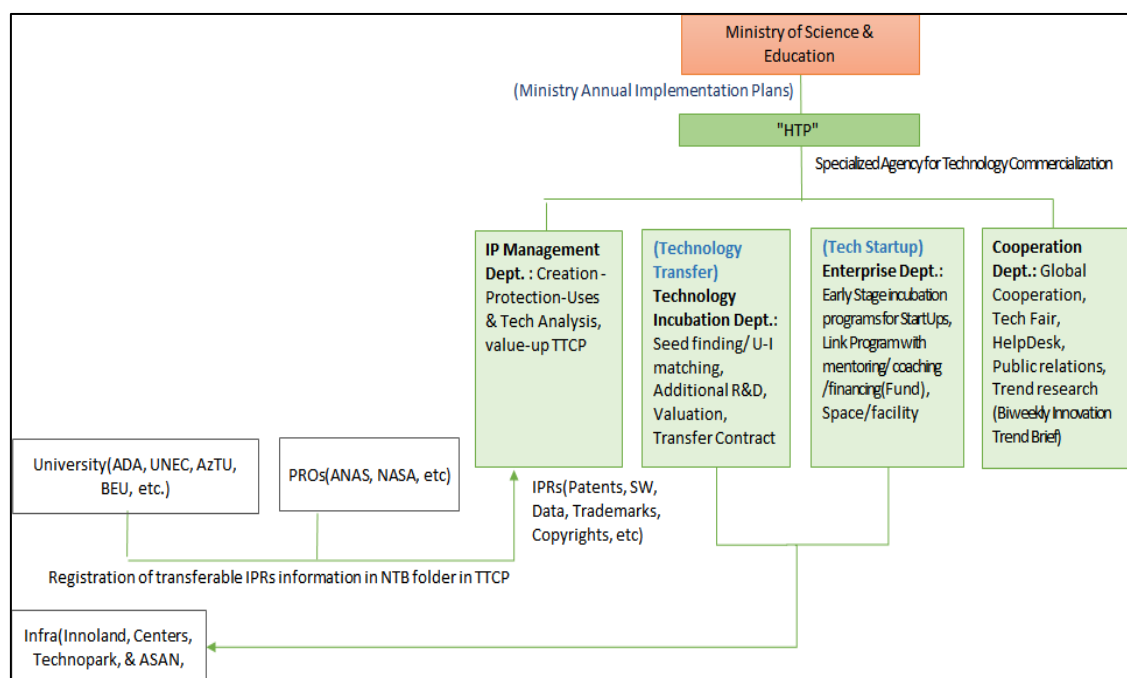
After a major change in policy governance in 2022, the existing Department of Education was expanded to form the Department of Science and Education, and the HTP was transferred from the existing ANAS umbrella to that of the Department of Science & Education. As it happens, the HTP requires rapid change and adaptation, and it is also important to grant it the appropriate status. To that end, the following measures need to be taken:

First, the Department of Science and Education must establish a basic plan for technology commercialization, in other words, a sketch of the vision and programs to be carried out for the year. The plan should also include the role of universities and research institutes, and it should be communicated and shared with other ministries. As such, is like a signal announcing the existence of the HTP.

Second, it is necessary to form partnerships and pipelines with domestic and foreign organizations. In order to achieve better outcomes, cooperation with related entities is needed, but not as an independent project. In particular, it is important to establish relationships with the TLOs of universities and research institutes because cooperative relations between universities, research institutes, and HTP are required throughout the entire commercialization cycle.

Third, it is essential to build upon and expand the expertise of human resources, and to increase the number of the organization's personnel and provide training to enhance their expertise.

[Figure 4-6] New Position of the HTP



Source: Author's elaboration

## 1.2.2 Program

### 1.2.2.1 Value-UP of TTCP

TTCP should be made into a platform for R&D achievements and technical information. Both those who seek information and those who provide information should be able to pay attention to TTCP. To achieve this, incentives are needed first of all. Basically, a great deal of information should be made available, which will be made possible through cooperation with universities and research institutes. The screening of information is also necessary, as accumulating high-quality data is the best way to provide useful information to those who seek it. Therefore, it is important for the HTP not only to collect R&D information, but also to manage it well. To It might consider writing a detailed description of the information. The easiest approach in this respect would be to provide an SMK (Sales Materials Kit) for each patent. To provide more detailed information about a patent, it should provide explanations of a technology product - including its competitiveness, market competition, industry

developments and risks - in an easily comprehensible manner<sup>46</sup>. Recently, a video of the prototype became available.

### **1.2.2.2 Start-UP for prospective entrepreneurs**

The HTP must provide the education and knowledge that are urgently required by prospective entrepreneurs. As such, entrepreneurship courses should cover a wide range of fields including technologies, markets, design, accounting, and business management.

Entrepreneurship competitions are also an important means of inducing motivation. Various competition models of more or less similar forms have already been introduced. What is important is the structure in which start-up accelerators can directly plan, select, and consult. To do this, excellent and experienced start-up accelerators are needed first of all. In the early stage of the program, it should be possible to consider a model in which foreign (including Korea) accelerators can participate, while follow-up support, such as space, network, capital and technology, should be provided to the winning teams.

It is also important to offer institutional support to entrepreneurs and start-ups, including services related to taxation, accounting, and law, as well as public procurement.

### **1.2.2.3 Make-UP for prototype manufacturing spaces**

Spaces for the manufacturing of prototypes are needed in order to expand technology transfer and support start-ups. Beyond a simple incubation space, such a space also includes the facilities and equipment needed by manufacturers. It is a space where technology products can be tested, check and designed. Professional technicians and prototype managers are offered residencies through the TiR (Technicians in Residence) system, which is designed to support budding entrepreneurs.

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<sup>46</sup> APEC(2017) , Guidebook on the IP-Business Cycle of SMEs.

## 2. Intellectual Property Management Policies

### 2.1 Introduction

This chapter proposes a number of options which the Azerbaijan government may consider in fine-tuning its IP management policies for technology transfer and the commercialization of home-grown technologies and intellectual properties (i.e. patents) which cover such technologies.

To this end, this chapter focuses on three main topics which are prerequisite to efficient technology transfer and the commercialization of such technologies and patents. These topics include the following:

- (1) Building an invention-friendly IP ecosystem by enhancing patent awareness among potential inventors (e.g. professors, graduate students, and researchers [either scientists or engineers] in Azerbaijan) as well as among government employees;
- (2) Determining whether or not to enact (or amend) laws or regulations regarding:
  - (a) ownership of inventions and patents arising from R&D sponsored by the government of Azerbaijan; and
  - (b) ownership of inventions and patents conceived by the employees of universities, research institutes or private companies (i.e. “employee inventions”);
- (3) Determining whether or not to establish an organization (i.e. a technology licensing office or technology transfer office) within universities and research institutes, where such an organization would be responsible for:
  - (a) acquiring inventions conceived by inventors working for those universities and research institutes;
  - (b) determining whether or not to seek patent protection for such inventions (e.g. filing patent applications, prosecuting those patent applications, paying maintenance fees, and such like);
  - (c) determining whether or not to pursue patent protection abroad;
  - (d) determining (if yes for c above) in which countries a patent should be obtained and maintained;

- (e) transferring technology to manufacturers or service providers; and
- (f) motivating more inventors to conceive more valuable inventions.

Each of the following sections relates to each of the three topics mentioned above. It is taken as understood throughout this chapter that:

- (1) the subject of technology transfer is patents granted to Azerbaijan-grown technologies;
- (2) technologies covered by such patents and developed by:
  - (a) professors, students (either graduate or undergraduate), and other engineers and researchers of Azerbaijan universities;
  - (b) researchers (either scientists or engineers) and other employees in Azerbaijan research institutes; and
  - (c) researchers (either scientists or engineers) and other employees in Azerbaijan companies.
- (3) the patented technologies which are covered by the patents are developed through R&D sponsored by the government of Azerbaijan;
- (4) the providers (or owners) of such technology-based patents are:
  - (a) Azerbaijani universities;
  - (b) Azerbaijani research institutes; and
  - (c) Azerbaijani companies.
- (5) the recipients (*e.g.*, the licensees) of such patents are manufacturing companies (or service providers), either domestic or international.

## 2.2 Building an Invention-Friendly IP Ecosystem

The successful transfer of a technology and its subsequent commercialization typically require many valuable patents, where new and competitive technologies (or inventions) must precede such valuable patents. However, it is almost impossible to obtain such valuable patents, unless the inventors or innovators come up with new and competitive technologies (or inventions).

This immediately raises the following questions: How many new and competitive technologies do you need for successful technology transfer? How many commercially valuable patents does one need for a “bonanza” commercialization?

In fact, one would need many new, innovative, and competitive technologies as well as numerous commercially valuable patents for successful technology transfer and bonanza commercialization. Why is this? Because our common sense gives us such a hint. Because the layman's statistics says so. In addition, because the *Pareto principle* states so. And more importantly, because those who have done that say so.

This section aims to provide the background to a virtuous IP ecosystem in which many competent inventors create lots of high-quality technologies (or inventions), in which many competent legal agents produce valuable patents from high-quality inventions, and in which numerous professionals commercialize such patents for profit.

Based on this background, this section introduces various projects that have been conducted by a Korean government agency (i.e. the Korea Invention Promotion Association [KIPA]), and derives from them a few policies which the Azerbaijan government may benchmark in building an invention-friendly IP ecosystem on Azerbaijan's soil.

### **2.2.1 Why We Need Lots (and Lots) of New Innovative Technologies and High-Quality Patents**

The process of commercializing a technological innovation is a complicated process. To achieve the successful commercialization of a given new technology, that technology must generally possess (1) technical viability, (2) marketability, and (3) patentability, to name just a few requirements.

#### **Many Technologies Fail due to the Complexity of the Technology Commercialization Process**

More importantly, technical viability, marketability, and patentability do not work in a parallel mode but rather in a serial mode. That is, successful commercialization requires that all three requirements must be met.

Therefore, even when a new technology satisfies two of the three abovementioned requirements, the attempt to commercialize it will fail if it does not satisfy any one of them.

**(A) Technical Viability of a New Technology**

First, a new technology must possess a feature that is superior to many competing, existent technologies. A new technology should also remain competitive, at least for the period of its commercialization.

Regardless of the superiority of a new technology, the licensee, e.g. the manufacturer, usually seeks to prove its viability. Therefore, when the licensee-manufacturer acquires a technology from a university etc. through licensing, the manufacturer must prove the viability of that technology by, for example, making a prototype and testing it, performing bench experiments, and so on.

Once a technology's technological viability has been proved, the manufacturer may proceed to scale up the bench process to a full-scale process for mass production. To that end, the licensee may have to invest heavily in buying land, building the factory and testing its facilities, and such like.

If a manufacturer fails in any of the above steps, the commercialization strategy will fail. For example, embodiment of the technology may prove to be too expensive, the facility for mass production may be too complicated, its reproducibility may be inconsistent, and so on. In other words, regardless of the superiority of a new technology, there is no guarantee that it will be commercially viable.

**(B) Marketability of Products that Embody a New Technology**

Even if a manufacturer comes up with an innovative scale-up process and succeeds in mass producing the related products based on a new and innovative technology through a viable mass-production process, the products must sell in the market.

As we know from past experience, many products that were technically superior to existing competing products nevertheless failed in the market for various reasons. One pertinent

example of this is what commentators refer to as a “vitamin or pain-killer” problem<sup>47,48</sup> or as a “vitamin, pain-killer, or cure” problem.<sup>49</sup> Another example may be a product that is based on a solid, new, and innovative technology but which happens to have a less attractive design, or deliver under-par performance, and such like.

Whatever the actual problems may be, a product which does not sell in the market is regarded as a failure of commercialization. Therefore, the marketability of a new and superior technology may be at least as important as its technical superiority.

### **(C) Patentability of a New Technology and Protection of the Patent**

Even in cases where a new technology is superior to competing technologies and has been successfully scaled up for mass production, and where the related products may already have succeeded in gaining a sizable market share, its commercialization may still fail if there is rampant copying of the products by copiers.

This is when the patent comes in to protect the manufacturer from copiers by allowing the manufacturer to exercise exclusive rights against the copiers and to protect its market share.

However, the exclusive rights come from the written disclosure and the claims of the patent which covers a new technology and which also covers any novel and non-obvious features of the mass-producing processes. Accordingly, when a patent is not properly drafted, the manufacturer cannot fend off such copying competitors. In the worst case scenario, the manufacturer’s patent could be invalidated, leaving it defenseless against the copies of competitors.

Furthermore, even when a manufacturer’s patent is properly written, the legal system within which the manufacturer operates may not be willing to enforce the patent to its full strength. As a result, the court may tolerate rampant copying of the manufacturer’s

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<sup>47</sup> George Deeb, Forbes (2014), ‘Is your start-up building a ‘vitamin’ or a ‘painkiller’?’].

<sup>48</sup> Hackernoon (2020), ‘Painkillers, vitamins and misunderstood start-up lessons’].

<sup>49</sup> O’Reilly Media, Inc. (2013), ‘Vitamin, painkiller, or cure?’, Ch. 2, ‘The agile start-up: quick and dirty lessons every entrepreneur should know’].



innovative technology, and the manufacturer may not be able to defend its market from competitors who sell their counterfeit copies at a ridiculously low price.

#### **(D) Human Factors**

Last but not the least in importance is human error. That is, regardless of the technical superiority and viability of a new technology, its huge marketability, and strong patentability, there is still a strong chance that the CEO of the manufacturer may make one or more fatal mistakes, which could result in the total failure of the commercialization of that technology.

As briefly discussed above, successful technology commercialization presupposes that a new technology and the patent covering it satisfy three key requirements. These requirements must be met one after another in a series mode. The managerial board of the manufacturer must also make reasonable and timely decisions. Considering all of the above, it naturally follows that any attempt to commercialize a new and innovative technology is more likely to fail than to succeed.

#### **Successful Technology Commercialization Is Exceedingly Rare**

Whether it is pure common sense or is based on hard science, the Pareto principle<sup>50</sup> states that for many outcomes, roughly 80% of the consequences come from 20% of the causes (the “vital few”). The Pareto principle is also known as the “80/20 rule”, the “law of the vital few” or the “principle of factor sparsity” to cite but a few of its other names. It then follows from this principle that about 80% of attempts to commercialize a new and innovative technology are doomed to fail from the outset, while only about 20% of such attempts succeed to some extent.

However, the Pareto principle actually seems to overestimate the chances of success of technology commercialization. In fact, many empirical studies indicate that the chances of successfully commercializing a new and ground-breaking technology to a potential licensee

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<sup>50</sup> Wikipedia (2022), “Pareto principle”.

are pretty slim.

One report suggests that “globally less than 5%” of the technologies developed in universities “make it to an actual commercial product that generates revenue for both the inventors and the university.”<sup>51</sup>

In addition, although Stanford University has received about 10,000 new inventions over the past forty-five years, “just three had generated multimillion-dollar licenses by 2016.”

While recognizing that technology commercialization is a complex series of processes and that most attempts to commercialize new, innovative technologies fail miserably, it should be noted that there is not only a statistical but also a simple and reasonable solution to achieving successful technology commercialization:

- ① Create as many innovative inventions as possible.
- ② Obtain as many high-quality patents as possible.
- ③ Contact as many potential licensees (*e.g.* Manufacturers) as possible.
- ④ Transfer as many technologies along with the related patents as possible.
- ⑤ Confirm whether the licensee manufacturer pays the correct amount of royalties on a certain day.

Of course, the strategy exemplified above may seem to be somewhat akin to a food recipe that does not specify which ingredients should be included in what amounts.

However, it still remains true that, all other things being equal, a society which generates more technologies is far more likely to have successful cases of technology commercialization than a society which generates fewer technologies.

In this context, the following section presents a few strategies which the government can adopt (1) to create more technologies, inventions, and patents on a macroscopic scale, and (2) to establish a virtuous IP ecosystem.

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<sup>51</sup> Stephen Rodda, Financial Review (2021), “Commercializing new technologies needs a two-way partnership”.

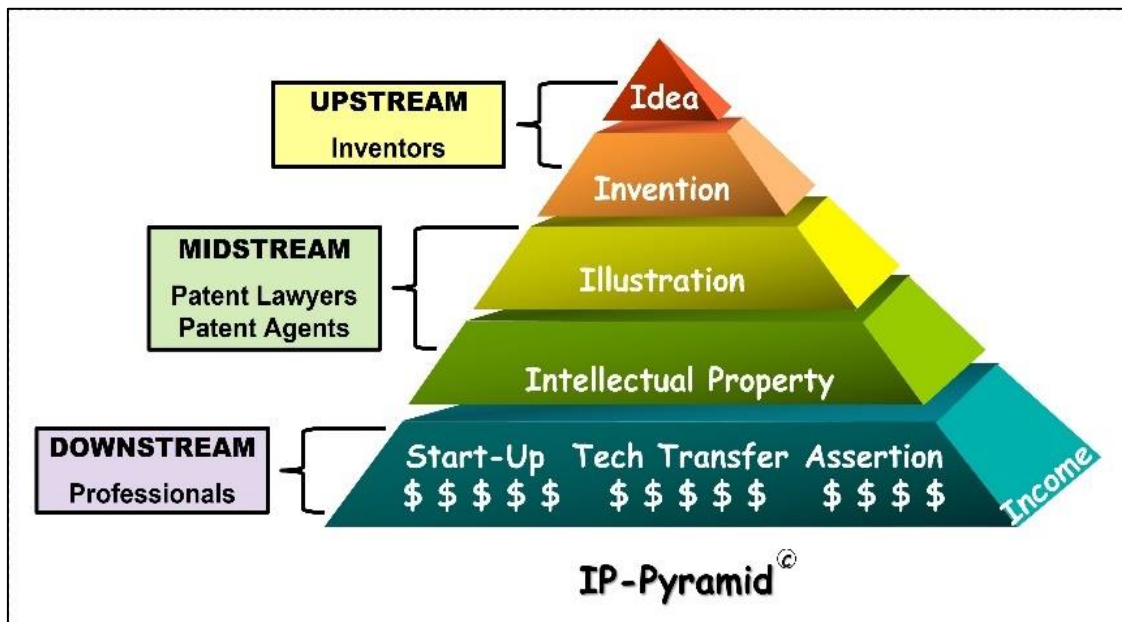
### Establishing an Invention-Rich Ecosystem to Generate Lots of Inventions

Innovative inventions and new technologies do not come from nowhere. That is to say, such inventions and technologies are neither created from scratch nor obtained at no cost. Rather, inventions and technology come from an ecosystem comprising several layers.

It has been noted that different people who are equipped with different skill sets usually play a major role in each such layer. In addition, these people tend to work sequentially so that, for instance, people in the second layer cannot start to perform their task until the people in the first layer have completed their task, while people in the third layer cannot start to perform their tasks until people in the second layer have finished their work, and so on.

To reflect these characteristics, one can set up a hypothetical intellectual property (IP) ecosystem, and then use the IP ecosystem in analyzing the characteristics of each layer as well as various interactions among the different layers.

[Figure 4-7] Graphical Presentation of an IP Ecosystem or an IP-Pyramid



Source: Y. Shim (2016), "The IP Silk Road"

[Figure 4-7] is a graphical representation of an exemplary intellectual property ecosystem, where such an ecosystem is referred to as an "IP-Pyramid." As illustrated in [Figure 4-7], the

ecosystem includes an upstream, a midstream, and a downstream, each of which is connected to the others in a series mode.

### **(A) Upstream of the IP Ecosystem**

The upstream of the IP ecosystem is characterized by the conception of an invention by an inventor. More particularly, the upstream starts when an inventor is struck with an idea about an innovative technology. The inventor then modifies his or her idea and ends up with plenty of related ideas. Eventually, the inventor sublimates his best idea into an invention (i.e. the inventor conceives the invention). Accordingly, there is no doubt that the inventor or the innovator is the main character in the upstream layer.

### **(B) Midstream of the IP Ecosystem**

The midstream of the IP ecosystem is characterized by the filing of a patent application and the issuance of a patent. More particularly, the inventor hires a patent lawyer (or a patent agent), who then prepares and files a patent application to a patent office in the country (or another country) of the inventor.

After filing the application, the patent lawyer deals with a patent examiner at the patent office, and assists his client in getting the patent application issued as a patent. Therefore, the patent lawyer or the patent agent is the main character in the midstream.

### **(C) Downstream of the IP Ecosystem**

Then, the downstream of the IP ecosystem is characterized by different modalities of commercializing (or monetizing) the patent. More particularly, the inventor (or patent owner) can resort to at least three different modalities to reap benefits from his invention.

(C-1) The 1<sup>st</sup> Modality of Commercialization = Self-Practicing the Patent

The first modality of commercialization is that the inventor applies his invention and makes his own product. To this end, the inventor may have to set up a start-up. However, it would be even better if the inventor himself were the owner of his own factory.

In this first modality of commercialization, the inventor creates sales revenue by selling his product in the market, and then takes a portion of the sales revenue as his profit. Of course, the commercialization fails when the inventor cannot sell his product in the market or when the balance stays in the red even after the inventor has generated sales revenues.

(C-2) The 2<sup>nd</sup> Modality of Commercialization = Technology (and Patent) Transfer

The second modality for commercialization is that the inventor transfers his technology and patent to a licensee for profit. Scientific or technological innovators are usually not entrepreneurs. As such, they like to work in the lab, not on the manufacturing front. Accordingly, most of these innovators tend to be looking constantly for a potential licensee-manufacturer. As the inventor finds and negotiates with a licensee-manufacturer and signs a licensing contract, the inventor can earn royalties in various timings, i.e. (1) an upfront royalty, (2) a running royalty, (3) a milestone royalty, and so forth.

The upfront royalty corresponds to the royalty which an inventor receives upon signing a contract for technology transfer. Conversely, the running royalty corresponds to a portion of the licensee's revenue. That is, when the licensee generates sales revenue, the licensee shares a certain portion of the revenue with the inventor, and that portion corresponds to the running royalty. The milestone royalty is the one which the licensee pays to the inventor when a certain condition is met or when a certain event happens.

It should be noted that the various types of royalty discussed above reflect different features of technology transfer and technology commercialization.

For example, a technology transfer that includes the upfront royalty, but not the running royalty, means that the real fruit of technology commercialization is kept

solely by the licensee. Therefore, once the contract for technology transfer has been signed, there will be very little, or no, incentive for the inventor to assist the licensee in carrying out the actual commercialization.

Conversely, a technology transfer that includes the upfront royalty as well as the running royalty means that the inventor and the licensee agree to share the real fruits of the technology commercialization.

Accordingly, compared with the upfront-only licensing outlined in the above paragraph, this type of technology transfer can provide plenty of incentives for an inventor to assist the licensee in carrying out and then successfully completing the actual commercialization.

### (C-3) The 3<sup>rd</sup> Modality of Commercialization = Asserting the Patent

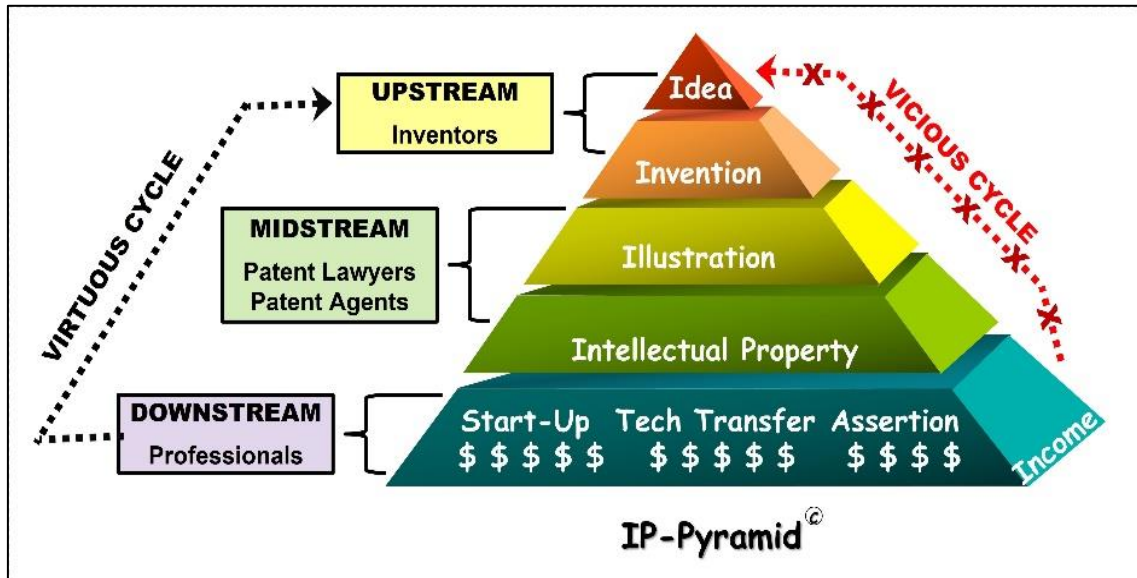
Some circumstances may force an inventor to go to the court to sue a competitor for its infringement of his or her patent. In one instance, an inventor self-practiced his patent, manufactured the products, and sold them in the market. However, a competitor made a copy-cat product, and sold the counterfeit product in the market. Due to the lack of (or lax) patent protection in the country, the inventor lost his market. Although the inventor had proposed to license the patent, the copying competitor refused. In this case, the inventor's only remaining option was to take the case to court, sue the copying competitor, win the lawsuit, and obtain damages from the copying manufacturer. In some instances, the inventor may not be interested in self-practicing his own patent at all. Instead, the inventor meets lots of manufacturers, advertise his new technology and patent, and propose a licensing deal with those manufacturers. However, the manufacturer chooses not to sign the contract, but to copy the patented technology of the inventor. Similar to the above example, the inventor's only remaining option is to take the case to court, sue the copying competitor, win the lawsuit, and then obtain damages from the copying manufacturer.

### **(D) A “Virtuous IP Ecosystem” Is Essential to All Players in All Layers**

As manifested in [Figure 4-7], an invention is transformed into a patent through a series of processes, where different people with different types of expertise play their roles in each of the upstream, midstream, and downstream elements of the IP ecosystem.

Some IP ecosystems can flourish, while others may perish. The main difference between these two ecosystems usually lies in their efficiency or productivity in generating profits and in distributing such profits to different players in different layers.

[Figure 4-8] Graphical Presentation of a Virtuous IP Ecosystem and a Vicious IP Ecosystem



Source: Y. Shim (2016), "The IP Silk Road".

[Figure 4-8] is a graphical representation of an IP ecosystem that flourishes in a virtuous cycle or operates in a vicious cycle and perishes. As shown in the figure, the new technologies and their patents are commercialized (or monetized) by various professionals in the downstream of the IP ecosystem. When a technology commercialization succeeds, the entrepreneur-inventor (of the 1<sup>st</sup> modality) can make profits from the sales revenue of the product, which embodies the patented technology and which he makes and sells.

A licensor-inventor (of the 2<sup>nd</sup> modality) can also make profits from the royalty he receives from the licensee-manufacturer. The plaintiff-inventor (of the 3<sup>rd</sup> modality) can make profits from the damages which the jury or judge awards to the inventor. According to the incentive theory of the patent system, as well as to our common sense, once an inventor gets his share of the profits, the henceforth financially rewarded and emotionally empowered inventor will go on to make the next generation of innovations, inventions, and patents. In

short, an IP ecosystem operating within the virtuous cycle always has channels through which profits generated in the downstream can be funneled back to the upstream. As a result, the inventors in the virtuous IP ecosystem are motivated to make inventions, to obtain patent protection, and to proceed with the commercialization of their technologies and patents. Because an IP ecosystem operating in such a virtuous cycle has a larger pool of serious inventors, the patent attorneys or agents serving those inventors can also make profits, as can the various professionals who assist them in commercializing their patents. As this virtuous cycle goes on and on, the IP ecosystem flourishes while adding width and depth thereto.

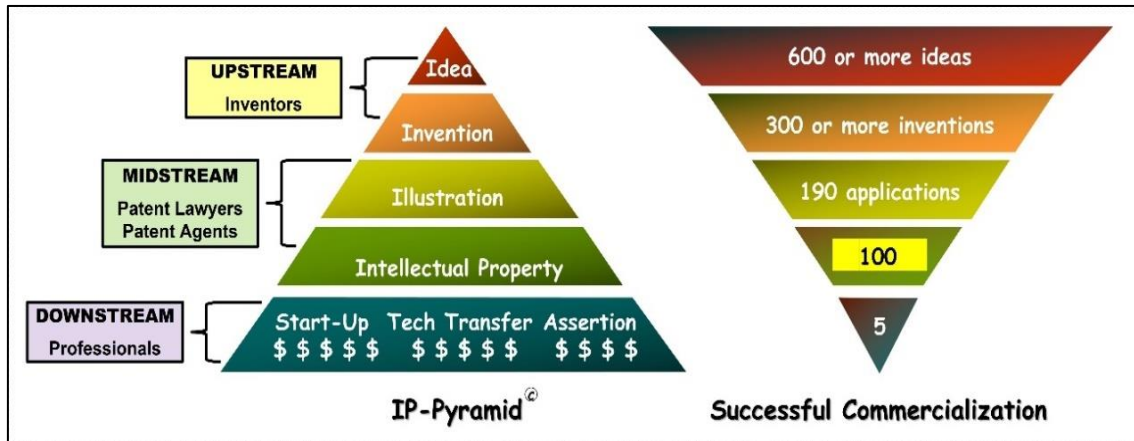
In an IP ecosystem which operates in the vicious cycle, however, everything goes in the exactly opposite direction. For example, without knowing of the “vicious feature”, the inventor invests his time, effort, and money in coming up with a decent invention. Then, the inventor spends more money in filing a patent application and acquiring a patent. However, the inventor cannot make any profits through commercialization due to, for example, a lack of (or lax) patent protection. Although the persistent inventor may go through this vicious cycle a few more times, it does not take long for the inventor to realize that making inventions and innovations only help his copycat competitors. The inventor will then determine to join the pool of copiers and, as a result, there will be no one in the pool of inventors and innovators.

#### **(E) Invention-Rich Ecosystem Starts with a Pool of Innovative Inventors that Leads to More Technology Transfer/Commercialization**

Even within an IP ecosystem which operates in the virtuous cycle, not all inventions can be commercialized. In fact, only a small fraction of the inventions can be transferred to a manufacturer. In addition, only a small portion of such transferred inventions can be commercialized. Furthermore, only a small portion of the commercialized inventions will find a way to occupy a meaningful market share.



[Figure 4-9] Chances of Successful Technology Commercialization



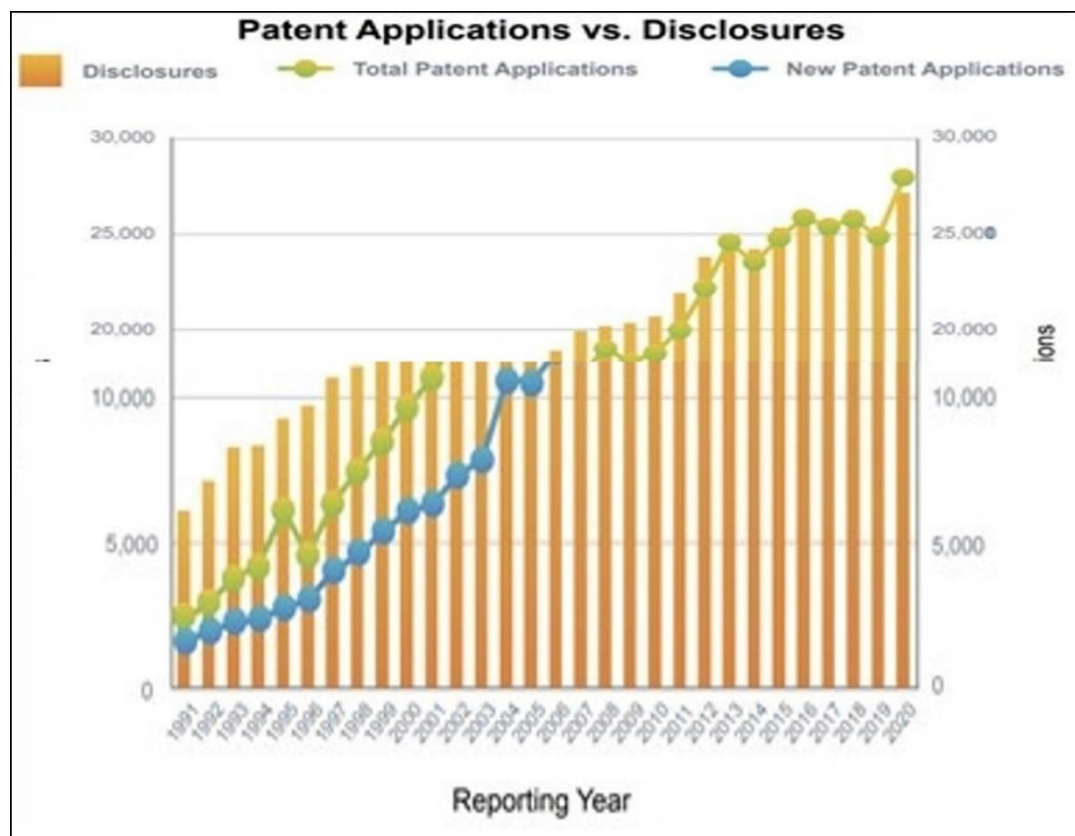
Source: Y. Shim (2016), "The IP Silk Road".

As mentioned above, at the global level less than 5% of the technologies developed in a university "make it into an actual commercial product that generates revenue for both the inventor and the university."<sup>52</sup>

Assuming that successfully commercialized university technologies are those which are also protected by patents, [Figure 4-9] shows a graphical representation of the chances of successful technology commercialization, where the numbers of ideas, inventions, and issued patents, each of which is required in order to reach the 5% level of successful commercialization, are also shown.

<sup>52</sup> Stephen Rodda, Financial Review (2021), [Commercializing new technologies needs a two-way partnership].

[Figure 4-10] Comparison of the Number of Invention Disclosures Received with the Number of Patent Applications Filed by Representative U.S. Universities



Source: AUTM (2021), "2021 AUTM Licensing Activity Survey".

In [Figure 4-9], this author assumes that about 190 patent applications are needed to obtain 100 issued patents. This assumption seems reasonable in that about 55% of patent applications are issued into patents in the U.S.

In [Figure 4-10], this author also assumes that about 300 invention disclosures are needed to get 190 patent applications. This assumption was derived from the recent data shown in Figure 0.4, such that U.S. universities file about 60% to 65% of the invention disclosures they receive from professors, researchers, graduate students, and others working for the representative U.S. universities.<sup>53</sup>

<sup>53</sup> AUTM (2021), [2021 AUTM Licensing Activity Survey].

This author further assumes in [Figure 4-9] that about 600 or more crude ideas would be needed in order to generate about 300 patent eligible inventions. Even though there are no statistical data to support this assumption, this 2:1 conversion ratio may even be too optimistic, considering that most inventors are sincere and, therefore, serial and addicted inventors, and that most inventors might even think up about 1,000 or more ideas before coming out and filing 300 invention disclosures with the Technology Licensing Offices of their universities. Whatever the actual conversion rates among crude ideas, inventions, patent applications, and issued patents may be, the bottom line is that the most important task of the government in promoting technology transfer and technology commercialization is to build a pool full of lots (and lots) of inventors and innovators.

## **2.2.2 “Tech. Transfer” vs. “Tech. Commercialization”: Which One is Easier to Accomplish? Which One is More Amenable to Quantitative Analysis and Evaluation? Which One Better Suits the Government Policy and Its Objective?**

Although the terms “technology transfer” and “technology commercialization” tend to be used interchangeably, they differ in terms of their definition and their scope. Therefore, one needs to carefully compare these two terms before taking one of them as the objective of the government policy.

### **2.2.2.1 “Short-Tailed” Technology Transfer vs. “Long-Tailed” Technology Commercialization**

Many treatises provide definitions of “technology transfer” and “technology commercialization”, and such definitions may vary from one to another depending on the source. For the purpose of comparison, this author consulted Wikipedia as the primary source of such definitions. The following is the definition of the term “technology transfer” given by Wikipedia:<sup>54</sup>

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<sup>54</sup> Noah Adam, PatentRebel (2019), [What percentage of patents are approved?].

Technology transfer, also called “transfer of technology”, is the process of transferring (disseminating) technology from the person or organization that owns or holds it to another person or organization, in an attempt to transform inventions and scientific outcomes into new products and services that benefit society.<sup>55,56</sup> Technology transfer is closely related to (and may arguably be considered a subset of) knowledge transfer. A current and comprehensive definition of technology transfer includes the notion of a collaborative process, as it has become increasingly clear that global challenges can only be resolved through the development of global solutions. Knowledge and technology transfer plays a crucial role in connecting innovation stakeholders and moving inventions from creators to public and private users.<sup>57</sup>

Alternatively, the following is the definition of the term “technology commercialization” taken from Wikipedia:<sup>58</sup>

(*Technology*) commercialization is the process of introducing a new product or production method into commerce, i.e. making it available on the market. The term often connotes entry into the mass market (as opposed to entry into earlier niche markets) in particular, but it also includes a move from the laboratory into (even limited) commerce. Many technologies begin in a research and development laboratory or in an inventor's workshop and may not be practical for commercial use in their infancy (as prototypes). The "development" segment of the "research and development" spectrum requires time and money as systems are engineered with a view to making a given product or method a paying commercial proposition. The product launch of a new product is the final stage of new product development - at this point advertising, sales promotion, and other marketing efforts encourage the commercial adoption of the product or method. Beyond commercialization (in which technologies enter the business world) may lie consumerization (in which they become consumer goods, as for example when computers go from the laboratory to the enterprise and then to the home,

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<sup>55</sup> AUTM (2020), [AUTM 2020 Licensing Activity Survey].

<sup>56</sup> Wikipedia (2022), [Technology transfer].

<sup>57</sup> Narry Bozeman, Research Policy (2000), [Technology transfer and public policy: a review of research and theory].

<sup>58</sup> Knowledge for Policy, Competence Center on Tech. Transfer (2022), [What is technology transfer?].

pocket, or body). (*emphasis added*).

As noted above, “technology transfer” refers to the transfer of the technology itself (along with, or without, the patents which are related to the technology). Therefore, technology transfer typically requires two parties, i.e. (1) the licensor-inventor or the owner of the technology or patent who sells or licenses the patent to a third party, and (2) the 3<sup>rd</sup> party, a licensee-manufacturer (or a licensee-service provider) that acquires the patent or obtains the (exclusive or non-exclusive) license from the licensor-inventor.

It then follows that the scope of “technology transfer” includes the process of selling or giving a license to the licensee-manufacturer, but does not include whether or not the licensee-manufacturer actually completes the complex process of manufacturing, distributing and selling the product, which itself embodies the acquired or licensed patented technology. On the other hand, “technology commercialization” presupposes that (1) the “technology” has been transferred and that (2) the licensee-manufacturer has embodied the acquired or licensed technology into a product. In addition, “technology commercialization” usually requires more participants than “technology transfer.” For example, setting up a start-up would require a CEO and managers. Even when an inventor sets up his own start-up, he should hire the managers anyway. In addition, when the inventor desires to sell or license the patent, he would need an experienced patent broker or intermediary. In other words, because “technology commercialization” encompasses a broader scope than does “technology transfer,” it is not surprising that more people have to be involved in the technology commercialization.

In this context, “technology transfer” may be viewed as having a short tail (in the relative sense), while “technology commercialization” may be viewed as having a long tail (also in the relative sense).

#### **2.2.2.2 Is It Easier to Accomplish and Quantitatively Analyze Short-Tailed “Technology Transfer”?**

Due to the difference in the scopes of technology transfer and commercialization, it is safe to assume that, other things being equal, accomplishing a certain target (e.g., in terms of a

number of technology transfer contracts signed) for technology transfer would be easier than fulfilling the same target for technology commercialization. In addition, assessing the efficacy of a certain government policy as well as conducting a quantitative analysis of that policy would be easier for technology transfer than for technology commercialization. For example, the quantitative assessment and analysis of technology transfer is rather straightforward, as all that one has to do is to find a number of signed contracts for assigning or licensing university technologies or patents to manufacturers or other parties that have paid money to the universities for such purchases or licenses. When desirable, one can easily calculate the total amount of income made by universities within a given period.

In contrary the quantitative assessment and analysis of technology commercialization can be rather complex. First of all, one has to establish the criteria for determining whether a technology commercialization has succeeded or failed. Should a technology commercialization be regarded as a success when a licensee-manufacturer (1) starts to sell the products in the market; (2) accomplishes a certain amount of sales revenue; (3) or occupies a certain market share? Or should a technology commercialization be regarded as a success if it (4) makes a net profit; (5) is acquired by a third party through an M&A deal; (6) is listed in the stock market of a certain country; (7) or survives in the market for longer than a certain period? and so forth. A government may establish and execute a policy designed to spur and motivate the creation of innovations and inventions, and to promote the transfer of university-born or research institute-born technologies and patents to the private sector. However, when the government desires to analyze the outcome of such a policy, it has to be careful in setting up the proper criteria for assessment and analysis of the outcome.

At least in this respect, assessing and analyzing the outcome of a technology transfer would be more amenable than assessing and analyzing the outcome of a technology commercialization.

### **2.2.3 Korean Example: Korea Invention Promotion Association**

KIPO (Korean Intellectual Property Office, formerly the Korean Industrial Property Office) has been very active in nurturing the invention-rich IP ecosystem in Korea. Although many

commentators have criticized its quantity-oriented strategies, there is no doubt that KIPO has been doing its share.

### 2.2.3.1 Brief History of KIPA (Korea Invention Promotion Association)

KIPO is affiliated with a dozen government-related institutions. KIPA (Korea Invention Promotion Association), one of the earliest such institutions, was established in 1973. It has contributed to developing the national economy and enhancing international competitiveness through technological innovation in the industry by systematically and efficiently promoting the invention promotion project and promoting the interests of inventors. [Table 4-2] provides a brief summary of the history of KIPA.

[Table 4-2] Brief History of KIPA (Korea Invention Promotion Association)

| Year | History                                                                                                                                                                      |
|------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1973 | Founded as the <i>Korean Patent Association</i> .                                                                                                                            |
| 1994 | Established the <i>Patent Technology Commercialization Center</i> and the <i>Korea Invention Promotion Association</i> (both based on Article 52, Invention Promotion Act).  |
| 1995 | Launched the <i>Center for Patent Technology Information</i> (later separated as the <i>Korea Institute of Patent Information</i> in 2001).                                  |
| 1997 | Launched the <i>Intellectual Property Research Center</i> (later separated as the <i>Korea Institute of Intellectual Property</i> in 2005).                                  |
| 2001 | Designated as a technology evaluation agency by the Ministry of Trade, Industry and Energy.                                                                                  |
| 2002 | Designated as a technology transaction agency by the Ministry of Trade, Industry and Energy.                                                                                 |
| 2003 | Opened the <i>Korea Intellectual Property Service Center (KIPS)</i> (28 floors).                                                                                             |
| 2006 | Designated as a remote education training center for innovative small and medium-sized enterprises by the Ministry of Education.                                             |
| 2007 | Designated as a public institution by the Ministry of Strategy and Finance.                                                                                                  |
| 2010 | Launched the software "system for measuring, analyzing and rating patent technology" (SMART) and established the <i>Center for Next-Generation Entrepreneurial Talents</i> . |
| 2012 | Founded the <i>Educational Research Institute for the Gifted in Invention</i> , launched the <i>Korea Intellectual Property Evaluation Center</i> .                          |
| 2015 | Designated as a general education training institute by the Ministry of Education.                                                                                           |

| Year | History                                                                                 |
|------|-----------------------------------------------------------------------------------------|
| 2018 | Acquired the nation's first certification of Intellectual Property Ability Test (IPAT). |
| 2020 | Launched the <i>IP Recovery Service Agency</i> .                                        |

Source: "KIPA: Intellectual Property No. 1 Partner."<sup>(13)</sup>

KIPA received a subsidy of U.S.\$ 102 million from the Korean government in 2021, followed by a further subsidy of U.S. \$111 million in 2022. KIPA had about 120 employees as of 2021, including both tenured and non-tenured employees.<sup>59</sup>

### 2.2.3.2 KIPA's Efforts to Establish an Invention-Rich Ecosystem to Generate Lots (and Lots) of Inventions

In its 2022 announcement, KIPA stated the following four tasks as its key features and roles:

- (1) Development of IP-based enterprises (support for the promotion of IP-based start-ups and the creation of intellectual properties, etc.);
- (2) Strategic use of IP (recovery and utilization of secured IPs, support for IP transactions, etc.);
- (3) Development of core IP human resources (inventive education promotion project, IP human resource training project, etc.); and
- (4) Dissemination and promotion of the IP culture (inventory day ceremony, development and dissemination of international IP contents, etc.).<sup>60</sup>

Among KIPA's abovementioned roles, the third and fourth are particularly worth analyzing, as such projects aim to establish an invention-rich ecosystem in Korea (1) by nurturing inventors of all ages and (2) by providing teaching and training on various levels of IP-related matters to students, scientists, engineers, and laymen. To this end, KIPA has been conducting diverse projects aimed at (1) establishing an invention-friendly intellectual property ecosystem in Korea, (2) nurturing inventors of all ages and from all walks of life, and (3) providing these inventors with the requisite IP education or training. The following

<sup>59</sup> KIPA (2022), 「KIPA: Intellectual Property No. 1 Partner」.

<sup>60</sup> All Public Information In-One (2022), 「Income and expenditure」.



are excerpts from “KIPA: Intellectual Property No. 1 Partner.”<sup>(13)</sup>

### (A) Invention Education

[Table 4-3] KIPA’s Invention Education Projects

| Program                                                                                                     | Details                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
|-------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| A-1 Operation of Invention Education Centers Nationwide                                                     | At more than 200 invention education centers in elementary, middle, and high schools nationwide, KIPA provides various educational programs to those interested in invention and offers balanced education throughout the country.                                                                                                                                                                                                                                                                                                            |
| A-2 Training Creative Invention Talents                                                                     | KIPA operates youth IP development programs and inventor programs aimed at training young people to acquire creative invention skills.                                                                                                                                                                                                                                                                                                                                                                                                        |
| A-2-1 Operating High Schools Specializing in Invention and Patent Education - IP Meister Program            | This program consisting of IP development education is designed to cultivate future technical experts by equipping students at IP Meister and specialized high schools with the invention skills required by various industries and by harnessing their creativity.                                                                                                                                                                                                                                                                           |
| A-2-2 Operating High Schools Specializing in Invention and Patent Education - Young Inventors Program (YIP) | This program is designed to foster future IP professionals and inventors by providing advanced invention and patent education and consultations to young learners with outstanding invention skills or those interested in invention, including winners of domestic or international invention contests; holders of patent or registered invention items; those who have completed intermediate/ advanced courses at invention education centers; and those who have completed an education program at the Center for Creative Entrepreneurs. |
| A-3 Training Next-Generation Entrepreneurial Talents                                                        | KIPA offers continuous, systematic training programs for students with the potential to grow into such creative entrepreneurs as Bill Gates (MS), Sergey Brin and Larry Page (Google) and harness the power of IP.                                                                                                                                                                                                                                                                                                                            |

Source: “KIPA: Intellectual Property No. 1 Partner.”<sup>(13)</sup>

## (B) Invention Contests

[Table 4-4] KIPA's Invention Contest Projects

| Program                                                                   | Details                                                                                                                                                                                                                                                                                                            |
|---------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| B-1 Youth Invention Contest<br>(Korea's Largest Youth Invention Festival) | The Youth Invention Festival, the largest of its kind in Korea, promotes invention awareness among young students. The festival expands invention education across the board by simultaneously hosting the Korea Student Invention Exhibition, Korean Student Creativity Championship, and Youth Inventor Program. |
| B-1-1 Korea Student Invention Exhibition                                  | This contest aims to cultivate the creativity of students, raise their awareness of invention, and develop them into future inventors by exhibiting outstanding student inventions and offering prizes, including the Presidential Award, along with special benefits to outstanding students.                     |
| B-1-2 Korean Student Creativity Championship                              | This contest aims to unearth potential inventors with outstanding creativity and ideas, and help young people to develop their creative and tenacious spirit as they solve tasks as part of a team.                                                                                                                |

Source: "KIPA: Intellectual Property No. 1 Partner." <sup>(13)</sup>

## (C) Training IP Personnel

[Table 4-5] KIPA's IP Personnel Training Projects (1)

| Program                                           | Details                                                                                                                                                                                                                                            |
|---------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| C-1 Professional Training Program                 | The Korea Invention Promotion Association is playing a leading role in the IP field by operating various education programs designed to train intellectual property talents, and by supplying outstanding personnel to universities and companies. |
| C-1-1 IPAT (IP Ability Test)                      | The nation's only IP competency test is conducted by the KIPA and is open to anyone who is interested in IP. The test evaluates basic and practical skills in all areas of IP.                                                                     |
| C-1-2 IP Education at Leading Universities        | KIPA trains IP professionals through independent IP education programs in universities.                                                                                                                                                            |
| C-1-3 IP Courses in Colleges and Graduate Schools | KIPA offers professional instructors, standardized curricula and textbooks to expand university-level IP education.                                                                                                                                |
| C-1-4 Campus Patent Universiade Competition       | This industry-academia cooperation program trains in-demand IP talents and supplies the creative ideas of colleges to the relevant industries.                                                                                                     |

Source: "KIPA: Intellectual Property No. 1 Partner." <sup>(13)</sup>

[Table 4-6] KIPA's IP Personnel Training Projects (2)

| Curriculum                           | Target                              | Major Contents                                                                                                                                                                                                                                                                                  |
|--------------------------------------|-------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| C-2 Offline IP Education (IP Campus) |                                     | KIPA provides practical training for patent officers, technology developers, and researchers of domestic companies and research institutes, as well as non-professionals in all areas such as patent, trademark, design, and copyrights, and runs invention science classes for young students. |
| C-2-1 General Education (B2C)        | Individuals                         | Systematic, specialized practical training for future IP professionals that covers the overall IP system, management practices, information search, infringement lawsuits, foreign patent laws, etc.                                                                                            |
| C-2-2 Commissioned Education (B2B)   | Companies<br>Institutions<br>Groups | Customized education based on planning and analysis of the educational needs of domestic companies, institutions, and universities.                                                                                                                                                             |
| C-2-4 Invention Education (B2C)      | Young<br>Students                   | Self-developed invention education program for elementary students (3rd-6th grades) aimed at training creative convergence talents of the future by enhancing their creative, scientific thinking and raising their invention awareness.<br>(Basic → Intermediate → Advanced)                   |

Source: "KIPA: Intellectual Property No. 1 Partner."

#### (D) On-Line IP Education

[Table 4-7] KIPA's On-line IP Education Projects

| Program                                                             | Details                                                                                                                                                                                                                                                               |
|---------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| D-1 IP Academy                                                      | IP Academy ( <a href="http://www.ipacademy.net">www.ipacademy.net</a> ), a national IP education platform, offers various contents and online education free of charge, raise awareness of the importance of IP, and provides life-long IP education to all citizens. |
| D-1-1 Education for All Parties Interested in Intellectual Property | Target: Anyone seeking online IP education (college students, company employees, etc.).<br>Curriculum: Protecting My Own IP, Prior Skills Search Know-how, Do's and Don'ts of Employee Invention, etc.                                                                |
| D-1-2 Online IP education for Groups                                | Specifically designed for employees of companies, research institutes, college students, and graduate students.<br>Target: companies, research institutes, organizations, associations, public institutions, and universities seeking group education.                |

| Program                                                                                             | Details                                                                                                                                                                                                                                                                                                                                                                                               |
|-----------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                                                                                     | Curriculum: Group B2B education including technology innovation utilizing TRIZ, IP flipped learning (on/offline) for corporations.                                                                                                                                                                                                                                                                    |
| D-1-3 Remote Educational Training                                                                   | Designed to boost teachers' invention education capabilities.<br>Target: elementary and secondary school teachers and public educational officials nationwide.<br>Curriculum: remote job training for 15 hours (1 credit) and 30 hours (2 credits).                                                                                                                                                   |
| D-1-4 E-learning Group Invention Education                                                          | Designed for elementary and secondary school students with the aim of training invention talents of the future.<br>Target: elementary/ middle/ high schools and invention education centers nationwide seeking online education on IP.<br>Curriculum: World-Changing Inventions, Creativity Sparkles, Science TikTok, etc.                                                                            |
| D-2 Intellectual Property Credit Bank System ( <a href="http://cb.academy.net">cb.academy.net</a> ) | A distance learning-based degree program in IP, which aims to train IP professionals and offer life-long education including theoretical and practical education in intellectual property studies.<br>Target: non-professionals and college students (universities with a credit exchange agreement).<br>Curriculum: 17 major courses including Introduction to Intellectual Property and Patent Law. |

Source: "KIPA: Intellectual Property No. 1 Partner."

## (E) Developing an Invention Culture at Home and Abroad

[Table 4-8] KIPA's Invention-Friendly Culture Development Projects

| Program                                                 | Details                                                                                                                                               |
|---------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------|
| E-1 Development of Invention Culture at Home and Abroad | By developing the IP culture at home and abroad through exhibitions and events, KIPA helps to promote and commercialize outstanding inventions.       |
| E-1-1 Invention Day Ceremony                            | KIPA awards outstanding inventors with the Order of Industrial Service Merit, Prime Minister's Award, and President's Award.                          |
| E-1-2 Korea Intellectual Property Exhibition            | Outstanding domestic and overseas IP-related products are displayed at the exhibition, including those protected by patents, designs, and trademarks. |
| E-2 Stimulating Employee Inventions                     | KIPA encourages companies to introduce the employee invention system by providing them with professional consultation and                             |

| Program                                                  | Details                                                                                                                                                                                                                                                    |
|----------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                                          | certifications if they offer compensation for employee inventions and support the voluntary compensation culture.                                                                                                                                          |
| E-2-1 Employee Invention System                          | The system allows companies to acquire new technologies and increase their profits as they are entitled to inventions created by their employees in return for sufficiently compensating staff for their work.                                             |
| E-3 Participation in International Invention Exhibitions | Since 1982, KIPA has sponsored the participation of outstanding entrepreneurs and individual inventors in international invention exhibitions to help them gain recognition in the global arena and develop new sales channels through promotion overseas. |

Source: “KIPA: Intellectual Property No. 1 Partner.”

As manifested in the above [Tables 4-3] to [4-8], KIPA may have been overly proactive in many frontiers of the IP ecosystem. For example, the IPAT (IP Ability Test), which is illustrated in C-1-1 of [Table 4-5], was ambitiously launched, but its score has never been recognized in any meaningful scope in Korea. The various IP contests explained in B-1, B-1-1, and B-1-2 of [Table 4-4] have also been used for purposes other than those which KIPA originally had in mind. For example, the parents of high school students take these contests as an opportunity to help their children to gain admission to a better university, by including a long list of awards in their résumés.

In addition, regarding “Training the next generation of entrepreneurial talents”, as shown in A-3 of [Table 4-3], KIPA originally planned to recruit elementary school students and then supervise them right through elementary, middle and high school. This corresponds to a 10-year supervisory program.

However, many commentators have criticized the old-fashioned curriculum and the conventional, non-intuitive training methods adopted by this program. Some people have even lamented that it would be the very kind of training program from which innovators like Bill Gates, Sergey Brin and Larry Page would run for their lives.

## **2.2.4 Policy Suggestions: Korean Precedents Worth Being Investigated and Benchmarked by the Government of Azerbaijan**

Despite such critiques of some of their ambitious programs, however, KIPA deserves much credit for developing various training programs year after year and for gradually and successfully creating a large pool of IP human resources in the Korean IP ecosystem. Accordingly, it is high time for Azerbaijan's government to carefully investigate which of KIPA's programs would help Azerbaijan in building a virtuous IP ecosystem. Of course there will inevitably be differences between the IP ecosystem built in Korea and the one that will be built in Azerbaijan. Regardless of differences in culture and infrastructure, however, every virtuous IP ecosystem has some common features. Based on such features, the following are just a few of the projects which KIPA has been consistently and persistently conducting in Korea and which, when adopted, will help Azerbaijan to develop a virtuous IP ecosystem and eventually obtain numerous successful cases of technology transfer and commercialization on Azerbaijan's soil.

### **2.2.4.1 Developing a Large Pool of Azerbaijani Inventors by Creating an Invention-Friendly Atmosphere**

As shown in [Figure 4-7], the IP ecosystem, whether it be a virtuous or a vicious one, always starts with inventors. As explained above, inventors are the main players in the upstream of the IP ecosystem.

Without inventors, the IP ecosystem cannot be developed to the midstream and the downstream. This leads one to the conclusion that successful technology transfer or successful technology commercialization has to start with inventors and their inventions. As explained in Figure 0.3, only a small fraction of the countless inventions that inventors come up with are ever commercialized, usually somewhere in the region of 5% or so. Here, the author has assumed that 600 or more ideas or 300 or more inventions would be required to obtain a single successful case of a commercialized technology. As such, a government could recruit a single, extremely clever inventor and ask him or her to come up with 600 ideas, and then push him or her to sublimate those 600 ideas into 300 patentable inventions. Of course, such a plan would be far more likely to fail than to succeed. Even if that single

inventor consented to this plan and worked very hard, there would inevitably be a limit to his or her creativity.

Accordingly, a better strategy would consist in developing a pool of inventors and motivating them to come up with lots of new and innovative ideas and then to convert such ideas into high-quality inventions. To develop such a big pool of inventors, then, Azerbaijan's government could launch many programs and create an invention-friendly atmosphere. When these programs work, more professors, graduate students, scientists, engineers, company researchers, and even ordinary citizens will be motivated to gladly enter Azerbaijan's IP ecosystem, and come up with ever more ideas and inventions.

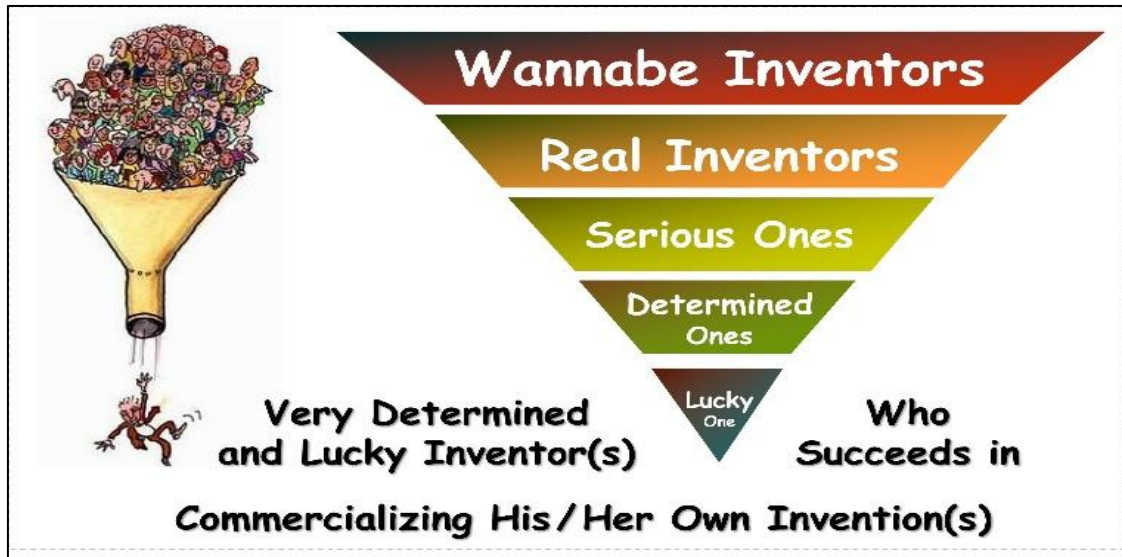
To this end, the Azerbaijan government could benchmark various of KIPA's projects, such as those listed in Table 0.3 (regarding invention contests hosted by KIPA) and those listed in Table 0.7 (regarding the development of an invention culture at home and abroad).

#### **2.2.4.2 Equipping Azerbaijani Inventors with IP Weapons through IP Education and Training**

There is a caveat, however, to creating a big pool of inventors and innovators. Regardless of the number of inventors in the pool, the *Pareto principle* predicts that most of them will either be novices or simply chasing their dreams, which may lack technical superiority, commercial marketability, and/or patentability.

In addition, when the assumption made in [Figure 4-9] is applied, only one inventor out of 600 candidates can succeed in commercializing his or her technological invention.

[Figure 4-11] Why a Virtuous IP Ecosystem needs lots of Inventors



Source: Y. Shim (2016), "The IP Silk Road".

[Figure 4-11] is a graphical representation of a large pool of inventors, where each inventor works hard to succeed in commercializing his or her own invention. As the figure shows, only a small fraction of the inventors succeed in commercializing their technology, while others end up losing all the time, effort, money, etc. that they have invested.

More importantly, small individual inventors or start-up entrepreneurs may be inflicted with financial difficulties during such a process, and many of them may never be able to overcome the financial and emotional damages. Once these many inventors drop out of the race, there will be fewer inventors in the IP ecosystem, and the ecosystem itself will fall into a vicious cycle.



[Figure 4-12] Human Wave Attacks



Source: Wikipedia (2022), "Human wave attack".

In this regard, scrounging novice (in the sense of IP expertise) inventors into the pool and forcing unprepared inventors to invent may amount to an “inventor wave attack”, in other words it is analogous to the reckless human wave attacks carried out during World War Two or the Korean War, as shown in [Figure 4-12].

Accordingly, it is essential to train potential inventors before sending them to join the pool of fellow inventors and to actually invent. To this end, the government may benchmark KIPA’s various projects, such as those listed [Table 4-5] (regarding the training of IP personnel).

**(A) *In Situ* IP Education and IP Training - For Professors, Graduate Students, and Researchers at Universities, Government Research Institutes, and Private Companies**

The Azerbaijan government could benchmark the IP education which is explained in C-1-2 of [Table 4-5]. For example, it might ask or even mandate professors, graduate students and other researchers at leading universities to undergo training about the basic patent system and patent protection. It would be even better if those professors, graduate students and other researchers were also conducting government-sponsored R&D.

After providing the IP education outlined in the above paragraph or simultaneously therewith, the Azerbaijan government could provide similar IP education to the college students who will become the next generation of inventors and innovators of Azerbaijan. It

should be noted that this IP education may be those explained in C-1-3 of [Table 4-5]. When feasible, Azerbaijan's government could provide such IP education as core courses for students in schools of engineering and science, and as elective courses for students in different fields. When it is not practical to include such IP courses in the regular curriculum due to administrative or other reasons, the government could urge universities or colleges to do so, while providing financial incentives, such as subsidizing the salaries of lecturers of IP courses, and the like.

**(B) *Ad Hoc* IP Training Tied to Government-Sponsored R&D, *a.k.a.*, No-Training-No-Funding Strategy**

The Azerbaijan government may (either alone or in combination with the IP education referred to in **Section (A)** above) mandate researchers (*e.g.* professors, graduate students or other researchers) to attend such training for a certain period of time, particularly when the researchers perform R&D with government funding. That is, the government may make R&D funding conditional upon the (successful) completion of the IP training. The government may also require such researchers to undergo IP training while they are conducting government-sponsored R&D. Alternatively, the government may provide the funding only to those researchers who finish such training. When desirable, the government may adjust the length of IP training according to the amount of funding which the researchers receive from the government or other factors. For example, the government could require a professor and his or her research group to undergo longer and more in-depth training if they receive more funding than other researchers. In another example, the Azerbaijan government could also adjust the length of IP training depending on the field of R&D. Accordingly, a professor and/or his or her research group in the field of physical science, electrical engineering or mechanical engineering (*a.k.a.*, the “predictable art”) could be required to undergo a simpler and shorter period of IP training. Conversely, those in the material science, chemistry, life science, genetic engineering, medicine, and pharmaceuticals sectors (*a.k.a.*, the “unpredictable art”) could be required to follow a longer period of more sophisticated IP training. For your information, many Korean IP experts have submitted such proposals to various ministries of the Korean government in the past. However, such proposals received negative responses from a majority of Korean professors and, therefore,

were rejected over and over again. The most common excuse proffered by those professors was that they were just too busy to find the time needed for such training. The second common excuse was that the professors had already assigned the IP management of a certain laboratory to a certain graduate student and thus did not see any extra reason why they and their graduate students should have to undergo additional IP training.

Regardless of the validity of the arguments advanced by Korean professors, it is only common sense that attaching an additional string to an existing and ongoing business will always be an uphill battle for an administrator. Accordingly, if the government of Azerbaijan wishes to adopt R&D funding conditioned upon the completion of IP training, it would be wise to implement such a provision as a new type of government-sponsored R&D or as R&D for which the scale of the funding may be of a greater order of magnitude than that set aside for ongoing projects.

### 2.3 Securing Employees' Intentions

As discussed above, successful technology transfer and its subsequent commercialization require commercially valuable patents, in which new and competitive technological inventions are the prerequisites for such valuable patents. That being the case, it is almost impossible to have commercially valuable patents, unless inventors can come up with new and competitive technological inventions. Therefore, a government may find it desirable to set up various policy goals and devise the related IP policies so that it can:

- ① Supervise inventors in their efforts to conceive innovative technological inventions in certain areas;
- ② Help inventors to file patent applications and then obtain commercially valuable patents, and
- ③ Supervise such patents for successful technology transfer and commercialization.

When a government succeeds in achieving (1) to (3) above, the IP ecosystem can vibrate with numerous successful technology transfers and their subsequent commercialization,

and a virtuous IP ecosystem can be built on that fertile soil in a shorter period of time. However, there are some technological inventions in which a government might not be able to intervene. For example, when an individual inventor (or a private company) spends his or her own money, comes up with such technological inventions, and obtains patents for them, those inventions and their patents are owned by the individual inventor. That is, such inventions and patents are private properties. Accordingly, the inventor may use his own inventions and patents as he sees fit. Any governmental intervention in such private properties, even with good intentions, may be or may amount to illegal interference in private properties. However, when an individual (inventor) receives R&D funds from a government and comes up with a technological invention, this is a completely different story, for the source of the government R&D fund is the taxes collected from the citizens of the country. Therefore, when an inventor conceives an invention while conducting government-sponsored R&D, the government may be tempted to claim either partial or complete ownership of any such inventions or patents, or may instead be tempted to intervene and participate in technology transfer and commercialization on the inventor's behalf. In addition to such *ex post* intervention in managing inventions and patents, the government may intervene *ex ante*, in other words, it may advise the inventor to conduct R&D in a certain area, to conceive valuable technological inventions in that area, and to manage such inventions and patents in certain ways.

In fact, many countries have passed special laws that allow their governments, either directly or indirectly, to guide inventors (or their employers) who conduct government-sponsored R&D to some extent. More particularly, those laws are designed to empower the employers of the inventors concerned in such a way that the employers themselves can manage any inventions and their patents - primarily for the purpose of systematically pursuing successful technology transfer and commercialization. In this respect, governmental policies and intervention based thereon can be regarded as indirect supervision of technological inventions and their patents by the government.

[Table 4-9] R&amp;D Expenditure as a Percentage of National GDP

| Country/Year | 2015  | 2016  | 2017  | 2018  |
|--------------|-------|-------|-------|-------|
| Azerbaijan   | 0.222 | 0.206 | 0.185 | 0.184 |
| China        | 2.06  | 2.10  | 2.12  | 2.14  |
| Israel       | 4.27  | 4.51  | 4.82  | 4.94  |
| Korea (Rep)  | 3.98  | 3.99  | 4.29  | 4.53  |
| Russia (Fed) | 1.10  | 1.10  | 1.11  | 0.98  |
| U.S.A.       | 2.72  | 2.76  | 2.82  | 2.83  |

Source: Global-economy.com (2020) "Azerbaijan: Research and development expenditure"

[Table 4-9] summarizes recent trends in R&D expenditure as a percentage of GDP of several representative countries. It is noted that expenditure as presented in [Table 4-9] includes both government and private sector expenditure, and that the average percentage of R&D expenditure of eighty-two countries in 2018 was 1.07%.<sup>61</sup> Although R&D expenditure as a percentage of GDP is very small for Azerbaijan, its government is responsible for 86% of all R&D expenditure.<sup>62</sup> In short, the Azerbaijan government has considerable capacity to spearhead government-sponsored R&D, the inventions created from such R&D, and the patents obtained from such inventions.

This section presents the general policy rationale behind the laws regarding "employee inventions," different laws and regulations on employee inventions adopted by other countries, and policy suggestions regarding how the Azerbaijan government could manage such employee inventions most efficiently so as to facilitate technology transfer and technology commercialization.

**2.3.1** presents a brief introduction to the Bayh-Dole Act of 1980 (U.S.A.) which gave rise to fundamental changes in the management of inventions and patents created by U.S. universities and U.S. government research institutes, particularly those inventions and patents created by their professors or researchers while conducting U.S. government-

<sup>61</sup> World Bank (2022), 「Research and development expenditure」.

<sup>62</sup> Global-economy.com (2022), 「Azerbaijan: Research and development expenditure」.

sponsored R&D. **2.3.1** also introduces the background to the technology licensing offices based at U.S. universities and government research institutes, which were also established based on the Bayh-Dole Act. Considering the *status quo* of the IP ecosystem under development in Azerbaijan, **2.3.2** provides (1) the basic rules on the ownership of an invention and a patent in **2.3.2.1**; (2) summarizes important provisions of the employee invention laws of the U.S. in **2.3.2.2**; (3) summarizes those of Korea in **2.3.2.3**; and (4) summarizes those of Germany in **2.3.2.4**.

Based thereon, **2.3.3** presents the following two policy suggestions for the Azerbaijan government: (1) the Azerbaijan government could indirectly manage the technological inventions and their patents created by the professors and researchers of domestic universities and research institutes while they are conducting government-sponsored R&D; and (2) the government could set up an infrastructure that enables Azerbaijan's universities and government research institutes to create, and then transfer or commercialize valuable inventions and patents, more efficiently.

### **2.3.1 Bayh-Dole Act (U.S.) and TLO (Technology Licensing Office)**

After World War II, the US government under President Truman decided to continue and even ramp up its spending on R&D, on the basis of Vannevar Bush's famous report titled "*Science, the Endless Frontier*", which stated that "scientific progress is one essential key to our security as a nation, to our better health, to more jobs, to a higher standard of living, and to our cultural progress."<sup>63</sup>

#### **2.3.1.1 The U.S. Government's Management of Patents Created through Government-Sponsored R&D before 1980**

Before the enactment of the Bayh-Dole Act in 1980, the U.S. government and its ministries and agencies owned the patents created through government-sponsored R&D. By 1980, the U.S. government was spending as much as \$75 billion each year, and had accumulated

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<sup>63</sup> Vannevar Bush, U.S. Government Printing Office, Washington (1945), [Science The Endless Frontier: A Report to the President by Vannevar Bush, Director of the Office of Scientific Research and Development].

28,000 patents, but fewer than 5% of those patents were commercially licensed.<sup>64</sup>

Both scholars and politicians sought different approaches to the patents obtained by government-sponsored R&D. For example, those with a Hamiltonian belief argued that the solution lay with a strong central government, which must take charge and actively manage these resources." Conversely, those with a Jeffersonian belief debated that the solution lay with the individual and that the best thing (the) government could do to provide incentives for success was to get out of the way of these individuals." The last group consisted of those who believed that "the government could only hurt and that the government should make sure that everyone benefited financially from the government's efforts."<sup>65</sup> However, the U.S. government did not have a unified patent policy governing all the agencies about government-funded R&D. The general policy was that government would retain the title to inventions and only license them nonexclusively. But this general policy encountered oppositions from the inside. It did not take long for government officers to realize that the government was not a suitable subject for spearheading technological transfer or commercialization. One reason for this may have been that, in their view, technological transfer and/or commercialization always involved the transfer of commercial value from one side to another, and that government officials were not trained for such tasks.

The above general policy also encountered opposition from the outside, such that the regulations and processes involved were too disorganized and complicated to lead to efficient technological transfer and commercialization. For example, the Government Accountability Office published a report stating that: "Those seeking to use government-owned technology found a maze of rules and regulations set out by the agencies in question because there was no uniform federal policy on patents for government-sponsored inventions or on the transfer of technology from the government to the private sector."<sup>66</sup> To alleviate those challenges, some government agencies and private institutions began to make compromises. For example, in 1968 the Department of Health, Education, and

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<sup>64</sup> Wikipedia (2022), [Bayh-Dole Act].

<sup>65</sup> Ashley Stevens (2004), [The enactment of Bayh-Dole] (2004).

<sup>66</sup> US General Accounting Office (1998), [Technology Transfer, Agencies' Rights to Federally Sponsored Biomedical Research].

Welfare introduced a uniform "Institutional Patent Agreement" to allow grantee nonprofit institutions to obtain the assignment of patentable inventions developed with federal funding for which the institutions concerned had decided to seek patents.<sup>67</sup> In addition, by 1978, over seventy universities and research organizations had negotiated "Institutional Patent Agreements" with the Department or with the National Science Foundation.<sup>68</sup>

In the 1970s, the faculty at Purdue University made important discoveries while in receipt of grants from the Department of Energy, which did not issue any kind of institutional patent agreement. The faculty complained to Senator Birch Bayh, whose staff investigated the matter. At the same time, Senator Robert Dole was made aware of similar issues, and the two senators agreed to collaborate on a bill along the lines of the "Jeffersonian belief" described above.<sup>69</sup>

### 2.3.1.2 Bayh-Dole Act of 1980 (U.S.)

The Bayh–Dole Act of 1980 grew out of the Congress's efforts to respond to such problems.<sup>70</sup> The main focus of the Bayh–Dole Act was how to manage inventions that were created with the more than \$75 billion a year invested in government-sponsored R&D. In short, the Bayh–Dole Act decentralized control of the federally-sponsored inventions and their patents, mainly by vesting the responsibility and authority to commercialize such inventions and patents with the institutions and companies in receipt of a federally-sponsored grant, with certain responsibilities falling to the government, the inventor, and the public. The following is a short list of the responsibilities of any institution or company that receives a government-sponsored R&D grant and which elects to retain the title to such inventions:

- ① Grant to the government a nonexclusive, nontransferable, irrevocable, and paid-up license to practice or to have practiced for, or on behalf of, the U.S. the subject invention throughout the world;

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<sup>67</sup> US Dept. of Health and Welfare (1968), [Institutional patent agreement governing grants and awards from the Department of Health, Education, and Welfare].

<sup>68</sup> Vicki Loise and Ashley J. Stevens, *les Nouvelles*, (2010) [The Bayh-Dole Act turns 30]

<sup>69</sup> Wikipedia (2022), [Bayh-Dole Act].

<sup>70</sup> US General Accounting Office (1998), [Technology Transfer, Agencies' Rights to Federally Sponsored Biomedical Research].



- ② File its initial patent application within one year of its election to retain the title;
- ③ Notify the government of its intention not to continue with the prosecution of an application or to let a patent lapse;
- ④ Convey to the Federal agency, by written request, the title to any subject invention in the event that the organization fails to file, does not continue with the prosecution of, or intends to allow its patent to lapse;
- ⑤ Include in each patent a statement that identifies the contract under which the invention has been made and notification of the government's rights concerning the invention;
- ⑥ Report on the utilization of a subject invention;
- ⑦ Require, in exclusive licenses to use or sell products in the United States, that such products will be manufactured substantially in the United States; and
- ⑧ Agree to allow the government to "march in" and require licenses to be granted, or to grant licenses, in certain circumstances, such as if an organization has not taken effective steps to achieve practical application of the invention.

It is appreciated that the above eight responsibilities are not mandatory and that there are many exceptions with which the institutions or companies concerned can find leeway's loopholes to avoid such responsibilities.

Many commentators have pointed out that the Bayh-Dole Act contributed to the birth of the biotechnology industry in the U.S.A. For example, the Baker Institute at Rice University published a report stating that the ability of U.S. universities to license patents has created strong incentives for many scientist-entrepreneurs to form companies around their inventions, that at least 50% of current biotech companies began as a result of university licenses, and that 76% of biotechnology companies have at least one license from a university.<sup>71</sup>

In addition to this strong contribution to the biotech industry, the Bayh-Dole Act made a crucial contribution to the formation of special offices in U.S. universities and government

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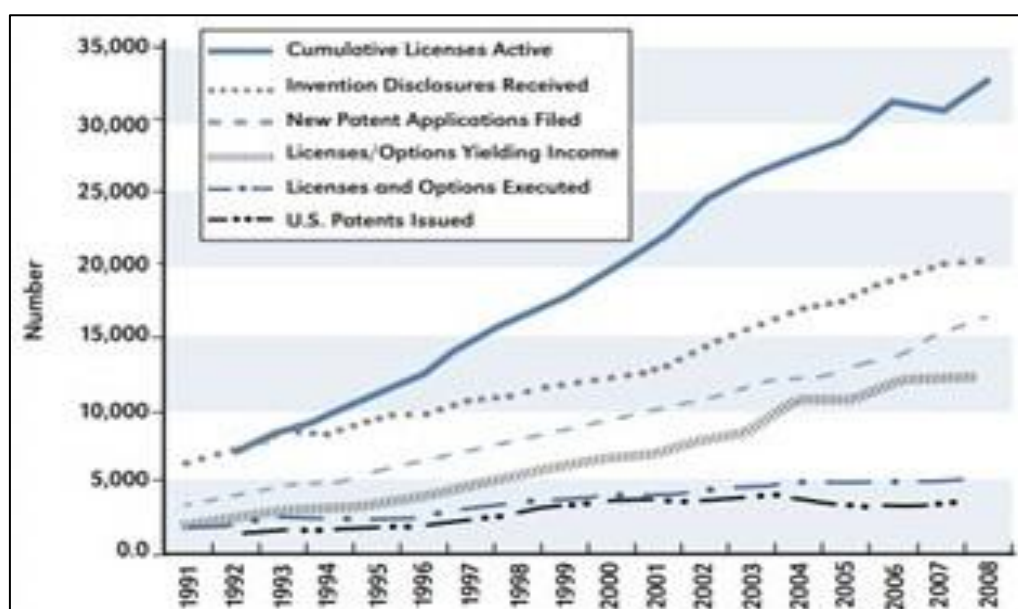
<sup>71</sup> Vicki Loise and Ashley J. Stevens, *Les Nouvelles*, (2010) [The Bayh-Dole Act turns 30].

research institutes, which were awarded the tasks of transferring or commercializing technological inventions and their patents. Although such offices may be referred to by different names, the terms “technology licensing office” (TLO), “technology transfer office” (TTL) or “office of technology transfer” (OTL) are those used most frequently.

### 2.3.1.3 Technology Licensing Offices (TLOs) of Universities and Government Research Institutes in the US

Before the Bayh-Dole Act of 1980, only twenty-three universities had TLOs. After its enactment, however, many more universities responded by establishing their own TLOs, although the initial response was somewhat slow.<sup>72</sup>

[Figure 4-13] Change in Core Measures of Technology Transfer Activity (U.S.)



Source: Vicki Loise and Ashley J. Stevens, *les Nouvelles* (2010), “The Bayh-Dole Act turns 30”.

Starting in 1983, the rate of establishment increased dramatically to the extent that, today, all major U.S. universities and research institutions have TLOs. [Figure 4-13] shows various core measures for the technology transfer activities of such TLOs.<sup>73</sup>

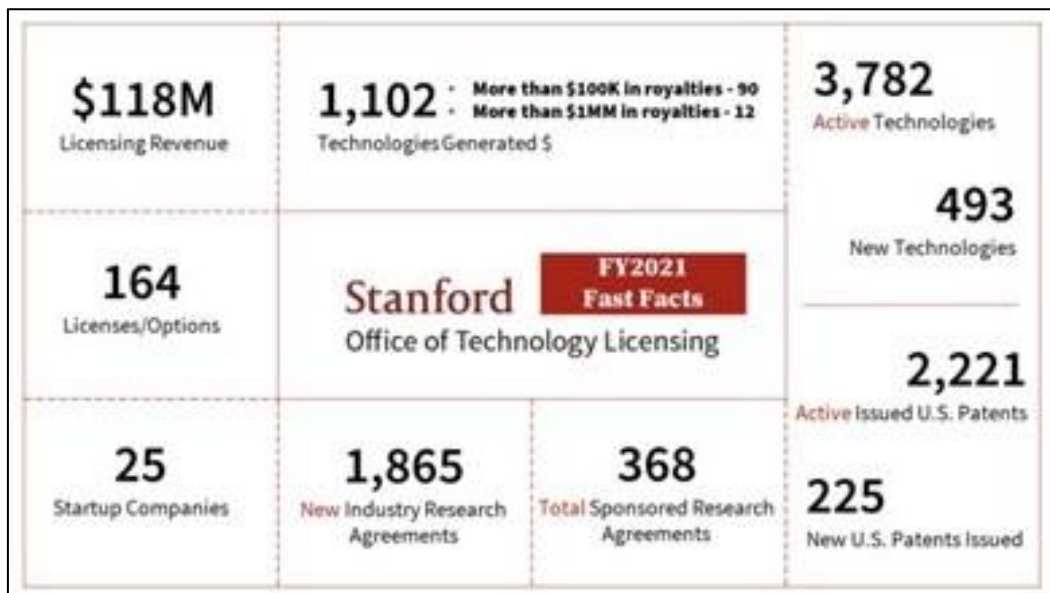
<sup>72</sup> Ashley Stevens (2004), [The enactment of Bayh-Dole] (2004)

<sup>73</sup> Baker Institute Blog (2016) [True impact of the Bayh-Dole Act]

As more U.S. universities and government research institutes establish their own TLOs, they become able to manage more employee inventions created by their professors, graduate students, and researchers through the TLOs, and can pursue technology transfer and commercialization more systematically.

One of the most famous university TLOs is the Office of Technology Licensing at Stanford University, which was established in 1977, i.e. three years before the enactment of the Bayh-Dole Act. The following chart explains the accomplishments of the Stanford OTL over the past forty-five years.<sup>74</sup>

[Figure 4-14] Accomplishments of Stanford Univ. OTL



Source: Stanford OTL (2022), "Fast Facts: Stanford OTL".

According to its home page, the Stanford OTL has fifty-one employees, of whom ten hold Ph.D. degrees in various science and technology areas, eight hold M.S. degrees, and eleven are patent or litigation lawyers. In contrast, one of the rising stars of the university TLOs is the University of Wisconsin's Wisconsin Alumni Research Foundation (WARF),<sup>75</sup> which

<sup>74</sup> Stanford OTL (2022), "Fast Facts: Stanford OTL".

<sup>75</sup> WARF (2022), "WARF, Wisconsin Alumni Research Foundation".

manages the IPs from thirteen state universities in the State of Wisconsin. In 2014, WARF filed a complaint to the effect that the Apple processors loaded into the iPhones 5s, 6, and Plus had infringed the “predictor circuit” patent (U.S. Pat. No. 5,781,752) filed by WARF. The district court initially awarded US \$234M to WARF in 2015, but WARF succeeded in getting the award increased to US \$506M in 2017. Apple appealed the district court verdict, and the Federal Circuit (the federal appellate court) reversed the judgement and held that Apple had not infringed WARF’s patent. Although WARF appealed the case to the U.S. Supreme Court, the Court did not take the case.<sup>76</sup>

According to its home page, the WARF has a total of seventy employees, of whom twelve hold Ph.D. degrees in various science and technology areas, fifteen hold M.S. degrees, seven are patent or litigation lawyers, and two are patent agents.<sup>77</sup> This does not mean that all the TLOs of U.S. universities and research institutes employ IP professionals and efficiently pursue technology transfer and commercialization. In fact, there are many university and research institute TLOs which do not and cannot function sufficiently well to promote technology transfer and commercialization, primarily because these organizations do not have qualified IP professionals, and are riddled with bureaucracy and the like.

Details as to how the Azerbaijan government can help the TLOs of domestic universities and research institutes and how such TLOs are operated are presented in **2.3.3**, while the following **2.3.2** covers the basic rules about the ownership of inventions and patents, and important provisions of the employee invention laws in the U.S., Korea, and Germany.

### 2.3.2 Employee Invention

An “employee invention” refers to an invention that is conceived by an employee (i.e. an employee-inventor) and falls within the scope of business of the employer of that employee. When all conditions are met, ownership shifts from the employee-inventor to the employer by way of law. The origins of this notion of employee invention can be found in the “shop

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<sup>76</sup> Malcolm Owen, *Apple Insider* (2019), [WARF fails to convince Supreme Court to reinstate \$506M award from Apple].

<sup>77</sup> WARF (2022), [Connect with WARF Employees].

right” of the medieval era. In general, the shop right refers to the right of an employer to use an employee's invention without compensating the employee for its use, in cases where an invention is made on the premises and during the hours of employment. In general, different countries adopt different laws or regulations about such employee inventions, but such differences are mainly attributable to different definitions about the scope of business of the employer.

### **2.3.2.1 Rules of Ownership of Inventions and Patents**

In the world of the inventions and patents, there are three rules regarding the ownership of inventions and patents. In general, an invention is an intangible property which does not exist on the Earth until an inventor conceives it. Accordingly, the first rule is that an inventor who conceives an invention is deemed to be the owner of that invention.

When a single inventor conceives an invention, that inventor becomes the sole owner of that invention. When two or more inventors conceive an invention together, those inventors are the joint owners or co-owners of that invention. A patent is also an intangible property which does not exist on the Earth until the owner of an invention files a patent application to a national patent office while citing that invention in at least one claim of the patent application. Accordingly, the second rule is that an inventor is deemed to be an owner of a patent when the patent includes at least one claim which cites an invention conceived by that inventor. When multiple joint inventors conceive an invention and at least one claim of the patent cites that invention, those joint inventors become the joint owners of the patent.

The third rule relates to the exceptions to the above first and second rules. That is, an inventor can assign the ownership rights of his or her invention to a third party through a personal contract; or, conversely, an inventor has to assign the ownership rights of his invention to his employer by way of law regarding the employee invention. In other words, when the above personal contract exists or when such a law is applicable, the ownership of an invention automatically shifts from the inventor to the third party or to the employer. The following are the relevant portions of the employee invention laws of the U.S., Korea,

and Germany. It is noted in the following that “employee” may correspond to a professor, a graduate student or a researcher, whereas “employer” corresponds to a technology licensing office (TLO) of a university or government research institute.

### 2.3.2.2 Employee Invention Laws in the U.S.

In the U.S., employee inventions are governed not by federal statutes but by state laws. Accordingly, there are as many as fifty different employee invention laws which may vary slightly from state to state. Fortunately, however, the regulations on employee inventions of the fifty states are more or less harmonized. In addition and interestingly enough, most states regulate “employee invention” as a part of the “labor law,” “consumer protection law” or “unfair and deceptive practices law.” Given that California is the state with the strictest labor law, the following is Article 3.5 of California’s Labor Code.<sup>78,79</sup>

California Labor Code  
Division 3. Employment Relations  
Chapter 2. Employer and Employee  
Article 3.5 Inventions Made by an Employee  
Section 2870.

(a) Any provision in an employment agreement which provides that an employee shall assign, or offer to assign, any of his or her rights to an invention to his or her employer shall not apply to an invention that the employee has developed entirely in his or her own time without using the employer’s equipment, supplies, facilities, or trade secret information, except for those inventions which either:

- relate at the time of their conception or reduction to the practice of the invention to the employer’s business, or which actually or demonstrably anticipate the research or development of the employer; or
- result from any work performed by the employee for the employer.

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<sup>78</sup> CA Government (2022), [California legislative information].

<sup>79</sup> The Business Professor (2022), [California’s Labor Code 2870 - Explained].

- (b) To the extent that a provision in an employment agreement purports to require an employee to assign an invention otherwise excluded from being required to be assigned under subdivision (a), such provision runs counter to the public policy of this state and is unenforceable.

It is noted that U.S. courts tend to analyze the conditions enumerated in subsection (a), although they may look trivial. Therefore, to prove that his or her invention does not belong to the company concerned, the employee must prove (1) that he/she has developed the invention entirely in his/her own time (i.e. not during work hours, but after work and/or during weekends or holidays); (2) that in developing the invention, he/she did not use any equipment (e.g. the company's copy machine or printer), supplies, facilities or trade secrets of the company. In addition, the employee must prove that his or her invention does not relate to the business and or to actual or demonstrably anticipated research or development of the company. Or the employee must prove that his or her invention does not result from any work performed by him/herself for the company. Although it may not be absolutely clear, the U.S. employee invention laws may classify an invention of an ex-employee as the employee's invention and as the property of the employer, in cases where the ex-employee conceives an invention after quitting (or being fired by) the employer's company, but that invention falls within the scope of actual or demonstrably anticipated research or development of the employer.

More importantly, the timing of the conception of an invention does not change the fact that an ex-employee's invention is an employee invention, in cases where the ex-employee conceives the invention after quitting (or being fired by) the employer's company, but where the ex-employee uses the trade secrets of the employer. It is also appreciated that the U.S. legal system strictly respects contracts executed between the parties. Accordingly, the U.S. state laws tend to not provide further details regarding additional rights and duties of employee and employer regarding employee inventions. In addition, the U.S. state laws do not expressly provide any provisions as to the compensation to be provided to employees who transfer the ownership rights to their employer.

### 2.3.2.3 Employee Invention Laws in Korea

Korean employee inventions are governed by Article 2 of the “Invention Promotion Act,” the relevant provisions of which are as follows:<sup>80</sup>

#### Invention Promotion Act

#### Article 2 (Definitions):

1. The term "invention" means inventions, designs, and creations subject to protection pursuant to the “Patent Act,” the “Utility Model Act,” or the “Design Protection Act.”
2. The term "employee invention" refers to an invention which is made by an employee, an officer of a corporation, or a public official (hereinafter referred to as "employee, etc."), and which is related to a task of the employee, in cases where the invention falls within the scope of the duties of an employer, corporation, or state or local government (hereinafter referred to as the "user, etc.") and where the act of conceiving the invention belongs to the present or past duties of the employee.
3. The term "individual inventor" means a person who has made an invention other than an employee invention.

The Invention Promotion Act also includes provisions on the additional rights and duties of the employee-inventor as well as the rights and duties of the employer-owner of such employee inventions. It is noted in general that a vast body of Korean law derives from Japanese law, which in turn derives from German law. Accordingly, it is presumed that those provisions related to employee inventions in the Invention Promotion Act are similar to those of the relevant Japanese law. In short, the main criteria for determining whether or not an invention of an employee-inventor is an employee invention in Korea are (1) the scope of the duties (or businesses) of the employer, and (2) whether or not the employee-inventor conceives the invention while performing his/her present or past duties.

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<sup>80</sup> Korean Invention Promotion Act (2022).



Similar to the U.S. employee invention laws, the Korean laws do not seem to require that the employee invention be one which is conceived by the employee during his or her employment. Accordingly, an invention conceived by an ex-employee can be regarded as an employee invention in which the act of conceiving the invention belongs to the past duties of the employee.

#### 2.3.2.4 Employee Invention Laws in Germany

On July 25, 1957, Germany enacted the separate “Employee Inventions Act (Gesetz über Arbeitnehmererfindungen). The following are English translations of the relevant provisions concerning the inventions of German employees.<sup>81</sup> It is appreciated that the German law classifies “employee invention” into (1) a “tied (or service) invention” and (2) a “free invention,” where the tied (or service) invention corresponds to the “employee invention” used in the U.S. and Korean laws.

##### Employee Inventions Act

##### Chapter 1. Applications and Definitions

##### Section 4. Service Inventions and Free Inventions

- (1) Employee inventions within the meaning of this Act may be either tied or free.
- (2) Tied inventions (service inventions) are those made during the term of employment which:
  - (i) have either resulted from than employee’s tasks in a private enterprise or in a public authority, or
  - (ii) are essentially based upon the experience or activities of the enterprise or public authority.
- (3) Other inventions of an employee shall be free inventions. They shall, however, be subject to the limitations set forth in Sections 18 and 19, below.
- (4) Subsections (1) to (3) shall apply mutatis mutandis to inventions made by civil servants and members of the armed forces.

When comparing with the respective parts of the U.S. and Korean laws, Section 4 of the

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<sup>81</sup> Deutsches Patent und Markenamt (2022), [Employee Inventions Act].

German Employee Inventions Act seems to define the scope of the tied invention very narrowly. However, that may not be the case when considering the above subsection (3) and the following Sections 18 and 19.

### Employee Inventions Act

#### Chapter 2. Inventions and Technical Improvement Proposals Made by Employees in Private Employment

1. Service Inventions
2. Free Inventions

#### Section 18. Duty to Notify

- (1) An employee who has made a free invention during the term of an employment contract shall notify his or her employer by making a statement thereof in text form without delay. He/she shall give the employer all the details concerning the invention and, if necessary, its realization - which the employer may need in order to judge whether it is in fact a free invention.
- (2) Where the employer does not contest that the invention notified to him is a free invention, by making a statement in text form to the employee within three months of receipt of the notification, the invention may no longer be claimed as a service invention (Section 6).
- (3) There shall be no obligation to notify the employer of a free invention if the invention is obviously not capable of being used in the employer's enterprise.

#### Section 19. Duty to Offer

- (1) Before exploiting a free invention further during the term of his or her employment contract, an employee must offer his or her employer at least a non-exclusive right to use the invention on reasonable terms, if the invention falls within the range of the actual or planned activities of the employer's enterprise at the time the offer is made. Such offer may be submitted together with the notification required by Section 18.
- (2) Where the employer does not accept the offer within three months, his/her prerogative shall lapse.

- (3) Where the employer states within the time provided by subsection (2) that he/she intends to acquire the rights offered to him/her, but claims that the terms offered to him/her are not reasonable, the court shall determine the terms upon a declaratory action by the employer or employee.
- (4) The employer or the employee may request a new determination of the terms, if the circumstances essential to the terms hitherto agreed or fixed have changed substantially.

Based on these Sections 18 and 19, the German Employee Inventions Act is unique in that an employee has to report most of his free inventions to his employer, and “must” offer the non-exclusive right to use the invention on reasonable terms. In short, the German Employee Inventions Act may be viewed as being as stringent as, or even more stringent than, its counterparts in other countries.

#### **2.3.2.5 Proactive Employee Invention Laws of Germany capable of Preventing Employer’s Moral Hazard and Maximizing the Efficiency of Technology Transfer and Commercialization**

Many countries have adopted employee invention laws in order to protect not only employees but also their employers. For example, such laws protect the rights of an employee to his free inventions by clarifying the scope of the employee inventions and by clarifying that any invention which falls outside the scope of “employee invention” are free inventions. In addition, the employee invention laws protect the rights of an employer as to the employer’s investment, tangible assets (e.g. equipment, facilities, and such like), and intangible assets (e.g. know-how and trade secrets). In this sense, the rights of an employee (e.g. the right to be compensated by an employer, the right to own his own free invention, etc.) correspond to the duties of the employer (e.g. the duty to compensate an employee who assigns his/her employee invention to his/her employer, a duty not to claim the employee’s invention as an employee invention, etc.).

Similarly, the duties of the employee (*e.g.* the duty to report an employee invention to his/her employer within a present period, a duty to give his/her employer a nonexclusive

licence to the employee invention when the latter has not included such a duty in a contract, etc.) correspond to the rights of the employer (e.g. the right to prevent the employee's fraud, the right of the employer to get a return on the investment, etc.). In practice, however, the employee invention laws of most countries cast a blind eye to a potential moral hazard of an employer. In other words, even after an employee transfers the rights of his employee invention to his employer, the employer may not utilize such employee inventions to the fullest extent. For example, where an employer obtains a title to an employee invention from an employee, but does not seek any patent protection for such an invention. Even when the employer seeks patent protection, the employer hires a cheap expert such that the resulting patent ends up having a very narrow scope of exclusive rights. Even when the employer obtains the patent for the employee invention, the employer does so only in his own country but not abroad. Accordingly, competitors around the world can practice the patented invention in other countries free of charge. In this respect, the German Employee Invention Act is exceptionally unique and intriguing in that it includes very interesting and reasonable provisions for preventing such moral hazard on the part of the employer, where representative provisions against the moral hazard on the part of the employer are provided in Sections 13 and 14 of the Act.<sup>82</sup>

#### Section 13. Application for Domestic Industrial Property Protection

- (1) An employer shall be under a duty - and shall be solely entitled - to apply for a domestic industrial property right for a service invention reported to him. Where the invention is capable of patent protection, he shall apply for a patent unless, on an evaluation of the industrial applicability of the service invention, protection as a utility model appears more appropriate. The application shall be filed without delay.
- (2) The employer's obligation to file an application shall not arise:
  - (i) where the service invention has become free (Section 8);
  - (ii) where the employee has agreed that no application is to be filed;
  - (iii) where the conditions contained in Section 17 are met.
- (3) Where, after making a claim to a service invention, an employer does not comply with his duty to apply for industrial property protection

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<sup>82</sup> Deutsches Patent und Markenamt (2022), [Employee Inventions Act].

and also fails to do so within a reasonable additional period fixed by the employee, the employee may file an industrial property application for the service invention in the employer's name and at the employer's expense.

- (4) Where a service invention has become free, only the employee shall be entitled to apply for industrial property protection therefor. Should his employer already have applied for industrial property protection for the service invention, his rights resulting from such application shall pass to the employee.

#### Section 14. Application for Industrial Property Protection Abroad

- (1) After making a claim to a service invention, an employer shall also be entitled to apply for industrial property protection abroad.
- (2) For foreign countries in which an employer does not desire to acquire industrial property rights, he shall release the service invention to the employee and shall, upon request, enable the employee to acquire such rights. The release must be effected in sufficient time for the employee to take advantage of the priority dates under international treaties in the field of industrial property.
- (3) At the time of releasing a service invention under subsection (2), an employer may reserve for himself a non-exclusive right to use the service invention in the foreign countries concerned, against reasonable compensation, and may require the employee, also against reasonable compensation, to respect the employer's obligations arising from contracts existing at the time of the invention's release, while the employee is exploiting the service invention released to him.

These Sections 13 and 14 seem to reflect the legislative intent of the German Congress that innovations do not happen often and, accordingly, that an employer who decides to obtain the ownership of an employee invention must fulfill his duty to protect the employee invention not only in Germany (Section 13) but also in other countries (Sub-Section (1) of Section 14). In addition, where the employer decides not to file a foreign patent application or decides to file a foreign patent application in some countries but not in others, the employer must inform the employee inventor sufficiently ahead of time and release the

rights to obtain patents in those other countries. Accordingly, the employee-inventor can invest his own resources, obtain the patents in those other countries where the employer does not obtain patent protection, and pursue technology transfer or commercialization in those other countries. It is appreciated that provisions such as Sections 13 and 14 are not found in the similar laws of other countries. Perhaps Sections 13 and 14 reflect German pragmatism such that:

- The adoptive parents (i.e. the employer) must do their best in raising an adopted child (i.e. protect and commercialize the employee invention);
- The adoptive parents (i.e. the employer) must send the adopted child (i.e. release the employee invention) back to his parents (i.e. back to the employee) when the adoptive parents (i.e. the employer) can no longer fulfil their duties (i.e. to protect and commercialize the employee invention).

### **2.3.2.6 Policy Suggestions: Adopt Employee Invention Laws Like the Bayh-Dole Act and Ensure Maximum Protection on Inventions Created through Government-Sponsored R&D**

In 1858, just three years before being elected as the President of the U.S.A., Abraham Lincoln gave a speech on the benefits of the patent system. Although not as famous as the one in Gettysburg, the following sentence exemplifies Lincoln's insight on the patent system:

"The patent system added the fuel of interest to the fire of genius, in the discovery and production of new and useful things."<sup>83</sup>

As Lincoln preached, the very basis of the patent system lies in the incentive system. Accordingly, the virtuous IP ecosystem can ignite the fire of a genius by guaranteeing to the genius the necessary fuel (which anyway is very precious these days). When the IP ecosystem is vibrating and kicking vibrant (i.e. the virtuous IP ecosystem), there is generally no need for a government to step in. Rather, the "invisible hand" of the ignited genius will take care of the details. However, when the IP ecosystem is going nowhere (i.e. the vicious

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<sup>83</sup> Abraham Lincoln (1858), at the Young Men's Association of Bloomington, Illinois [Discoveries and Inventions].

IP ecosystem), the government can play a role in converting the vicious IP ecosystem into a virtuous one. For example, the government may invite the genius, supply the fuel to the genius, ignite the fuel in the hands of the genius, and so on. By supervising them but not intervening in-between, the virtuous IP ecosystem can be established in a shorter period of time and, ultimately, more successful technology transfer and commercialization can be accomplished. To this end, the following are just a few policies which, when adopted, could help Azerbaijan in guaranteeing the systematic creation of technological inventions and in systematically and efficiently converting such inventions into patents in Azerbaijan as well as in other strategically important countries. By providing such a systematic and efficient inflow of high-quality inventions into the IP ecosystem, the Azerbaijan government could speed up the construction of the virtuous IP ecosystem and eventually accomplish successful technology transfer and commercialization on Azerbaijan's soil.

### **2.3.2.7 The Need for Bayh-Dole Act-Like Employee Invention Laws**

In this respect, Azerbaijan's government could enhance the creation of technological inventions and their patents by enacting new Bayh-Dole-like employee invention laws or amending the current laws. During such enactment or amendment it has been noted that (1) well-thought-out policies and laws are tied to government-sponsored R&D such that those policies and laws can assist universities or research institutes in Azerbaijan with systematically creating as many inventions as possible and (2), when managed properly, the TLOs of such universities or research institutes can efficiently convert such inventions into patents issued not only in Azerbaijan but also in strategically important countries. More particularly, the Azerbaijan government is in a good position to do this as it is responsible for 86% of the country's entire R&D expenditure.<sup>84</sup> Accordingly, the execution of such policies and laws in conjunction with government-sponsored R&D can guarantee a consistent line in producing technology inventions. In addition, by providing in situ IP education and training for inventors as well as ad hoc IP training tied to government-sponsored R&D, as proposed in 2.2.4.2, the quality of inventions created through

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<sup>84</sup> Global-economy.com (2022), [Azerbaijan: Research and development expenditure].

government-sponsored R&D can be improved as well.

### **2.3.2.8 Pragmatic Supervision or Intervention May Be the Key to Successful Technology Transfer and Commercialization**

Regardless of how many inventions may be created through government-sponsored R&D, such inventions should be converted into high-quality patents in order to improve the chances of success of technology transfer and commercialization. As will be explained in **Section 3** below, however, even many TLOs of universities and government research institutes in the U.S. suffer from below-average management of inventions and patents. To make matters worse, such subpar TLOs tend to turn into big bureaucratic organizations; and, as one may expect, such bureaucratic offices never perish once formed. Therefore, whether the Azerbaijan government enacts the new employee invention law or amends the current law, it must be pragmatic. That is, the new or amended law had better provide supplementary or alternative channels by which either or both universities (and research institutes) and inventors can pursue technology transfer and commercialization. To this end, the pragmatic provisions of the German Employee Invention Act could be fully incorporated.

## **2.4 The Best Player to Supervise or Manage University-Born Inventions and Patents for Technology Transfer or Commercialization**

The aim of this chapter is to propose a few options that would enable Azerbaijan's government to fine-tune its IP management policies and laws and establish a virtuous IP ecosystem in which home-grown technologies are transferred to and commercialized by the end users. To this end, it was suggested in **Section 1** that Azerbaijan's government should take the initiative in raising patent awareness among inventors (*e.g.* professors, graduate students, and researchers) and government employees, mainly through IP education and training spearheaded by the government. In addition, when necessary, it was also suggested in **Section 1-4-2** that the government could tie such IP education and training with government-sponsored R&D. It was also suggested in **Section 2** that the government should take a closer look at its employee invention laws, and either enact a new law or



amend an existing law in such a way that the Azerbaijan IP ecosystem can enjoy a steady (ever-increasing) influx of technological inventions. To this end, the government was recommended to take a closer look at the pragmatic provisions of the German Employee Invention Act in such a way that the new or amended Azerbaijan employee invention law would provide supplementary or alternative channels through which universities (and research institutes) and inventors could pursue technology transfer and commercialization, either individually or together. The above **Section 2** and **Section 3** logically lead to this **Section 3** where attention is given to determining who would be the best candidate to manage university-born or research institute-born inventions and patents. In addition, attention should also be given to what would be the objective of such management. In this regard, Section 3.1 provides the licensing activities of the top 10 TLOs of U.S. universities, while Section 3.2 provides the licensing activities of the top 10 TLOs of Korean universities. Section 3.3 then compares the (licensing) activities and licensing incomes of the TLOs of U.S. universities with those of Korean universities and presents the underlying reasons for such differences in those TLOs, particularly the inherent deficiencies in governance and poor staffing problems of many TLOs of Korean universities as well as the undesirable influence of the Korean Ministry of Education. Based thereupon, Section 3.4 provides a few policy suggestions to the Azerbaijan government.

### 2.4.1 Licensing Activities and Impressive Royalties of the Top 10 U.S. Universities

As discussed above, some twenty-three U.S. universities had established TLOs even before the enactment of the Bayh-Dole Act in 1980,<sup>85</sup> while many U.S. universities followed suit after its enactment.

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<sup>85</sup> Vicki Loise and Ashley J. Stevens, *Les Nouvelles*, (2010) [The Bayh-Dole Act turns 30].

[Table 4-10] Licensing Activities and Royalties of the Top 10 U.S. Universities (2013)

| Arial 11.5                      |       | R&D Budget | Licensing | Start-up | Inv. Disc. | Pat. Filed    | Royalty       | Running Royalty |
|---------------------------------|-------|------------|-----------|----------|------------|---------------|---------------|-----------------|
|                                 |       | (US \$M)   | (ea.)     | (ea.)    | (ea.)      | (ea.)         | (US \$M)      | (US \$M)        |
| 01. Northwestern University     |       | 668.1      | 44        | 11       | 213        | 118           | 257.3         | N/A             |
| 02. New York University         |       | 469.3      | 40        | 8        | 172        | 76            | 214.2         | 211.9           |
| 03. Columbia University         |       | 769.4      | 89        | 14       | 372        | 271           | 146.3         | 139.0           |
| 04. Princeton University        |       | 199.1      | 12        | 1        | 107        | 150           | 134.0         | 133.7           |
| 05. U. of California System     |       | 5,695.4    | 261       | 66       | 1,595      | 1,268         | 104.8         | 69.8            |
| 06. U. of Washington Foundation |       | 1,012.5    | 260       | 17       | 410        | 215           | 99.5          | 87.1            |
| 07. WARF (Wisconsin)            |       | 1,123.5    | 63        | 7        | 386        | 167           | 94.2          | 94.2            |
| 08. Stanford University         |       | 873.8      | 123       | 9        | 502        | 296           | 87.0          | 65.6            |
| 09. U. of Pennsylvania          |       | 903.7      | 130       | 26       | 391        | 136           | 86.6          | 58.4            |
| 10. M.I.T.                      |       | 1,606.0    | 103       | 14       | 672        | 449           | 69.7          | 46.1            |
| All 1 - 10                      | Total | 13,320.8   | 1,125     | 173      | 4,820      | 3,148         | 1,293.6       | 905.8           |
|                                 | Mean  | 1,332.1    | 112.5     | 17.3     | 482        | 314.8 (65.3%) | 129.4 (9.71%) | 90.6 (70.0%)    |

Source: AUTM (2013), "2013 AUTM Licensing Activity Survey".

[Table 4-10] summarizes the licensing activities and royalty incomes of the top 10 U.S. universities in 2013. Each column of the table shows the total R&D budget of each of university, the number of licensing agreements, the number of start-ups originating from each university's technology, the total number of invention disclosures, the total number of patent applications filed (which is almost always smaller than the number of employee inventions disclosed), the total amount of a royalty, the total amount of a running royalty, and so forth.

The "total number of invention disclosures" in the table means the total number of employee inventions which were disclosed by employee-inventors to the university. It is

noted in Table 0.9 that each of the top 10 U.S. universities recouped 9.71% of its total R&D budget spent. Given that each university spent some of its R&D budget on the liberal arts or other areas in which technical transfer or commercialization is not feasible, that 9.71% is a pretty impressive number.

In addition, each university earned US \$1.15M per licensing agreement (*i.e.*  $129.4 \times 100 / 112.5$ ). It should be noted, however, that although Table 0.8 lists the royalty income for 2013 and the number of licensing agreements signed in 2013, there is no doubt that some royalties must have been attributed to licensing agreements signed before 2013. That is, some royalties do not correspond to the licensing agreements executed in the year 2013. That said, these numbers may still be used as the representative averages of the activities of such TLOs. It is also appreciated that “royalty” can be classified into an “upfront royalty” and a “running royalty,” where the former refers to the royalty which a licensee pays to a licensor (*e.g.* the university or its TLO) as a commemoration of signing the agreement, while the latter is a portion of a sales revenue or a profit which the licensee earns by making and selling products which incorporate the licensed technology. In this sense, when a university receives a running royalty from a licensee, it means that (1) the licensee-manufacturer has embodied the licensed technology into its product, (2) the licensee has sold the product and generated revenues or made profits, and (3) the licensee has paid a portion of the sales revenue or profit to the TLO according to the contract. That is, the “running royalty” is a token signifying that the licensee-manufacturer has embodied the licensed technology into its product, and thus it is the token of a successful technology transfer that has resulted in a successful technology commercialization. It is extremely notable in this respect that 70.0% of the total royalties earned by each of the top 10 U.S. university TLOs are running royalties.

## 2.4.2 Licensing Activities and Non-Impressive Royalties of the Top 10 Korean Universities

### 2.4.2.1 Licensing Activities and Royalties of the Top 10 Korean Universities

[Table 4-11] summarizes the licensing activities and royalty incomes of the top 10 Korean universities in 2012.<sup>86</sup> Each column of the table shows the total R&D budget of each university, the number of licensing agreements, the number of start-ups originating from a university's technology, the total number of domestic patent applications filed, the total number of foreign patent applications filed, the total amount of a royalty, the total amount of a running royalty, and so forth.

[Table 4-11] Licensing Activities and Royalties of the Top 10 Korean Universities (2012)

| Ariel 11.5                    | R&D Budget | Licensing | Start-up | Pat. Filed | Pat. Filed | Royalty  | Running Royalty |
|-------------------------------|------------|-----------|----------|------------|------------|----------|-----------------|
|                               | (US \$M)   | (ea.)     | (ea.)    | (Korea)    | (Foreign)  | (US \$M) | (US \$M)        |
| 01. Hanyang University        | 214.3      | 30        | 7        | 462        | 101        | 3.91     | N/A             |
| 02. Yonsei University         | 312.1      | 56        | 4        | 622        | 149        | 3.38     | N/A             |
| 03. Seoul Nat'l University    | 431.7      | 77        | 0        | 744        | 250        | 3.31     | N/A             |
| 04. Seongkyun-gwan University | 200.7      | 72        | 1        | 452        | 96         | 3.24     | N/A             |
| 05. KyungHee University       | 113.0      | 37        | 0        | 366        | 71         | 2.79     | N/A             |
| 06. K.A.I.S.T.                | 221.5      | 48        | 14       | 1,139      | 242        | 2.78     | N/A             |
| 07. Korea University          | 211.6      | 106       | 0        | 613        | 119        | 2.63     | N/A             |
| 08. G.I.S.T.                  | 40.2       | 22        | 0        | 155        | 111        | 2.26     | N/A             |
| 09. POSTECH                   | 148.9      | 29        | 0        | 315        | 211        | 1.54     | N/A             |
| 10. Busan University          | 120.6      | 56        | N/A      | N/A        | N/A        | 1.32     | N/A             |
| All 1 ~ 10                    | Total      | 1,996.5   | 533      | 26         | 4,888      | 1,350    | 27.1            |
|                               | Mean       | 199.7     | 53.3     | 2.9        | 543        | 135      | 2.71 (1.36%)    |

Source: 「2012 Korean University-Industry Collaboration Activities Report」 (2013).<sup>[39]</sup>

<sup>86</sup> KISED (2013), 「2012 Korean University-Industry Collaboration Activities Report」.

It is noted that, although the table does not list the “total number of invention disclosures,” it is customary for Korean university TLOs to file a domestic patent application for almost every invention disclosure. The major reason for this almost 1:1 correspondence between these numbers may be attributed to the fact that one of the quantitative indices for determining the success or failure of government-sponsored R&D is the number of patent applications filed. As a result, even when the TLO staff determines that a disclosed invention lacks technical superiority, commercial viability, or marketability, he or she hires a cheaper patent firm, and files a domestic patent application anyway. It is also noted in Table 0.10 that each of the top 10 Korean universities recouped 1.36% of its total R&D budget spent. This corresponds to 14% (or about one-seventh) of the recoup percentage of their U.S. counterparts.

#### **2.4.2.2 Korean University TLOs May Have Apparently Succeeded with Technology Transfer, but They Have Definitely Failed with Technology Commercialization**

Although [Table 4.11] does not give the exact numbers, it is customary that a substantially major portion of the royalties earned by Korean universities does not come from running royalties but rather from upfront royalties. Considering that the “running royalty” is the token of a successful technology transfer and commercialization, the fact that the running royalty is the type of royalty which Korean universities have mainly earned indicates that Korean universities may well have succeeded with technology transfer, but they have failed miserably where technology commercialization is concerned. There may be many reasons for this apparently successful rate of technology transfer coupled with the miserable failure of technology commercialization.

One salient reason may be that Korean manufacturers are more savvy than university TLOs, which enables the manufacturers to successfully convince the TLOs that the upfront royalty is better than the running royalty, to pay less money (*i.e.* as the upfront royalty) to the universities, and then to keep all the profits they make by selling the products. Conversely, another reason may be that the university TLOs are just as savvy as the manufacturers, but they do not like the running royalty because they do not like to examine the accounting

books in order to confirm exactly how much running royalties they should receive. In a sense, these lazy TLOs are just satisfied by taking the upfront royalty and that is it.

Yet another reason is a somewhat disappointing one, for this one is caused by government regulations. For example, the government sometimes offers a sponsored R&D fund to a team from a university and a company, hoping that university and company will earnestly collaborate as one team, and that the resulting technology can be successfully transferred from the university to the company. To judge whether or not the university and the company earnestly collaborate and succeed in this project, the government adopts a quantitative index such as, for example, whether the company licenses the resulting technology from the university and how much royalty the company pays to the university. When the company learns that the technology developed by the university is useless and, therefore, does not license that technology, the government may view this project as a failure. And because the collaboration has failed, neither the university nor the company can get additional R&D funding from the government. However, when the company pays money to the university and licenses the technology from the university, regardless of how useless or poor that technology may be, the government may view this collaboration as a successful one. And then not only the university but also the company can apply for and get additional government-sponsored R&D. Through these additional R&D projects, the company can successfully recoup its wasted investment in the useless or poor technology. And that university and that company can repeat the same routine over and over again.

As a result, there may be plenty of successful cases of government-sponsored R&D projects as well as plenty of successful cases of technology transfer. But because these technology transfers are just decoys or pure counterfeits, there cannot be plenty of successful cases of technology commercialization. Just as good intentions do not always bear good results, this sincere hope on the part of the government may be wasted by fixing between universities and companies.

### 2.4.3 Korean University TLOs in the Hands of Non-Technical and Non-Business-Minded Bureaucrats

Korea's first TLO was established in Seoul National University in 2003 under the special act of the Ministry of Commerce, Industry, and Energy. In the same year, however, the Ministry of Education competitively set up a very similar law. Based thereon, all other Korean universities set up their own TLOs in 2004 and thereafter.

It is noted that the Ministry of Education has wielded absolute power over every Korean university ever since the first Korean government was set up in 1948. The Ministry has absolute discretion in distributing government subsidies to those universities as it sees fit. Just as a side note, most Korean universities receive about 30-40% of their budgets in the form of government subsidies or, more particularly, from the Ministry of Education. With that absolute power, the Ministry can strictly determine how many students each university can admit each year, how many faculty members each university can hire, how much salary each of them should be paid, how many courses each university, each school, and each department can offer every semester, to name but a few of its powers. If a university does not (or does not seem to) strictly obey the guidelines delivered by the Ministry (regardless of their merit or stupidity), the Ministry can cut the government subsidy, or cut down the number of incoming students, and so forth, and each such decision can be a fatal blow to the universities. In addition, many officers from the Ministry are sent to the universities as permanent employees or, conversely, the universities endeavor to recruit those officers from the Ministry, just to show off their friendliness with the Ministry.

But let's get back to the TLOs. After learning that the Ministry of Commerce, Industry, and Energy had enacted a law allowing Seoul National University to set up the first TLO in Korea, the Ministry of Education did not want other universities to follow suit. Accordingly, in 2003, the Ministry of Education enacted a new law entitled the "Act on Promoting Industrial Education and Industry-University Cooperation," which stipulated that each university could set up a TLO. And how was the Ministry of Education able to help Korean universities create these TLOs? It was not from scratch: It was actually an easy task or a "piece of cake" for the Ministry, as it simply ordered the universities to change the name of their "then Office of

Research Affairs” to “University-Industry Foundation,” which is the formal name of the TLO in Korea. In essence, the “University-Industry Foundation” performs all the tasks which the “Office of Research Affairs” used to perform, where examples of such tasks include (1) the accounting and management of all R&D projects, (2) supporting the facilities and operations of the universities, and such like. On top of that, the new Act also empowered the “University-Industry Foundation” to carry out (1) tasks related to employee inventions, (2) tasks related to acquiring and managing patents, (3) tasks related to technology transfer and commercialization, (4) tasks related to start-ups, and (5) tasks related to promoting entrepreneurship among others.

Most importantly, the new Act empowered the “University-Industry Foundation” to set up a TLO as a subordinate department. Strictly abiding by this provision, the tenured, but non-technical and non-IP-knowledgeable personnel of the “Office of Research Affairs,” *a.k.a.* the “University-Industry Foundation,” hire a few patent agents as subordinate, at-will employees. Although those patent agents are specialists, they do not have any power to appoint or fire their assistants. To make matters worse, many personnel of the “University-Industry Foundation” take turns as the managers of the patent agents, while wasting the valuable time of the patent agents by making them explain the basic patent law to the non-technical, non-patent-knowledgeable managers. To make matters far worse, those staffs of the “University-Industry Foundation” take turns every two years. As a result, the ever-busy, ever-neglected patent agents have to give the same lecture on basic patent laws to those new managers over and over again.

In essence, the TLOs of Korean universities had been established on the wrong ground. It is appreciated that the “then Office of Research Affairs” was the key department of every Korean university, for no university could function properly without it. But the “then Office of Research Affairs” was not suited to either technology transfer or technology commercialization, not to mention the management of inventions and patents. Because the foundation does not just fit the very function (*i.e.* patent business such as technology/patent transfer and technology/patent commercialization) of the TLO, all Korean TLOs were forced to walk on the wrong foot. The same problems persist over and over in almost all Korean TLOs, and the TLOs of Korean universities (as well as those of research institutes) have



become mired in a bureaucratic, vicious cycle. The worst news of all is that there seems to be no end to such bureaucratic governance of the TLOs of Korean universities, and that the Ministry of Education and its employees have imprisoned the TLOs in its stronghold.

#### **2.4.4 Policy Suggestions - Do Azerbaijan Universities Need the TLOs? If Yes, What Kind of TLOs? What Kind of Governance?**

As Azerbaijan's universities and research institutes gradually accumulate employee inventions, the government of Azerbaijan may have to decide who will drive the campaign for successful technology transfer and commercialization. As such, the government is strongly recommended to consider the following factors in deciding which body or organization will drive such a campaign:

- Set up a TLO and assign to it all the tasks related to invention acquisition, patent protection (domestic or abroad), and technology transfer/commercialization;
- Alternatively, assign all the tasks related to invention acquisition, patent protection, and technology transfer/commercialization to the inventors; or
- Assign the tasks to a combination of (a) and (b).

As explained in this section and in the example of Korean university TLOs, setting up a new TLO is not an easy decision. For one thing, many experts are required to run a TLO properly before setting it up, but it is impossible to train and nurture experts in technology transfer and commercialization without a TLO.

In this respect, whether or not to set up a new TLO is similar to the “Which came first, the chicken or the egg?” problem, but should be strongly emphasized that starting a TLO on the wrong foot may result in a constantly degrading vicious cycle.

## 2.5 Implications

The Azerbaijan government will be required to play various roles in promoting technology transfer and commercialization.

As proposed in **Section 2.2**, Azerbaijan's government will have to participate in or even take the initiative of providing IP education and training to potential inventors of different age groups and from different walks of life. When desirable or necessary, the government may have to tie government-sponsored R&D to such IP education or training. By providing inventors with practical knowledge of patent law, the quality of the end-products developed by such inventors will improve and their number may also double, triple or quadruple accordingly. This will also contribute to laying the firm groundwork for the virtuous IP ecosystem in Azerbaijan.

To ensure that a steady stream of new inventions flows into the Azerbaijan IP ecosystem, the Azerbaijan government may enact a new employee invention act or revise the current Employee Invention Act. As proposed in **Section 2.3**, the main emphasis of enacting or revising the Act would be to ensure that the inventions originating from government-sponsored R&D are systematically assigned to their employers, i.e. Azerbaijan's universities or government research institutes.

As employee inventions are accumulated by Azerbaijan's universities and research institutes, the government may have to decide who will drive the campaign the successful technology transfer and technology commercialization. As such, the government is strongly recommended to consider the following factors in deciding which body or organization will drive such a campaign:

- (1) Set up a TLO and assign to it all the tasks related to invention acquisition, patent protection (domestic or abroad), and technology transfer/commercialization;
- (2) Alternatively, assign all the tasks related to invention acquisition, patent protection, and technology transfer/commercialization to the inventors; or
- (3) Assign the tasks to a combination of (a) and (b).

As explained in Section 3 and in the example of Korean university TLOs, setting up a new TLO is not an easy decision. For one thing, many experts are required to run a TLO properly before setting it up, but it is impossible to train and nurture experts in technology transfer and commercialization without a TLO.

In this respect, whether or not to set up a new TLO is similar to the “Which came first, the chicken or the egg?” problem, but it should be strongly emphasized that starting a TLO on the wrong foot may result in a constantly degrading vicious cycle.

Accordingly, more discussions will be required and more data will have to be gathered and reviewed to accurately assess the status of Azerbaijan’s IP ecosystem and, more particularly, the guidelines or regulations on IPs or patents related to Azerbaijan government-sponsored R&D, as well as the guidelines and regulations of Azerbaijan’s universities and research institutes regarding inventions and patents, the availability of technology transfer or commercialization in Azerbaijan, and so forth.

Although not included in the current version of this Chapter 0, further discussions may be needed to come up with additional policy suggestions, if any, regarding Azerbaijan’s patent office, patent court systems, etc.

### 3. Start-up Programs

#### 3.1 Korea's start-up programs and government support budget

##### 3.1.1 Korea's start-up programs and government support budget

As of 2022, according to the announcement of the Ministry of SMEs and Start-ups, the total budget for central and local government start-up support projects is reported to be about 3.66 trillion won (USD \$ 3.66 billion). Fourteen central government agencies, seventeen metropolitan municipalities, and sixty-three basic municipalities are participating as supporting organizations in Korea's start-up program support. Currently there are around 378 start-up programs in Korea, with a total support budget of 3.66 trillion won, the highest figure ever.

[Figure 4-15] 2022 Start-up Support Project Integration Announcement, Ministry of SMEs and Start-ups



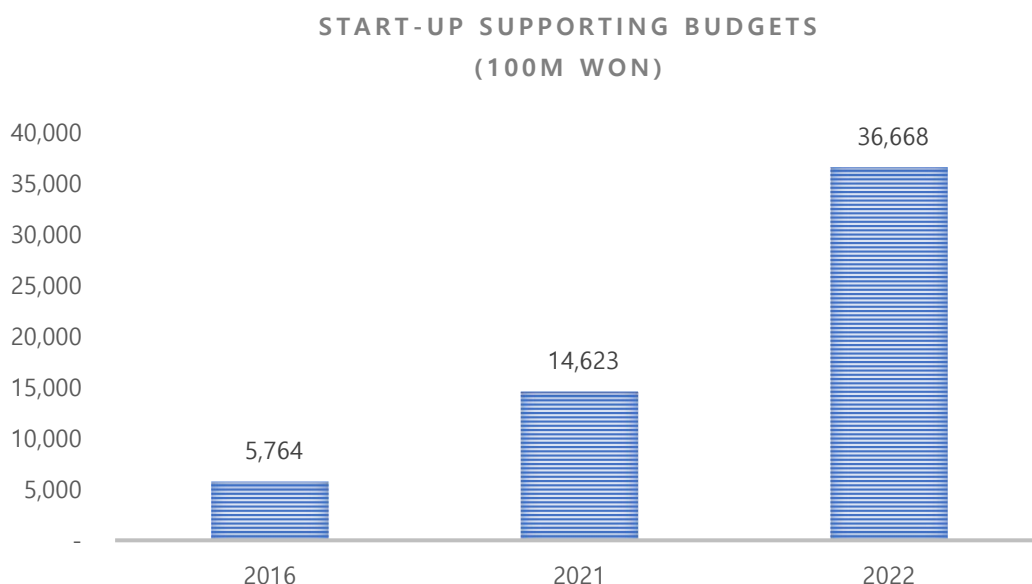
Source: Ministry of SMEs and Start-ups (2022)

### 3.1.2 Start-up support budget of each supporting institution

Central government: If we classify the start-up support provided by the central government based on the amount of support, the Ministry of SMEs and Start-ups, as the ministry in charge of start-up companies, accounted for about 93.1% of the total budget at 3.31 trillion won, followed by the Ministry of Culture, Sports and Tourism at 1.8%, or 62.68 billion won, and the Ministry of Science, Technology, Information and Communication at 1.5%, or about 53.37 billion won. This represents an increase of about 55% for the Ministry of SMEs and Start-ups compared to 2021, and an increase of 363% and 602% for the Ministry of Culture, Sports and Tourism and the Ministry of Science and ICT, respectively.

As for Metropolitan municipalities, when comparing the scale of the start-up support provided by regional municipalities, Gyeonggi Province had the highest budget among all the regional municipalities, accounting for about 17.5% of the total, or 15.52 billion won, followed by Seoul at 12.4%, or 11.01 billion won, and Jeollanam-do at 10.2%, or KRW 8.9 billion won. Finally, when comparing the size of the start-up support provided by basic local governments, Jeonbuk Iksan City had the biggest budget at 8.7% of the total, or 1.790 billion won, followed by Ulju-gun, Ulsan, at 5.8%, or 120 billion won, and Ansan, Gyeonggi-do, at 5.4%, or 11.10 billion won

[Figure 4-16] 2022 Start-up Support Budgets, Ministry of SMEs and Start-ups, 2022



Source: Ministry of SMEs and Start-ups (2022)

### 3.1.3 Recent trends in venture and start-up-related investment and status by type of support project

The amount of venture investment in 2020 reached 4.304.5 trillion won, breaking the record high of 4.27 trillion won recorded in 2019. In 2020, they broke through 4,000 for the first time, respectively, breaking new record highs.

Nevertheless, venture investment continued to grow by more than 10% every year from 2017, but in the aftermath of the COVID-19 crisis, the growth rate of venture and growth capital in 2020 compared to 2019 fell sharply, reaching only about 0.63%. In particular, it appears that investment in the distribution, service, video, performance, and record industries, which were damaged in the aftermath of the COVID-19 pandemic, decreased compared to 2019.

[Table 4-12] Venture investment by growth stage in 2009 to 2020

(Unit: 100 Million Won)

| Stage     | 2009  | 2010    | 2011    | 2012    | 2013    | 2014    | 2015    | 2016    | 2017    | 2018    | 2019   | 2020   |
|-----------|-------|---------|---------|---------|---------|---------|---------|---------|---------|---------|--------|--------|
| Early     | 247.6 | 319.2   | 372.2   | 369.6   | 369.9   | 504.5   | 647.2   | 790.9   | 779.6   | 981.0   | 1390.1 | 1320.5 |
| Expansion | 260.1 | 290.4   | 329.6   | 313.7   | 325.9   | 406.9   | 582.8   | 615.6   | 664.1   | 1,193.5 | 1766.2 | 1726.8 |
| Later     | 359.4 | 481.4   | 559.0   | 550.0   | 688.7   | 727.9   | 855.8   | 743.8   | 936.6   | 1,250.4 | 1121.4 | 1257.2 |
| Total     | 867.1 | 1 091.0 | 1 260.8 | 1 233.3 | 1 384.5 | 1 639.3 | 2 085.8 | 2 150.3 | 2 380.3 | 3,424.9 | 4277.7 | 4304.5 |

Source: Korea Venture Capital Association (2022)

Venture investment increased sharply again in 2021, and based on the results of an analysis of the venture investment performance of the Ministry of SMEs and Start-ups, in 2021, the venture investment performance in 2020. It was found to have hit a record high of 7.68 trillion won, a new record of 3.4 trillion won.

Investments in all sectors increased as sectors where investment declined in 2020 recovered. In particular, the information and communication technology (ICT) service, distribution/service, and biotechnology (bio)/medical sector, which emerged as promising industries in the era of the Coronavirus, doubled by more than 2.5 trillion won to reach 4.119 billion won in 2020, showing a significant increase in venture investment.

Venture investment in the first half of 2022 is estimated to reach KRW 4.61 trillion, an increase of 24.3% (+782.1 billion won) compared to the previous record high of 3,224 billion won recorded in the first half of 2021, and exceeding 4 trillion won for the first time. However, as oil prices surged due to the Russian-Ukrainian War and other adverse events, such as interest rate hikes, venture investment is expected to enter a downward trend in the second half of 2022.

Looking at the total support budget for central government and local government start-up support projects, and start-up-related support projects by type, project cost loans accounted for the highest share at about 2.22 trillion won, or 55.1% of the total, while in terms of the number of projects, 172 were related to technology commercialization,

accounting for 45.5% of the total.

Among these, the budget for commercialization support, technology development, start-up incubation facilities, and start-up education directly supported by start-up companies, excluding the budget support for business loans, amounted to about 1.589 trillion won, accounting for about 43.3% of the total.

[Table 4-13] Start-up support budgets by growth stage in 2022

(Unit: 100 Million Won)

| Growth Stage | Loan   | Business Incubation | R&D   | Business Infrastructure | Start-up Education | Others | Total  |
|--------------|--------|---------------------|-------|-------------------------|--------------------|--------|--------|
| Budget       | 20,220 | 9,132               | 4,639 | 1,549                   | 569                | 570    | 36,679 |
| %            | 55.1%  | 24.9%               | 12.6% | 4.2%                    | 1.6%               | 1.6%   | 100.0% |

Source: Ministry of SMEs and Start-ups (2022)

Compared to the previous year (excluding basic local government and loan projects), the above figures increased by about 82.9 billion won in the commercialization area, up 10.1% in terms of the support budget, followed by the technology development support projects at 43.2 billion won, an increase of 10.3%.

If we look at the reorganization of this by each ministry, excluding project cost loan projects, from the budget of each government department, the following budget was prepared in 2022, yet the Ministry of SMEs and Start-ups still recorded 1.29 trillion won (about 78.5% of the total) by individual government ministries. In the case of the Ministry of Science, Technology and Information and Communication, which is closely related to technology start-ups centred on the laboratory technologies of universities, the budget support amounts to about 53.37 billion won (3.2% of the total).



[Table 4-14] Start-up support budgets by government department

(Unit: 100 Million Won)

| Government department | Ministry of SMEs and Start-ups | Ministry of Culture, Sports and Tourism | Ministry of Science and Technology Information and Communication | Others  | Total    |
|-----------------------|--------------------------------|-----------------------------------------|------------------------------------------------------------------|---------|----------|
| budgets               | 12,911.2                       | 626.8                                   | 533.7                                                            | 2,376.3 | 16,448.0 |
| %                     | 78.5%                          | 3.8%                                    | 3.2%                                                             | 14.4%   | 100.0%   |

Source: Ministry of SMEs and Start-up (2022)

### 3.1.4 Government investment for supporting public R&D-based start-ups

There may be various criteria for classifying Korea's start-up programs, but if we look at the classification based on the budget criteria of the government's major start-up support projects, there are three types of start-ups: technology start-ups, youth start-ups, and local start-ups centred on the Ministry of SMEs and Start-ups.

**Technology Start-up:** Support provided for the establishment of a start-up ecosystem centred on innovative talents equipped with technological prowess.

We aim to promote customized support projects, and the most representative program is the TIPS program of the Ministry of SMEs and Start-ups. In 2022, the TIPS program received a budget of about 29.349 billion won, an increase of about 109.05 billion won from 2021. In addition, KRW 280.87 billion was provided for the preliminary start-up package, initial start-up package, and leap-up start-up package, which are representative support programs that differentiate the necessary funds and support programs for each start-up business and provide step-by-step support.

**Youth Entrepreneurship:** We support idea-oriented start-ups by young people who seek solutions based on various problems encountered in daily life. The most representative project is the 'Youth Entrepreneurship Academy' project, through which young people can receive support for the entire process of starting a business, from preparing to start a

business to entering the market. This project provides business start-up support for youth and university students in the agricultural, forestry and cultural fields, specialized training in the AI field (start-up artificial intelligence (AI) technical manpower training, KRW 2.7 billion), and tax accounting consulting for young people within three years of founding a start-up (start-up support service voucher, 121.3 100 million), etc.

[Table 4-15] TIPS programs in Korea

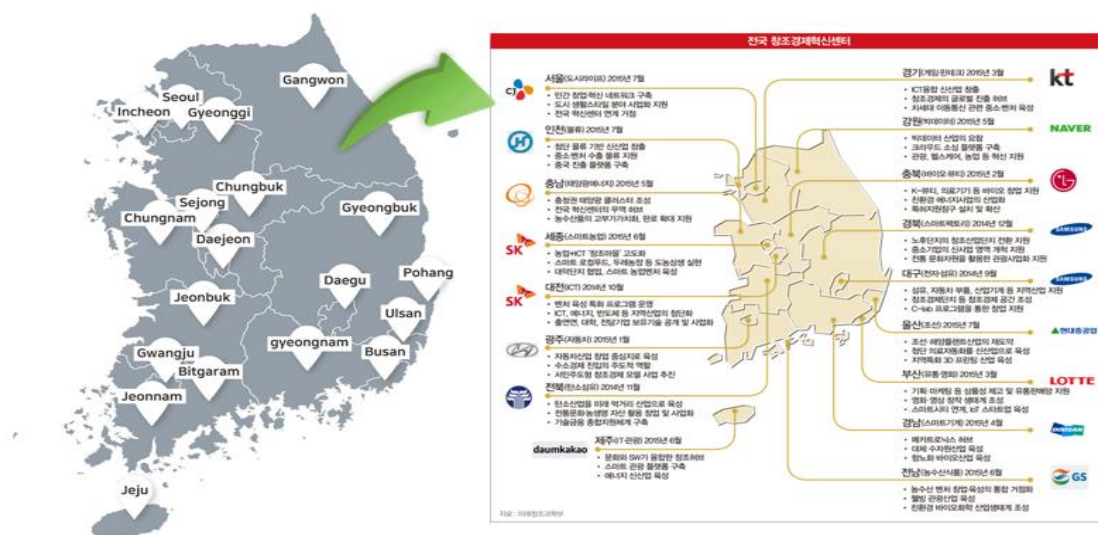
|                     | Incubation terms | Start-up fund                          | R&D fund              |                          |            | Extra benefits                                                                 |
|---------------------|------------------|----------------------------------------|-----------------------|--------------------------|------------|--------------------------------------------------------------------------------|
|                     |                  | Angel Capital fund (GP)                | Government fund       | Private fund             |            |                                                                                |
|                     |                  |                                        |                       | Cash                     | Asset      |                                                                                |
| Start-up (Per Team) | 2-3 years        | Avg. 100K won (20% of government fund) | Max. 500K won         | Over 50% of private fund | The amount | Start-up fund matching (Up to 100K won)                                        |
|                     |                  |                                        | Below 80% of R&D fund | 20% of R&D fund          |            | Angel matching fund (up to 200K won)<br>Global marketing fund (up to 100K won) |

Source: TIPS (2022)

Regional start-up: Regional start-up infrastructure creation projects designed to promote the revitalization of regional-based innovative start-ups in each region across the country are becoming the focus. The most representative project program is the Center for Creative Economy Innovation(CCEI), which provides mentoring, education, and networking (networking) services for start-ups in nineteen regions across the country.

Each Creative Economy Innovation Center is designed to focus on nurturing specific technological fields. For example, Samsung supports smart factories, electronics and textiles in Gyeongsangbuk-do and Daegu; LG supports bio industry and cosmetics-related fields in Chungcheongbuk-do; and Hyundai Motor supports automobiles in Gwangju. The companies participating in the nineteen regions are major corporations that represent Korea in their respective industries.

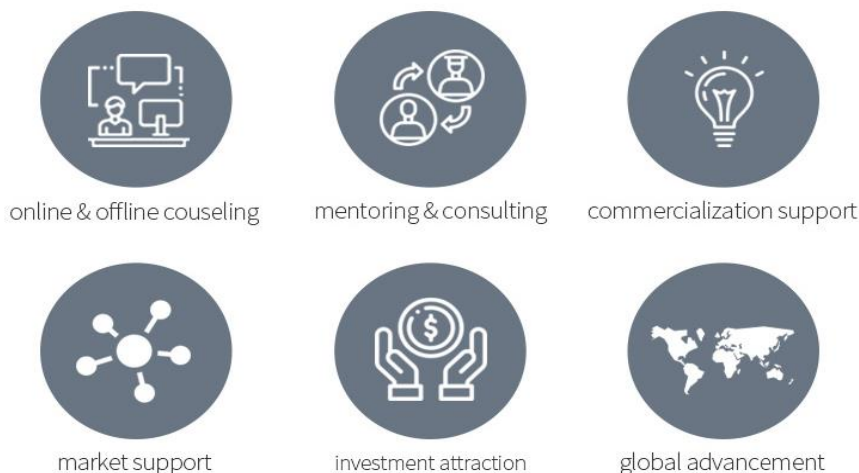
[Figure 4-17] Nationwide centers for creative economy innovation in Korea



Source: Korea Policy Briefing<sup>87</sup> (2015)

[Figure 4-18] Startup support service of CCEI

For these purposes, the Center for Creative Economy and Innovation provides the start-up support services including online & offline counseling, mentoring & consulting, commercialization support, market support, investment attraction, and global advancement.



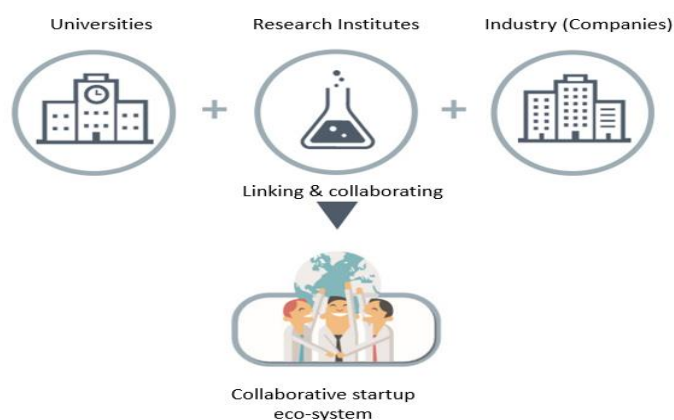
Source: Creative Korea<sup>88</sup> (2022).

<sup>87</sup> <https://www.korea.kr/news/policyNewsView.do?newsId=148798616>

<sup>88</sup> <https://ccei.creativekorea.or.kr/eng/center/info.do>

[Figure 4-19] Networking service of CCEI

In addition, it is also building an innovative start-up ecosystem by linking & collaborating with local partner companies and related organizations.



Source: Creative Korea<sup>89</sup> (2022)

In addition, financial support of about KRW 6.9 billion is being provided to local value entrepreneurs' (local creators) support projects that commercialize local culture and values.

Among the three types mentioned above, especially in relation to universities, there is a difference in the role of each government department regarding the support target and program, depending on the characteristics of the start-up item that is the basis of each start-up. In this regard, the representative start-up support projects and programs of the major government ministries are summarized as follows.

As for Ministry of SMEs and Start-ups, it supports diverse start-ups ranging from the first idea start-up to the technology start-up. There are start-up support programs that focus on omnidirectional support from university students to the general public and customized programs for each stage. In addition, Korea Innovation Centers (KICs) have been established in Silicon Valley, Washington, D.C., Berlin, Germany, and Beijing, China to support the overseas expansion of Korean start-up companies. For example, Silicon Valley KIC runs the "Soaring Program", whose main purpose is to provide education and support for local technology commercialization.

<sup>89</sup> <https://ccei.creativekorea.or.kr/eng/center/info.do>

[Figure 4-20] Korea innovation centers around the world

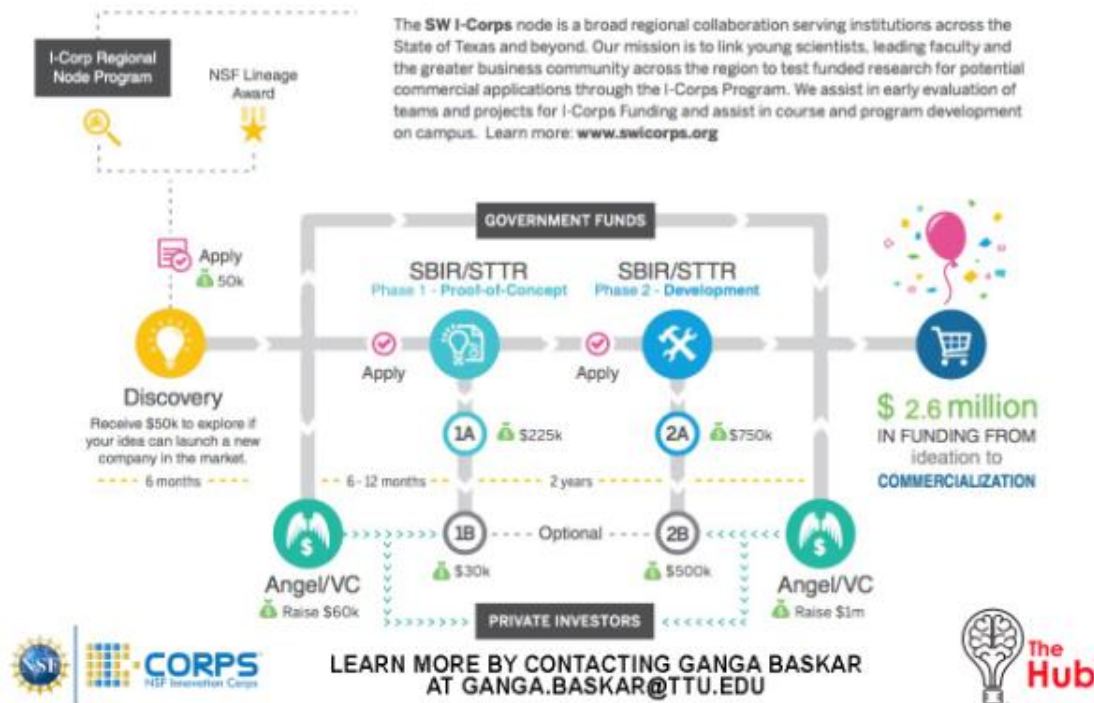


Source: Korea Innovation Center<sup>90</sup> (2022)

In the case of the Ministry of Science and ICT, it provides R&D support based on public technology and IP creation, and mainly deals with laboratory start-up support centred on public institutions, such as universities and research institutes, its major support targets and program categories. In particular, among the projects of the Ministry of Science and ICT, the representative projects related to laboratory start-ups include the public technology-based market-linked start-up search project (the so-called Korean I-Corps program), the laboratory-specialized start-up-leading university support project, and university technology management.

<sup>90</sup> <https://www.kicsv.org/>

[Figure 4-21] Diagram of the NSF I-Corps entrepreneurial ecosystem

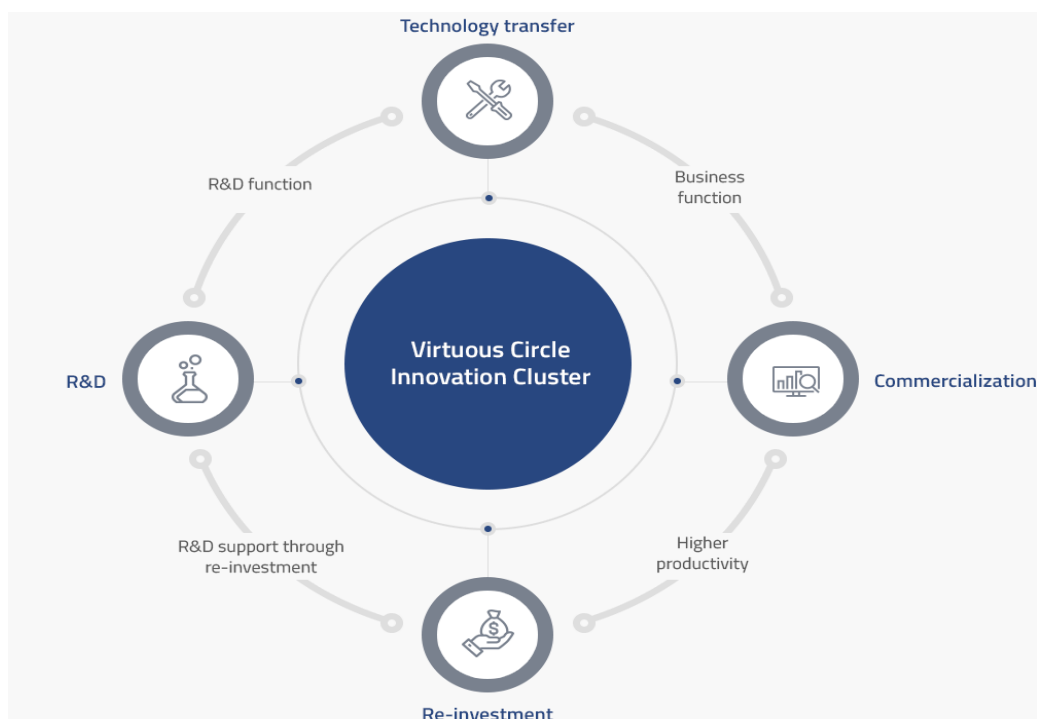


Source: Texas Tech University<sup>91</sup> (2022)

Furthermore, in order to provide R&D technology commercialization support between universities and state-funded research institutes, the Korea Innovation Foundation (research and development special zone promotion foundation, Innopolis) is leading another IP R&BD innovation cluster model to support research and development companies and to plan and support programs for nurturing technology start-ups in progress.

<sup>91</sup> <https://www.depts.ttu.edu/research/research-park/prototype-build/nsf-icorps.php>

[Figure 4-22] Innovative cluster model of the Korea Innovation Foundation



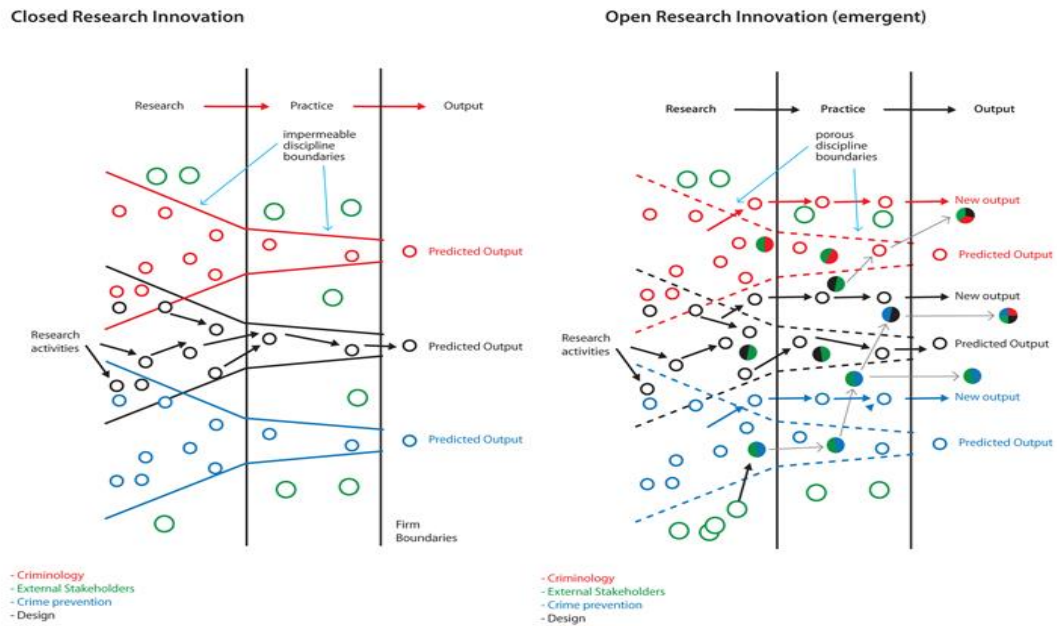
Source: Innopolis (2022)

As for the Ministry of Education, it provides start-up support targeted at undergraduate and graduate students of universities but, overall, entrepreneurship and start-up education are the mainstays rather than direct support for start-ups. We are also supporting such projects as the fostering project (LINC 3.0), the university creative IP R&D asset commercialization support project (BRIDGE+), the laboratory-specialized start-up leading university project, and the university start-up fund.

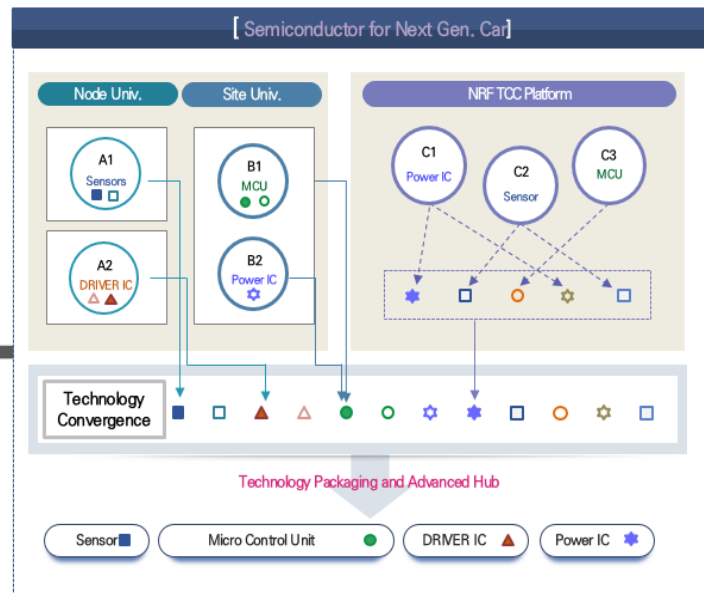
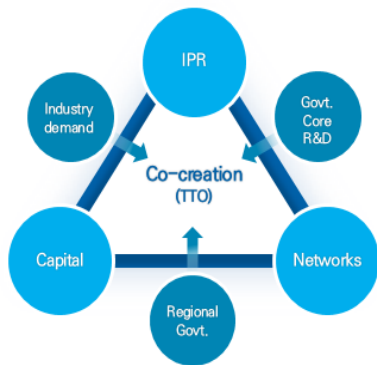
In particular, the BRIDGE (“Beyond Research and Innovation for Good Enterprise”) project of the Ministry of Education is based on the triple helix and open innovation model. It supports the research and development work of universities aiming to respond to companies’ demand for high-tech, and supports various programs and technology convergence to promote technology commercialization. Nationwide, twenty-four universities and university technology holding companies are participating in the project, which has been evaluated as the most successful of the government's IP R&D projects in Korea.



[Figure 4-23] BRIDGE Project of the Ministry of Education



- The Korean government will designate **10 core-R&D/technologies**, and promote the sharing and collaborative technology commercialization thru the co-creation networks as a triple helix among stakeholders.



Source: NRF (2022)

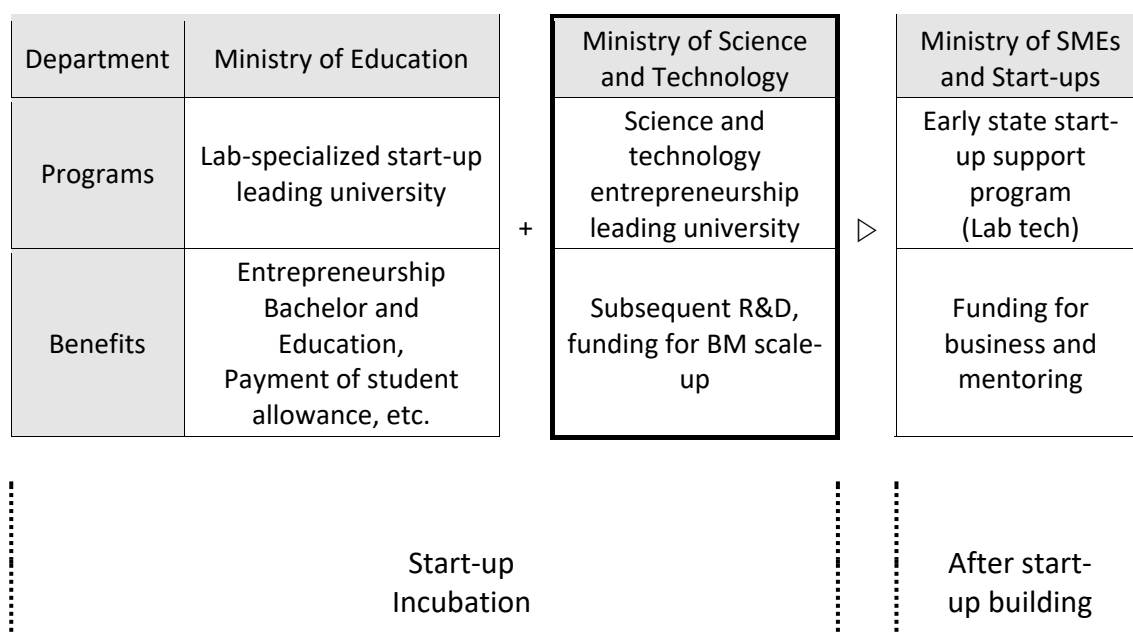
Also, in the case of Korea, the Co-Creation model is used to create and secure links between the start-up-related programs of each government department.



As an example;

- The Ministry of Education is required to discover technology seeds through laboratory-specialized, start-up-leading university projects such as LINC 3.0, BRIDGE+, University Start-up Fund, etc.
- The Ministry of Science and ICT induces the advancement of BM by supporting science and technology start-up-leading university projects and Korean-style I-Corps projects (officially named the “Tech Frontier” program) as well as TMC (Technology Management Center) projects.
- The Ministry of SMEs and Start-ups uses a method of linking the initial start-up package with programs such as TIPS (Technology Incubator Program for Start-up Korea) for technology start-up teams that have been nurtured in this way by the Ministry of SMEs and Start-ups.

[Table 4-16] Start-up Co-creation model in Korea

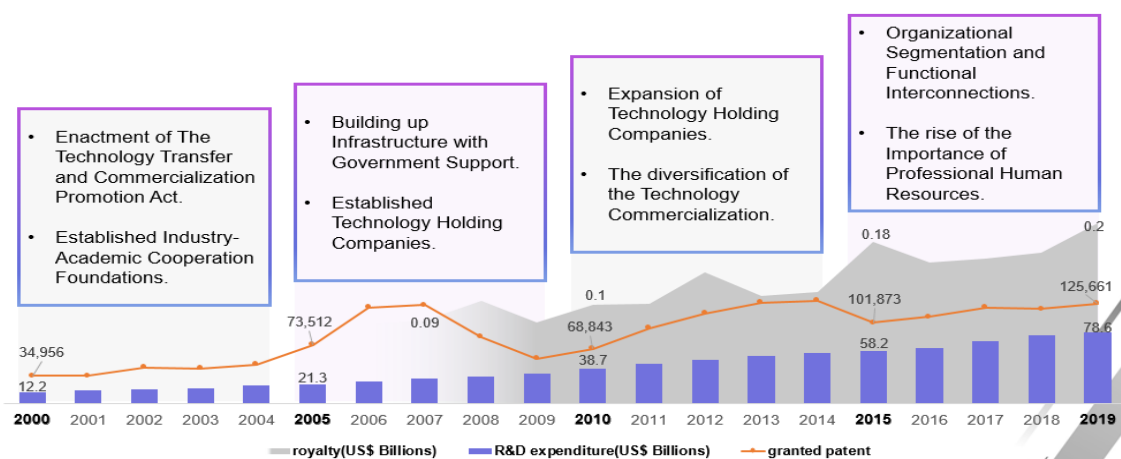


Source: NRF (2022) “Start-up Co-Creation Model”

### 3.1.5 Lab-to-Market promotion projects in Korea

Korea's R&D investment budget amounted to 89.04 trillion won as of 2019, of which the government's R&D investment was 20.5 trillion won, or 23.0% of the total R&D budget in 2019; followed by 24.1 trillion won in 2020; 27.2 trillion won in 2021; and 29.7 trillion won in 2022, thus increasing by trillions of dollars per year. However, in terms of performance, Korea's commercialization success rate was just 20%, which is significantly lower than the rates of other major developed countries (UK: 70.7%; US: 69.3%; Japan: 54.1%), ranking 27th in the world. In particular, in the case of universities, research productivity, which represents the technology transfer revenue per 1 billion KRW of all domestic university research expenses, was just 1.41% as of 2019, which is less than 1/3 of the 4.66% recorded in the United States (2017).

[Figure 4-24] Growth of university TTOs in Korea



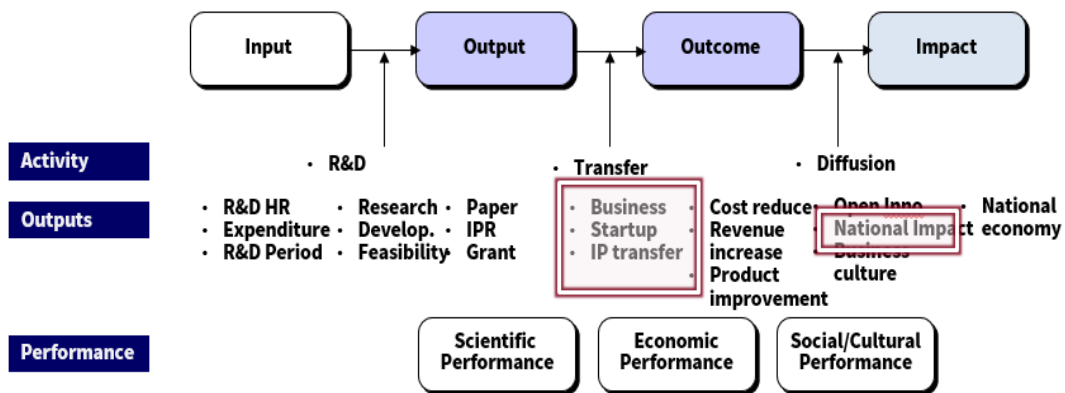
Source: Korea Association of University Technology Transfer (2021)

The ratio of private and public investment in total R&D is 8:2.

- However, since performance information is not usually provided for private R&D, the data are focused on public R&D.
- Regarding the technology transfer of Korea's public R&D achievements, the total number of technology transfers from government research institutes and universities came to about 12,592 (ratio of 50:50).

- The total technology transfer outcome was about KRW 336.1 billion (USD \$ 330 M), to which universities contributed about 29.7%.
- 73.1% of the total number of cases of technology transfer were for SME segments.
- About 61.8% of all technology transfer contracts were for license-based technology transfers.

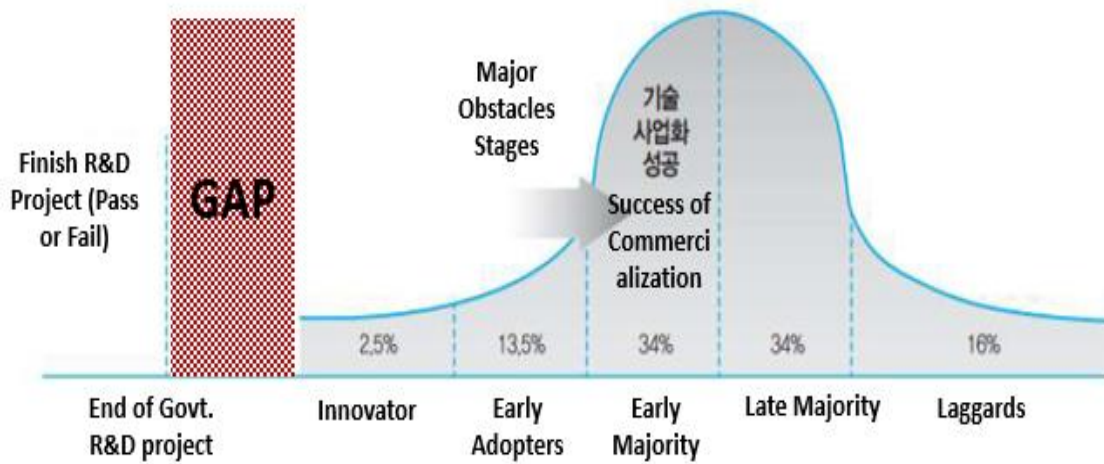
[Figure 4-25] IP R&D vs. Outcome Issue, Ministry of Education, 2011



Source: Ministry of Education (2011)

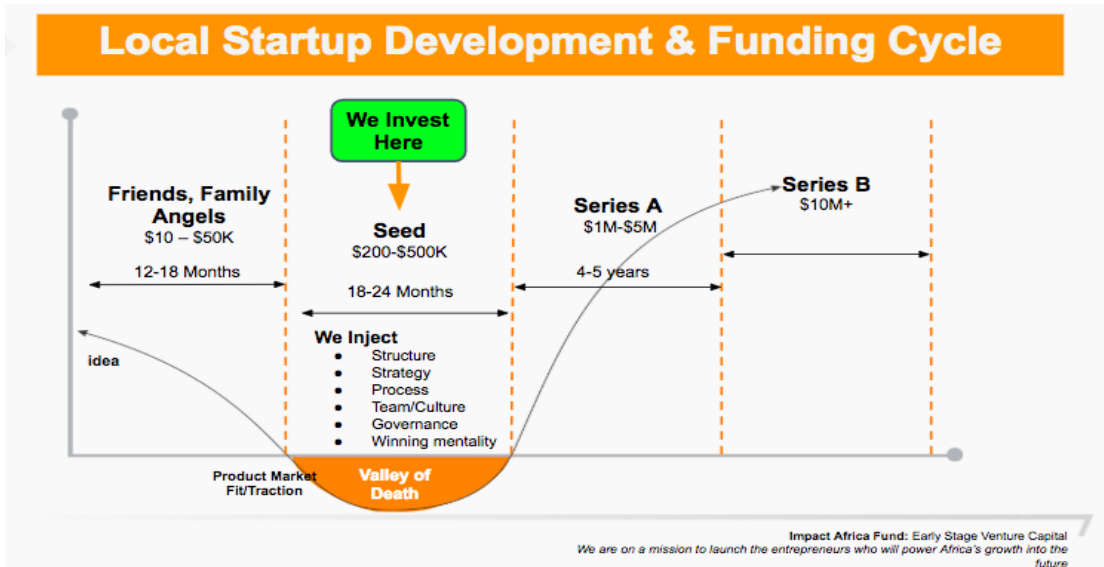
Park Jong-bok (2011) pointed out a structural problem in that, in relation to the technology commercialization transfer model of RAND (2003), in Korea the success of technology commercialization is determined by patents, papers, and prototypes created through government R&D. In other words, in the case of Korea, a separate budget or management system is still insufficient for the characteristics of R&D government tasks paid for by public taxes and for tracking the social and economic success judgment after commercialization. Therefore, it is necessary to support the process of commercialization necessary for actual R&D tasks in order to create good social and economic outcomes by utilizing patents or technical outputs after the completion of R&D tasks. In this process, it is essential to accompany fund investment with the necessary support so as to overcome the gap in the technology commercialization process represented by the so-called 'Valley of the Devil' or the 'Valley of Death'.

[Figure 4-26] Gap between R&D success and technology commercialization



Source: Park JB (2011)

In the case of the United States, the situation is not much different. In 2010, during the Obama administration, a fund of about 7 billion dollars (approx. 8 trillion KRW) was invested in the basic research of the National Science Foundation (NSF). In the case of federal R&D funding, even more than that, i.e. about \$ 172 billion (approx. 180 trillion won) was invested every year, and yet the technology transfer rate remained low and few technology start-ups were established.

[Figure 4-27] Local start-up development and funding cycle, Impact Africa Fund<sup>92</sup>, 2020

Source: Impact Africa Fund "Early Stage Venture Capital" (2020)

In this regard, according to Professor Jong-seong Kim (2017) of Boston University, in the case of the United States, the study by Professor Gary P. Pisano (2006) of Harvard Business School was introduced as an alternative to adjust the gap between low R&D productivity and technology commercialization. Professor Pisano argued that there is a fundamental difference between science and technology and business as a result of years of research by companies in the field of biotechnology, and argued that it is necessary to educate scientists on how to develop business in companies. They argued that business feasibility would be improved only by finding and collaborating on "broad research" with experts in other fields, rather than by collaborating with colleagues in the same field.

In other words, in the case of the United States, there is SBIR (Small Business Innovation Research), a representative scientific start-up model that was launched in 1982 by the Reagan government and has continued for the past forty years. The SBIR fund system has been recognized for its effectiveness to the extent that 2.5% of all R&D budgets of the central government are required to be used annually for start-up support, and this rate has

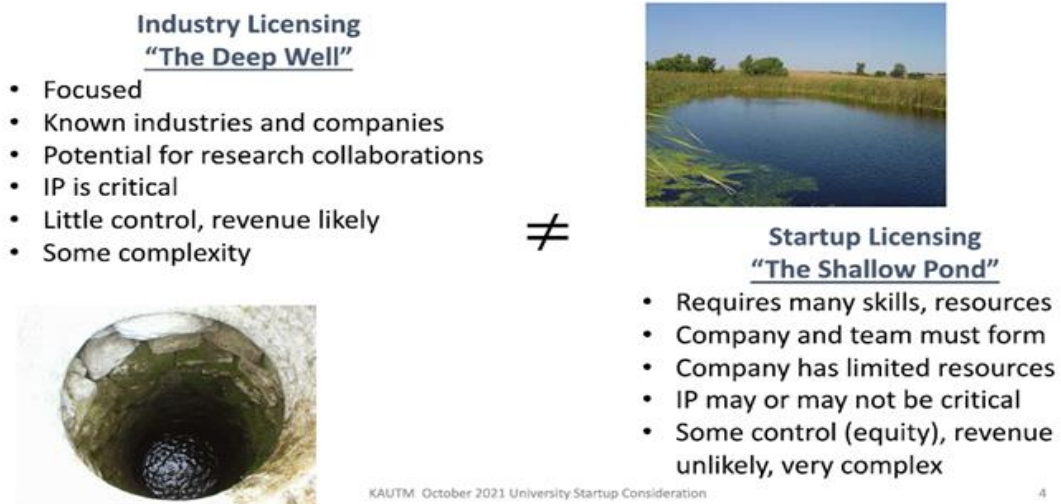
<sup>92</sup> <https://42place.innovationacademy.kr/archives/2516>.

been raised to 3.2% since 2017. Therefore, in order to ensure the effectiveness of seed funds for technology-based start-ups in directly managing the program in accordance with the guidelines in which government agencies are participating and which are presented to the parliament,

- ① Not a loan (even if the results of R&D fail) There is no need to repay
- ② No strings attached (funds are given without specific conditions)
- ③ No equity dilution (there is no demand for corporate equity for investment) so-called “3-No operation method” is adopted.

[Figure 4-28] The difference between the industry licensing and start-up licensing models

## University Research Commercialization – TTO Skill Sets



Source: Korea Association of University Technology Transfer (2021)

The US SBIR support method consists of three phases.

- ① (Phase I) The possibility of technology development is tested and the commercialization of this technology is verified.
- ② (Phase II) The development and production of the products to be commercialized are carried out.
- ③ (Phase III) Commercialization of the product, which is centred on attracting

private investment, and no government financial support is provided. In addition to the above three-step support program, the Fast Track and Phase II-Bridge programs are operated.

Although the SBIR's Phase I selection rate is low at about 16%, there are relatively few restrictions on eligibility requirements, and it consists of 20 to 30 private company CEOs, CTOs, or professors.

[Table 4-17] SBIR programs in the USA

| Stage | Fast Track | Phase I | Phase II | Phase II-Bridge | Direct Phase II | CRP |
|-------|------------|---------|----------|-----------------|-----------------|-----|
| %     | 19%        | 16%     | 37%      | 40%             | 28%             | 29% |

Source: SBIR (2019)

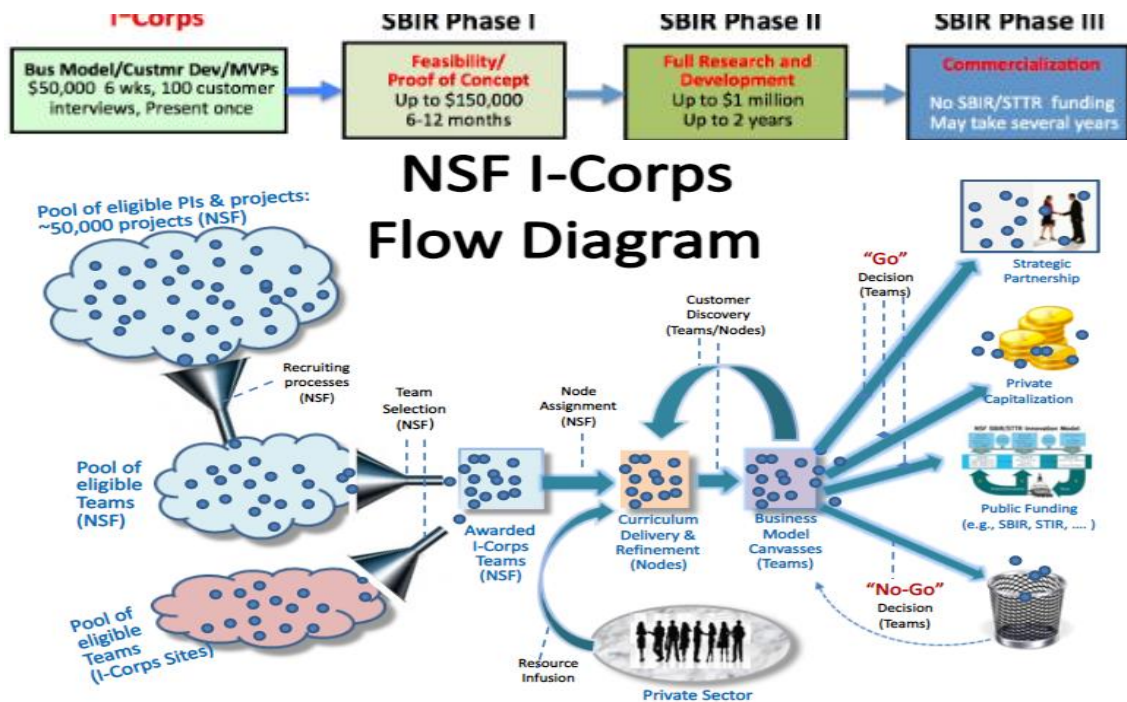
According to the SBIR (2019), the main selection criteria are largely composed of the following five categories.

- ① (Significance) The importance of the problem.
- ② (Investigator) The researcher's performance capability.
- ③ (Innovation) The innovativeness of the technology proposed to solve the problem.
- ④ (Approach) The strategy for solving the problem.
- ⑤ (Environment) The research infrastructure and environment for execution of the task.

However, despite these active support measures, when entrepreneurship by scientists did not meet expectations, the Obama administration asked, in 2009: "Why are scientists reluctant to start a business?" The research began in earnest, resulting in the creation in 2011 of the I-Corps education and training program led by the National Science Foundation (NSF). The main target of the education was to provide entrepreneurship education by selecting 400 people, or about 1%, from among the 40,000 or so responsible researchers supported by the NSF fund.



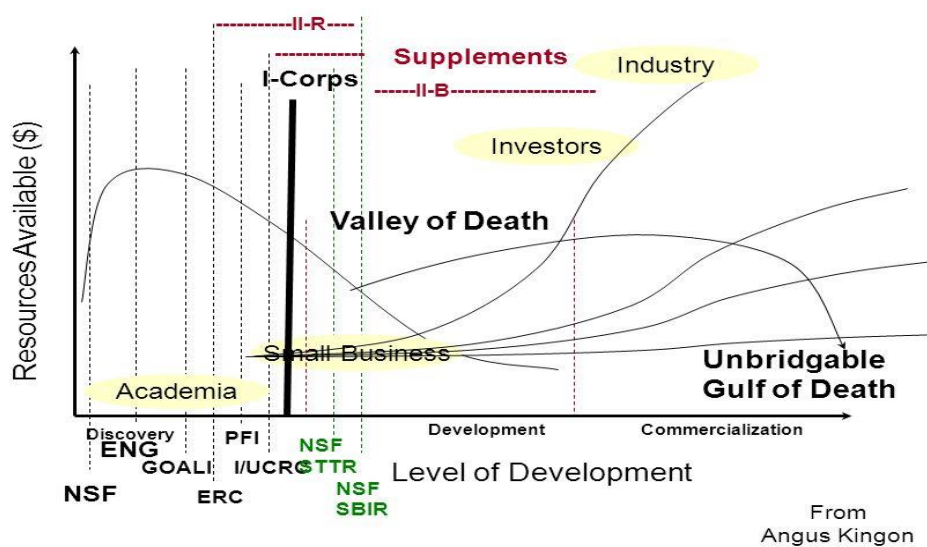
[Figure 4-29] NSF I-Corps Flow Diagram



Source: NSF (2019)

[Figure 4-30] Technology Innovation Spectrum

# Technology Innovation Spectrum

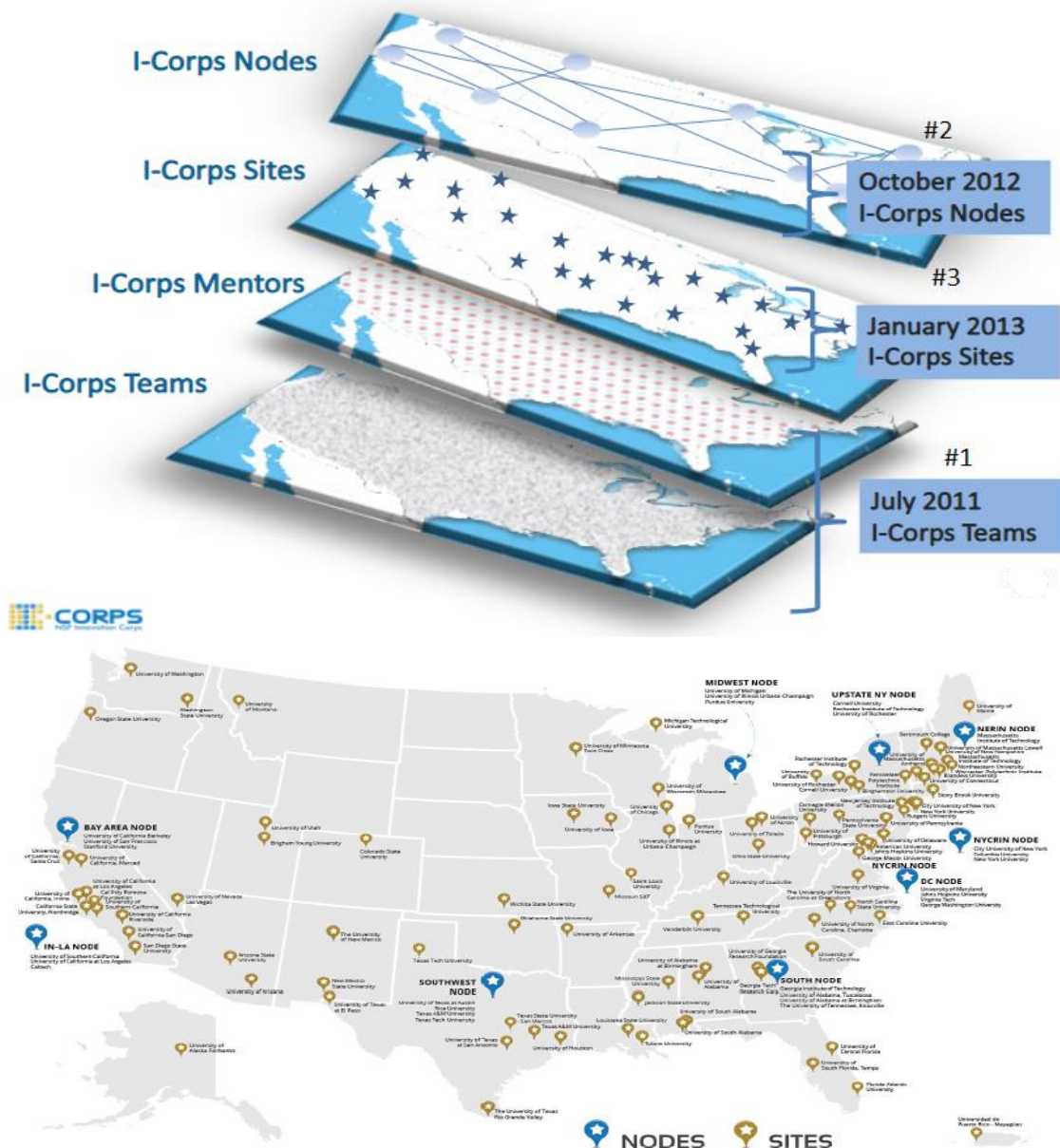


Source: NSF (2019)



In addition, by promoting continuous education programs to spread this to universities, the NSF has been conducting technology start-up education since 2011. By the end of 2021, there will be 9 node universities across the United States, and approximately 280 universities in the United States through approximately 99 site universities.

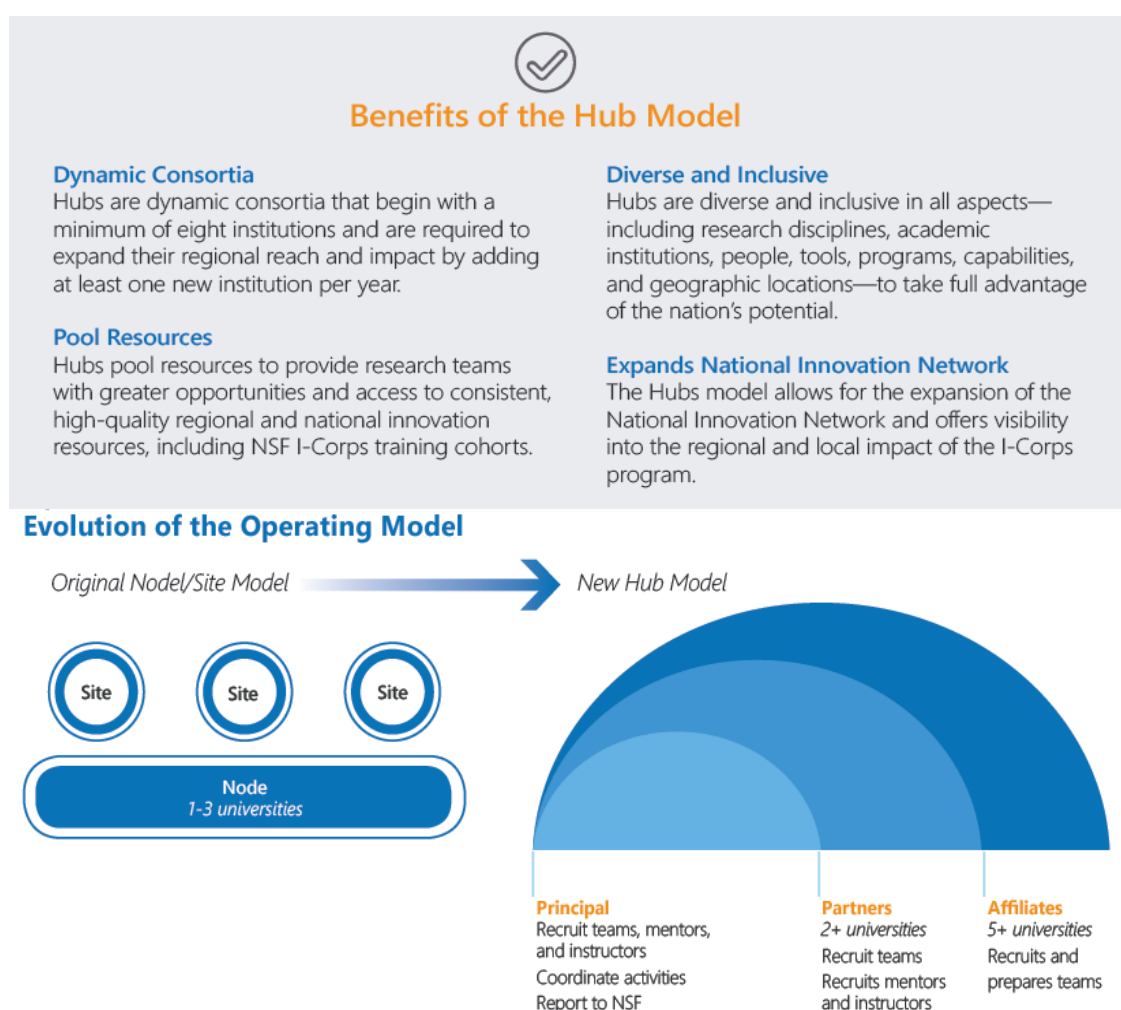
[Figure 4-31] Nation-wide NSF I-Corps Nodes and Sites, 2021



Source: NSF (2021)

In the fall of 2021, rather than funding two separate types of awards, Nodes and Sites, to support the National Innovation Network, NSF started to fund Hubs. This new, more integrated award (NSF Solicitation 20-529) incorporates the lessons learned from the program's initial success and represents an operational change to the way institutions will receive funding from the NSF. The new operational model provides increased and level (rather than previously declining) funding, and is more scalable, to support the expansion of I-Corps.

[Figure 4-32] New I-Corps Hubs of the NSF



Source: NSF (2021)

The I-Corps Hubs will serve national needs in the following five areas of responsibility:

- ① Create and implement tools, resources and training activities that enhance the nation's innovation capacity.
- ② Identify, develop, and support promising research that can generate economic value.
- ③ Gather, analyse, evaluate, and utilize data and insights resulting from the experiences of those participating in local, regional, and national I-Corps programs.
- ④ Provide opportunities to diverse communities of innovators.
- ⑤ Share and leverage effective innovation practices on a national scale to improve the quality of life across the nation.

The NSF plans to issue a new I-Corps Hubs solicitation in the fall of 2021 to continue to expand opportunities for entrepreneurship education and technology translation throughout the United States. The five I-Corps Hubs announced today, and the participating institutions, are listed below.

NSF I-Corps Hub: Great Lakes region — NSF 2048612

- University of Michigan — Lead
- Iowa State University
- Michigan Technological University
- Missouri University of Science & Technology
- Purdue University
- The University of Akron
- University of Chicago
- University of Illinois Urbana-Champaign
- University of Minnesota Twin Cities
- The University of Toledo

- University of Wisconsin — Milwaukee

**NSF I-Corps Hub: Mid-Atlantic region — NSF 2044502**

- University of Maryland, College Park — Lead
- Carnegie Mellon University
- The George Washington University
- Howard University
- Johns Hopkins University
- North Carolina State University
- Penn State
- The University of North Carolina at Chapel Hill
- University of Pennsylvania
- Virginia Tech

**NSF I-Corps Hub: New York region — NSF 2048498**

- The City University of New York — Lead
- Columbia University
- Icahn School of Medicine at Mount Sinai
- New York University
- Rockefeller University
- Stevens Institute of Technology
- Stony Brook University
- University of Massachusetts Medical School
- University at Albany

## NSF I-Corps Hub: Northeast region — NSF 2048602

- Princeton University — Lead
- Delaware State University
- Lehigh University
- New Jersey Institute of Technology
- Rowan University
- Rutgers University
- Temple University
- University of Delaware

## NSF I-Corps Hub: West region — NSF 2048703

- University of Southern California — Lead
- California Institute of Technology
- Colorado School of Mines
- University of California, Los Angeles
- University of California, Riverside
- University of Colorado Boulder
- The University of New Mexico
- The University of Utah

The I-Corps Hubs build on the lessons learned during the first decade of the I-Corps program. Specifically, in the program's initial phase, I-Corps Nodes, consortia of universities responsible for delivering a standardized curriculum, and I-Corps Sites, universities providing internal institutional support to scientists and engineers, were funded separately. Under the new approach, the I-Corps Hubs offer a more integrated approach that incorporates the lessons learned from the program's initial success. The new operational model provides increased funding and enables the scaling of I-Corps across the entire

country. The NSF-supported Sites and Nodes will continue to participate in the National Innovation Network while the NSF transitions to the Hubs.

In Korea, the I-Corps program was started by the Ministry of Science and ICT (now renamed as the Ministry of Science and Technology). The Ministry of Science and Technology introduced I-Corps, a representative education program for science and technology start-ups, following negotiations with the US NSF in 2014.

Starting with the two base universities of KAIST and POSTECH in 2015, the program was expanded to four base universities with the addition of Korea University and Sungkyunkwan University in 2018, while GIST, UNIST, and Ewha Womans University (a base university specializing in advanced education courses for women) were added in 2021.

### The 7 I-Corps Nodes in Korea

- Northern metropolitan and Gangwon Node: Korea University
- Southern metropolitan Node: Seongkyun-kwan University
- Chungcheong Node: KAIST
- Honam Node: GIST
- Daegu and Gyeongsangbuk-do Node: POSTECH
- Gyeongnam Node: UNIST
- Women-specific Node: Ewha Womans University

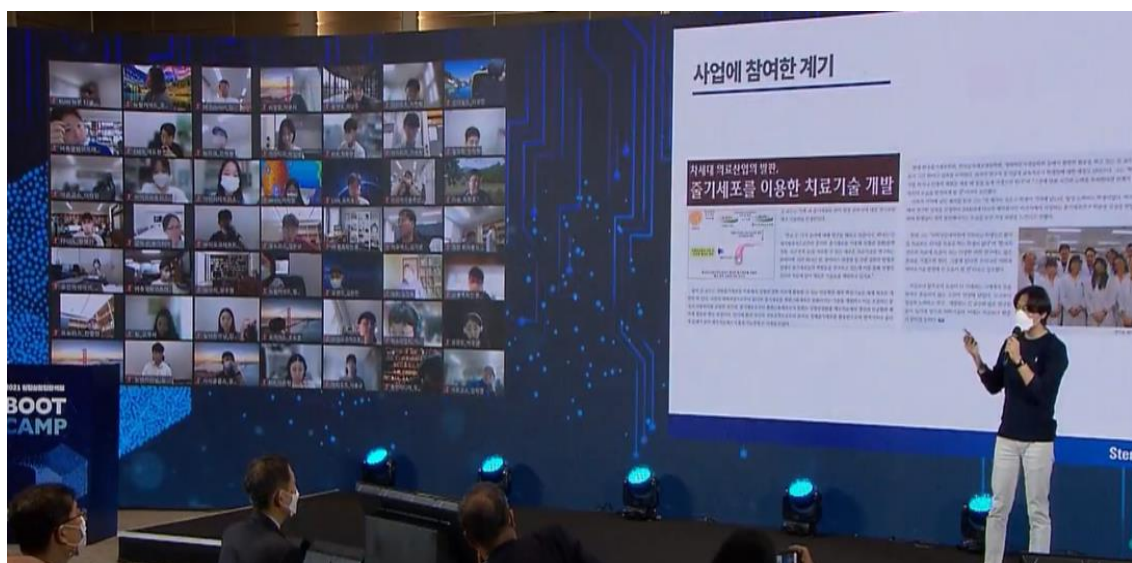
[Figure 4-33] Korean I-Corps - Regional Node Universities



Source: NRF (2022)

After becoming operational, the I-Corps program was operated under the current seven Node universities, training a total of 148 universities and 11 government research institutes' start-up preliminary teams (including duplicates), and ultimately establishing itself as a representative science and technology start-up education program of Korean universities.

[Figure 4-34] Lab to Start-up Boot Camp



Source: Korea Entrepreneurship Foundation, 2021.



In Korea, a total of 435 teams have completed the course since 2015, the year in which the Korean-style I-Corps project started, including 125 teams selected in 2021, and an additional 137 teams have been selected for the first time in 2022. Regarding the start-up rate, 186 (42.7%) of the 435 teams that completed the training have established start-up companies, which is close to the 50% rate of the NSF I-Corps in the United States.

Of those start-up companies, 122 start-up companies, about 65.5%, have survived, showing excellent performances of about 70% for the 3-year survival rate and 44.5% for the 5-year survival rate. This can be interpreted as a better performance than the 5-year survival rate of Korean start-ups, which is 31.2% (Statistics Office, 2020).

In terms of job creation, analysis has shown that 1,003 new jobs were created, with an average of 5.4 people being hired by each start-up company. In the case of instructors (Korea I-Corps Instructors), from 2017 to 2022, about eight I-Corps Instructors were nurtured every year under an agreement with George Washington University in the United States, and, as of 2022, a total of about forty-six I-Corps Instructors have the ability to lead a Korean technology start-up team. The figures shown in the table below are based on the findings of a report on a comparative study of the performance of the Korea I-Corps Program with that of the US I-Corps program from 2017 to 2020 (Shim et al., 2022).

[Table 4-18] Comparison of NSF I-Corps vs. Korea I-Corps program

|                                                                         | NSF I-Corps   | NIH I-Corps  | DoE I-Corps  | Korean I-Corps |
|-------------------------------------------------------------------------|---------------|--------------|--------------|----------------|
| Teams <sup>(a)</sup>                                                    | 1102          | 144          | 75           | 275            |
| Start-ups <sup>(b)</sup>                                                | 593           | 143          | 8            | 132            |
| Budgets <sup>(c)</sup>                                                  | \$133,444,206 | \$10,080,000 | \$10,525,000 | \$22,481,960   |
| VC Funds <sup>(d)</sup>                                                 | \$260,791,253 | \$61,533,568 | \$64,414,361 | \$36,124,606   |
| % of start-ups established <sup>(e)*</sup><br>(against teams)           | 53.8%         | 99.3%        | 10.7%        | 48.0%          |
| No. of start-ups by budgets <sup>(f)**</sup><br>(Per USD \$ 1 Millions) | 4.5           | 14.2         | 0.8          | 5.9            |
| VC Funds <sup>(g)***</sup>                                              | 2x            | 6.1x         | 6.1x         | 1.6x           |



|                                                | NSF I-Corps   | NIH I-Corps  | DoE I-Corps  | Korean I-Corps |
|------------------------------------------------|---------------|--------------|--------------|----------------|
| (against budgets)                              |               |              |              |                |
| Public Funds                                   | \$147,406,659 | \$60,033,568 | \$13,378,324 | \$2,119,740    |
| SBIR/STTR                                      | \$96,493,958  | \$26,599,791 | \$183,000    | -              |
| Other federal funding [non-SBIR/STTR]          | \$40,264,700  | \$33,258,777 | N/A          | -              |
| State funding                                  | \$10,648,001  | \$175,000    | \$0          | -              |
| Private Funds                                  | \$113,384,594 | \$1,500,000  | \$51,036,037 | \$34,004,867   |
| Venture capital                                | \$99,055,020  | \$130,000    | \$0          | -              |
| Angel or individual investors                  | \$9,038,330   | \$0          | N/A          | -              |
| Private industry, corporate funding, or others | \$5,291,244   | \$1,370,000  | \$47,002,037 | -              |

$$* \textcircled{e} = (\textcircled{b} / \textcircled{a}) \times 100$$

$$** \textcircled{f} = (\textcircled{b} / \textcircled{c}) \times 1 \text{ Million}$$

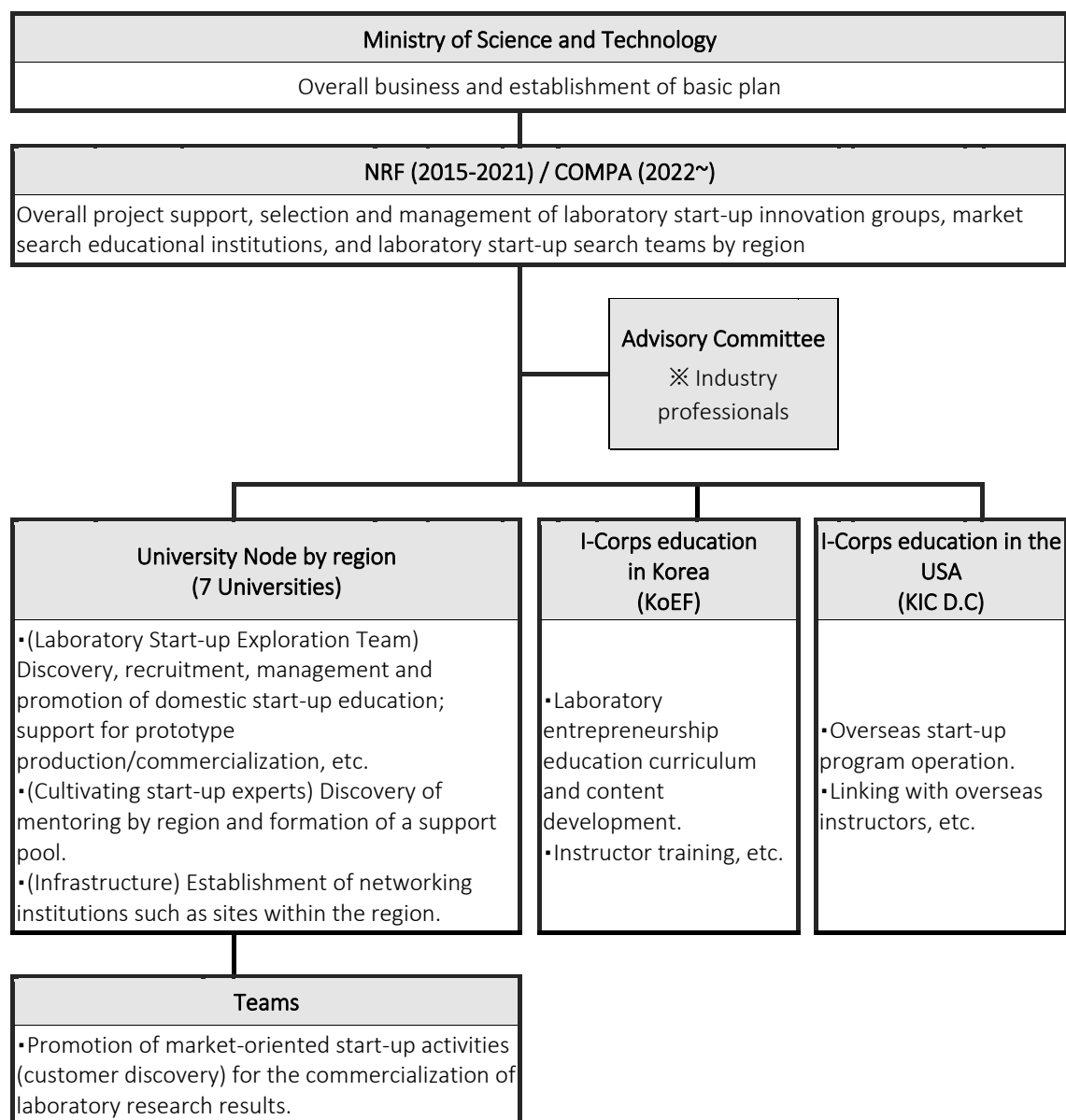
$$*** \textcircled{g} = \textcircled{d} / \textcircled{c}$$

Source: NRF (2022) "NSF I-Corps program vs. Korea I-Corps program (2017-2020)"

### 3.1.6 Structure of the Korea I-Corps program for R&D Start-ups

The government promotion department of the I-Corps project in Korea is the Ministry of Science, Technology and Information and Communication. Regarding the operation of domestic education programs, the Korea Youth Entrepreneurship Foundation (KoEF) is in charge of domestic education, and in the case of overseas education, KIC D.C operates overseas education courses in connection with George Washington University and UC Berkeley University in the United States.

[Figure 4-35] Structure of the Korea I-Corps program, NRF (2022)



The second phase of the Korean I-Corps project, which started in 2021, consists of two phases. From the first to the second phase, seven regional laboratory start-up innovation groups (Node University) and market search educational institutions were selected via a phase-based evaluation. The business is being promoted in a way.

[Table 4-19] Korea I-Corps program support plan by year

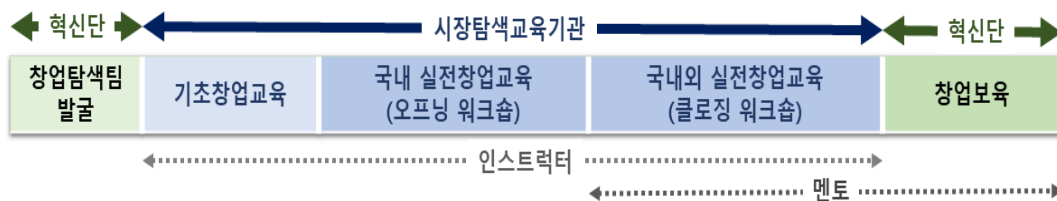
| Year   | Phase I              |                      | Phase II             |                      |                      |
|--------|----------------------|----------------------|----------------------|----------------------|----------------------|
|        | 1 <sup>st</sup> year | 2 <sup>nd</sup> year | 3 <sup>rd</sup> year | 4 <sup>th</sup> year | 5 <sup>th</sup> year |
| Period | 2021.03 -<br>2022.02 | 2022.03 -<br>2023.02 | 2023.03 -<br>2024.02 | 2024.03 -<br>2025.02 | 2025.03 -<br>2026.02 |

Source: NRF (2022) “Korea I-Corps program”

The roles of each major entity in operating the business are as follows.

- University’s Lab to Start-up support team: Start-up team recruitment and incubation.
- Instructor: Training teams with I-Corps methodology and “lessons learned” sessions.
- Mentor: Team coaching and running office hours.

[Figure 4-36] Korea I-Corps stakeholders and the role of each group, NRF (2022)



Source: Korea I-Corps program, NRF, 2022.

If we look at this program in detail, it is organized according to the following roles and responsibilities.

[Table 4-20] Korea I-Corps program support plan by year

|              | Group             | Role                            | R&R                                                                                                                                                                                                                                                                                                                                                        |
|--------------|-------------------|---------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Stakeholders | Node Universities | Start-up incubation and support | <ul style="list-style-type: none"> <li>• Discovery and nurturing of a start-up search team (intensive education, pivoting, etc.).</li> <li>• Establishment of a mentor pool for each region and support for mentoring.</li> <li>• Establishment of a site within the region and discovery of overseas market research educational institutions.</li> </ul> |

|               | Group      | Role                         | R&R                                                                                                                                                                                                                                                                                                |
|---------------|------------|------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|               | KoEF       | Education (Local)            | <ul style="list-style-type: none"> <li>Development of domestic entrepreneurship education curriculum.</li> <li>Domestic common start-up education, basic and practical start-up education for domestic start-up search teams.</li> <li>Discovery, nurturing and support of instructors.</li> </ul> |
|               | KIC D.C    | Education (Global)           | <ul style="list-style-type: none"> <li>Basic and practical start-up education for overseas start-up search teams.</li> <li>After overseas training, support for additional mentoring and incubation by overseas start-up experts.</li> <li>Links with international instructors.</li> </ul>        |
| Professionals | Mentor     | Scale-up (Pivoting)          | <ul style="list-style-type: none"> <li>Pivoting and supplementing the directions for start-up items.</li> <li>Assistance with discovering customers and sharing start-up know-how.</li> </ul>                                                                                                      |
|               | Instructor | I-Corps methodology training | <ul style="list-style-type: none"> <li>Customer Discovery Methodology Training.</li> <li>Training of start-up search teams (enhancing entrepreneurial mindset, etc.).</li> </ul>                                                                                                                   |

Source: NRF (2022) "Korea I-Corps stakeholders and roles of each group"

In addition, each base university, called the Node University, discovers and nurtures the technology start-up teams of other universities within the region, and conducts education and team discovery activities for preliminary technology start-up teams as an accelerated course through the operation of the mini-i-corps program.

[Table 4-21] Roles of Node and Site Universities

|              |           |                                                                                                                                                                                  |                                                                                                                                                                                                                                                                                                                     |
|--------------|-----------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <Structure > |           |                                                                                                                                                                                  |                                                                                                                                                                                                                                                                                                                     |
|              |           |                                                                                                                                                                                  |                                                                                                                                                                                                                                                                                                                     |
|              |           | Condition                                                                                                                                                                        | <ul style="list-style-type: none"> <li>Regional university as a business incubation agency</li> <li>Belongs to the region's Node University</li> </ul>                                                                                                                                                              |
|              |           | Major role                                                                                                                                                                       | <ul style="list-style-type: none"> <li>Recruitment of technology based start-ups for the region</li> <li>Start-up incubation program (around 5 months)</li> <li>Recruitment of technology based start-ups and introduction of the teams to the Node University</li> <li>Mini I-Corps program (2-5 weeks)</li> </ul> |
|              | Follow up | <ul style="list-style-type: none"> <li>Follow-up for excellent teams with other start-up programs</li> <li>Follow-up for excellent teams to enter the I-Corps program</li> </ul> |                                                                                                                                                                                                                                                                                                                     |

Source: NRF (2022) "Korea I-Corps program"

### 3.1.7 I-Corps program localization for the Korea region

The seven-week U.S. NSF I-Corps training program is largely composed of basic theory education, discovery of customers, inferring implications based on the results of customer discovery (lessons learned), and coaching with a dedicated team mentor (office hours). Upon completion of the 7-week course, the program is promoted in such a way that excellent teams are selected through an IR pitching-type start-up competition.

The Korean I-Corps education program mainly consists of a boot camp, a pre-session, a basic course, and a practical course conducted over a period of 2 months.

In the case of laboratory start-up programs in Korea, the I-Corps trademark cannot be used officially. This is because the original program, named I-Corps, is an educational program created by the taxpayers of the United States. Therefore, in Korea, the official name is a “public technology-based market-linked start-up search support project”, and in the case of the US I-Corps program, the name “Tech Frontier” is officially used. However, in this research report, the Korean program also uses the name I-Corps informally to help the reader to understand the start-up program.

[Table 4-22] Korea I-Corps Training Course

| Boot Camp<br>(half day)                           | Pre-session<br>(3 nights/4 days)                                   | Course work<br>(basic to main sessions,<br>3/7 weeks) |
|---------------------------------------------------|--------------------------------------------------------------------|-------------------------------------------------------|
| Orientation                                       | Introductory session for learning Lean Start-up and BMC (9 Blocks) | Introductory session (3 to 7 weeks)                   |
| Introduction of each I-Corps group                |                                                                    |                                                       |
| Assignment of instructors to each Node University | Value proposition                                                  | Team introduction with business item                  |
| Ice-breaking among teams                          |                                                                    |                                                       |
| Initial business item review                      | Customer segments                                                  | Learning from Best Practices                          |
| Case study                                        | Key partners                                                       | Value proposition (w/ case study)                     |
| Lecture (Entrepreneurship)                        | Customer relations and channels (Revenue streams)                  | Customer segments (w/case study)                      |
| Networking among teams                            | Key activities, pricing structure                                  | Customer discovery (w/case study)                     |

## 2022 K-Innovation Partnership Program with Azerbaijan

|  |                                   |                                                             |
|--|-----------------------------------|-------------------------------------------------------------|
|  | Practical session on building BMC | Interview plan (w/case study)                               |
|  | Case study                        | How to use BCB/LaunchPad                                    |
|  | Debriefing session                | Interview plan and design                                   |
|  | Customer discovery                | Case study                                                  |
|  | Interview plan                    | Customer discovery                                          |
|  |                                   | Special session: Entrepreneurship                           |
|  |                                   | Team pitching (10min), feedback (5min)                      |
|  |                                   | Customer discovery                                          |
|  |                                   | Team pitching (10min), feedback (5min)                      |
|  |                                   | Comparing BMCs (before/after customer discovery)            |
|  |                                   | Customer discovery (lessons learned)                        |
|  |                                   | Special session: Understanding Start-ups and Stakeholders   |
|  |                                   | Customer discovery (lessons learned)                        |
|  |                                   | Special session: Channels (B2C, B2B, G2B, O2O, M2E, others) |
|  |                                   | Customer discovery                                          |
|  |                                   | Team pitching (10min), feedback (5min)                      |
|  |                                   | Customer discovery                                          |
|  |                                   | Office hours                                                |
|  |                                   | Customer discovery                                          |
|  |                                   | Team pitching (10min), feedback (5min)                      |
|  |                                   | Office hours                                                |
|  |                                   | Customer discovery                                          |
|  |                                   | Team pitching (10min), feedback (5min)                      |
|  |                                   | Team pitching (video-based)                                 |
|  |                                   | Office hours                                                |
|  |                                   | IR Pitching                                                 |

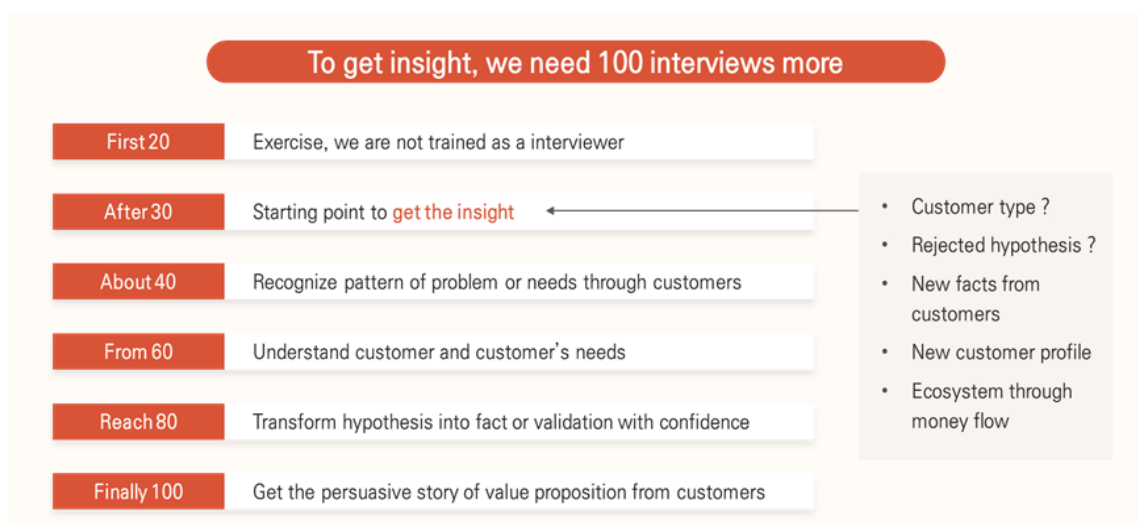
Source: NRF (2022) "Korea I-Corps program"

Furthermore, non-face-to-face education methods were introduced in addition to the existing face-to-face educational methods during the COVID-19 pandemic in 2020. improved through interviews.

The activity of the Korean-style I-Corps education program can be defined as customer discovery in the same way as in the United States. Customer discovery is the process of finding out whether a business hypothesis established by a technology-based start-up team is meaningful to the target customer segment in the real-world market through in-depth interviews with real people.

With the Korean I-Corps program, as in the US, 100 customer discovery interviews are held, i.e. 15 for the Basic Course, an additional 20 between the Basic Course and the Practical Course, and 65 for the Practical Course. These 100 interviews are a process of finding out whether the R&D and laboratory technology possessed by the technology start-up team is an urgent and important issue for segmented customers, and is usually referred to as “Product Market Fit” (PMF). In addition, approaches to solving the problems of potential target customers, with the focus on customers, are taught at the same time. This is a process of exploring the application fields of a laboratory technology as a solution to the problems of potential customers, and is commonly called “Problem Solution Fit” (PSF)

[Figure 4-37] Korea I-Corps customer discovery, KoEF (2021)



Source: Korea Entrepreneurship Foundation (2021)

The main content is the current Korean content, which includes detailed laboratory start-up education in the educational process, whereby students who have majored in engineering are first familiarized with the I-Corps model of hypothesis testing and the market exploration in-depth interview, etc., and then learn about the market validation process. It is necessary to operate through the flip-learning method as an after-sales education, which is a stage that can be accepted through the verification of business hypotheses, and some curriculum systems have been developed in a way that combines the advantages of online and offline methods, which were promoted in the past through policy research related to the end of 2021. Since 2022, it has been operated in five stages as follows:

[Table 4-23] Korea I-Corps and blended learning for 2022

| BootCamp<br>(half day)                                                                                                                                                                                                                                                                 | Pre-session<br>(2 nights/3days)                                                                                                                                                                                                                     | Basic course<br>(2 weeks)                                                                                                                                                                                                                                                                                                                                                                                                                              | Practical course<br>(3-4weeks)                                                                                                                                                                                                                                                                                                                                                        | Follow-up session<br>(1 week)                                                                                                                                                                                                            |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <ul style="list-style-type: none"> <li>• Orientation</li> <li>• Introduction of project participants.</li> <li>• Senior team(case) presentation.</li> <li>• (Commissioning ceremony, networking with the innovation group).</li> </ul>                                                 | <ul style="list-style-type: none"> <li>• <b>Offline workshop</b></li> <li>• VPC, PSF, PMF-centered training among BMCs.</li> <li>• Customer Discovery (CD) Methodology Training.</li> </ul>                                                         | <ul style="list-style-type: none"> <li>• Program and Team Introduction.</li> <li>• Opening Special Lecture</li> <li>• How to use BCB.</li> <li>• BMC VPC, PSF, PMF-focused lectures.</li> <li>• Introduction to Assignment Templates.</li> <li>• Interview planning guidance.</li> <li>• Lesson learned.</li> <li>• Office Hour.</li> </ul>                                                                                                            | <ul style="list-style-type: none"> <li>• Program and Team Introduction.</li> <li>• Opening Special Lecture.</li> <li>• How to use BCB.</li> <li>• BMC VPC, PSF, PMF-focused lectures.</li> <li>• Introduction to Assignment Templates.</li> <li>• Interview planning and networking methodology establishment guidance.</li> <li>• Lesson learned.</li> <li>• Office Hour.</li> </ul> | <ul style="list-style-type: none"> <li>• Use of standard textbook-centered online recorded video lectures and workbooks provided in the first year of the second phase of the project according to a blended learning method.</li> </ul> |
| <ul style="list-style-type: none"> <li>• Special Lecture on Entrepreneurship.</li> <li>• Item sharing between teams (non-face-to-face: metaverse, or face-to-face: presentation).</li> <li>• It is necessary to provide a platform for continuous networking between teams.</li> </ul> | <ul style="list-style-type: none"> <li>• Conduct offline or real-time lectures as much as possible.</li> <li>• Promote team understanding by providing a library of some of the existing online content (focused on essential content*).</li> </ul> | <ul style="list-style-type: none"> <li>• After basic training, only the overseas training team promotes a CD list writing workshop for 2 days and 1 night.</li> <li>• Quantitative presentation goal: <b>15 CDs</b> in total (1 week: 5, 2 week: 10).</li> <li>• (In the period of basic, practical transfer course: +20).</li> <li>• Use of workbooks as assignment templates.</li> <li>• Extra CDs before practical course: <b>20 CDs</b></li> </ul> | <ul style="list-style-type: none"> <li>• Same as previous course.</li> <li>• Divided into global and domestic teams.</li> <li>• Quantitative presentation goals.</li> <li>• <b>65 CDs</b> in total (100 performed in total).</li> <li>• Use of workbooks as assignment templates.</li> </ul>                                                                                          | <ul style="list-style-type: none"> <li>• Improvement of the system: the content of the workbook is presented as an assignment online and must be performed before moving on to the next content.</li> </ul>                              |
| May to June                                                                                                                                                                                                                                                                            | June                                                                                                                                                                                                                                                | July                                                                                                                                                                                                                                                                                                                                                                                                                                                   | July to August                                                                                                                                                                                                                                                                                                                                                                        | August                                                                                                                                                                                                                                   |

Source: Source: NRF (2022) "Korea I-Corps program"

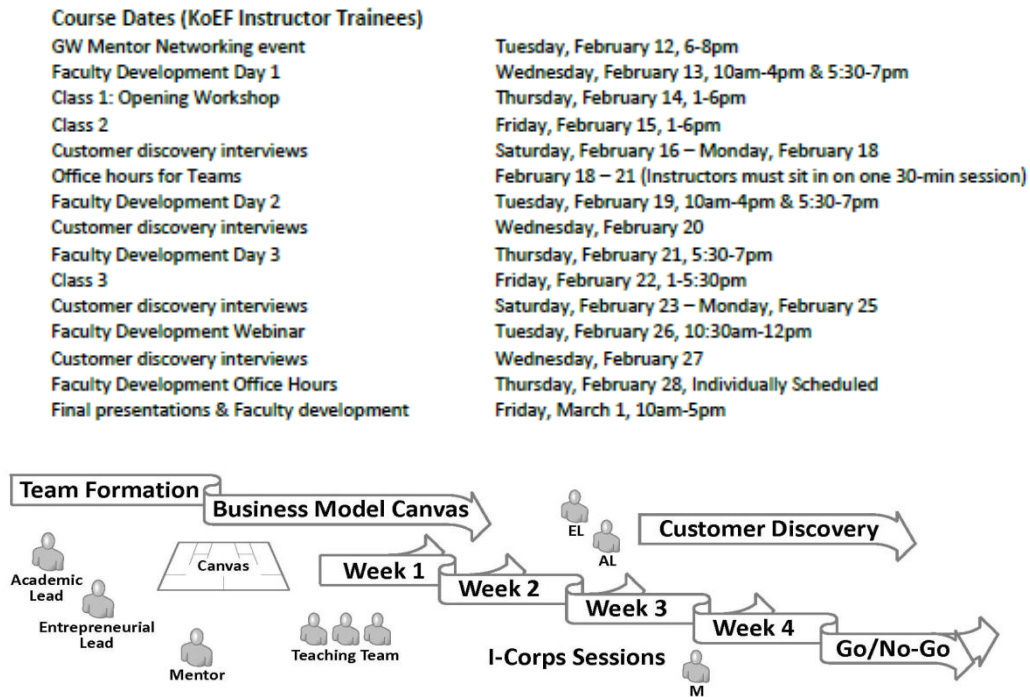


## **3.2 “Train the Trainer” start-up program for Azerbaijan**

### **3.2.1 Train the Trainer(TTT) program**

In the case of Korea, in order to internalize the Korean-style I-Corps program launched in 2015, Korean start-up experts have been dispatched to George Washington University, a representative NSF I-Corps Node University in the United States, for education since 2017. Instructors conduct interviews in Korean and English after reviewing documents in advance, but applicants must have at least three years of educational experience related to start-up and professional activities, such as consulting and mentoring in the field of start-ups. The evaluation committee is composed of experts in the field of public technology commercialization, and the interview evaluation is conducted based on the applicant's understanding of I-Corps. Through the Korean I-Corps program, approximately 6-8 instructors with outstanding educational capabilities have been selected each year since 2017, and a total of 46 will be produced by 2022. The selected Korean I-Corps instructors participate in the Pre-session and Basic Course in Korea immediately after being selected, thereby deepening their understanding of the basic I-Corps curriculum program and their basic experience in guiding a start-up team. Upon completing the education course in Korea, the participants attend the Instructors Course at George Washington University (US), in which the NSF I-Corps instructors participate, for about three weeks. At this time, the instructors on the U.S. curriculum conduct customer discovery, which the actual start-up team experiences, in two-cohort teams of four people. The main curriculum can be summarized as follows:

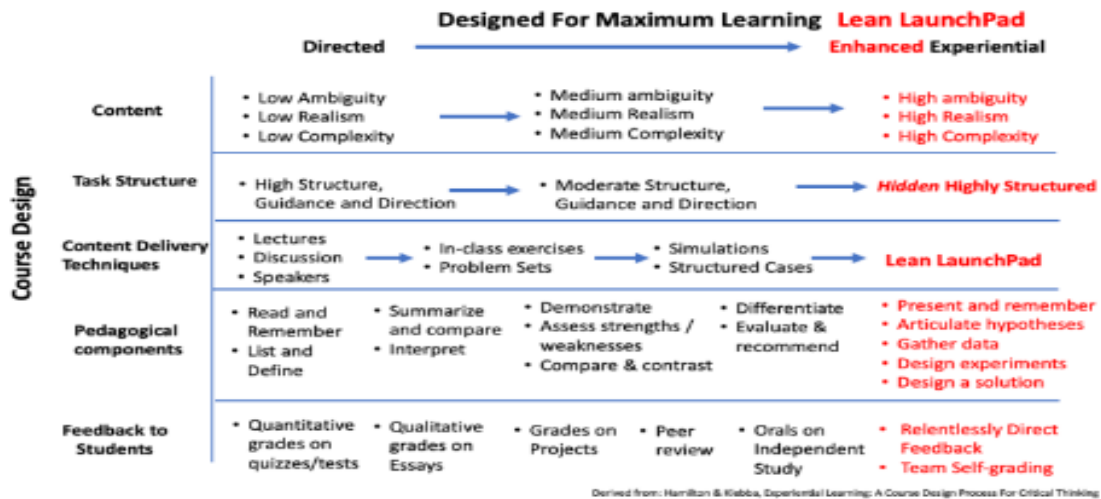
[Figure 4-38] TTT program, George Washington University, KIC D.C (2019)



Source: KIC D.C. (2019)

The core of the training process is customer discovery of the Lean Start-up method, which repeatedly tests business hypotheses in the market. In addition, various online-based tools are used in the curriculum, of which the most characteristic element consists in using a Cloud-based Web Solution called LaunchPad (<https://www.glidr.io/launchpad#accelerators>). Through this, feedback on the customer discovery results and points to be noted during the training process can be shared in real time during the training course run by the US instructor. In particular, LaunchPad provides video-recorded lectures (mostly on Lean Start-up methodologies, customer discovery, interview tips) by Professor Steve Blank of Stanford University, who designed the NSF I-Corps program. The lectures are linked to each Business Model 9 Block, and feedback on the results of customer discovery is provided simultaneously by multiple mentors. In addition, all the participants can communicate with each other via LaunchPad's collaborative features, which allow them to receive and/or send a messages.

[Figure 4-39] LaunchPad, GLODR (2022)



Source: GLODR (2022)

### 3.2.2 “Train the Trainer” (TTT) program syllabus for Korean Instructors

GW (George Washington University) Mentor Networking Event: A session based around personal exchanges and networking between Korean instructors, American instructors, and mentors.

#### GW Mentor Networking Event

Tuesday, February 12 from 6-8pm

Location: GW Science & Engineering Hall, Basement Level 1, 800 22nd St NW, Washington, DC 20052

| Time                   | Topic                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
|------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 2 hrs<br>6:00 – 8:00pm | <b>GW Mentor Networking Event</b><br>A regularly scheduled GW event for our students and mentors to network with each other.<br><b>Registration Link:</b> <a href="https://www.eventbrite.com/e/gw-nvc-mentor-mingle-tickets-53696379316">https://www.eventbrite.com/e/gw-nvc-mentor-mingle-tickets-53696379316</a><br>(Bring a photo ID with you to be let into the event)<br><br><i>-REQUIRED for KoEF instructor trainees</i><br><i>-OPTIONAL for non-KoEF instructor trainees</i> |

Faculty Development Day 1.: Training on the basic mind-set, lean start-up methodology, and Business Model Canvas (BMC) required of instructors.

## 2022 K-Innovation Partnership Program with Azerbaijan

### Faculty Development Day 1

Wednesday, February 13, 10am-4pm & 5:30-7pm

Location: Multiple, noted below

| Time    |                                                                | Topic                                                                                                                                                                                                                                                                                                                                                                                            |
|---------|----------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 6 hrs   | <b>10am-4pm</b><br>Room 201,<br>1922 F Street NW               | <b>Faculty Development Workshop</b><br>Instructor participants get classroom training on Lean Startup pedagogy.                                                                                                                                                                                                                                                                                  |
| 1.5 hrs | <b>5:30-7:00pm</b><br>Gelman Library Room 219,<br>2130 H St NW | <b>Business Model Canvas Workshop</b><br><b>Registration Link:</b> <a href="https://www.eventbrite.com/e/business-model-canvas-workshop-tickets-48873388621">https://www.eventbrite.com/e/business-model-canvas-workshop-tickets-48873388621</a><br>(Bring a photo ID with you to be let into the event)<br>-REQUIRED for KoEF instructor trainees<br>-OPTIONAL for non-KoEF instructor trainees |

Class 1, Course Opening Workshop: The key concepts of the BMC 9 blocks are taught in detail in the form of a lecture, and learners are encouraged to participate by taking part in quizzes from time to time.

### Class 1, Course Opening Workshop

Thursday, February 14, 1-6pm

Location: GW Science & Engineering Hall, Room B1270, 800 22nd St NW, Washington, DC 20052

| Time   |                        | Topic                                                                                                                                                                                                                                                                                     |
|--------|------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 15 min | <b>1-1:15pm</b>        | <b>Registration/Surveys</b>                                                                                                                                                                                                                                                               |
| 45 min | <b>1:15 – 2:00pm</b>   | <b>Introductions</b> <ul style="list-style-type: none"> <li>• Introduction to Lean Startup</li> <li>• Class Goals and Expectations</li> <li>• Class introductions</li> </ul>                                                                                                              |
| 30 min | <b>2-2:30pm</b>        | <b>Customer Segments</b> <ul style="list-style-type: none"> <li>• Customer Segments, Sub-Segments, Types</li> <li>• Workshop</li> </ul>                                                                                                                                                   |
| 60 min | <b>2:30pm – 3:30pm</b> | <b>Value Proposition Design</b> <ul style="list-style-type: none"> <li>• Introduction to Value Proposition Canvas</li> <li>• Workshop</li> </ul>                                                                                                                                          |
| 15 min | <b>3:30pm – 3:45pm</b> | <b>Break</b>                                                                                                                                                                                                                                                                              |
| 60 min | <b>3:45 – 4:45pm</b>   | <b>Hypothesis Development</b> <ul style="list-style-type: none"> <li>• Workshop</li> </ul>                                                                                                                                                                                                |
| 30 min | <b>4:45 - 5:15pm</b>   | <b>Assignments</b> <ul style="list-style-type: none"> <li>• LaunchPad Central (enter VP &amp; CS, associated hypotheses)</li> <li>• Hypothesis testing assignment</li> <li>• Interview lead generation (Identify at least 20 people you want to interview)</li> <li>• LinkedIn</li> </ul> |
| 45 min | <b>5:15-6:00pm</b>     | <b>Faculty and TTT Debrief</b>                                                                                                                                                                                                                                                            |

Class 2, Team Presentation, Workshop, Assignment: A team called Cohort is formed to give a team presentation on business items that will be used for customer discovery in advance for each team. In addition to the BMC 9 Blocks, through the topic of advanced topics, training on issues that the start-up team should consider in order to verify business hypotheses other than BMC is conducted in the form of a workshop. Each team is given a task to complete, and approximately ten customer discovery interviews are conducted during a given period (the tasks are usually presented at intervals of 3 days to 1 week).

### Class 2

Friday, February 15, 1-6pm

Location: Duques Hall, Room 151, 2201 G St NW, Washington, DC 20052

| Time    |               | Topic                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
|---------|---------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 45 min  | 1 – 1:45pm    | <b>Team Presentations</b> <ul style="list-style-type: none"> <li>Value Proposition &amp; Customer Segments Presentation</li> <li>Primary hypotheses to test</li> </ul> <b>Slide Template -</b><br>Slide 1 must include: <ul style="list-style-type: none"> <li>One Ecosystem Hypothesis to test</li> <li>Describe experiment/interviewees related to that hypothesis</li> </ul> Slide 2 must include: <ul style="list-style-type: none"> <li>One Customer Hypothesis to test</li> <li>Describe experiment/interviewees related to that hypothesis</li> </ul> |
| 45 min  | 1:45 – 2:30pm | <b>Best Practices for Customer Discovery</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
| 15 min  | 2:30 – 2:45pm | <b>Break</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
| 105 min | 2:45 – 4:30pm | <b>Workshop</b> <ul style="list-style-type: none"> <li>Competition Research</li> <li>Cold calling</li> <li>Interview &amp; email script</li> <li>Interviewing</li> <li>Interview scheduling</li> </ul>                                                                                                                                                                                                                                                                                                                                                       |
| 15 min  | 4:30 – 4:45pm | <b>AccelerateGW Grant Discussion</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
| 15 min  | 4:45 – 5:00pm | <b>Assignments</b> <ul style="list-style-type: none"> <li>Schedule and conduct 10 interviews before next class</li> <li>Presentation due to GW Box by 11:59PM before the next session (template in the session syllabus information)</li> </ul>                                                                                                                                                                                                                                                                                                              |
| 1 hr    | 5:00 – 6:00pm | <b>Faculty Debrief &amp; TTT</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |

Customer discovery interview: Customer discovery is not about selling the skills the Cohort team possesses; rather, it is about designing an interview that can help people to identify customer pain or customer gain points in their minds.

## 2022 K-Innovation Partnership Program with Azerbaijan

### Customer Discovery Interviews

Between Saturday, February 16 and Monday, February 18, 2019

Location: Determined by interviews

| Time                                      | Topic                                                                                                                                      |
|-------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------|
| Individually Scheduled throughout the day | <b>Customer Discovery Interviews</b><br><i>-REQUIRED for KoEF instructor trainees</i><br><i>-OPTIONAL for non-KoEF instructor trainees</i> |

Office hours: After customer discovery, teams receive coaching through Mentor and online meeting tools such as Zoom or WebEx. Helping to become customer discovery thereof.

### Office Hours (REQUIRED) - Teams

Between Monday, February 18 and Friday, February 21, 2019

Location: Virtual (webex) or conference call

| Time   | Topic                                                                                                                                                |
|--------|------------------------------------------------------------------------------------------------------------------------------------------------------|
| 30 min | <b>Office Hours</b><br><i>-REQUIRED for KoEF instructor trainees to join <u>one</u> session</i><br><i>-OPTIONAL for non-KoEF instructor trainees</i> |

Class 3, Team Presentation, What's Next: The above training courses are repeated each week. When participants move on to the second half, they proceed with the process of pushing to complete the Product Market Fit or Problem Solution Fit of their assignment.

### Class 3

Friday, February 22, 1-5:30pm

Location: Duques Hall, Room 151, 2201 G St NW, Washington, DC 20052

| Time    | Topic                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
|---------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 120 min | <b>1-3:00pm</b><br><b>Team Presentations</b><br>12-minutes team presentations using the following template:<br><br><b>Cover Slide</b> , including: <ul style="list-style-type: none"> <li>▪ (Team member names, Team name, Team number)</li> <li>▪ Business thesis (Who is your customer, what problem do they have, what is your solution?)</li> <li>▪ Total # of customer interviews completed</li> </ul> <b>Slide 2 – n:</b> What have you learned through your customer discovery? Focus on customer segments and value propositions. Include answers to the following points<br><i>(slides updated from the assignments in Classes 1 &amp; 2):</i> <ul style="list-style-type: none"> <li>▪ Hypothesis: Here's What We Thought</li> <li>▪ -Experiments: So, Here's What We Did</li> <li>▪ -Results: So, Here's What We Found</li> <li>▪ -Iterate: So, Here's What We Are Going to Do Next</li> </ul> <b>Final Slide:</b> Current business model canvas with any changes marked |
| 15 min  | <b>3:00 – 3:15pm</b><br><b>Break</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
| 45 min  | <b>3:15 – 4:00pm</b><br><b>What's Next</b> <ul style="list-style-type: none"> <li>• GW Office of Innovation and Entrepreneurship</li> <li>• I-Corps National Teams Program, I-Corps Go, SBIR</li> <li>• Venture Creation Support</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |

Final presentation and completion ceremony.

#### Class 4, Final Presentations (individually scheduled)

Friday, March 1, 10am-5pm

Location: Old Main, Rooms 201 & 208 (TTT); Rooms 403C & 403 B (Teams), 1922 F Street NW, Washington, DC 20052

#### For Instructor Trainees:

| Time  |                  | Topic                                                                                                                                                                                                                                                                                                 |
|-------|------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 6 Hrs | 10:00am - 4:00pm | <b>Team Final Presentations &amp; TTT Final Presentations</b><br>Closing lessons learned presentations for customer discovery interviews and Train-the-Trainer program.<br><br><i>-FULL DAY REQUIRED for KoEF instructor trainees</i><br><i>-AT LEAST 3 REQUIRED for non-KoEF instructor trainees</i> |
| 1 hr  | 4:00 – 5:00pm    | <b>Final TTT Closing &amp; Debrief</b>                                                                                                                                                                                                                                                                |

Most of the textbooks required during the TTT program are provided as slides in the PDF format, but the following textbooks are recommended as reference materials for TTT trainees.

#### Textbooks

##### **Value Proposition Design**

Alexander Osterwalder, Yves Pigneur, Greg Pernarda & Alan Smith

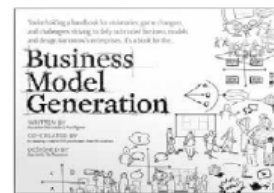
- A free download of the first chapter of the book is available at <https://strategyzer.com/value-proposition-design? ga=1.152090042.2059273423.1389715841>



##### **Talking to Humans**

Giff Constable

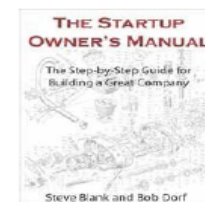
- A free download of the book is available at <http://www.talkingtohumans.com/>



##### **Business Model Generation – pages 14-51**

Alexander Osterwalder & Yves Pigneur

- A free download of the first chapter of the book is available at <http://businessmodelgeneration.com/book>



##### **The Startup Owner's Manual – pages 195-199**

Steve Blank & Bob Dorf

##### **12 Tips for Early Customer Development Interviews**

Giff Constable

- A free download is available at: <http://giffconstable.com/2010/07/12-tips-for-early-customer-development-interviews>

### **3.3 Implications and Policy Suggestions**

#### **3.3.1 Suggestion for enhancing Azerbaijan's start-up eco-systems**

As was the case with Korea's industrialization growth process, to create an innovative ecosystem in Azerbaijan, it will be necessary to strengthen its IP R&D capabilities in order to develop public technologies that fit the strategic direction of the country. In addition, after securing intellectual properties and technologies that are suited to the regional and industrial characteristics of Azerbaijan, the implementation of I-Corps-type technology start-up education to build an innovative ecosystem comprising start-ups will also be helpful in promoting socio-structural change.

#### **3.3.2 Industry fields for Korean-Azerbaijan start-up cooperation through ODA projects**

Boss ([www.boss.az](http://www.boss.az)) is the largest local job posting site. According to Vadim, it is not easy for foreign nationals to start a local business given the poor business environment in Azerbaijan, including the closed economic structure and the language barrier. However, in view of the recent development of the tourism industry and the good image of Korea, it is considered that tourism and trade could be promising fields for start-ups in the short term.

- ① Tourism: As the number of Korean tourists to the three countries of the Caucasus increases, the demand for local travel services will increase accordingly.
- ② Trading industry: Demand for Korean products is steady due to the good level of brand recognition of Korea, and the import business of Azerbaijani fruits and vegetables is also promising. However, as the poor business environment delays the introduction of advanced management know-how and services, certain common business items in other countries could become new and profitable business items here. If sufficient market research and preparation is undertaken, local start-ups could also be promising.
- ③ Recently opened household goods stores such as Miniso are operating with great success, and a substitute driver is operating at the suggestion of an international student who studied in Korea.



- ④ Recently, Hezar Dalga TV has begun to air Korean TV dramas during prime time (9pm), and exports of related Hallyu products are expected to increase.

### 3.3.3 “Train the trainer” start-up program for Azerbaijan via the K-Innovation program

I-Corps is a globally-known US approach to commercializing research outputs. Unfortunately, there are no similar educational programs in Azerbaijan as yet, so instructors and mentors work with start-ups with an ad-hoc based approach. Fortunately, “Azerbaijan 2026: Social and Economic Development Strategy of the Republic of Azerbaijan in 2022-2026” covers certain areas of research commercialization which resemble particular aspects of NSF I-Corps (though they are not necessarily similar programs).

For example, the third pillar of the Strategy aims to support the commercialization of scientific research and R&D projects and increase such expenditures up to 0.1 percent of Azerbaijan’s GDP, as well as to increase the added-value of high- and medium technology industries up to 15% in terms of aggregate national added-value by 2026. Furthermore, the Strategy will form the Science-Industry 4.0 Cluster and increase the level of participation by 15 percent per year .

Therefore, as in the case of Korea, promoting innovative technology-based start-ups will be useful in promoting changes in the industrial structure and innovation as a major policy.

In this regard, it is recommended as a good short-term direction to transfer the I-Corps TTT (Train the Trainer) course and the related experience currently being accumulated in Korea to train professors at major universities in Azerbaijan and instructors at major start-up institutions. In this way, it should be possible to nurture instructors who have learned the I-Corps type of educational methodology in a relatively short period of time. It is expected that Azerbaijan's own I-Corps program education and operation system will be prepared by directly operating the training course in the form of co-teaching with a few Korean instructors for the locally discovered technology start-up team in the initial stages.

In the event that we co-develop Azerbaijan’s own instructors who can teach the I-Corps

methodology, it is expected that it will be possible to operate a program that can autonomously advance technology start-ups and idea-based start-ups even after the end of the ODA program.

## **3.4 Expected Impacts**

### **3.4.1 Impact on building up the R&D based start-up training and ecosystem**

The I-Corps project in Korea can be evaluated as having contributed to an important opportunity to transform the undergraduate-centred idea-based start-up culture into laboratory start-ups centred on the major hub Node universities in many universities. In terms of its specific and quantitative performance, Korea's I-Corps program is a representative technology start-up program. Among the laboratory start-up search teams that have fostered some 325 teams nationwide, 161 start-ups (49.5%) and 723 new jobs have been created (4.6 people per company on average); the 3-year survival rate of companies started in the first phase is about 70%, the 5-year survival rate is 53.8%, and the average survival rate as of the current year is 75.8%. Analysed to have created a distinct innovative performance. As such, the project has been evaluated as having achieved a high degree of innovative technologies.

This means that the technology efforts of university labs developed with the government's R&D budget have shifted from the output of patents, theses, and prototypes, which can be called “research achievements”, to actual laboratory start-ups, which can be defined as “outcomes”. As such, it can be considered that a foothold has been established. If this program is transferred to Azerbaijan, it could present an important opportunity to establish a technology start-up ecosystem.

### 3.4.2 Fostering R&D based start-up instructors and a mentoring pool

As mentioned above, in the case of Korea, instructors trained in teaching and learning methods through major NSF I-Corps node universities in the United States are intensively nurtured from the beginning of the project in order to spread the technology start-up ecosystem through I-Corps. As such, instructors who have learned the I-Corps-type educational methodology can be nurtured in a relatively short period of time. It is expected that Azerbaijan's own I-Corps program education and operation system will be prepared by directly operating the training course in the form of co-teaching with Korean instructors for the locally discovered technology start-up team. In the event that we co-develop Azerbaijan's own instructors to teach the I-Corps methodology, it is expected that it will be possible to operate a program that can autonomously advance technology start-ups and idea-based start-ups even after the ODA program ends.

## 3.5 Conclusion

As was the case with Korea's industrialization process, to create an innovative ecosystem in Azerbaijan, it is necessary to strengthen its IP R&D capabilities in order to develop public technologies that are suited to the strategic direction of the country. Belongs with the intellectual properties and technology capabilities suited to the regional and industrial characteristics of Azerbaijan, I-Corps-type technology start-up education that can build an innovative ecosystem using start-ups will be helpful in promoting social and structural change. As with the I-Corps project in Korea, it could be evaluated as contributing to an important opportunity to transform the undergraduate-centred idea-based start-up culture into laboratory start-ups centred on the major hub Node University in many universities.

This means that the technology work done by university labs with the government's R&D budget will shift from the output of patents, theses, and prototypes, which can be defined as "research achievements", to actual laboratory start-ups, which can be defined as "outcomes". In this way it can be considered that a foothold has been established. If this program is transferred to Azerbaijan, it is believed that it will provide an important opportunity to establish a technology start-up ecosystem. To achieve this type of goal and

take the opportunity to promote social innovation, in the case of Korea, instructors trained in teaching and learning methods through major NSF I-Corps node universities in the United States are intensively nurtured from the beginning of the project in order to spread the technology start-up ecosystem through I-Corps.

Therefore, as in the case of Korea, promoting innovative technology-based start-ups will be useful in promoting changes in the industrial structure and innovation as a policy. In this regard, it is recommended as a good short-term to transfer the I-Corps TTT (Train the Trainer) course and the experience currently being accumulated in Korea to train professors at major universities in Azerbaijan and instructors at major start-up institutions. In this way, instructors who have learned the I-Corps-type educational methodology can be nurtured in a relatively short period of time. It is expected that Azerbaijan's own I-Corps program education and operation system will be prepared by directly operating the training course in the form of co-teaching with a few Korean instructors for the locally discovered technology start-up team in the initial stages. Instructors who have learned the I-Corps-type educational methodology can be nurtured in a relatively short period of time. It is expected that Azerbaijan's own I-Corps program education and operation system will be prepared by directly operating the training course in the form of co-teaching with the Korean instructors for the locally discovered technology start-up team.

In conclusion, if Azerbaijan develops its own instructors who can teach the I-Corps methodology, it is expected that it will be possible to operate a program that can autonomously advance technology start-ups and idea start-ups even after the ODA program ends.



# CHAPTER 5

Strategic Plans for  
Improving the  
Technology  
Commercialization  
System and Developing  
the HTP Program



## Chapter 5. Strategic Plans for Improving the Technology Commercialization System and Developing the HTP Program

### 1. Strategic Plan for Improving the Technology Commercialization System

#### 1.1 Technology commercialization system and its management

A large number of decisions (laws, decrees, orders, acts, etc.) have been adopted with the aim of establishing the knowledge economy and the innovation ecosystem in the Republic of Azerbaijan. However, the legislative mechanism and resources for supporting the organization and development of industrially relevant scientific research, as well as the related science commercialization processes, are still weak. In this respect, it should be noted that the ecosystem in Azerbaijan is currently in the initial stage of formation.

In the Republic of Azerbaijan, there are diverse organizational infrastructures (higher education institutions, research institutes, technological parks, agro-parks, business and industrial enterprises, business incubation and acceleration centers, laboratories, etc.) that directly play an important role in the innovation ecosystem. However, it should be noted that the commercialization of science and technology transfer mechanisms among the operational mechanisms of the infrastructural facilities do not fully meet current and prospective expectations. However, the activities carried out for the purpose of developing intellectual capital are becoming more and more relevant, thus allowing us to make optimistic predictions about the innovation ecosystem of Azerbaijan, including the commercialization of science.

All sides of (participants in the innovation ecosystem in Azerbaijan (scientific-pedagogical, management, market, educators and learners, etc.) are sympathetic to innovative

approaches. In our opinion, the main problems here are the lack of mass to show stability and durability, as well as tolerance to the processes of formation of innovations as value. The occurrence of such situations in the formative stage of the ecosystem can be seen in the global experience as a whole.

The Republic of Korea provides support (both financial and intellectual) for the formation and development of an ecosystem to promote technology transfer and the commercialization of science in Azerbaijan. These forms of support are based on the advanced experience of Korea, and are implemented after taking into account the needs of the local market in Azerbaijan. We can note that the support provided through the ODA, KOICA and other programs can benefit the economy of Azerbaijan, especially the innovation ecosystem.

## **1.2 Strengths and Weaknesses of Technology Commercialization**

### **1.2.1 Strengths**

- In the last decade, especially in the last few years, the relevance of the issue has increased in the private sector and at the state level.
- The population is interested in investing in education and professional development. In particular, families prefer their children to receive a high level of education.
- Adolescents and young people make up about 35% of the total population.
- The new generation shows great interest in smart technologies.
- Even with small achievements in technology transfer and the commercialization of science, there are opportunities to gain fame.
- The country has currency and resources.
- The population has a high interest in business entrepreneurship.



### **1.2.2 Weaknesses**

- Technology transfer and commercialization of science are still not priorities in the education system.
- Free market opportunities are weak.
- Macroeconomic indicators are dominated by a resource-based economy.
- Public condemnation has a negative effect on entrepreneurship.
- The legislation and financial resources supporting technology transfer and science commercialization processes are weak.
- The domestic market is very small.
- Local manufacturing capabilities remain weak in terms of global competition.
- Progressive approaches to the management of research and higher education institutions are still insufficient.

## **1.3 Proposals on the Effective Management of the Technology Transfer and Science Commercialization System**

### **1.3.1 Education and professional development**

- Promote the involvement of persons with decision-making and executive functions in training and ensure their professional development accordingly.
- Conduct organizational activities (conferences, panels, forums, business trips, etc.) to support the application of progressive management mechanisms in scientific research and educational institutions.
- Enable scientific researchers and educators to teach and apply TT and TC skills in order to ensure academic independence at this level.
- Modernize the educational curricula.

### **1.3.2 Mentoring services, resource assessment and business networking**

- Organize competitions and provide mentoring support to teams with promising projects.
- Valuate existing resources and shape success stories with progressive approaches.
- Promote science-education-business cooperation.
- Support global networking.

### **1.3.3 Innovation capital and its formation**

- Organize activities to support the formation of venture capital.
- Provide access to venture capital.
- Organize incubation and acceleration services, including improvement of this field in Azerbaijan based on Korean experience.

### **1.3.4 State policy and ecosystem support mechanisms**

- Organize lobbying activities in order to achieve effective organization of state policy in the direction of TT and TC.
- Establish long-term relations with international markets at the state level.
- Share STEPI's experiences in Azerbaijan.

### **1.3.5 Formation and active operation of the digital platform**

- Support the strengthening and popularization of the Technology Transfer and Science Commercialization Platform (ttcp.az).
- Strengthen ties with global markets by benefiting from Korean experience.

### **1.3.6 Increasing the relevance of intellectual property rights**

- Organize activities to increase the relevance of intellectual property rights.
- Modernize domestic patenting procedures.
- Develop relationships with global patent centers.

### **1.3.7 Formation of free market opportunities and provision of global market access**

- Raise awareness of the benefits that free market opportunities can bring to the economy.
- Support private sector entrepreneurship.
- Establish the “Azerbaijan-Korea Innovation Home” and carry out its activities in order to increase access to global markets.

## 2. Strategic Plan for Developing the Technology Commercialization Program

### 2.1 Current Situation

In the current era, high technologies form the basis for development of the world economy. High technology refers to modern technologies used in unique, highly scientific technological processes, devices and materials, which together determine the international level of fundamental and applied research. Biotechnologies, laser techniques and technologies, aviation, space research, nuclear energy, genetic engineering, which allow high technologies to produce competitive products that are widely used in the priority directions of the country's socio-economic development, including the improvement of defence capabilities, from the modern scientific-technical and intellectual potential, computer and micro-opto-electronic technologies, chemistry, telecommunications and information technologies are related. The intensification of production and the utilization of the results of scientific and technical progress are accelerating the rate of innovation of products and services. As mentioned above, the role of high technologies is decisive in determining the development of the new economy, its stability and its sustainability.

A techno park is a structural (territorial) or legal entity equipped with advanced technologies that comprehensively creates and adopts new technologies, produces modern equipment, devices and facilities and promotes their commercial application. Techno parks are one of the main tools for the formation of a new type of economy based on innovation and knowledge.

Applying the successful results of international experience in the creation, organization and management of innovative technology parks to the structures of the technology parks being formed in Azerbaijan is considered one of the most urgent issues of the current period. For Azerbaijan, the main developmental directions consist in developing innovative entrepreneurship, creating a favourable environment for the development of new types of activities and products, strengthening the measures for transferring and adopting advanced technologies, and creating techno parks and innovation zones for the development and application of science-intensive products and technologies.

Azerbaijan's National Academy of Sciences of High Technologies was established by Decree No. 2425, dated November 8, 2016, of the President of the Republic of Azerbaijan Mr. Ilham Aliyev. The main goals of creating the HTP are to increase state support for the sustainable development of the economy, raise competitiveness, expand the innovation and high-tech areas based on modern scientific and technological achievements, conduct scientific research and create modern complexes for the development of new technologies.

## 2.2 Main Objectives and Targets of the HTP Program

The main goal of this strategic plan is to create favourable conditions for residents of Azerbaijan who are engaged in scientific research and practical-constructive work and commercialize their results in industry, services, and other fields through business and technological incubators, funds, technological transfer and scientific research centers. The plan aims to develop the HTP, which ensures their creation and support.

In order to achieve the main goal, the goals of the HTP according to the development concept are as follows:

- Development of competitive science-intensive production based on innovative and high technologies; creation of fertile conditions for service provision, and support for entrepreneurship in this field.
- Promotion of innovation, scientific research, experimental work, inventions and discoveries; commercialization of technologies and the application thereof to production.
- Reduction of economic dependence on oil and gas by exploiting scientific and innovative potential.
- Formation of an environment conducive to scientific and technologically based ideas that can lead to the development of innovative products.
- Commercialization and promotion of potentially marketable inventions, start-ups and business ideas developed by residents.

- Raising the competitiveness of science-intensive, high-tech products in the international arena.
- Support for the development of the ecosystem in Azerbaijan.
- Development of innovative small businesses and creation of new innovation enterprises.
- Establishment of a favorable environment for the development of small and medium enterprises in the field of high technology.
- Attraction of science-intensive high technologies and technological products to the market.
- Transfer of high technology; increasing the specific weight of science-intensive products in the economy.
- Establishment of commercial relations between high-technology developers and investors.
- Involvement of investments in technological projects at different stages of development.
- Increase of employment among the working population.
- Formation of relationships with higher education institutions and scientific-research institutes, including support for the implementation of joint projects.
- Promotion of the scientific research and creative process with the aim of obtaining innovative scientific knowledge, conducting research, acquiring expertise, and applying/promoting the results of innovative scientific research.
- Scientific innovation activities, creation of progressive innovations and intellectual property objects (scientific results, inventions, discoveries, systems, original technologies, brands, trade and trademarks, etc.) that are formed on the basis of scientific research; improvement of the management system and increased competitiveness, and support and encouragement of activities aimed at their implementation.

## 2.3 Necessary Conditions for the Development of the HTP Program

In order to develop competitive science-efficient production based on innovative and high technologies, and to create fertile conditions for providing services and support entrepreneurship in this field, the following measures are necessary:

- Obtain greater concessions in the legislation;
- Improve the territory, traffic, management, and social infrastructure of the HTP;
- Construct start-up and business incubators, as well as modern offices for residents' activities.

To promote innovation, scientific research, experimental work, inventions and discoveries, and the commercialization and application of technology to HTP production, it is necessary to create the following conditions:

- Create favorable conditions for the development of innovative projects by large companies in order to promote rapid industrial assimilation of the results of scientific research and experimental projects, inventions and discoveries.

To increase the competitiveness of science-intensive high-tech products in the international arena, the following measures are necessary:

- Gain access to international markets;
- Study the knowledge-based, innovative development experience and problems of developed countries, and apply that knowledge to the implementation of the HTP development strategy.

To support the development of the ecosystem in Azerbaijan, the following measures are necessary:

- Exchange scientific and technological innovative experience with different countries in order to develop the ecosystem in Azerbaijan.

- Create a favorable environment for the development of small and medium enterprises in the field of high technology.
- Support the creation and development of small and medium-sized innovative enterprises that are residents of the techno park.

To attract investments to technological projects in various stages of development, the following measures are necessary:

- Establish a favorable business investment environment.
- Coordinate the implementation of innovative projects, develop test-expertise and prototypes, and commercialize science activities with international cooperation and grant programs, including the involvement of potential donors.
- Create favorable conditions for financing innovation projects, attract and promote investments, and carry out the relevant organizational work.

To support the transfer of high technology from its developers to various sectors of industry, and to develop partnerships between the public and private sectors of the economy, the following measures are necessary:

- Establish a mechanism for issuing state orders concerning products produced in the HTP.
- Promote cooperation with higher education institutions and scientific research institutes in order to support the implementation of joint projects, as well as fostering relations with higher education institutions and scientific research institutes.

In order to promote the scientific research and creative process with the aim of obtaining innovative scientific knowledge, conducting research, acquiring expertise, and applying/promoting the results of innovative scientific research, the following measures are necessary:

- Support the implementation of scientific and technical projects and programs aimed at creating science-metric technologies and competitive products.



- Support scientific research works of a high level and great practical importance, and which can contribute to the economic development of the country and lead to important progress through the application of new technologies.

The non-existence of the conditions listed above will either make it impossible to achieve the main goals and objectives, or lead to poor or ineffective results.

## 2.4 Necessary Steps for the Development of the HTP Program

- ① Coordinate university-industry cooperation; create opportunities for the application of science, including intellectual property issues; provide mentoring services to technological entrepreneurs, and ensure that research conducted in academia and universities receives more support from financial sources; protect and commercialize knowledge obtained in academia; establish a Technology Transfer Office staffed by professionals to support academic entrepreneurship.
- ② Create favorable conditions for attracting capital and other investments for venture financing.
- ③ Take necessary measures to optimize tax rates, mandatory state social insurance fees and customs duties.
- ④ Take necessary measures to simplify the certification and licensing procedure:

Simplify the certification and licensing procedure of high-tech products and equipment produced by HTP residents, as well as products and equipment necessary for their production and imported from abroad, as this will have a positive effect on the dynamics of production acceleration in this area.

- ⑤ Offer favorable rental prices to residents (by HTTP):

In order to minimize the expenses of residents engaged in entrepreneurial activities and stimulate the production of science-efficient technological products, favorable rental prices will be offered by the HTP.

- ⑥ Apply extraterritorial residency in the HTP:

Operating the HTP according to the extraterritorial principle should be one of the main priority directions. Legal and natural persons to be registered in the HTP may use all relevant benefits regardless of the location of their place of business.

- ⑦ Develop business incubators and start-up projects, and create model enterprises: The state budget, donations from donor organizations, as well as funds from local and foreign investors should be used to improve the operation of the existing business incubator in the HTP, and to create new incubators. The development of appropriate financing mechanisms should be considered to ensure access to initial financial resources for other start-ups and those who wish to establish a new business participating in the HTP business incubator. Start-up projects can serve as a platform for entrepreneurs who want to start a business or execute their new ideas. Steps related to ensuring the participation of the private sector, donor organizations and various financial institutions in the financing of start-up projects should be determined and various projects should be reviewed.
- ⑧ Create innovation funds - legal entities with non-commercial status that carry out targeted financing of scientific research in the form of grants: Innovation funds make it possible to increase the quality and efficiency of scientific research in the HTP, and to provide targeted financing for applied and innovative scientific programs and projects. The development of innovative infrastructure and the movement of local innovations to international markets will be ensured; and the transfer, exchange and development of innovative technologies, as well as their commercialization and application, will be realized.
- ⑨ Create and develop an information portfolio on innovative research works in higher education institutions and scientific research institutes, as well as the involvement of scientific applied works in the HTP.
- ⑩ Improve the territory of the HTP; restore serviceable buildings and facilities for adaptation to new projects; demolish non-serviceable buildings and facilities and construct new buildings in their place; construct new roads and restore existing ones. To improve the social welfare of residents' employees and families, create

a complex of residential, social, healthcare, educational, cultural, catering, sports, entertainment and other facilities within the HTP.

- ⑪ Sale of products produced in the HTP on the basis of state orders: This results in the improvement and maintenance of the infrastructure of the HTP, including the development of start-ups, business incubation, and the technology transfer and commercialization processes in accordance with its mission, along with internal sources, innovation or venture funds, budget funds allocated by the state, including direct subsidies provided by the state.

## **2.5 Concept of Incubation and the Technology Transfer Centre of the HTP**

### **2.5.1 Main purposes of the concept**

- To assist in the commercialization of scientific research activities carried out in scientific institutions.
- To create economic, social and cultural values as a result of this process.
- To establish cooperation between universities and private sector organizations.
- To increase existing cooperation in order to produce the information and technology needed by industry.
- To coordinate technology transfer and the development of concrete results between universities and industry.

Reestablishment of the center is expected to contribute to the following areas: research, the development of and support for research and development projects, the registration and protection of intellectual and industrial property rights, marketing, commercialization, support for venture capital, the establishment of an incubation center, business management, and consulting and training services.

## 2.5.2 Vision

- To bring research innovations to the market with an effective technology transfer strategy, develop high value-added projects between universities and industry, actively pursue academic entrepreneurship research, and expand internationalization activities.
- To create a world-renowned, effective, enterprising, advanced and sustainable technology transfer office with a rich technology portfolio, and to realize successful commercialization activities and strong collaboration between universities and industry.

## 2.5.3 Responsibilities

- ① Develop and implement marketing and commercialization strategies necessary for transferring the portfolio of intellectual and industrial rights to industry, including patents, utility models, and industrial designs belonging to the Academy;
- ② Conduct preliminary research on the patentability and commercialization of reported inventions.
- ③ Manage the relationships, negotiations and contractual processes with potential buyers and investors for the licensing or transfer of intellectual and industrial property rights with commercialization potential.
- ④ Expand the national and international cooperation network of the Incubation and Technology Transfer Center of ANAS HTP and organize events aimed at increasing cooperation, potential, and promotion.
- ⑤ Represent the Incubation and Technology Transfer Center of the HTP of ANAS in front of companies, investors, government agencies and other scientific and educational institutions in the country and overseas.
- ⑥ Carry out project writing and project management training.

## **2.6 Planned Work**

### **2.6.1 Awareness, advocacy, information and training services**

- Orient researchers in accordance with the needs of private sector organizations, promote researchers and infrastructure resources for private sector representatives.
- Inform researchers and private sector representatives on research and development, innovation and intellectual property rights, and raise awareness of/provide training on these topics.

### **2.6.2 Services to take advantage of support programs**

- Provide project development, implementation and monitoring services to national and international financial institutions
- Inform private sector representatives and researchers about research and development and innovation financing, facilitate their access to funds, and direct them to appropriate funds

### **2.6.3 Project development / management services (university-industry cooperation activities)**

- Advise the private sector and researchers on access to information, technology and collaboration opportunities; provide them with guidance, and mediate national/international partnership agreements.
- Present project results.
- Create a project team for multi-partner projects, including the implementation and coordination of activities.

#### **2.6.4 Intellectual and industrial rights management and licensing services**

- Identify intellectual property rights assets, provide advice and services on compliance with the procedures for the legal protection of assets.
- Manage the commercialization processes by transferring the rights arising from intellectual and industrial property rights to other structures in the form of licensing, in particular to industry.

#### **2.6.5 Corporate and entrepreneurial services**

- Provide support services such as legal operations, business development and marketing to help establish spin-off companies.
- Provide fixed capital, incubation and business management services to mediate the creation of spin-off companies.

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# CHAPTER 6

## Conclusion









## Chapter 6. Conclusion

### 1. Conclusion

The 2022 K-Innovation Partnership Program aims to strengthen the operation and potential competitiveness of the HTP through the Technology Commercialization Knowledge Lab (TCK-Lab) and training program, and to support the formation and improvement of an innovative economic ecosystem in Azerbaijan.

Firstly, TCK-Lab has provided training for the members of the beneficiary group of Azerbaijan through seven online lectures on the commercialization of technologies, intellectual properties, start-up and incubation, and university-industry cooperation, which were presented in the format of a "knowledge laboratory" by Korean experts.

The 3-day training workshop was carried out online due to the COVID-19 situation. In the workshop, Korean experts from universities, PRIs, and enterprises delivered lectures to twenty Azerbaijan experts with the aim of equipping them with more practical knowledge. In particular, the Korean experts shared their experience in areas requiring improvement in Azerbaijan, such as industry-academia-research cooperation platforms and technology tech valuation.

The “dissemination forum” of the 2022 project was held with the support of Azerbaijan's Ministry of Science and Education. This forum introduced STEPI and the K-Innovation Partnership Program, and served to inform technology commercialization stakeholders about the importance of the HTP's role in improving Azerbaijan's technology commercialization system and creating an ecosystem.

Lastly, theories on, and Korea's experience of, technology commercialization governance, intellectual property management, and start-up programs contributed decisively to establishing strategic plans for improving the technology commercialization system in Azerbaijan and the HTP program development plans.

Concerning the activities of the project for the year 2022, the following expectations have been defined:

- Capacity building of the stakeholders in the Azerbaijan innovation ecosystem.
- Contribution to the formation of human capital in Azerbaijan.
- Expansion of Azerbaijan-Korea scientific, technology and innovation (STI) cooperation.
- Improvement of the operation program of the HTP.
- Further improvement of the process of technology commercialization should be observed with the transformation of knowledge into economic value.
- Formation of a new entrepreneurial environment based on Korea's experience of start-up and business incubation

We hope that the training and other methods of skills development carried out by well-known experts of the Republic of Korea and local consultants will have a significant and meaningful impact on the development of representatives of the High Technology Park and other organizations represented in the beneficiary group.

## 2. Follow-up Activity Plan for the 2023 Project

### 2.1 Determination of Needs and Resources

- a) Preparation of the draft version of the project.
- b) Discussion with and consent of STEPI.
- c) Discussion and consent of the obligations of STEPI.
- d) Identification and consent of local partners for the project.

### 2.2 Organization of Institutional Development of TTCP

- a) Preparation of the TTCP development plan for 2023-2024.
- b) Organization of local and international partnerships.
- c) Organization of events aimed at popularizing services in TTCP and ensuring the functionality of online services.
- d) Organization of face-to-face events; organization of 3 panels, 2 forums and 1 conference with the participation of representatives from TTCP service areas.
- e) Participation in affiliate programs of a partnership/marketing nature.
- f) Preparation of the TTCP development plan for the next 3-year period, and organization of partnership with potential research, and assessment of possible resources along with partnerships.

### 2.3 Organization of Trainer Training

- a) Determination of motivational tools of the participants involved in trainer training; issuance of trainer certificates.
- b) Preparation of the rules on the issuance of certificates.
- c) Preparation of a “participant certificate” as per the regulations.

- d) Preparation of a “certificate of successful completion” as per the regulations.
- e) Issuance of a reference letter based on the successful results of participation in trainer training.
- f) Organization of mass training sessions in the presentation of successful attendees; involvement of successful attendees in TTCP activities; determination of the possibilities of involving successful participants in the work of the "Korea-Azerbaijan Innovation House" to be established in the future, and announcement thereof to the participants.
- g) Decision on and announcement of appropriate rewards or gift(s) for high-performing attendees.
- h) Preparation of training directions and syllabuses.
- i) Development of the general implementation plan of the exercises; organization of training.
- j) Reproduction and distribution of training materials to participants.
- k) Evaluation of training results; presentation of certificates and other (predetermined) gifts to attendees.

## **2.4 Organization of the Start-up Competition**

### **1) Organizational work of the competition**

- Formation of the organizing committee and organization of its work.
- Formation of the jury and organization of its work.

### **2) Preparation of the competition regulations, including rules and conditions**

- Preparation of the rules and conditions.
- Definition of the criteria.
- Preparation of the registration form.
- Development of the assessment methodologies and methods.

- Determination of the motivational tools (e.g. seed investment for competition winners, incubation services and other prizes).

### **3) Announcement of competition and organization/registration of projects**

- Organization of a press conference on the launch of the competition.
- Organization of the online registration of projects on the platform.
- Holding of information sessions (at least 5 face-to-face info sessions, at least 3 online info sessions).

### **4) Organization of preliminary assessment of projects and selection of semi-finalists**

### **5) Organization of the development program for semi-finalists**

- Determining the needs of semi-finalist projects.
- Organization of training and mentoring services according to trainees' needs.
- Training and mentoring on the presentation of trainees of trainer training.
- Mentorship on the presentation of Korean experts.

### **6) Organization of the semi-final stage**

- Organization of the presentations of the Semi-finalists before the jury.
- Determination of the Finalists.

### **7) Organization of a Demo Day within the final stage**

- Finalists' commitment to the Demo Day.
- Organization of the final program.
- Presentation of awards to the winners (at the Conference).

## **2.5 Organization of the Conference on "The Current Situation of TT and TC in Azerbaijan and the Republic of Korea's Support for its formation"**

- a) Opening event (delegation from Korea, high-ranking officials from the relevant state institutions of Azerbaijan (at least two ministers) and other representatives, beneficiaries, scientists, investors, etc.).
- b) Announcement of the founding of the Korea-Azerbaijan Innovation House.

- c) Exhibition (presentation of the projects of all semi-finalists in the Start-up Competition).
- d) Demo Day for the presentations of the finalists in the Start-up Competition.
- e) Announcement and presentation of awards to the winners of the Start-up Competition.

## **2.6 Organization of the Incubation Program**

- a) Organization of incubation services for the winners of the Start-up Competition.
- b) Organization of the winners of the Start-up Competition to benefit from Korea's experience.
- c) Organization of mentoring services for other projects.

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